Phy-101 Important Mcq's
For Final Term !!
Solve By Vu-Topper RM!!



Question No:1 **(Marks:1)** **Vu-Topper RM**

The image produced by a convex mirror of an erect object in front of the mirror is always:

Virtual, erect, and larger than the object

Virtual, erect, and smaller than the object

Real, erect, and larger than the object

Real, erect, and smaller than the object

Question No:2 **(Marks:1)** **Vu-Topper RM**

In some movies, you sometimes see an actor looking in a mirror and you can see his face in the mirror. During the filming of this scene, what does the actor see in the mirror?

His face

Your face

The movie camera (Correct)

The director's face

Question No:3 **(Marks:1)** **Vu-Topper RM**

A femtometer is:

10-9 m

10-12 m

10-15 m

10-18 m

Question No:4 **(Marks:1)** **Vu-Topper RM**

The "strength" of a lens is measured in _____.

Meters

Diopters

~~**Lense meter (Correct)**~~

Provatans

Question No:5 **(Marks:1)** **Vu-Topper RM**

In constructing a thermometer it is NECESSARY to use a substance that:

Undergoes some change when heated or cooled

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Question No:6 **(Marks:1)** **Vu-Topper RM**

When an object (iron/tungsten) is heated, it emits light. It emits infrared (IR) light when it is heated at a temperature _____.

Below 800 degree C

Between 800 degree C to 1600 degree C

Between 1600 degree C to 2500 degree C

above 2500 degree C

Question No:7 **(Marks:1)** **Vu-Topper RM**

The reason there are two slits, rather than one, in a Young's experiment is:

to create a path length difference

to increase the intensity

one slit is for frequency, the other for wavelength

one slit is for Electric fields , the other is for Magnetic fields

Question No:8 **(Marks:1)** **Vu-Topper RM**

An electron is moving at 0.6c.If we calculate its kinetic energy using (1/2) mv2, we get a result that is:

Just right

Just half enough

Twice the correct value

About 28% too low (Correct)

Question No:9 **(Marks:1)** **Vu-Topper RM**

During a slow adiabatic expansion of a gas:

The pressure remains constant

Energy is added as heat

Work is done on the gas

No energy enters or leaves as heat

Question No:10 **(Marks:1)** **Vu-Topper RM**

The theoretical upper limit for the frequency of electromagnetic waves is:

just slightly greater than that of red light

there is no upper limit

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Question No:11 **(Marks:1)** **Vu-Topper RM**

One revolution is the same as:

1 rad

57 rad

π/2 rad

π rad

2π rad

Question No:12 **(Marks:1)** **Vu-Topper RM**

For a body to be in equilibrium under the combined action of several forces:

All the forces must be applied at the same point

all the forces must be applied at the same point

all of the forces form pairs of equal and opposite forces

any two of these forces must be balanced by a third force

Question No:13 **(Marks:1)** **Vu-Topper RM**

Capacitors C1 and C2 are connected in series. The equivalent capacitance is

given by

C1C2/(C1 + C2)

(C1 + C2)/C1C2

1/(C1 + C2)

C1/C2

Question No:14 **(Marks:1)** **Vu-Topper RM**

If the potential difference across a resistor is doubled:

only the current is doubled

only the current is halved

only the resistance is doubled

only the resistance is halved

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Question No:15 (Marks:1) Vu-Topper RM

By using only two resistors, R1 and R2, a student is able to obtain resistances of 3Ω, 4Ω, 12Ω, and 16 Ω. The values of R1 and R2 (in ohms) are:

3, 4

2, 12

3, 16

4, 12

Question No:16 (Marks:1) Vu-Topper RM

Faraday's law states that an induced emf is proportional to:

the rate of change of the electric field

the rate of change of the magnetic flux

the rate of change of the electric flux

the rate of change of the magnetic field

Question No:17 (Marks:1) Vu-Topper RM

A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

1000V

500V

250V

100V

Question No:18 (Marks:1) Vu-Topper RM

Which of the following electromagnetic radiations has photons with the greatest energy?

blue light

yellow light

x rays

radio waves

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Question No:19 (Marks:1) Vu-Topper RM

We desire to make an LC circuit that oscillates at 100 Hz using an inductance of
2.5H. We also need a capacitance of:
1 F
1mF
1 μF
100 μF

Question No:20 (Marks:1) Vu-Topper RM

The wavelength of red light is 700 nm. Its frequency is _____.
4.30 * 104 Hertz
4.30 * 103 Hertz 4.29 10^14 SEE PGE NO.92
4.30 * 105 Hertz
4.30 * 102 Hertz

Question No:21 (Marks:1) Vu-Topper RM

Which of the following statements is NOT TRUE about electromagnetic
waves?
Electromagnetic waves satisfy the Maswell's Equation.
Electromagnetic waves can not travel through space.
The receptions of electromagnetic waves require an antenna.
The electromagnetic radiation from a burning candle is unpolarized.

Question No:22 (Marks:1) Vu-Topper RM

Radio waves and light waves are
Longitudinal waves
Transverse waves
Electromagnetic and transverse both

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Question No:23 **(Marks:1)** **Vu-Topper RM**

An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

a change in entropy

a decrease of internal energy

an increase of pressure

a change in temperature

Question No:24 **(Marks:1)** **Vu-Topper RM**

The Stern-Gerlach experiment makes use of:

a strong uniform magnetic field

a strong non-uniform magnetic field

a strong uniform electric field

a strong non-uniform electric field

Question No:25 **(Marks:1)** **Vu-Topper RM**

A net torque applied to a rigid object always tends to produce:

linear acceleration

rotational equilibrium

angular acceleration

rotational inertia

Question No:26 **(Marks:1)** **Vu-Topper RM**

As the wavelength of a wave in a uniform medium increases, its speed will _____.

Decrease

Increase

Remain the same

None of these

Question No:27 **(Marks:1)** **Vu-Topper RM**

The current in a car headlamp is 2.0 A. The headlamp is switched on for 3.0 minutes. How much charge passes through the headlamp?

360c

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Question No:28 **(Marks:1)** **Vu-Topper RM**

A gas which strictly obeys the gas laws under all conditions of temperature and pressure is called:

Perfect gases

Question No:29 **(Marks:1)** **Vu-Topper RM**

In RL series circuit phase angle is given by:

Tan-1(wl/R)

Question No:30 **(Marks:1)** **Vu-Topper RM**

Which of the following have the greatest wavelength?

Radio waves

Question No:31 **(Marks:1)** **Vu-Topper RM**

When a charge is projected perpendicular to a uniform magnetic field, its path is:

A circular path

Question No:32 **(Marks:1)** **Vu-Topper RM**

When two objects come to common temperature, the body is said to be in:

Thermal equilibrium

Question No:33 **(Marks:1)** **Vu-Topper RM**

Which method(s) of heat transfer can take place in vacuum?

Electromagnetic waves RADIATION

Question No:34 **(Marks:1)** **Vu-Topper RM**

A step-down transformer is used to:

Decrease the voltage

Question No:35 **(Marks:1)** **Vu-Topper RM**

In an adiabatic process:

The work done by the environment on the system equals the change in internal energy.

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Question No:36 **(Marks:1)** **Vu-Topper RM**

The focal length of a spherical mirror is N times its radius of curvature where N is:

F=R/2

Question No:37 **(Marks:1)** **Vu-Topper RM**

Capacitive reactance Xc is:

1/2 π fc

Question No:38 **(Marks:1)** **Vu-Topper RM**

Force per unit charge is:

Electric field.

Question No:39 **(Marks:1)** **Vu-Topper RM**

If the kinetic energy of a free particle is much less than its rest energy then its kinetic energy is proportional to:

The square of the magnitude of its momentum

Question No:40 **(Marks:1)** **Vu-Topper RM**

Which physical quantity is produced by a calculation where a charge is multiplied by a potential difference (p.d)?

Energy

Question No:41 **(Marks:1)** **Vu-Topper RM**

Work done on a charged particle moving in uniform magnetic field is:

Zero

Question No:42 **(Marks:1)** **Vu-Topper RM**

Which of the following is equivalent to 1 V?

1 Joule/1 coulomb = 1 J/C

Question No:43 **(Marks:1)** **Vu-Topper RM**

If Io is the peak value of an AC supply, then its rms value is given as Irms:?phy101]

IO/square root2

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Question No:44 **(Marks:1)** **Vu-Topper RM**

Heat has the same units as:

Work

Question No:45 **(Marks:1)** **Vu-Topper RM**

Which of the following types of electromagnetic radiation at the vacuum?

All of these travel at the same speed

Question No:46 **(Marks:1)** **Vu-Topper RM**

Inductive reactance XL of an inductor is pi

2(pi)Fl

Question No:47 **(Marks:1)** **Vu-Topper RM**

The sum of electric and magnetic force is called

Lorentz force

Question No:48 **(Marks:1)** **Vu-Topper RM**

Radio waves and light waves are _____.

Electromagnetic and transverse

Question No:49 **(Marks:1)** **Vu-Topper RM**

Kirchhoff's 2nd rule is the manifestation of the law of conservation of

Energy

Question No:50 **(Marks:1)** **Vu-Topper RM**

The total energy in an circuit is 5.0 * 10-6 J. if L=25mH the maximum current is:

20mA

Question No:51 **(Marks:1)** **Vu-Topper RM**

The potential difference between the ends of a conductor is 12V. How much electrical energy is converted to other forms of energy in the conductor when 100C of charge flows through it?

1200J

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Question No:52 **(Marks:1)** **Vu-Topper RM**

The Focal length of convex lens is:

Positive

Question No:53 **(Marks:1)** **Vu-Topper RM**

The AC circuit in which current and voltage are in phase the power factor is

1 0.5

Question No:54 **(Marks:1)** **Vu-Topper RM**

The wavelength of red light is 700 nm. Its frequency is _____.

4.29*10^14 hertz

Question No:55 **(Marks:1)** **Vu-Topper RM**

The known laws of electromagnetism before James Clerk Maxwell were following

EXCEPT

Faraday's Law deborglie law

Question No:56 **(Marks:1)** **Vu-Topper RM**

When some dielectric is inserted between the plates of a capacitor, then capacitance;

Increased

Question No:57 **(Marks:1)** **Vu-Topper RM**

Which of the following is NOT the characteristics of electromagnetic waves?

They can travel with different speeds in vacuum depending on0.4m

Question No:58 **(Marks:1)** **Vu-Topper RM**

The quantity of deltaV/deltar is called

Potential gradient

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Question No:59 (Marks:1) Vu-Topper RM

Which electromagnetic radiation transmits the highest photon energy?

Gamma rays

Question No:60 (Marks:1) Vu-Topper RM

The reaction of an particle at 50 Hz is 10.0 its reactance at 100Hz becomes

5.0

Question No:61 (Marks:1) Vu-Topper RM

Work done on a charged particle moving in uniform magnetic field is

Zero

Question No:62 (Marks:1) Vu-Topper RM

Select the correct statement

Gamma Rays have higher frequency than infrared waves

Question No:63 (Marks:1) Vu-Topper RM

Potentiometer can be used as

Potential divider

Question No:64 (Marks:1) Vu-Topper RM

All of the followings are the different types of mental images; people generally experience them while close their eyes Except:

Solving an algebra equations

Question No:65 (Marks:1) Vu-Topper RM

The nation that motivational state of the organism is understood and explained in terms of positive and negative environmental stimuli BEST connects to;

Incentive approaches

Question No:66 (Marks:1) Vu-Topper RM

If one wants to increase the capacity of short-term memory, through which process more items can be held?

Chunking

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Question No:67 **(Marks:1)** **Vu-Topper RM**

If potential difference across two plates of a capacitor is doubled, then

Four times

Question No:68 **(Marks:1)** **Vu-Topper RM**

How could the unit of potential difference the volt also be written?

J/C

Question No:69 **(Marks:1)** **Vu-Topper RM**

If magnetic field is doubled then magnetic energy density becomes

Four times

Question No:70 **(Marks:1)** **Vu-Topper RM**

The primary of a 3.1 step-up transformer is connected to a source and the secondary this situation is P. If R is connected directly to the source it will dissipate a power of:

P/9

Question No:71 **(Marks:1)** **Vu-Topper RM**

A particle having 2e charge falls through a potential difference of 5.V energy acquired by it is

10 eV

Question No:72 **(Marks:1)** **Vu-Topper RM**

A generator supplies 100V to the primary call of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is

1000V

Question No:73 **(Marks:1)** **Vu-Topper RM**

The combined effect of resistance and reactance is known as

Impedance

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Question No:74 **(Marks:1)** **Vu-Topper RM**

A transparent refracting medium bounded by the two curved surfaces called

Prism LENS

Question No:75 **(Marks:1)** **Vu-Topper RM**

A student rubs a rod held in his hand. Which action causes the rod to gain a large electrostatic charge

Rubbing a polythene rod with a woolen duster

Question No:76 **(Marks:1)** **Vu-Topper RM**

A charge of 30 C flows through an electric appliance in 2.0 minutes. What is the average current in the appliance?

0.25A

Question No:77 **(Marks:1)** **Vu-Topper RM**

Total flux through a closed surface depends on

Charge and -N CHARGE ENCLOSED

Question No:78 **(Marks:1)** **Vu-Topper RM**

Electromagnetic waves emitted from antenna are

Transverse

Question No:79 **(Marks:1)** **Vu-Topper RM**

Two parallel straight wires carrying current in same direction will.

Attract each other

Question No:80 **(Marks:1)** **Vu-Topper RM**

Charge carriers in electrolysis are:

Positive and negative ions

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Question No:81 **(Marks:1)** **Vu-Topper RM**

A detector is used to detect the speed at a particular wave if the frequency of an electromagnetic signal is 400 MHz, the speed define the detector is

300000 km/s

Question No:82 **(Marks:1)** **Vu-Topper RM**

The product of resistance and capacitance is called

Time

Question No:83 **(Marks:1)** **Vu-Topper RM**

Greenhouse gases absorb ____ radiation

Infrared

Question No:84 **(Marks:1)** **Vu-Topper RM**

Electromagnetic waves do not transport

Charge

Question No:85 **(Marks:1)** **Vu-Topper RM**

Electromagnetic and transverse both

Longitudinal waves

Transverse waves

Electromagnetic and transverse both

Electromagnetic and longitudinal both

Question No:86 **(Marks:1)** **Vu-Topper RM**

An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

A. 200 Hz

B. 300 Hz

C. 400 Hz

D. 600 Hz

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Question No:87 **(Marks:1)** **Vu-Topper RM**

Room temperature is about 20 degrees on the:

Kelvin scale

Celsius scale

Fahrenheit scale

Absolute scale

Question No:88 **(Marks:1)** **Vu-Topper RM**

A particle with zero mass and energy E carries momentum:

Ec

Ec2

vEc

E/c

Question No:89 **(Marks:1)** **Vu-Topper RM**

J.J.Thompson's measurement of e/m for electrons provides evidence of the:

Wave nature of matter

Particle nature of matter

Wave nature of radiation

Particle nature of radiation

Question No:90 **(Marks:1)** **Vu-Topper RM**

During a slow adiabatic expansion of a gas:

The pressure remains constant

Energy is added as heat

Work is done on the gas

No energy enters or leaves as heat

Question No:91 **(Marks:1)** **Vu-Topper RM**

A force of 5000N is applied outwardly to each end of a 5.0-m long rod with a radius of 34.0 cm and a Young's modulus of 125 × 108 N/m2. The elongation of the rod is:

0.55mm

1.42mm

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Question No:92 **(Marks:1)** **Vu-Topper RM**

An object attached to one end of a spring makes 20 complete oscillations in 10 s. Its period is: Select correct option:

2Hz

0.5 s

0.5Hz

2 s

Question No:93 **(Marks:1)** **Vu-Topper RM**

back and forth in a leftward and rightward direction. This type of wave is known as a

Mechanical

Electromagnetic

Transverse

Longitudinal

Question No:94 **(Marks:1)** **Vu-Topper RM**

The flow rate of blood through the average human aorta is about 9.0 ? 101 cm3/s. If the aorta has a radius of 1.0 cm, what is the velocity of the blood flow?

14 cm/s

32 cm/s

37 cm/s

29 cm/s

Question No:95 **(Marks:1)** **Vu-Topper RM**

Metal pipes, used to carry water, sometimes burst in the winter because:

Metal contractst more han water

Outside of the pipe contracts more than the inside

Metal becomes brittle when cold

Water expands when it freezes

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Question No:96 **(Marks:1)** **Vu-Topper RM**

The reason there are two slits, rather than one, in a young's experiment is:

To create a path length difference

Question No:97 **(Marks:1)** **Vu-Topper RM**

Which of the following types of electromagnetic radiation travels at the greatest speed in vaccum?

(Radio, gamma, x)rays .

Question No:98 **(Marks:1)** **Vu-Topper RM**

Electric energy is measured :

Watt

Question No:99 **(Marks:1)** **Vu-Topper RM**

Of the following which might NOT vanish over one cycle of a cyclic process?

The work done by the substance

Question No:100 **(Marks:1)** **Vu-Topper RM**

The most refined form of matter is:

Light

Question No:101 **(Marks:1)** **Vu-Topper RM**

How could the unit of potential difference, the volt, also be written?

J/C

Question No:102 **(Marks:1)** **Vu-Topper RM**

In Compton scattering from stationary particles the maximum change in wavelength can be made smaller by using.

More massive particles

Question No:103 **(Marks:1)** **Vu-Topper RM**

Hotness or coldness of an object is represented in terms of:

Temperature

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Question No:104 **(Marks:1)** **Vu-Topper RM**

The unit of thermal conductivity might be:

Cal/(cm*s*C?)

Question No:105 **(Marks:1)** **Vu-Topper RM**

Which physical quantity is produced by a calculation where a charge is multiplied by a potential difference (p.d)?

Energy

Question No:106 **(Marks:1)** **Vu-Topper RM**

A transparent refracting medium bounded by the two curved surfaces called:

Prism LENS

Question No:107 **(Marks:1)** **Vu-Topper RM**

It is more di? Cult to measure the coe? Cient of volume expansion of a liquid than that of a solid because?

The containing evessel aslo xpands

Question No:108 **(Marks:1)** **Vu-Topper RM**

The product of resistance and capacitance is called

Time

Question No:109 **(Marks:1)** **Vu-Topper RM**

The potential difference between the ends of a conductor is 12 V. How much electrical energy is converted to other forms of energy in the conductor when 100C of charge flows through it?

1200

Question No:110 **(Marks:1)** **Vu-Topper RM**

When we accelerate the charge, which types of waves are produced?

Electromagnetic waves

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Question No:111 (Marks:1) Vu-Topper RM

Due to relative motion of observer and the frames of reference of events, time always:

Statics itself

Question No:112 (Marks:1) Vu-Topper RM

The allowed energy states of atoms or molecules can be described by a set of _____.

Quantum numbers

Question No:113 (Marks:1) Vu-Topper RM

Probability values are assigned on a scale of

0 to 1

Question No:114 (Marks:1) Vu-Topper RM

The magnitude of the momentum of a particle can never exceed:

E/c, where E is its total energy

Question No:115 (Marks:1) Vu-Topper RM

The ratio of the size of image to the size of object.

Angular magnification

Question No:116 (Marks:1) Vu-Topper RM

When white light passes through a prism, it splits into _____ colors.

7

Question No:117 (Marks:1) Vu-Topper RM

Electromagnetic waves do not transport:

Charge

Question No:118 (Marks:1) Vu-Topper RM

A heat of transformation of a substance is:

The energy per unit mass absorbed as heat during a phase transformation

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Question No:119 (Marks:1) Vu-Topper RM

Which of the following electromagnetic radiations has photons with the greatest energy?

X rays

Question No:120 (Marks:1) Vu-Topper RM

When two objects come to common temperature, the body is said to be in:

Thermal equilibirium

Question No:121 (Marks:1) Vu-Topper RM

If electric and gravitational force on an electron in a uniform electric field balance each other, then the intensity of electric field will be:

Mg/q

Question No:122 (Marks:1) Vu-Topper RM

Electromagnetic waves do not transport:

Charge

Question No:123 (Marks:1) Vu-Topper RM

During a slow adiabatic expansion of a gas No energy enters or leaves as heat

52.

Question No:124 (Marks:1) Vu-Topper RM

The electron states that constitute a single shell for an atom all have:

The same value of n

Question No:125 (Marks:1) Vu-Topper RM

An isothermal process for an ideal gas is represented on a p-V diagram by:

A portion of a hyperbola

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Question No:126 **(Marks:1)** **Vu-Topper RM**

The concept of an electric field lines is introduced by:

Faraday

Question No:127 **(Marks:1)** **Vu-Topper RM**

The current in a car headlamp is 2.0A. The headlamp is switched on for 30 minutes. How much charge passes through the headlamp?

360 C

Question No:128 **(Marks:1)** **Vu-Topper RM**

In some movies, you sometimes see an actor looking in a mirror and you can see his face in the mirror. During the filming of this scene, what does the actor see in the mirror?

The movie camera

Question No:129 **(Marks:1)** **Vu-Topper RM**

The pauli exculusion principle is obeyed by:

All particles with spin quantum numbers of ½

Question No:130 **(Marks:1)** **Vu-Topper RM**

The focal length of convex lens is:

Positive

Question No:131 **(Marks:1)** **Vu-Topper RM**

An electron (m=9.11 * 10-31kg) has a speed of 0.95C. its kinetic energy is;

1.8*10-31J

Question No:132 **(Marks:1)** **Vu-Topper RM**

An electron (m=9.11 x 10-31kg) has a speed of 4.0.95C. Its kinetic energy is:

1.8x10-13J

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Question No:133 (Marks:1) Vu-Topper RM

An electron (m=9.11 x 10-31kg) has a momentum of 4.0x10-22kg*m/s. Its kinetic energy:

6.3*10-14J

Question No:134 (Marks:1) Vu-Topper RM

Force on a charged particle is zero when projected at angle with the magnetic field

0 degree

Question No:135 (Marks:1) Vu-Topper RM

If potential difference across two plates of a capacitor is doubled, then energy stored in it will be:

Four times

Question No:136 (Marks:1) Vu-Topper RM

In case of metallic conductors, the charge carriers are.

Electrons

Question No:137 (Marks:1) Vu-Topper RM

Room temperature is about 20 to 22 degrees on the:

Celcius scale

Question No:138 (Marks:1) Vu-Topper RM

As compared to distance measured by an observer on Earth, the distance from earth star measured by an observer in a moving spaceship would seem:

Same

Question No:139 (Marks:1) Vu-Topper RM

Evidence that a gas consists mostly of empty space is the fact that The density of a gas becomes much greater when it is

liquefied

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Question No:140 **(Marks:1)** **Vu-Topper RM**

For an electron, the rest mass energy is:

0.511 MeV

Question No:141 **(Marks:1)** **Vu-Topper RM**

Antiparticle of electron is:

Position

Question No:142 **(Marks:1)** **Vu-Topper RM**

According to the first law of thermodynamics, applied to a gas, the increase in the internal energy during any process

Equals the heat input plus the work done on the gas

Question No:143 **(Marks:1)** **Vu-Topper RM**

The approach to personality stresses that people possess a basic goodness, and have a natural tendency to grow to higher levels of functioning is.

Humanistic

Question No:144 **(Marks:1)** **Vu-Topper RM**

In a photoelectric experiment at a frequency above cut off, the stopping potential is proportional to :

The energy of the most energetic electron after it is ejected

Question No:145 **(Marks:1)** **Vu-Topper RM**

There is a temperature at which the reading on the kelvin scale is numerically:

Equal to that on the Celsius scale

Question No:146 **(Marks:1)** **Vu-Topper RM**

When you stand in front of a plane mirror, your image is:

Virtual, erect and the same size as you

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Question No:147 **(Marks:1)** **Vu-Topper RM**

The process for which entropy remains constant is:

A reversible process

Question No:148 **(Marks:1)** **Vu-Topper RM**

The appearance of colors in thin film is due to:

Interference

Question No:149 **(Marks:1)** **Vu-Topper RM**

Greenhouse gases absorb _____ radiation

Infrared

Question No:150 **(Marks:1)** **Vu-Topper RM**

The theoretical uppr limit for the frequency of electromagnetic waves is:

There is no upper limit

Question No:151 **(Marks:1)** **Vu-Topper RM**

Heat has the same units as:

Work

Question No:152 **(Marks:1)** **Vu-Topper RM**

Capacitance of a capacitor in vaccum is given by:

A eo /d

Question No:153 **(Marks:1)** **Vu-Topper RM**

AN electron in a K shell of an atom has the principal quantum number:

N =1

Question No:154 **(Marks:1)** **Vu-Topper RM**

Bending of light around the edges of an obstacle called:

Diffraction

Question No:155 **(Marks:1)** **Vu-Topper RM**

In RL series circuit phase angle is given by:

Tan-1(wl/R)

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Question No:156 **(Marks:1)** **Vu-Topper RM**

Interface of light is evidence that:

Light is a wave phenomenon

Question No:157 **(Marks:1)** **Vu-Topper RM**

In a photoelectric e? ect experiment at a frequency above cut o? the number of electrons ejected is proportional to:

The number of photons that hit the sample

Question No:158 **(Marks:1)** **Vu-Topper RM**

In a photoelectric e? ect ecperiment the stopping potential is:

The electric potential that causes the electron current to vanish

Question No:159 **(Marks:1)** **Vu-Topper RM**

In an adiabatic process:

The work done by the environment on the system equals the change in internal energy

Question No:160 **(Marks:1)** **Vu-Topper RM**

A free electron and a free proton have the same kinetic energy . This means that, compared to the matter wave associated with the proton, the matter wave associated with the electron has:

A longer wavelength and the same frequency

Question No:161 **(Marks:1)** **Vu-Topper RM**

Polarization experiments provide evidence that light is:

A transverse wave

Question No:162 **(Marks:1)** **Vu-Topper RM**

If light of low wavelength is used in young's double slit experiment, then width of the fringe will:

Decreases

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Question No:163 (Marks:1) Vu-Topper RM

Lenz's law is in accordance with law of conservation of

Energy…

Question No:164 (Marks:1) Vu-Topper RM

The known laws of electromagnetism before James Clerk Maxwell were following EXCEPT:

Faraday's Law deborglie law

Question No:165 (Marks:1) Vu-Topper RM

A particle having 2e charge falls through a potential difference of 5V energy acquired by it is.

10eV

Question No:166 (Marks:1) Vu-Topper RM

According to relatively theory a particle of mass m with a momentum of 2mc has a speed of:

c/2

Question No:167 (Marks:1) Vu-Topper RM

If the kinetic energy of a free particle is much less than its rest energy then its kinetic energy is proportional to:

The square of the magnitude of its momentum

Question No:168 (Marks:1) Vu-Topper RM

The two metallic strips that constitute some thermostats must di? Er in:

Coefficient of linear expansion

Question No:169 (Marks:1) Vu-Topper RM

Phase difference between V and I of an AC through resistor is:

0 degree

Question No:170 (Marks:1) Vu-Topper RM

Average value of current and voltage over complete cycle is:

Zero

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Question No:171 **(Marks:1)** **Vu-Topper RM**

A detector is used to detect the speed of a particular wave. If the frequency of an electromagnetic signal is 400 MHz, the speed detected by detector is _____.

300000 km/s



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As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, a distance of 0.75m from the origin. The z component of the angular momentum around the origin is:

- **6.0kg · m2/s**
- 9.0kg · m2/s
- 11 kg · m2/s
- 14 kg · m2/s

A net torque applied to a rigid object always tends to produce:

- linear acceleration
- rotational equilibrium
- **angular acceleration**
- rotational inertia

An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

- 1.57 rad/s
- **2.0 rad/s**
- 6.3 rad/s

In simple harmonic motion, the restoring force must be proportional to the:

- amplitude
- frequency
- velocity

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► **displacement**

Mercury is a convenient liquid to use in a barometer because:

► it is a metal

► it has a high boiling point

► it expands little with temperature

► **it has a high density**

The units of the electric field are:

► **J/m**

► J/(C·m)

► J/C

► J·C

A farad is the same as a

► **J/V**

► V/J

► C/V

► V/C

We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5H. We also need a capacitance of:

► 1 F

► 1mF

► 1 µF

► **100 µF**

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The wavelength of red light is 700 nm. Its frequency is _____.

4.29 10^14

- 4.30 * 104 Hertz
- 4.30 * 103 Hertz
- **4.30 * 105 Hertz**
- 4.30 * 102 Hertz

4.29 10^14 SEE PGE NO.92

Which of the following statements is NOT TRUE about electromagnetic waves?

- Electromagnetic waves satisfy the Maswell's Equation.
- Electromagnetic waves can not travel through space.
- The receptions of electromagnetic waves require an antenna.
- **The electromagnetic radiation from a burning candle is unpolarized.**

Radio waves and light waves are _____.

- Longitudinal waves
- Transverse waves
- **Electromagnetic and transverse both**
- Electromagnetic and longitudinal both

Wien's Law states that, *lmax* = _____ K.

- 2.90 * 10-3 Hertz
- 2.90 * 10-3 s
- 2.90 * 10-3 kg
- **2.90 * 10-3 m**

Interference of light is evidence that:

- the speed of light is very large

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- light is a transverse wave
- **light is a wave phenomenon**
- light is electromagnetic in character

Fahrenheit and Kelvin scales agree numerically at a reading of:

- **-40**
- 0
- 273
- 574

According to the theory of relativity:

- **moving clocks run fast**
- energy is not conserved in high speed collisions
- the speed of light must be measured relative to the ether
- none of the above are true

Light from a stationary spaceship is observed, and then the spaceship moves directly away from the observer at high speed while still emitting the light. As a result, the light seen by the observer has:

- higher frequency and a longer wavelength than before
- **lower frequency and a shorter wavelength than before**
- higher frequency and a shorter wavelength than before
- lower frequency and a longer wavelength than before

How fast should you move away from a 6.0 × 1014 Hz light source to observe waves with a frequency of 4.0 × 1014 Hz?

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- 20c
- **38c**
- 45c
- 51c

The quantum number n is most closely associated with what property of the electron in a hydrogen atom?

- **Energy**
- Orbital angular momentum
- Spin angular momentum
- Magnetic moment

The quantum number ms is most closely associated with what property of the electron in an atom?

- Magnitude of the orbital angular momentum
- **Energy**
- z component of the spin angular momentum
- z component of the orbital angular momentum

As the wavelength of a wave in a uniform medium increases, its speed will _____.

- Decrease
- Increase
- **Remain the same**
- None of these

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The number of significant figures in 0.00150 is:

- 5
- 4
- 3
- 2

For a body to be in equilibrium under the combined action of several forces:

- All the forces must be applied at the same point
- all of the forces form pairs of equal and opposite forces
- any two of these forces must be balanced by a third force
- the sum of the torques about any point must equal zero

A bucket of water is pushed from left to right with increasing speed across a horizontal surface. Consider the pressure at two points at the same level in the water.

- It is the same
- It is higher at the point on the left
- It is higher at the point on the right
- At first it is higher at the point on the left but as the bucket speeds up it is lower there

An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

- 200 Hz
- 300 Hz

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- 400 Hz
- **600 Hz**

Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by

- C1C2/(C1 + C2)
- (C1 + C2)/C1C2
- 1/(C1 + C2)
- C1/C2

If the potential difference across a resistor is doubled:

- only the current is doubled
- only the current is halved
- only the resistance is doubled
- only the resistance is halved

By using only two resistors, R1 and R2, a student is able to obtain resistances of 3 _, 4_, 12_, and 16 _. The values of R1 and R2 (in ohms) are:

- 3, 4
- 2, 12
- 3, 16
- 4, 12

Faraday's law states that an induced emf is proportional to:

- the rate of change of the electric field
- the rate of change of the magnetic flux

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- the rate of change of the electric flux
- the rate of change of the magnetic field

A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

- 1000V
- 500V
- 250V
- 100V

Which of the following electromagnetic radiations has photons with the greatest energy?

- blue light
- yellow light
- x rays
- radio waves

A virtual image is one:

- toward which light rays converge but do not pass through
- from which light rays diverge as they pass through
- toward which light rays converge and pass through
- from which light rays diverge but do not pass through

What is the unit of magnification factor?

- meter.Kelvin
- radian.Kelvin

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- degree.Kelvin
- no units

During an adiabatic process an object does 100 J of work and its temperature decreases by 5K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is.

- 20 J/K
- 100 J/K
- 15 J/K
- 5 J/K

An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

- a change in entropy
- a decrease of internal energy
- an increase of pressure
- a change in temperature

The Stern-Gerlach experiment makes use of:

- a strong uniform magnetic field
- a strong non-uniform magnetic field
- a strong uniform electric field
- a strong non-uniform electric field

A large collection of nuclei are undergoing alpha decay. The rate of decay at any instant is proportional to:

- the number of undecayed nuclei present at that instant

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- the time since the decays started
- the time remaining before all have decayed
- the half-life of the decay

Consider Gauss's law: $E \cdot d A = q/0$. Which of the following is true?

- E must be the electric field due to the enclosed charge
- If q = 0, then E = 0 everywhere on the Gaussian surface
- **If the three particles inside have charges of +q, +q, and −2q, then the integral is zero**
- on the surface E is everywhere parallel to d A

A charged point particle is placed at the center of a spherical Gaussian surface. The electric flux ΦE is changed if:

- the sphere is replaced by a cube of the same volume
- the sphere is replaced by a cube of one-tenth the volume
- the point charge is moved off center (but still inside the original sphere)
- **the point charge is moved to just outside the sphere**

Choose the INCORRECT statement:

- Gauss' law can be derived from Coulomb's law
- Gauss' law states that the net number of lines crossing any closed surface in an outward direction is proportional to the net charge enclosed within the surface
- Coulomb's law can be derived from Gauss' law and symmetry

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► **According to Gauss' law, if a closed surface encloses no charge, then the electric field must vanish everywhere on the surface**

The outer surface of the cardboard center of a paper towel roll:

► is a possible Gaussian surface

► cannot be a Gaussian surface because it encloses no charge

► cannot be a Gaussian surface since it is an insulator

► **cannot be a Gaussian surface because it is not a closed surface**

A physics instructor in an anteroom charges an electrostatic generator to 25 µC, then carries it into the lecture hall. The net electric flux in N · m2/C through the lecture hall walls is:

► 0

► 25 × 10−6

► 2.2 × 105

► **2.8 × 106**

A point particle with charge q is placed inside the cube but not at its center. The electric flux through any one side of the cube:

► is zero

► is q/0

► is q/60

► **cannot be computed using Gauss' law**

A particle with charge 5.0-µC is placed at the corner of a cube. The total electric flux in N · m2 /C through all sides of the cube is:

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- ▶ 0
- ▶ 7.1 × 104
- ▶ 9.4 × 104
- ▶ **5.6 × 105**

Questions:

1. **Which weighs more, a liter of ice or a liter of water?**
2. **Will the current in a light bulb connected to a 220-V source be greater or less than when the same bulb is connected to 110-V source?**
3. **How is the wavelength of light related to its frequency?**
4. **We don't notice the de Broglie wavelength for a pitched baseball. Is this Because the wavelength is very large or because it is very small?**
5. **Does every magnet necessarily have a north and south pole? Explain**
6. **In a cool room, a metal or marble table top feels much colder to the touch than does a wood surface even though they are at the same temperature. Why?**
7. **If a water wave oscillates up and down three times each second and the distance between wave crests is 2 m, what is its frequency? What is its wavelength? What is its wave speed?**
8. **A transformer has N1 = 350 turns and N2 = 2 000 turns. If the input voltage is coil?**

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9. **Why do astronomers looking at distant galaxies talk about looking backward in time?**
10. **Some distant astronomical objects, called quasars, are receding from us at half the speed of light (or greater). What is the speed of the light we receive from these quasars?**

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14. A point particle with charge q is at the center of a Gaussian surface in the form of a cube. The electric flux through any one face of the cube is:

- ► q/0
- ► q/4π0
- ► q/30
- ► **q/60**

The table below gives the electric flux in N·m2/C through the ends and round surfaces of four Gaussian surfaces in the form of cylinders. Rank the cylinders according to the charge inside, from the most negative to the most positive. Left end right end rounded surface

cylinder 1: +2 × 10−9 +4 × 10−9 −6 × 10−9

cylinder 2: +3 × 10−9 −2 × 10−9 +6 × 10−9

cylinder 3: −2 × 10−9 −5 × 10−9 +3 × 10−9

cylinder 4: +2 × 10−9 −5 × 10−9 −3 × 10−9

- ► 1, 2, 3, 4
- ► 4, 3, 2, 1
- ► 3, 4, 2, 1
- ► **4, 3, 1, 2**

A conducting sphere of radius 0.01m has a charge of 1.0 × 10−9 C deposited on it. The magnitude of the electric field in N/C just outside the surface of the sphere is:

- ► 0

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- 450
- **900**
- 4500

A round wastepaper basket with a 0.15-m radius opening is in a uniform electric field of 300N/C, perpendicular to the opening. The total flux through the sides and bottom, in N · m2 C, is:

- 0
- 4.2
- **21**
- 280

10C of charge are placed on a spherical conducting shell. A particle with a charge of −3C is placed at the center of the cavity. The net charge on the inner surface of the shell is:

- −7C
- −3C
- 0C
- **+3C**

10C of charge are placed on a spherical conducting shell. A particle with a charge of −3C is placed at the center of the cavity. The net charge on the outer surface of the shell is:

- −7C
- −3C

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- 0C
- **+7C**

Charge Q is distributed uniformly throughout an insulating sphere of radius R. The magnitude of the electric field at a point R/2 from the center is:

- Q/4π0R2
- Q/π0R2
- 3Q/4π0R2
- **Q/8π0R2**

Positive charge Q is distributed uniformly throughout an insulating sphere of radius R, centered at the origin. A particle with positive charge Q is placed at x = 2R on the x axis. The magnitude of the electric field at x = R/2 on the x axis is:

- Q/4π0R2
- Q/8π0R2
- **Q/72π0R2**
- 17Q/72π0R2

Charge Q is distributed uniformly throughout a spherical insulating shell. The net electric flux in N · m2 /C through the inner surface of the shell is:

- **0**
- Q/0
- 2Q/0
- Q/4π0

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Charge Q is distributed uniformly throughout a spherical insulating shell. The net electric flux in N · m2 /C through the outer surface of the shell is:

- ▶ 0
- ▶ **Q/0**
- ▶ 2Q/0
- ▶ Q/40

A 3.5-cm radius hemisphere contains a total charge of 6.6 × 10−7 C. The flux through the rounded portion of the surface is 9.8 × 104 N · m2 /C. The flux through the flat base is:

- ▶ 0
- ▶ +2.3 × 104 N · m2 /C
- ▶ **−2.3 × 104 N · m2 /C**
- ▶ −9.8 × 104 N · m2 /C

Charge is distributed uniformly along a long straight wire. The electric field 2 cm from the wire is 20N/C. The electric field 4 cm from the wire is:

- ▶ 120N/C
- ▶ 80N/C
- ▶ 40N/C
- ▶ **10N/C**

Positive charge Q is placed on a conducting spherical shell with inner radius R1 and outer radius R2. A point charge q is placed at the center of the cavity. The

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magnitude of the electric field at a point outside the shell, a distance r from the center, is:

- zero
- Q/4π0r2
- q/4π0r2
- **(q + Q)/4π0r2**

Positive charge Q is placed on a conducting spherical shell with inner radius R1 and outer radius R2. A point charge q is placed at the center of the cavity. The magnitude of the electric field produced by the charge on the inner surface at a point in the interior of the conductor, a distance r from the center, is:

- 0
- Q/4vπ0R2 1
- Q/4π0R2 2
- **q/4π0r2**
- Q/4π0r2

A long line of charge with λ charge per unit length runs along the cylindrical axis of a cylindrical shell which carries a charge per unit length of λc. The charge per unit length on the inner and outer surfaces of the shell, respectively are:

- λ and λc
- **–λ and λc + λ**
- –λ and λc – λc
- λ + λc and λc – λ

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Charge is distributed uniformly on the surface of a large flat plate. The electric field 2 cm from the plate is 30N/C. The electric field 4 cm from the plate is:

- 120N/C
- 80N/C
- **30N/C**
- 15N/C

A particle with charge Q is placed outside a large neutral conducting sheet. At any point in the interior of the sheet the electric field produced by charges on the surface is directed:

- toward the surface
- away from the surface
- **toward Q**
- away from Q

A hollow conductor is positively charged. A small uncharged metal ball is lowered by a silk thread through a small opening in the top of the conductor and allowed to touch its inner surface. After the ball is removed, it will have:

- a positive charge
- a negative charge
- **no appreciable charge**
- a charge whose sign depends on what part of the inner surface it touched

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A spherical conducting shell has charge Q. A particle with charge q is placed at the center of the cavity. The charge on the inner surface of the shell and the charge on the outer surface of the shell, respectively, are:

- 0, Q
- q, Q − q
- **−q, Q + q**
- −q, 0

A particle with a charge of 5.5×10−8C is 3.5 cm from a particle with a charge of −2.3×10−8 C. The potential energy of this two-particle system, relative to the potential energy at infinite separation, is:

- 3.2 × 10−4 J
- **−3.2 × 10−4 J**
- 9.3 × 10−3 J
- −9.3 × 10−3 J

A particle with a charge of 5.5 × 10−8C is fixed at the origin. A particle with a charge of −2.3×10−8 C is moved from x = 3.5 cm on the x axis to y = 4.3 cm on the y axis. The change in potential energy of the two-particle system is:

- 3.1 × 10−3 J
- −3.1 × 10−3 J
- **6.0 × 10−5 J**
- −6.0 × 10−5 J

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A particle with a charge of 5.5 × 10−8 C charge is fixed at the origin. A particle with a charge of −2.3 × 10−8 C charge is moved from x = 3.5 cm on the x axis to y = 3.5 cm on the y axis. The change in the potential energy of the two-particle system is:

- ▶ 3.2 × 10−4 J
- ▶ −3.2 × 10−4 J
- ▶ 9.3 × 10−3 J
- ▶ **0**

Three particles lie on the x axis: particle 1, with a charge of 1×10−8 C is at x = 1 cm, particle 2, with a charge of 2 × 10−8 C, is at x = 2 cm, and particle 3, with a charge of −3 × 10−8 C, is at x = 3 cm. The potential energy of this arrangement, relative to the potential energy for infinite separation, is:

- ▶ +4.9 × 10−4 J
- ▶ **−4.9 × 10−4 J**
- ▶ +8.5 × 10−4 J
- ▶ −8.5 × 10−4 J

Two identical particles, each with charge q, are placed on the x axis, one at the origin and the other at x = 5 cm. A third particle, with charge −q, is placed on the x axis so the potential energy of the three-particle system is the same as the potential energy at infinite separation. Its x coordinate is:

- ▶ **13 cm**
- ▶ 2.5 cm

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- 7.5 cm
- 10 cm

Choose the correct statement:

- The potential of a negatively charged conductor must be negative
- If E = 0 at a point P then V must be zero at P
- If V = 0 at a point P then E must be zero at P
- **None of the above are correct**

If 500 J of work are required to carry a charged particle between two points with a potential difference of 20V, the magnitude of the charge on the particle is:

- **12.5C**
- 20C
- cannot be computed unless the path is given
- none of these

The potential difference between two points is 100V. If a particle with a charge of 2C is transported from one of these points to the other, the magnitude of the work done is:

- **200 J**
- 100 J
- 50 J
- 100 J

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Two large parallel conducting plates are separated by a distance d, placed in a vacuum, and connected to a source of potential difference V . An oxygen ion, with charge 2e, starts from rest on the surface of one plate and accelerates to the other. If e denotes the magnitude of the electron charge, the final kinetic energy of this ion is:

- eV/2
- eV/d
- eV d
- **2eV**

An electron volt is :

- the force acting on an electron in a field of 1N/C
- the force required to move an electron 1 meter
- **the energy gained by an electron in moving through a potential difference of 1 volt**
- the energy needed to move an electron through 1 meter in any electric field

An electron has charge –e and mass me. A proton has charge e and mass 1840me. A "proton volt" is equal to:

- **1 eV**
- 1840 eV
- (1/1840) eV
- √1840 eV

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Two conducting spheres are far apart. The smaller sphere carries a total charge Q. The larger sphere has a radius that is twice that of the smaller and is neutral. After the two spheres are connected by a conducting wire, the charges on the smaller and larger spheres, respectively, are:

- Q/2 and Q/2
- **Q/3 and 2Q/3**
- 2Q/3 and Q/3
- zero and Q

A conducting sphere with radius R is charged until the magnitude of the electric field just outside its surface is E. The electric potential of the sphere, relative to the potential far away, is:

- zero
- E/R
- E/R2
- **ER**

A 5-cm radius conducting sphere has a surface charge density of 2 ×10−6 C/m2 on its surface. Its electric potential, relative to the potential far away, is:

- **1.1 × 104 V**
- 2.2 × 104 V
- 2.3 × 105 V
- 3.6 × 105 V

A hollow metal sphere is charged to a potential V . The potential at its center is:

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- V
- 0
- −V
- 2V

Questions:

1. **Consider a lamp hanging from a chain. What is the tension in the chain?**
2. **Aproton travels with a speed of 3.00 * 106 m/s at an angle of 37.0° with the direction of a magnetic field of 0.300 T in the + y direction. What are (a) the magnitude of the magnetic force on the proton and (b) its acceleration?**
3. **Light from the Sun takes approximately 8.3 min to reach the Earth. During this time interval the Earth has continued to rotate on its axis. How far is the actual direction of the Sun from its image in the sky?**
4. **Do all current-carrying conductors emit electromagnetic waves? Explain**
5. **Explain solar convection zone. What is its other name?**
6. **If a charged particle moves in a straight line through some region of space, can you say that the magnetic field in that region is zero?**
7. **Can all gas molecules in the vessel have the same speed?**
8. **What are the properties of wave function?**
9. **A bike accelerates uniformly from rest to a speed of 7.10 m/s over a distance of 35.4 m. Determine the acceleration of the bike.**

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10.A flat loop of wire consisting of a single turn of cross-sectional area 8.00 cm2 is perpendicular to a magnetic field that increases uniformly in magnitude from 0.500 T to 2.50 T in 1.00 s. What is the resulting induced current if the loop has a resistance of 2.00 W?

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Positive charge is distributed uniformly throughout a non-conducting sphere. The highest electric potential occurs:

- **at the center**
- at the surface
- halfway between the center and surface
- just outside the surface

A total charge of 7×10−8 C is uniformly distributed throughout a non-conducting sphere with a radius of 5 cm. The electric potential at the surface, relative to the potential far away, is about:

- −1.3 × 104 V
- **1.3 × 104 V**
- 7.0 × 105 V
- −6.3 × 104 V

Eight identical spherical raindrops are each at a potential V , relative to the potential far away. They coalesce to make one spherical raindrop whose potential is:

- V/8
- V/2
- 2V
- **4V**

A metal sphere carries a charge of 5 × 10−9 C and is at a potential of 400V, relative to the potential far away. The potential at the center of the sphere is:

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- **400V**
- −400V
- 2 × 10−6 V
- 0

A 5-cm radius isolated conducting sphere is charged so its potential is +100V, relative to the potential far away. The charge density on its surface is:

- +2.2 × 10−7 C/m2
- −2.2 × 10−7 C/m2
- +3.5 × 10−7 C/m2
- **+1.8 × 10−8 C/m2**

A conducting sphere has charge Q and its electric potential is V , relative to the potential far away. If the charge is doubled to 2Q, the potential is:

- V
- **2V**
- 4V
- V/4

The potential difference between the ends of a 2-meter stick that is parallel to a uniform electric field is 400V. The magnitude of the electric field is:

- zero
- 100V/m
- 200V/m
- **800V/m**

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In a certain region of space the electric potential increases uniformly from east to west and does not vary in any other direction. The electric field:

- points east and varies with position
- **points east and does not vary with position**
- points west and varies with position
- points west and does not vary with position

If the electric field is in the positive x direction and has a magnitude given by E = Cx2, where C is a constant, then the electric potential is given by V =:

- 2Cx
- −2Cx
- Cx3/3
- **−Cx3/3**

The work required to carry a particle with a charge of 6.0C from a 5.0-V equipotential surface to a 6.0-V equipotential surface and back again to the 5.0-V surface is:

- **0**
- 1.2 × 10−5 J
- 3.0 × 10−5 J
- 6.0 × 10−5 J

The equipotential surfaces associated with a charged point particles are:

- radially outward from the particle
- vertical planes

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- horizontal planes
- **concentric spheres centered at the particle**

The electric field in a region around the origin is given by E = C(xˆi + yˆj), where C is a constant. The equipotential surfaces in that region are:

- **concentric cylinders with axes along the z axis**
- concentric cylinders with axes along the x axis
- concentric spheres centered at the origin
- planes parallel to the xy plane

A particle with charge q is to be brought from far away to a point near an electric dipole. No work is done if the final position of the particle is on:

- the line through the charges of the dipole
- **a line that is perpendicular to the dipole moment**
- a line that makes an angle of 45◦ with the dipole moment
- a line that makes an angle of 30◦ with the dipole moment

Equipotential surfaces associated with an electric dipole are:

- spheres centered on the dipole
- cylinders with axes along the dipole moment
- planes perpendicular to the dipole moment
- **none of the above**

The units of capacitance are equivalent to:

- J/C
- V/C

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- J2/C
- **C2/J**

A farad is the same as a:

- J/V
- V/J
- **C/V**
- V/C

A capacitor C "has a charge Q". The actual charges on its plates are:

- Q, Q
- Q/2, Q/2
- **Q, –Q**
- Q/2, –Q/2

Each plate of a capacitor stores a charge of magnitude 1mC when a 100-V potential difference is applied. The capacitance is:

- 5 µF
- **10 µF**
- 50 µF
- 100 µF

To charge a 1-F capacitor with 2C requires a potential difference of:

- **2V**
- 0.2V
- 5V

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- 0.5V

The capacitance of a parallel-plate capacitor with plate area A and plate separation d is given by:

- 0d/A
- 0d/2A
- **0A/d**
- 0A/2d

The capacitance of a parallel-plate capacitor is:

- **proportional to the plate area**
- proportional to the charge stored
- independent of any material inserted between the plates
- proportional to the potential difference of the plates

The capacitance of a parallel-plate capacitor can be increased by:

- increasing the charge
- decreasing the charge
- increasing the plate separation
- **decreasing the plate separation**

If both the plate area and the plate separation of a parallel-plate capacitor are doubled, the capacitance is:

- doubled
- halved
- **unchanged**

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- tripled

If the plate area of an isolated charged parallel-plate capacitor is doubled:

- the electric field is doubled
- **the potential difference is halved**
- the charge on each plate is halved
- the surface charge density on each plate is doubled

If the plate separation of an isolated charged parallel-plate capacitor is doubled:

- the electric field is doubled
- the potential difference is halved
- the charge on each plate is halved
- **none of the above**

Pulling the plates of an isolated charged capacitor apart:

- increases the capacitance
- **increases the potential difference**
- does not affect the potential difference
- decreases the potential difference

If the charge on a parallel-plate capacitor is doubled:

- the capacitance is halved
- the capacitance is doubled
- the electric field is halved
- **the electric field is doubled**

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A parallel-plate capacitor has a plate area of 0.2m2 and a plate separation of 0.1mm. To obtain an electric field of 2.0 × 106 V/m between the plates, the magnitude of the charge on each plate should be:

- ▶ 8.9 × 10−7 C
- ▶ 1.8 × 10−6 C
- ▶ 3.5 × 10−6 C
- ▶ **7.1 × 10−6 C**

A parallel-plate capacitor has a plate area of 0.2m2 and a plate separation of 0.1 mm. If the charge on each plate has a magnitude of 4 × 10−6 C the potential difference across the plates is approximately:

- ▶ 0
- ▶ 4 × 10−2 V
- ▶ 1 × 102 V
- ▶ **2 × 102 V**

The capacitance of a spherical capacitor with inner radius a and outer radius b is proportional to:

- ▶ a/b
- ▶ b − a
- ▶ b2 − a2
- ▶ **ab/(b − a)**

The capacitance of a single isolated spherical conductor with radius R is proportional to:

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- **R**
- R2
- 1/R
- 1/R2

The capacitance of a cylindrical capacitor can be increased by:

- decreasing both the radius of the inner cylinder and the length
- **increasing both the radius of the inner cylinder and the length**
- increasing the radius of the outer cylindrical shell and decreasing the length
- only by decreasing the length

A battery is used to charge a series combination of two identical capacitors If the potential difference across the battery terminals is V and total charge Q flows through the battery during the charging process then the charge on the positive plate of each capacitor and the potential difference across each capacitor are:

- Q/2 and V/2, respectively
- Q and V , respectively
- Q/2 and V , respectively
- **Q and V/2, respectively**

A battery is used to charge a parallel combination of two identical capacitors. If the potential difference across the battery terminals is V and total charge Q flows through the battery during the charging process then the charge on the

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positive plate of each capacitor and the potential difference across each capacitor are:

- ▶ Q/2 and V/2, respectively
- ▶ Q and V , respectively
- ▶ **Q/2 and V , respectively**
- ▶ Q and V/2, respectively

A 2-µF and a 1-µF capacitor are connected in series and a potential difference is applied across the combination. The 2-µF capacitor has:

- ▶ twice the charge of the 1-µF capacitor
- ▶ half the charge of the 1-µF capacitor
- ▶ twice the potential difference of the 1-µF capacitor
- ▶ **half the potential difference of the 1-µF capacitor**

A 2-µF and a 1-µF capacitor are connected in parallel and a potential difference is applied across the combination. The 2-µF capacitor has:

- ▶ **twice the charge of the 1-µF capacitor**
- ▶ half the charge of the 1-µF capacitor
- ▶ twice the potential difference of the 1-µF capacitor
- ▶ half the potential difference of the 1-µF capacitor

Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in series must have the same:

- ▶ **Q only**
- ▶ V only

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- U only
- Q and U only

Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in parallel must have the same:

- Q only
- **V only**
- U only
- Q and U only

Capacitors C1 and C2 are connected in parallel. The equivalent capacitance is given by:

- C1C2/(C1 + C2)
- (C1 + C2)/C1C2
- C1/C2
- **C1 + C2**

Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by:

- **C1C2/(C1 + C2)**
- (C1 + C2)/C1C2
- 1/(C1 + C2)
- C1/C2

Capacitors C1 and C2 are connected in series and a potential difference is applied to the combination. If the capacitor that is equivalent to the

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combination has the same potential difference, then the charge on the equivalent capacitor is the same as:

- **the charge on C1**
- the sum of the charges on C1 and C2
- the difference of the charges on C1 and C2
- the product of the charges on C1 and C2

Questions:

1. **An ideal gas is contained in a vessel at 300 K. If the temperature is increased to 900 K, by what factor does each one of the following change?**
   - **(a) The average kinetic energy of the molecules.**
   - **(b) The rms molecular speed.**
   - **(c) The average momentum change of one molecule in a collision with a wall.**
   - **(d) The rate of collisions of molecules with walls.**
   - **(e) The pressure of the gas.**
2. **Who discover the nucleus? Write the experimental setup that he follows.**
3. **In an analogy between electric current and automobile traffic flow, what would correspond to charge? What would correspond to current?**
4. **(a)When can you expect a body to emit blackbody radiation?**
   (b) Which law is obeyed by Sun and other stars, briefly explain it.

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5. **Two people are carrying a uniform wooden board that is 3.00 m long and weighs 160 N. If one person applies an upward force equal to 60 N at one end, at what point does the other person lift? Begin with a free-body diagram of the board.**
6. **If a charged particle moves in a straight line through some region of space, can you say that the magnetic field in that region is zero?**
7. **You want to explore the shape of a cel1ain molecule by scattering electrons of momentum p from a gas of the molecules and studying the deflection of the electrons.**
8. **You will be able to see finer details in the molecules by (a) increasing p; (b) decreasing p: (c) not worrying what p is.**
9. **A vessel is filled with gas at some equilibrium pressure and temperature. Can all gas molecules in the vessel have the same speed?**
10. **What are the properties of wave function?**

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As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, a distance of 0.75m from the origin. The z component of the angular momentum around the origin is:

- **6.0kg · m2/s**
- 9.0kg · m2/s
- 11 kg · m2/s
- 14 kg · m2/s

A net torque applied to a rigid object always tends to produce:

- linear acceleration
- rotational equilibrium
- **angular acceleration**
- rotational inertia

An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

- 1.57 rad/s
- **2.0 rad/s**
- 6.3 rad/s

In simple harmonic motion, the restoring force must be proportional to the:

- amplitude
- frequency
- velocity

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► **displacement**

Mercury is a convenient liquid to use in a barometer because:

► it is a metal

► it has a high boiling point

► it expands little with temperature

► **it has a high density**

The units of the electric field are:

► **J/m**

► J/(C·m)

► J/C

► J·C

A farad is the same as a

► **J/V**

► V/J

► C/V

► V/C

We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5H. We also need a capacitance of:

► 1 F

► 1mF

► 1 µF

► **100 µF**

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The wavelength of red light is 700 nm. Its frequency is _____4.29 10^14_____.

- 4.30 * 104 Hertz
- 4.30 * 103 Hertz

4.29 10^14 SEE PGE NO.92

- **4.30 * 105 Hertz**
- 4.30 * 102 Hertz

Which of the following statements is NOT TRUE about electromagnetic waves?

- Electromagnetic waves satisfy the Maswell's Equation.
- Electromagnetic waves can not travel through space.
- The receptions of electromagnetic waves require an antenna.
- **The electromagnetic radiation from a burning candle is unpolarized.**

Radio waves and light waves are _____.

- Longitudinal waves
- Transverse waves
- **Electromagnetic and transverse both**
- Electromagnetic and longitudinal both

Wien's Law states that, *lmax* = _____ K.

- 2.90 * 10-3 Hertz
- 2.90 * 10-3 s
- 2.90 * 10-3 kg
- **2.90 * 10-3 m**

Interference of light is evidence that:

- the speed of light is very large

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- light is a transverse wave
- **light is a wave phenomenon**
- light is electromagnetic in character

Fahrenheit and Kelvin scales agree numerically at a reading of:

- **-40**
- 0
- 273
- 574

According to the theory of relativity:

- ~~**moving clocks run fast**~~
- energy is not conserved in high speed collisions
- the speed of light must be measured relative to the ether
- none of the above are true

Light from a stationary spaceship is observed, and then the spaceship moves directly away from the observer at high speed while still emitting the light. As a result, the light seen by the observer has:

- higher frequency and a longer wavelength than before
- **lower frequency and a shorter wavelength than before**
- higher frequency and a shorter wavelength than before
- lower frequency and a longer wavelength than before

How fast should you move away from a 6.0 × 1014 Hz light source to observe waves with a frequency of 4.0 × 1014 Hz?

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- 20c
- **38c**
- 45c
- 51c

The quantum number n is most closely associated with what property of the electron in a hydrogen atom?

- **Energy**
- Orbital angular momentum
- Spin angular momentum
- Magnetic moment

The quantum number ms is most closely associated with what property of the electron in an atom?

- Magnitude of the orbital angular momentum
- **Energy**
- z component of the spin angular momentum
- z component of the orbital angular momentum

As the wavelength of a wave in a uniform medium increases, its speed will _____.

- Decrease
- Increase
- **Remain the same**
- None of these

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The number of significant figures in 0.00150 is:

- 5
- 4
- 3
- 2

For a body to be in equilibrium under the combined action of several forces:

- All the forces must be applied at the same point
- all of the forces form pairs of equal and opposite forces
- any two of these forces must be balanced by a third force
- the sum of the torques about any point must equal zero

A bucket of water is pushed from left to right with increasing speed across a horizontal surface. Consider the pressure at two points at the same level in the water.

- It is the same
- It is higher at the point on the left
- It is higher at the point on the right
- At first it is higher at the point on the left but as the bucket speeds up it is lower there

An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

- 200 Hz
- 300 Hz

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- 400 Hz
- **600 Hz**

Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by

- C1C2/(C1 + C2)
- (C1 + C2)/C1C2
- 1/(C1 + C2)
- C1/C2

If the potential difference across a resistor is doubled:

- only the current is doubled
- only the current is halved
- only the resistance is doubled
- only the resistance is halved

By using only two resistors, R1 and R2, a student is able to obtain resistances of 3 _, 4_, 12_, and 16 _. The values of R1 and R2 (in ohms) are:

- 3, 4
- 2, 12
- 3, 16
- 4, 12

Faraday's law states that an induced emf is proportional to:

- the rate of change of the electric field
- the rate of change of the magnetic flux

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- the rate of change of the electric flux
- the rate of change of the magnetic field

A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

- 1000V
- 500V
- 250V
- 100V

Which of the following electromagnetic radiations has photons with the greatest energy?

- blue light
- yellow light
- x rays
- radio waves

A virtual image is one:

- toward which light rays converge but do not pass through
- from which light rays diverge as they pass through
- toward which light rays converge and pass through
- from which light rays diverge but do not pass through

What is the unit of magnification factor?

- meter.Kelvin
- radian.Kelvin

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- degree.Kelvin
- no units

During an adiabatic process an object does 100 J of work and its temperature decreases by 5K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is.

- 20 J/K
- 100 J/K
- 15 J/K
- 5 J/K

An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

- a change in entropy
- a decrease of internal energy
- an increase of pressure
- a change in temperature

The Stern-Gerlach experiment makes use of:

- a strong uniform magnetic field
- a strong non-uniform magnetic field
- a strong uniform electric field
- a strong non-uniform electric field

A large collection of nuclei are undergoing alpha decay. The rate of decay at any instant is proportional to:

- the number of undecayed nuclei present at that instant

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- the time since the decays started
- the time remaining before all have decayed
- the half-life of the decay

Consider Gauss's law: $E \cdot d A = q/0$. Which of the following is true?

- E must be the electric field due to the enclosed charge
- If q = 0, then E = 0 everywhere on the Gaussian surface
- **If the three particles inside have charges of +q, +q, and −2q, then the integral is zero**
- on the surface E is everywhere parallel to d A

A charged point particle is placed at the center of a spherical Gaussian surface. The electric flux ΦE is changed if:

- the sphere is replaced by a cube of the same volume
- the sphere is replaced by a cube of one-tenth the volume
- the point charge is moved off center (but still inside the original sphere)
- **the point charge is moved to just outside the sphere**

Choose the INCORRECT statement:

- Gauss' law can be derived from Coulomb's law
- Gauss' law states that the net number of lines crossing any closed surface in an outward direction is proportional to the net charge enclosed within the surface
- Coulomb's law can be derived from Gauss' law and symmetry

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► **According to Gauss' law, if a closed surface encloses no charge, then the electric field must vanish everywhere on the surface**

The outer surface of the cardboard center of a paper towel roll:

► is a possible Gaussian surface

► cannot be a Gaussian surface because it encloses no charge

► cannot be a Gaussian surface since it is an insulator

► **cannot be a Gaussian surface because it is not a closed surface**

A physics instructor in an anteroom charges an electrostatic generator to 25 µC, then carries it into the lecture hall. The net electric flux in N · m2/C through the lecture hall walls is:

► 0

► 25 × 10−6

► 2.2 × 105

► **2.8 × 106**

A point particle with charge q is placed inside the cube but not at its center. The electric flux through any one side of the cube:

► is zero

► is q/0

► is q/60

► **cannot be computed using Gauss' law**

A particle with charge 5.0-µC is placed at the corner of a cube. The total electric flux in N · m2 /C through all sides of the cube is:

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- ▶ 0
- ▶ 7.1 × 104
- ▶ 9.4 × 104
- ▶ **5.6 × 105**

14. A point particle with charge q is at the center of a Gaussian surface in the form of a cube. The electric flux through any one face of the cube is:

- ▶ q/0
- ▶ q/4π0
- ▶ q/30
- ▶ **q/60**

The table below gives the electric flux in N·m2/C through the ends and round surfaces of four Gaussian surfaces in the form of cylinders. Rank the cylinders according to the charge inside, from the most negative to the most positive. Left end right end rounded surface

cylinder 1: +2 × 10−9 +4 × 10−9 −6 × 10−9

cylinder 2: +3 × 10−9 −2 × 10−9 +6 × 10−9

cylinder 3: −2 × 10−9 −5 × 10−9 +3 × 10−9

cylinder 4: +2 × 10−9 −5 × 10−9 −3 × 10−9

- ▶ 1, 2, 3, 4
- ▶ 4, 3, 2, 1
- ▶ 3, 4, 2, 1
- ▶ **4, 3, 1, 2**

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A conducting sphere of radius 0.01m has a charge of 1.0 × 10−9 C deposited on it. The magnitude of the electric field in N/C just outside the surface of the sphere is:

- ▶ 0
- ▶ 450
- ▶ **900**
- ▶ 4500

A round wastepaper basket with a 0.15-m radius opening is in a uniform electric field of 300N/C, perpendicular to the opening. The total flux through the sides and bottom, in N · m2 C, is:

- ▶ 0
- ▶ 4.2
- ▶ **21**
- ▶ 280

10C of charge are placed on a spherical conducting shell. A particle with a charge of −3C is placed at the center of the cavity. The net charge on the inner surface of the shell is:

- ▶ −7C
- ▶ −3C
- ▶ 0C
- ▶ **+3C**

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10C of charge are placed on a spherical conducting shell. A particle with a charge of −3C is placed at the center of the cavity. The net charge on the outer surface of the shell is:

- −7C
- −3C
- 0C
- **+7C**

Charge Q is distributed uniformly throughout an insulating sphere of radius R. The magnitude of the electric field at a point R/2 from the center is:

- Q/4π0R2
- Q/π0R2
- 3Q/4π0R2
- **Q/8π0R2**

Positive charge Q is distributed uniformly throughout an insulating sphere of radius R, centered at the origin. A particle with positive charge Q is placed at x = 2R on the x axis. The magnitude of the electric field at x = R/2 on the x axis is:

- Q/4π0R2
- Q/8π0R2
- **Q/72π0R2**
- 17Q/72π0R2

Charge Q is distributed uniformly throughout a spherical insulating shell. The net electric flux in N · m2 /C through the inner surface of the shell is:

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► **0**

► Q/0

► 2Q/0

► Q/4π0

Charge Q is distributed uniformly throughout a spherical insulating shell. The net electric flux in N · m2 /C through the outer surface of the shell is:

► 0

► **Q/0**

► 2Q/0

► Q/40

A 3.5-cm radius hemisphere contains a total charge of 6.6 × 10−7 C. The flux through the rounded portion of the surface is 9.8 × 104 N · m2 /C. The flux through the flat base is:

► 0

► +2.3 × 104 N · m2 /C

► **−2.3 × 104 N · m2 /C**

► −9.8 × 104 N · m2 /C

Charge is distributed uniformly along a long straight wire. The electric field 2 cm from the wire is 20N/C. The electric field 4 cm from the wire is:

► 120N/C

► 80N/C

► 40N/C

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► **10N/C**

Positive charge Q is placed on a conducting spherical shell with inner radius R1 and outer radius R2. A point charge q is placed at the center of the cavity. The magnitude of the electric field at a point outside the shell, a distance r from the center, is:

► zero

► Q/4π0r2

► q/4π0r2

► **(q + Q)/4π0r2**

Positive charge Q is placed on a conducting spherical shell with inner radius R1 and outer radius R2. A point charge q is placed at the center of the cavity. The magnitude of the electric field produced by the charge on the inner surface at a point in the interior of the conductor, a distance r from the center, is:

► 0

► Q/4vπ0R2 1

► Q/4π0R2 2

► **q/4π0r2**

► Q/4π0r2

A long line of charge with λ charge per unit length runs along the cylindrical axis of a cylindrical shell which carries a charge per unit length of λc. The charge per unit length on the inner and outer surfaces of the shell, respectively are:

► λ and λc

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- **–λ and λc + λ**
- –λ and λc – λc
- λ + λc and λc – λ

Charge is distributed uniformly on the surface of a large flat plate. The electric field 2 cm from the plate is 30N/C. The electric field 4 cm from the plate is:

- 120N/C
- 80N/C
- **30N/C**
- 15N/C

A particle with charge Q is placed outside a large neutral conducting sheet. At any point in the interior of the sheet the electric field produced by charges on the surface is directed:

- toward the surface
- away from the surface
- **toward Q**
- away from Q

A hollow conductor is positively charged. A small uncharged metal ball is lowered by a silk thread through a small opening in the top of the conductor and allowed to touch its inner surface. After the ball is removed, it will have:

- a positive charge
- a negative charge
- **no appreciable charge**

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► a charge whose sign depends on what part of the inner surface it touched

A spherical conducting shell has charge Q. A particle with charge q is placed at the center of the cavity. The charge on the inner surface of the shell and the charge on the outer surface of the shell, respectively, are:

► 0, Q

► q, Q − q

► **−q, Q + q**

► −q, 0

A particle with a charge of 5.5×10−8C is 3.5 cm from a particle with a charge of −2.3×10−8 C. The potential energy of this two-particle system, relative to the potential energy at infinite separation, is:

► 3.2 × 10−4 J

► **−3.2 × 10−4 J**

► 9.3 × 10−3 J

► −9.3 × 10−3 J

A particle with a charge of 5.5 × 10−8C is fixed at the origin. A particle with a charge of −2.3×10−8 C is moved from x = 3.5 cm on the x axis to y = 4.3 cm on the y axis. The change in potential energy of the two-particle system is:

► 3.1 × 10−3 J

► −3.1 × 10−3 J

► **6.0 × 10−5 J**

► −6.0 × 10−5 J

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A particle with a charge of 5.5 × 10−8 C charge is fixed at the origin. A particle with a charge of −2.3 × 10−8 C charge is moved from x = 3.5 cm on the x axis to y = 3.5 cm on the y axis. The change in the potential energy of the two-particle system is:

- ▶ 3.2 × 10−4 J
- ▶ −3.2 × 10−4 J
- ▶ 9.3 × 10−3 J
- ▶ **0**

Three particles lie on the x axis: particle 1, with a charge of 1×10−8 C is at x = 1 cm, particle 2, with a charge of 2 × 10−8 C, is at x = 2 cm, and particle 3, with a charge of −3 × 10−8 C, is at x = 3 cm. The potential energy of this arrangement, relative to the potential energy for infinite separation, is:

- ▶ +4.9 × 10−4 J
- ▶ **−4.9 × 10−4 J**
- ▶ +8.5 × 10−4 J
- ▶ −8.5 × 10−4 J

Two identical particles, each with charge q, are placed on the x axis, one at the origin and the other at x = 5 cm. A third particle, with charge −q, is placed on the x axis so the potential energy of the three-particle system is the same as the potential energy at infinite separation. Its x coordinate is:

- ▶ **13 cm**
- ▶ 2.5 cm

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- 7.5 cm
- 10 cm

Choose the correct statement:

- The potential of a negatively charged conductor must be negative
- If E = 0 at a point P then V must be zero at P
- If V = 0 at a point P then E must be zero at P
- **None of the above are correct**

If 500 J of work are required to carry a charged particle between two points with a potential difference of 20V, the magnitude of the charge on the particle is:

- **12.5C**
- 20C
- cannot be computed unless the path is given
- none of these

The potential difference between two points is 100V. If a particle with a charge of 2C is transported from one of these points to the other, the magnitude of the work done is:

- **200 J**
- 100 J
- 50 J
- 100 J

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Two large parallel conducting plates are separated by a distance d, placed in a vacuum, and connected to a source of potential difference V . An oxygen ion, with charge 2e, starts from rest on the surface of one plate and accelerates to the other. If e denotes the magnitude of the electron charge, the final kinetic energy of this ion is:

- eV/2
- eV/d
- eV d
- **2eV**

An electron volt is :

- the force acting on an electron in a field of 1N/C
- the force required to move an electron 1 meter
- **the energy gained by an electron in moving through a potential difference of 1 volt**
- the energy needed to move an electron through 1 meter in any electric field

An electron has charge –e and mass me. A proton has charge e and mass 1840me. A "proton volt" is equal to:

- **1 eV**
- 1840 eV
- (1/1840) eV
- √1840 eV

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Two conducting spheres are far apart. The smaller sphere carries a total charge Q. The larger sphere has a radius that is twice that of the smaller and is neutral. After the two spheres are connected by a conducting wire, the charges on the smaller and larger spheres, respectively, are:

- Q/2 and Q/2
- **Q/3 and 2Q/3**
- 2Q/3 and Q/3
- zero and Q

A conducting sphere with radius R is charged until the magnitude of the electric field just outside its surface is E. The electric potential of the sphere, relative to the potential far away, is:

- zero
- E/R
- E/R2
- **ER**

A 5-cm radius conducting sphere has a surface charge density of 2 ×10−6 C/m2 on its surface. Its electric potential, relative to the potential far away, is:

- **1.1 × 104 V**
- 2.2 × 104 V
- 2.3 × 105 V
- 3.6 × 105 V

A hollow metal sphere is charged to a potential V . The potential at its center is:

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► **V**

► 0

► −V

► 2V

Positive charge is distributed uniformly throughout a non-conducting sphere. The highest electric potential occurs:

► **at the center**

► at the surface

► halfway between the center and surface

► just outside the surface

A total charge of 7×10−8 C is uniformly distributed throughout a non-conducting sphere with a radius of 5 cm. The electric potential at the surface, relative to the potential far away, is about:

► −1.3 × 104 V

► **1.3 × 104 V**

► 7.0 × 105 V

► −6.3 × 104 V

Eight identical spherical raindrops are each at a potential V , relative to the potential far away. They coalesce to make one spherical raindrop whose potential is:

► V/8

► V/2

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► 2V

► **4V**

A metal sphere carries a charge of 5 × 10−9 C and is at a potential of 400V, relative to the potential far away. The potential at the center of the sphere is:

► **400V**

► −400V

► 2 × 10−6 V

► 0

A 5-cm radius isolated conducting sphere is charged so its potential is +100V, relative to the potential far away. The charge density on its surface is:

► +2.2 × 10−7 C/m2

► −2.2 × 10−7 C/m2

► +3.5 × 10−7 C/m2

► **+1.8 × 10−8 C/m2**

A conducting sphere has charge Q and its electric potential is V , relative to the potential far away. If the charge is doubled to 2Q, the potential is:

► V

► **2V**

► 4V

► V/4

The potential difference between the ends of a 2-meter stick that is parallel to a uniform electric field is 400V. The magnitude of the electric field is:

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- zero
- 100V/m
- 200V/m
- **800V/m**

In a certain region of space the electric potential increases uniformly from east to west and does not vary in any other direction. The electric field:

- points east and varies with position
- **points east and does not vary with position**
- points west and varies with position
- points west and does not vary with position

If the electric field is in the positive x direction and has a magnitude given by E = Cx2, where C is a constant, then the electric potential is given by V =:

- 2Cx
- −2Cx
- Cx3/3
- **−Cx3/3**

The work required to carry a particle with a charge of 6.0C from a 5.0-V equipotential surface to a 6.0-V equipotential surface and back again to the 5.0-V surface is:

- **0**
- 1.2 × 10−5 J
- 3.0 × 10−5 J

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- 6.0 × 10−5 J

The equipotential surfaces associated with a charged point particles are:

- radially outward from the particle
- vertical planes
- horizontal planes
- **concentric spheres centered at the particle**

The electric field in a region around the origin is given by E = C(xˆi + yˆj), where C is a constant. The equipotential surfaces in that region are:

- **concentric cylinders with axes along the z axis**
- concentric cylinders with axes along the x axis
- concentric spheres centered at the origin
- planes parallel to the xy plane

A particle with charge q is to be brought from far away to a point near an electric dipole. No work is done if the final position of the particle is on:

- the line through the charges of the dipole
- **a line that is perpendicular to the dipole moment**
- a line that makes an angle of 45◦ with the dipole moment
- a line that makes an angle of 30◦ with the dipole moment

Equipotential surfaces associated with an electric dipole are:

- spheres centered on the dipole
- cylinders with axes along the dipole moment
- planes perpendicular to the dipole moment

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► **none of the above**

The units of capacitance are equivalent to:

► J/C

► V/C

► J2/C

► **C2/J**

A farad is the same as a:

► J/V

► V/J

► **C/V**

► V/C

A capacitor C "has a charge Q". The actual charges on its plates are:

► Q, Q

► Q/2, Q/2

► **Q, –Q**

► Q/2, –Q/2

Each plate of a capacitor stores a charge of magnitude 1mC when a 100-V potential difference is applied. The capacitance is:

► 5 µF

► **10 µF**

► 50 µF

► 100 µF

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To charge a 1-F capacitor with 2C requires a potential difference of:

- **2V**
- 0.2V
- 5V
- 0.5V

The capacitance of a parallel-plate capacitor with plate area A and plate separation d is given by:

- 0d/A
- 0d/2A
- **0A/d**
- 0A/2d

The capacitance of a parallel-plate capacitor is:

- **proportional to the plate area**
- proportional to the charge stored
- independent of any material inserted between the plates
- proportional to the potential difference of the plates

The capacitance of a parallel-plate capacitor can be increased by:

- increasing the charge
- decreasing the charge
- increasing the plate separation
- **decreasing the plate separation**

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If both the plate area and the plate separation of a parallel-plate capacitor are doubled, the capacitance is:

- doubled
- halved
- **unchanged**
- tripled

If the plate area of an isolated charged parallel-plate capacitor is doubled:

- the electric field is doubled
- **the potential difference is halved**
- the charge on each plate is halved
- the surface charge density on each plate is doubled

If the plate separation of an isolated charged parallel-plate capacitor is doubled:

- the electric field is doubled
- the potential difference is halved
- the charge on each plate is halved
- **none of the above**

Pulling the plates of an isolated charged capacitor apart:

- increases the capacitance
- **increases the potential difference**
- does not affect the potential difference
- decreases the potential difference

If the charge on a parallel-plate capacitor is doubled:

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- the capacitance is halved
- the capacitance is doubled
- the electric field is halved
- **the electric field is doubled**

A parallel-plate capacitor has a plate area of 0.2m2 and a plate separation of 0.1mm. To obtain an electric field of 2.0 × 106 V/m between the plates, the magnitude of the charge on each plate should be:

- 8.9 × 10−7 C
- 1.8 × 10−6 C
- 3.5 × 10−6 C
- **7.1 × 10−6 C**

A parallel-plate capacitor has a plate area of 0.2m2 and a plate separation of 0.1 mm. If the charge on each plate has a magnitude of 4 × 10−6 C the potential difference across the plates is approximately:

- 0
- 4 × 10−2 V
- 1 × 102 V
- **2 × 102 V**

The capacitance of a spherical capacitor with inner radius a and outer radius b is proportional to:

- a/b
- b − a

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- b2 – a2
- **ab/(b – a)**

The capacitance of a single isolated spherical conductor with radius R is proportional to:

- **R**
- R2
- 1/R
- 1/R2

The capacitance of a cylindrical capacitor can be increased by:

- decreasing both the radius of the inner cylinder and the length
- **increasing both the radius of the inner cylinder and the length**
- increasing the radius of the outer cylindrical shell and decreasing the length
- only by decreasing the length

A battery is used to charge a series combination of two identical capacitors If the potential difference across the battery terminals is V and total charge Q flows through the battery during the charging process then the charge on the positive plate of each capacitor and the potential difference across each capacitor are:

- Q/2 and V/2, respectively
- Q and V , respectively
- Q/2 and V , respectively

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► **Q and V/2, respectively**

A battery is used to charge a parallel combination of two identical capacitors. If the potential difference across the battery terminals is V and total charge Q flows through the battery during the charging process then the charge on the positive plate of each capacitor and the potential difference across each capacitor are:

► Q/2 and V/2, respectively

► Q and V , respectively

► **Q/2 and V , respectively**

► Q and V/2, respectively

A 2-µF and a 1-µF capacitor are connected in series and a potential difference is applied across the combination. The 2-µF capacitor has:

► twice the charge of the 1-µF capacitor

► half the charge of the 1-µF capacitor

► twice the potential difference of the 1-µF capacitor

► **half the potential difference of the 1-µF capacitor**

A 2-µF and a 1-µF capacitor are connected in parallel and a potential difference is applied across the combination. The 2-µF capacitor has:

► **twice the charge of the 1-µF capacitor**

► half the charge of the 1-µF capacitor

► twice the potential difference of the 1-µF capacitor

► half the potential difference of the 1-µF capacitor

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Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in series must have the same:

- **Q only**
- V only
- U only
- Q and U only

Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in parallel must have the same:

- Q only
- **V only**
- U only
- Q and U only

Capacitors C1 and C2 are connected in parallel. The equivalent capacitance is given by:

- C1C2/(C1 + C2)
- (C1 + C2)/C1C2
- C1/C2
- **C1 + C2**

Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by:

- **C1C2/(C1 + C2)**
- (C1 + C2)/C1C2

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- 1/(C1 + C2)
- C1/C2

Capacitors C1 and C2 are connected in series and a potential difference is applied to the combination. If the capacitor that is equivalent to the combination has the same potential difference, then the charge on the equivalent capacitor is the same as:

- **the charge on C1**
- the sum of the charges on C1 and C2
- the difference of the charges on C1 and C2
- the product of the charges on C1 and C2

Questions:

1. **Which weighs more, a liter of ice or a liter of water?**
2. **Will the current in a light bulb connected to a 220-V source be greater or less than when the same bulb is connected to 110-V source?**
3. **How is the wavelength of light related to its frequency?**
4. **We don't notice the de Broglie wavelength for a pitched baseball. Is this Because the wavelength is very large or because it is very small?**
5. **Does every magnet necessarily have a north and south pole? Explain**
6. **In a cool room, a metal or marble table top feels much colder to the touch than does a wood surface even though they are at the same temperature. Why?**

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7. If a water wave oscillates up and down three times each second and the distance between wave crests is 2 m, what is its frequency? What is its wavelength? What is its wave speed?
8. A transformer has N1 = 350 turns and N2 = 2 000 turns. If the input voltage is coil?
9. Why do astronomers looking at distant galaxies talk about looking backward in time?
10. Some distant astronomical objects, called quasars, are receding from us at half the speed of light (or greater). What is the speed of the light we receive from these quasars?
11. Consider a lamp hanging from a chain. What is the tension in the chain?
12. Aproton travels with a speed of 3.00 * 106 m/s at an angle of 37.0° with the direction of a magnetic field of 0.300 T in the + y direction. What are (a) the magnitude of the magnetic force on the proton and (b) its acceleration?
13. Light from the Sun takes approximately 8.3 min to reach the Earth. During this time interval the Earth has continued to rotate on its axis. How far is the actual direction of the Sun from its image in the sky?
14. Do all current-carrying conductors emit electromagnetic waves? Explain
15. Explain solar convection zone. What is its other name?
16. If a charged particle moves in a straight line through some region of space, can you say that the magnetic field in that region is zero?

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17. Can all gas molecules in the vessel have the same speed?
18. What are the properties of wave function?
19. A bike accelerates uniformly from rest to a speed of 7.10 m/s over a distance of 35.4 m. Determine the acceleration of the bike.
20. A flat loop of wire consisting of a single turn of cross-sectional area 8.00 cm2 is perpendicular to a magnetic field that increases uniformly in magnitude from 0.500 T to 2.50 T in 1.00 s. What is the resulting induced current if the loop has a resistance of 2.00 W?
21. An ideal gas is contained in a vessel at 300 K. If the temperature is increased to 900 K, by what factor does each one of the following change?
    (a) The average kinetic energy of the molecules.
    (b) The rms molecular speed.
    (c) The average momentum change of one molecule in a collision with a wall.
    (d) The rate of collisions of molecules with walls.
    (e) The pressure of the gas.
22. Who discover the nucleus? Write the experimental setup that he follows.
23. In an analogy between electric current and automobile traffic flow, what would correspond to charge? What would correspond to current?
24. (a)When can you expect a body to emit blackbody radiation?
    (b) Which law is obeyed by Sun and other stars, briefly explain it.

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25. Two people are carrying a uniform wooden board that is 3.00 m long and weighs 160 N. If one person applies an upward force equal to 60 N at one end, at what point does the other person lift? Begin with a free-body diagram of the board.
26. If a charged particle moves in a straight line through some region of space, can you say that the magnetic field in that region is zero?
27. You want to explore the shape of a cel1ain molecule by scattering electrons of momentum p from a gas of the molecules and studying the deflection of the electrons.
28. You will be able to see finer details in the molecules by (a) increasing p; (b) decreasing p: (c) not worrying what p is.
29. A vessel is filled with gas at some equilibrium pressure and temperature. Can all gas molecules in the vessel have the same speed?
30. What are the properties of wave function?
31. A bike accelerates uniformly from rest to a speed of 7.10 m/s over a distance of 35.4 m. Determine the acceleration of the bike.
32. Who discover the nucleus? Write the experimental setup that he follows.

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FINAL TERM MCQS

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1. The units of the Planck constant h are those of:

A. energy

B. power

C. momentum

D. angular momentum

E. frequency

2. Which of the following is true of all virtual images?

A. They can be seen but not photographed

B. They are ephemeral

C. They are smaller than the objects

D. They are larger than the objects

E. None of the above

ans: E

3. In constructing a thermometer it is NECESSARY to use a substance that:

A. expands with rising temperature

B. expands linearly with rising temperature

C. will not freeze

D. will not boil

E. undergoes some change when heated or cooled

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4. A 60-watt light bulb carries a current of 0.5 A. The total charge passing through it in one hour is:

A. 120 C

B. 3600 C

C. 3000 C

D. 2400 C

E. 1800 C

5. If the charge on a parallel-plate capacitor is doubled:

A. the capacitance is halved

B. the capacitance is doubled

C. the electric field is halved

D. the electric field is doubled

6. the wavelength of red light is 700 nm. Its frequency is: 4.29 10^14

A. $4.30*10^4$ Hertz 4.29 10^14 4.29 10^14 SEE PGE NO.92

B. $4.30*10^5$ Hertz NOT SURE

C. $4.30*10^3$ Hertz

D. $4.30*10^2$ Hertz

7. A capacitor C "has a charge Q". The actual charges on its plates are:

A. Q, Q

B. Q/2, Q/2

C. Q , −Q

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D. Q/2, −Q/2

8. How fast should you move away from a 6.0×10^{14} Hz light source to observe waves with a frequency of 4.0×10^{14} Hz?

A. 0.20c

B. 0.38c

C. 0.45c

D. 0.51c

E. 0.76c

9. The units of the electric field are:

A. J/(C·m)

B. J/C

C. J·C

D. J/m

10. If two light waves are coherent:

A. their amplitudes are the same
B. their frequencies are the same
C. their wavelengths are the same
D. their phase difference is constant

11. In a photoelectric effect experiment at a frequency above cut off, the stopping potential is proportional to:

A. the energy of the least energetic electron before it is ejected

B. the energy of the least energetic electron after it is ejected

C. the energy of the most energetic electron before it is ejected

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D. the energy of the most energetic electron after it is ejected

12. The total energy in an LC circuit is 5.0 × 10−6 J. If L = 25 mH the maximum current is:

A. 10 mA

B. 14 mA

C. 20 mA

D. 28 mA

13. The quantum number ms is most closely associated with what property of the electron in an atom?

A. Magnitude of the orbital angular momentum

B. Energy

C. z component of the spin angular momentum

D. z component of the orbital angular momentum

14. If 500 J of work are required to carry a charged particle between two points with a potential difference of 20 V, the magnitude of the charge on the particle is:

A. 0.040 C

B. 12.5 C

C. 20 C

D. cannot be computed unless the path is given

15. The smallest particle of any chemical element that can exist by itself and yet retain the qualities that distinguish it as that element is

: A. an electron

B. a proton

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C. a neutron

D. an atom

E. a molecule

16. A particle with a charge of 5.5×10−8 C is 3.5 cm from a particle with a charge of −2.3×10−8 C. The potential energy of this two-particle system, relative to the potential energy at infinite separation, is:

A. 3.2 × 10−4 J

B. −3.2 × 10−4 J

C. 9.3 × 10−3 J

D. −9.3 × 10−3 J

E. zero

17. A farad is the same as a:

A. J/V

B. V/J

C. C/V

D. V/C

E. N/C

18. 1 weber is the same as:

A. 1 V/s

B. 1 T/s

C. 1 T/m

D. 1 T · m2

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19. coherent sources of light mean

A. They must have same speeds

B. They must have same frequency

C. They must have same amplitude

D. They must have fixed phase relative to each other
not sure

20. Room temperature is about 20 degrees on the:

A. Kelvin scale

B. Celsius scale

C. Fahrenheit scale

D. absolute scale

E. C major scale

21. The appearance of color in the thin film is due to:

- Interference

22. When you stand in front of a plane mirror, your image is:

A. real, erect, and smaller than you

B. real, erect, and the same size as you

C. virtual, erect, and smaller than you

D. virtual, erect, and the same size as you

23. In a photoelectric effect experiment at a frequency above cut off, the number of electrons ejected is proportional to:

A. their kinetic energy

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B. their potential energy

C. the work function

D. the frequency of the incident light

E. the number of photons that hit the sample

24. In a photoelectric effect experiment at a frequency above cut off, the stopping potential is proportional to:

A. the energy of the least energetic electron before it is ejected

B. the energy of the least energetic electron after it is ejected

C. the energy of the most energetic electron before it is ejected

D. the energy of the most energetic electron after it is ejected

E. the electron potential energy at the surface of the sample

25. The units of magnetic dipole moment are:

A. ampere

B. ampere·meter

C. ampere·meter2

D. ampere/meter

26. The primary of a 3 : 1 step-up transformer is connected to a source and the secondary is connected to a resistor R. The power dissipated by R in this situation is P. If R is connected directly to the source it will dissipate a power of:

A. P/9

B. P/3

C. P

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D. 3P

E. 9P

27. We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5 H. We also need a capacitance of:

A. 1 F

B. 1 mF

C. 1 μF

D. 100 μF

28. An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

A. a change in entropy

B. an increase of pressure

C. a change in temperature

D. a decrease of internal energy

30. A step-down transformer is used to:

A. increase the power

B. decrease the power

C. increase the voltage

D. decrease the voltage

31. Which of the following electromagnetic radiations has photons with the greatest energy?

A. blue light

B. yellow light

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C. x rays

D. radio waves

33. The Stern-Gerlach experiment makes use of:

A. a strong uniform magnetic field

B. a strong non-uniform magnetic field

C. a strong uniform electric field

D. a strong non-uniform electric field

34. Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in series must have the same:

A. Q only

B. V only

C. U only

D. Q and U only

35. Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in parallel must have the same:

A. Q only

B. V only

C. U only

D. Q and U only

36. . “The sum of the currents into a junction equals the sum of the currents out of the junction” is a consequence of:

A. Newton’s third law

B. Ohm’s law

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C. Newton's second law

D. conservation of energy

E. conservation of charge

ans: E

37. "The sum of the emf's and potential differences around a closed loop equals zero" is a consequence of:

A. Newton's third law

B. Ohm's law

C. Newton's second law

D. conservation of energy

E. conservation of charge

38. Evidence for the wave nature of matter is:

A. electron diffraction experiments of Davisson and Germer

B. Thompson's measurement of e/m

C. Young's double slit experiment

D. the Compton effect

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39. A conducting sphere with radius R is charged until the magnitude of the electric field just outside its surface is E. The electric potential of the sphere, relative to the potential far away, is:

A. zero

B. E/R

C. E/R2

D. ER

40. radio waves and light waves are ;

- Electromagnetic and transverse

41. An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

A. a change in entropy

B. an increase of pressure

C. a change in temperature

D. a decrease of internal energy

42. Capacitors A and B are identical. Capacitor A is charged so it stores 4 J of energy and capacitor B is uncharged. The capacitors are then connected in parallel. The total stored energy in the capacitors is now:

A. 16 J

B. 8 J

C. 4 J

D. 2 J

43. Capacitors C1 and C2 are connected in parallel. The equivalent capacitance is given by:

A. C1C2/(C1 + C2)

B. (C1 + C2)/C1C2

C. 1/(C1 + C2)

D. C1/C2

E. C1 + C2

ans: E

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44. Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by:

A. C1C2/(C1 + C2)

B. (C1 + C2)/C1C2

C. 1/(C1 + C2)

D. C1/C2

E. C1 + C2

45. During an adiabatic process an object does 100 J of work and its temperature decreases by 5 K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is:

A. 20 J/K

B. 24 J/K

C. 5 J/K

D. 15 J/K

46. A virtual image is one:

A. toward which light rays converge but do not pass through

B. from which light rays diverge but do not pass through

C. from which light rays diverge as they pass through

D. toward which light rays converge and pass through

47. A generator supplies 100 V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

A. 1000 V

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B. 500 V

C. 250 V

D. 100 V

48. The rate at which electrical energy is used may be measured in:

A. watt/second

B. watt·second

C. watt

D. joule·second

QUESTION: A vessel is filled with gas at some equilibrium pressure and temperature. Can all gas molecules in the vessel have same speed?

Answer: They can, as this possibility is not contradicted by any of our descriptions of the motion of gases. If the vessel contains more than a few molecules, it is highly improbable that all will have the same speed. Collisions will make their speeds scatter according to the Boltzmann distribution law."

QUESTION: Electric potential energy:

Answer:

The electric potential is the energy of a unit charge in an electric field. So in our MKS units the unit of potential is 1Joule/Coulombs=1Volt.Another useful unit is 1Electron Volt.

QUESTION: All objects radiate energy then why are we not able to see all the objects in the dark? Our eyes are not sensitive to infrared radiation so we do not see it. Cold bodies radiate mostly in the infrared zone (invisible to the human eye), but as the temperature increases the body will emmit higher frequencies with more intensity. So room temperature objects will not be seen due to black body radiation.

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QUESTION: In a cool room, metal and marble is cool but wood is not col when we touch. Why?

Answer: Because metal and stone conduct heat better than wood. Skin temperature is higher than room temperature, so when we touch metal, we give heat more quickly to the metal than wood. Another reason is that metal and stone store more energy than a like sized object of wood

QUESTION: what causes electric shock or voltage?

ANS: one clock is upstairs of the bedroom and second clock is in downstairs of the kitchen. Then which clock runs slowly?
Really, neither will run appreciably slower than the other. In the strictest sense, the clock in the upstairs bedroom is moving a few centimeters per year faster than the clock downstairs in the kitchen, because it is a few feet further from the Earth's center than the clock in the upstairs.

QUESTION: Difference between reflection and refraction?
ANS:

Reflection involves a change in direction of waves when they bounce off a barrier. Refraction of waves involves a change in the direction of waves as they pass from one medium to another. Refraction, or the bending of the path of the waves, is accompanied by a change in speed and wavelength of the waves.

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Question:

If an electron and proton have same De Broglie's wavelength, which particle has greater speed

The de Broglie wavelength of a particle is inversely proportional to its momentum p = m v; since a proton is about 1800 times more massive than an electron, its momentum at the same speed is 1800 times that of an electron, and therefore its wavelength 1800 times smaller. The electron has the longer wavelength.

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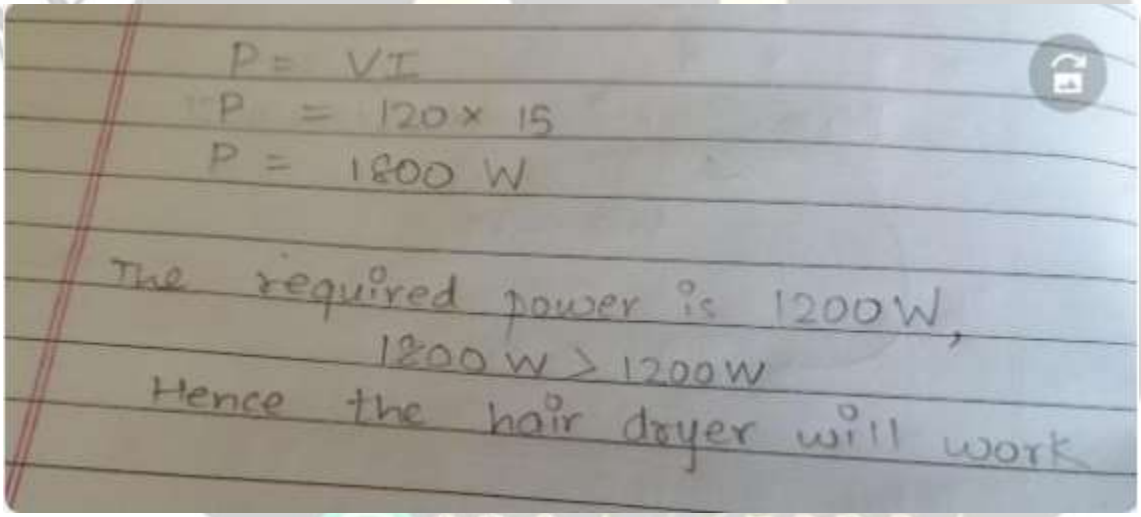
QUESTION: black and black body radiation?

ANS:

An object that absorbs all radiation falling on it, at all wavelengths, is called a black body. When a black body is at a uniform temperature, its emission has a characteristic frequency distribution that depends on the temperature. Its emission is called black-body radiation.

For examples include the heating element of a toaster and the filament of a light bulb.

QUESTION: If a 120 line to shocked is limited 15a by a safety fuse will it operate a 1200w hair dryer?



QUESTION: Define statically mechanics

Ans: - The study of heat, considered as arising from the random motion of the basic constituents of matter, is an area of physics called statistical mechanics.

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QUESTION: is it true to say that Centre of mass and Centre of graving are the same? Either yes or no explain in each case.

Answer: Many people assume that the terms "Centre of mass" and "Centre of gravity" are synonymous, but this is not the case. The center of mass and the Centre of gravity of an object are in the same position if the gravitational field in which the object exists is uniform.

QUESTION: What is the work done? If Force is 500N, height of downwards cliff is 30m Email and angle with land is 30 degrees.?

QUESTION: Magnetic Flux and Magnetic Flux density and when Flux is maximum and when it is zero?

ANS:

Because the induced EMF is proportional to the time rate of change of the magnetic flux, not to the magnetic flux itself, as shown in Faraday's law. In a sinusoidal wave, the maximum rate of change occurs at the zero crossing; in that point, the slope of the line tangent to the wave is the greatest, so the rate of change is the greatest (recall the derivative of a invariable function gives the slope of such function at each point). At the maximum or minimum value, the slope of the tangential line is zero, since it's a horizontal line, so the rate of change is zero.

The magnetic flux as a function of time is

$\Phi(t)=B\cdot A(t)=BA\cos(wt)$

$\Phi(t)=B\cdot A(t)=BA\cos(\omega t)\Phi(t)=B\cdot A(t)=BA\cos\ (\omega t)$
BB =the magnetic field and
A(t)A(t) =the area vector as a function of time
ωtωt =the angle between the field and the area vector as a function of time.
Cos(90)=0

But

Then the rate of change of the flux as a function of time is

$\Phi'(t) = -BA\omega\sin(\omega t)$

$\Phi'(t) = -BA\omega\sin(\omega t) \Phi'(t) = -BA\omega\sin(\omega t)$
the angle between(the area vector vector and the magnetic field) =zero.

flux =nonzero and equal to AB, its maximum,but the rate of change of the flux vanishes because

Sin(0)=0,sin(90)=1.

QUESTION: How Isotopes change Atomic Mass, Magnetic spin moment and Chemical behavior of Elements

ANS:

Isotopes are atoms that have the same number of protons and electrons, but a different number of neutrons. Changing the number of neutrons in an atom does not change the element. Atoms of elements with different numbers of neutrons are called "isotopes" of that element. Isotopes behave similarly to other isotopes of the same element in chemical reactions because such reactions don't depend on the properties of atomic nuclei. Rather they depend on the number and arrangement of an atom's electrons.

QUESTION: Similarity and difference of gland meter and electric motor

ANS:

The motor requires Electricity, but the generator produces electricity. The shaft of the motor is driven by the magnetic force developed between the armature and the field winding, while in the case of the generator the shaft is connected to the rotor and driven by mechanical force

QUESTION: Fundamental source of Electromagnetic radiation

ANS:

A moving charge particle is the fundamental source of electromagnetic radiation.

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QUESTION: A photon of green light has a wavelength of 530nm. Find the photos frequency, magnitude of momentum and electron volts. Express the energy in joles and electron volts.

ANS:

TAKE IDEA FROM THIS LINK

https://socratic.org/questions/calculate-the-energy-in-joules-of-a-photon-of-green-light-having-a-wavelength-of

https://www.numerade.com/questions/a-photon-of-green-light-has-a-wavelength-of-520-nm-find-the-photons-frequency-magnitude-of-momentum-/

QUESTION: Find the difference between two waves which interfere constructively and destructively.

ANS:

There are two types of interference, constructive and destructive. In constructive interference, the amplitudes of the two waves add together resulting in a higher wave at the point they meet. In destructive interference, the two waves cancel out resulting in a lower amplitude at the point they meet.

QUESTION: Similarity and difference of glvano meter and electric motor

ANS:

Galvanometers and electric motors are both based on the same basic principle of **an external magnetic field exerting** a force on a current carrying wire. The difference is in how far the device is allowed to move. Motors must be able to move freely and rotate through complete 360o turns.

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QUESTION:

A 2000kg car is moving with a velocity of 20 m/s collides and locks with a 1500 kg car at rest a stop sign. Show the momentum is conserved.

A 2000-kg car moving with a speed of 20 m/s collides with and sticks to a 1500-kg car at rest at a stop sign. Show that because momentum is conserved in the rest frame, momentum is also conserved in a reference frame moving with a speed of 10 m/s in the direction of the moving car

QUESTION:

If you hold water in a paper cup over flame, can you bring the water to boil, if then how?

Just fill a **paper cup** with **water** and start trying to burn it from the outside. The **water will boil** vigorously inside the **cup**, but the **paper** itself **wills** not combust. ... The **boiling water can**'t get **above** 212 degrees, and the **paper** is thin enough that this **water** keeps its temperature from climbing high enough to combust.

Question: What is Corona?

Ans:

A corona discharge is **an electrical discharge caused by the ionization of a fluid** such as air surrounding a conductor carrying a high voltage.A corona occurs at locations where the strength of the electric field (potential gradient) around a conductor exceeds the dielectric strength of the air.

Question: Greenhouse effect explain.

Ans:

The greenhouse effect is a natural process that warms the Earth's surface. When the Sun's energy reaches the Earth's atmosphere, some of it is

reflected back to space and the rest is absorbed and re-radiated by greenhouse gases. The absorbed energy warms the atmosphere and the surface of the Earth

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PHY101 FINAL TERM SOLVED PAPER

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Question No: 1 (Marks: 1) - Please choose one

The number of significant figures in 0.00150 is:

- ► 5
- ► 4
- ► 3
- ► 2

Question No: 2 (Marks: 1) - Please choose one

One revolution is the same as:

2π rad

- ► 1 rad
- ► 57 rad
- ► π/2 rad
- ► π rad
- ► 2π rad

Question No: 3 (Marks: 1) - Please choose one

For a body to be in equilibrium under the combined action of several forces:

- ► all the forces must be applied at the same point

all the forces must be applied at the same point

- ► all of the forces form pairs of equal and opposite forces
- ► any two of these forces must be balanced by a third force
- ► the sum of the torques about any point must equal zero

Question No: 4 (Marks: 1) - Please choose one

A bucket of water is pushed from left to right with increasing speed across a horizontal surface. Consider the pressure at two points at the same level in the water.

- ► It is the same

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- It is higher at the point on the left
- It is higher at the point on the right
- At first it is higher at the point on the left but as the bucket speeds up it is lower there

Question No: 5 (Marks: 1) - Please choose one

An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

- A. 200 Hz
- B. 300 Hz
- C. 400 Hz
- D. 600 Hz

Question No: 6 (Marks: 1) - Please choose one

Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by

- $C_1C_2/(C_1 + C_2)$
- $(C_1 + C_2)/C_1C_2$
- $1/(C_1 + C_2)$
- C_1/C_2

Question No: 7 (Marks: 1) - Please choose one

If the potential difference across a resistor is doubled:

- only the current is doubled
- only the current is halved
- only the resistance is doubled
- only the resistance is halved

Question No: 8 (Marks: 1) - Please choose one

By using only two resistors, R1 and R2, a student is able to obtain resistances of 3 Ω, 4Ω, 12Ω, and 16 Ω. The values of R1 and R2 (in ohms) are:

- 3, 4
- 2, 12
- 3, 16
- 4, 12

Question No: 9 (Marks: 1) - Please choose one

Faraday's law states that an induced emf is proportional to:

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► the rate of change of the electric field
► the rate of change of the magnetic flux
► the rate of change of the electric flux

► the rate of change of the magnetic field

Question No: 10 (Marks: 1) - Please choose one

A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

► 1000V
► 500V
► 250V
► 100V

Question No: 11 (Marks: 1) - Please choose one

The wavelength of red light is 700 nm. Its frequency is 4.29 10^14 .

4.29 10^14 SEE PGE NO.92

► 4.30 * 10^4 Hertz
► 4.30 * 10^3 Hertz
► 4.30 * 10^5 Hertz
► 4.30 * 10^2 Hertz

4.29 10^14

Question No: 12 (Marks: 1) - Please choose one

In some movies, you sometimes see an actor looking in a mirror and you can see his face in the mirror. During the filming of this scene, what does the actor see in the mirror?

► His face

► Your face

► The movie camera

► The director's face

Question No: 13 (Marks: 1) - Please choose one

A laser in a compact disc player generates light that has a wavelength of 780 nm in air. The light then enters into the plastic of a CD. If the index of refraction of plastic is 1.55, the speed of this light once enter the plastic is_____.

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► $3.00 * 10^8$ m/s
► $1.94 * 10^8$ m/s
► $4.29 * 10^8$ km/h
► $3.00 * 10^8$ km/h

Question No: 14 (Marks: 1) - Please choose one

Which of the following electromagnetic radiations has photons with the greatest energy?

► blue light
► yellow light
► x rays
► radio waves

Question No: 15 (Marks: 1) - Please choose one

A virtual image is one:

► toward which light rays converge but do not pass through
► from which light rays diverge as they pass through
► toward which light rays converge and pass through
► from which light rays diverge but do not pass through

Question No: 16 (Marks: 1) - Please choose one

What is the unit of magnification factor?

► meter.Kelvin
► radian.Kelvin
► degree.Kelvin
► no units

Question No: 17 (Marks: 1) - Please choose one

During an adiabatic process an object does 100 J of work and its temperature decreases by 5K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is.

► 20 J/K

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► 100 J/K
► 15 J/K
► 5 J/K

Question No: 18 (Marks: 1) - Please choose one

An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

► a change in entropy
► a decrease of internal energy
► an increase of pressure
► a change in temperature

Question No: 19 (Marks: 1) - Please choose one

The Stern-Gerlach experiment makes use of:

► a strong uniform magnetic field
► a strong non-uniform magnetic field
► a strong uniform electric field
► a strong non-uniform electric field

Question No: 20 (Marks: 1) - Please choose one

A large collection of nuclei are undergoing alpha decay. The rate of decay at any instant is proportional to:

► the number of undecayed nuclei present at that instant
► the time since the decays started
► the time remaining before all have decayed
► the half-life of the decay

Question No: 21 (Marks: 1)

Which weighs more, a liter of ice or a liter of water?

Question No: 22 (Marks: 1)

Will the current in a light bulb connected to a 220-V source be greater or less than when the same bulb is connected to 110-V source?

Question No: 23 (Marks: 1)

How is the wavelength of light related to its frequency?

Question No: 24 (Marks: 1)

We don't notice the de Broglie wavelength for a pitched baseball. Is this because

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the wavelength is very large or because it is very small?

Question No: 25 (Marks: 2)

Does every magnet necessarily have a north and south pole? Explain

Question No: 26 (Marks: 2)

In a cool room, a metal or marble table top feels much colder to the touch than does a wood surface even though they are at the same temperature. Why?

Question No: 27 (Marks: 3)

If a water wave oscillates up and down three times each second and the distance between wave crests is 2 m, what is its frequency? What is its wavelength? What is its wave speed?

Question No: 28 (Marks: 3)

A transformer has N_1 = 350 turns and N_2 = 2 000 turns. If the input voltage is □ v(t) = (170 V) cos □□ t, what rms voltage is developed across the secondary coil?

Question No: 29 (Marks: 3)

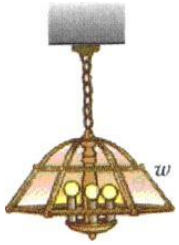
Why do astronomers looking at distant galaxies talk about looking backward in time?

Question No: 30 (Marks: 3)

Some distant astronomical objects, called quasars, are receding from us at half the speed of light (or greater). What is the speed of the light we receive from these quasars?

Question No: 31 (Marks: 5)

Consider a lamp hanging from a chain. What is the tension in the chain?



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Question No: 32 (Marks: 5)

A proton travels with a speed of 3.00 * 10^6 m/s at an angle of 37.0° with the direction of a magnetic field of 0.300 T in the + y direction. What are (a) the magnitude of the magnetic force on the proton and (b) its acceleration?

Question No: 33 (Marks: 5)

1. Light from the Sun takes approximately 8.3 min to reach the Earth. During this time interval the Earth has continued to rotate on its axis. How far is the actual direction of the Sun from its image in the sky?
2. Do all current-carrying conductors emit electromagnetic waves? Explain

2. Yes all current carrying conductors emit electromagnetic waves, and these are at the right angle of the current passes thorough as right hand rule of Fleming's explains it.

Question No: 34 (Marks: 5)

Explain solar convection zone. What is its other name?

Question No: 35 (Marks: 10)

a) Explain why you can't just open your refrigerator to cool your kitchen on a hot day. Why is it that turning on a room air conditioner will cool down the room but opening a refrigerator door will not?

b) On a humid day, water vapor condenses on a cold surface. During condensation, the entropy of the water(a) in-creases, (b) remains constant, (c) decreases, (d) may decreases or remain unchanged. Give its reason.

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PHY101 Physics Final Term Solved MCQs Mega File (All in One)

1. A total charge of 6.3×10−8 C is distributed uniformly throughout a 2.7-cm radius sphere. The volume charge density is:
A. 3.7 × 10−7 C/m3
B. 6.9 × 10−6 C/m3
C. 6.9 × 10−6 C/m2
D. 2.5 × 10−4 C/m3
E. 7.6 × 10−4 C/m3
ans: E

2. Charge is placed on the surface of a 2.7-cm radius isolated conducting sphere. The surface charge density is uniform and has the value 6.9 × 10−6 C/m2. The total charge on the sphere is:
A. 5.6 × 10−10 C
B. 2.1 × 10−8 C
C. 4.7 × 10−8 C
D. 6.3 × 10−8 C
E. 9.5 × 10−3 C
ans: D



3. A spherical shell has an inner radius of 3.7 cm and an outer radius of 4.5 cm. If charge is distributed uniformly throughout the shell with a volume density of 6.1×10−4 C/m3 the total charge is:
A. 1.0 × 10−7 C
B. 1.3 × 10−7 C
C. 2.0 × 10−7 C
D. 2.3 × 10−7 C
E. 4.0 × 10−7 C
ans: A

4. A cylinder has a radius of 2.1 cm and a length of 8.8 cm. Total Charge.1×10−7 C is distributed uniformly throughout. The volume charge density is:
A. 5.3 × 10−5 C/m3
B. 5.3 × 10−5 C/m2
C. 8.5 × 10−4 C/m3
D. 5.0 × 10−3 C/m3
E. 6.3 × 10−2 C/m3
ans: D

5. When a piece of paper is held with one face perpendicular to a uniform electric field the flux through it is 25N · m2 /C. When the paper is turned 25◦ with respect to the field the flux through it is:
A. 0
B. 12N · m2/C
C. 21N · m2/C
D. 23N · m2/C
E. 25N · m2/C
ans: D

6. The flux of the electric field (24N/C)ˆi + (30N/C)ˆj + (16N/C)ˆk through a 2.0m2 portion of the yz plane is:

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A. 32N · m2 /C
B. 34N · m2 /C
C. 42N · m2 /C
D. 48N · m2 /C
E. 60N · m2 /C
ans: D

7. Consider Gauss's law: □ □ E · d□ A = q/□0. Which of the following is true?
A. □ E must be the electric field due to the enclosed charge
B. If q = 0, then □ E = 0 everywhere on the Gaussian surface
C. If the three particles inside have charges of +q, +q, and −2q, then the integral is zero
D. on the surface □ E is everywhere parallel to d□ A
E. If a charge is placed outside the surface, then it cannot affect □ E at any point on the surface
ans: C

8. A charged point particle is placed at the center of a spherical Gaussian surface. The electric flux ΦE is changed if:
A. the sphere is replaced by a cube of the same volume
B. the sphere is replaced by a cube of one-tenth the volume
C. the point charge is moved off center (but still inside the original sphere)
D. the point charge is moved to just outside the sphere
E. a second point charge is placed just outside the sphere
ans: D

9. Choose the INCORRECT statement:
A. Gauss' law can be derived from Coulomb's law
B. Gauss' law states that the net number of lines crossing any closed surface in an outward direction is proportional to the net charge enclosed within the surface
C. Coulomb's law can be derived from Gauss' law and symmetry
D. Gauss' law applies to a closed surface of any shape
E. According to Gauss' law, if a closed surface encloses no charge, then the electric field must vanish everywhere on the surface
ans: E



10. The outer surface of the cardboard center of a paper towel roll:
A. is a possible Gaussian surface
B. cannot be a Gaussian surface because it encloses no charge
C. cannot be a Gaussian surface since it is an insulator
D. cannot be a Gaussian surface because it is not a closed surface
E. none of the above
ans: D

11. A physics instructor in an anteroom charges an electrostatic generator to 25 μC, then carries it into the lecture hall. The net electric flux in N · m2/C through the lecture hall walls is:
A. 0
B. 25 × 10−6
C. 2.2 × 105
D. 2.8 × 106
E. can not tell unless the lecture hall dimensions are given
ans: D

12. A point particle with charge q is placed inside the cube but not at its center. The electric flux through any one side of the cube:
A. is zero
B. is q/□0
C. is q/4□0
D. is q/6□0
E. cannot be computed using Gauss' law
ans: E

13. A particle with charge 5.0-μC is placed at the corner of a cube. The total electric flux in N · m2 /C through all sides of the cube is:
A. 0
B. 7.1 × 104
C. 9.4 × 104

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D. 1.4 × 105
E. 5.6 × 105
ans: E

14. A point particle with charge q is at the center of a Gaussian surface in the form of a cube. The electric flux through any one face of the cube is:
A. q/□0
B. q/4π□0
C. q/3□0
D. q/6□0
E. q/12□0
ans: D

15. The table below gives the electric flux in N·m2/C through the ends and round surfaces of four Gaussian surfaces in the form of cylinders. Rank the cylinders according to the charge inside, from the most negative to the most positive. left end right end rounded surface
cylinder 1: +2 × 10−9 +4 × 10−9 −6 × 10−9
cylinder 2: +3 × 10−9 −2 × 10−9 +6 × 10−9
cylinder 3: −2 × 10−9 −5 × 10−9 +3 × 10−9
cylinder 4: +2 × 10−9 −5 × 10−9 −3 × 10−9

A. 1, 2, 3, 4
B. 4, 3, 2, 1
C. 3, 4, 2, 1
D. 3, 1, 4, 2
E. 4, 3, 1, 2
ans: E

16. A conducting sphere of radius 0.01m has a charge of 1.0 × 10−9 C deposited on it. The magnitude of the electric field in N/C just outside the surface of the sphere is:
A. 0
B. 450
C. 900
D. 4500
E. 90, 000
ans: C



17. A round wastepaper basket with a 0.15-m radius opening is in a uniform electric field of 300N/C, perpendicular to the opening. The total flux through the sides and bottom, in N · m2 C, is:
A. 0
B. 4.2
C. 21
D. 280
E. can not tell without knowing the areas of the sides and bottom
ans: C

18. 10C of charge are placed on a spherical conducting shell. A particle with a charge of −3C is placed at the center of the cavity. The net charge on the inner surface of the shell is:
A. −7C
B. −3C
C. 0C
D. +3C
E. +7C
ans: D

19. 10C of charge are placed on a spherical conducting shell. A particle with a charge of −3C is placed at the center of the cavity. The net charge on the outer surface of the shell is:
A. −7C
B. −3C
C. 0C
D. +3C
E. +7C
ans: E

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20. A 30-N/C uniform electric field points perpendicularly toward the left face of a large neutral conducting sheet. The surface charge density in C/m2 on the left and right faces, respectively, are:
A. −2.7 × 10−9 C/m2; +2.7 × 10−9 C/m2
B. +2.7 × 10−9 C/m2; −2.7 × 10−9 C/m2
C. −5.3 × 10−9 C/m2; +5.3 × 10−9 C/m2
D. +5.3 × 10−9 C/m2; −5.3 × 10−9 C/m2
E. 0; 0
ans: A

23. Charge Q is distributed uniformly throughout an insulating sphere of radius R. The magnitude of the electric field at a point R/2 from the center is:
A. Q/4π□0R2
B. Q/π□0R2
C. 3Q/4π□0R2
D. Q/8π□0R2
E. none of these
ans: D

24. Positive charge Q is distributed uniformly throughout an insulating sphere of radius R, centered at the origin. A particle with positive charge Q is placed at x = 2R on the x axis. The magnitude of the electric field at x = R/2 on the x axis is:
A. Q/4π□0R2
B. Q/8π□0R2
C. Q/72π□0R2
D. 17Q/72π□0R2
E. none of these
ans: C

25. Charge Q is distributed uniformly throughout a spherical insulating shell. The net electric flux in N · m2 /C through the inner surface of the shell is:
A. 0
B. Q/□0
C. 2Q/□0
D. Q/4π□0
E. Q/2π□0
ans: A



26. Charge Q is distributed uniformly throughout a spherical insulating shell. The net electric flux in N · m2 /C through the outer surface of the shell is:
A. 0
B. Q/□0
C. 2Q/□0
D. Q/4□0
E. Q/2π□0
ans: B

27. A 3.5-cm radius hemisphere contains a total charge of 6.6 × 10−7 C. The flux through the rounded portion of the surface is 9.8 × 104 N · m2 /C. The flux through the flat base is:
A. 0
B. +2.3 × 104 N · m2 /C
C. −2.3 × 104 N · m2 /C
D. −9.8 × 104 N · m2 /C
E. +9.8 × 104 N · m2 /C
ans: C

28. Charge is distributed uniformly along a long straight wire. The electric field 2 cm from the wire is 20N/C. The electric field 4 cm from the wire is:
A. 120N/C
B. 80N/C
C. 40N/C
D. 10N/C
E. 5N/C
ans: D

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29. Positive charge Q is placed on a conducting spherical shell with inner radius R1 and outer radius R2. A point charge q is placed at the center of the cavity. The magnitude of the electric field at a point outside the shell, a distance r from the center, is:
A. zero
B. Q/4π□0r2
C. q/4π□0r2
D. (q + Q)/4π□0r2
E. (q + Q)/4π□0(R2
1 − r2)
ans: D

30. Positive charge Q is placed on a conducting spherical shell with inner radius R1 and outer radius R2. A point charge q is placed at the center of the cavity. The magnitude of the electric field produced by the charge on the inner surface at a point in the interior of the conductor, a distance r from the center, is:
A. 0
B. Q/4 vπ□0R2 1
C. Q/4π□0R2 2
D. q/4π□0r2
E. Q/4π□0r2
ans: D

31. A long line of charge with λ□ charge per unit length runs along the cylindrical axis of a cylindrical shell which carries a charge per unit length of λc. The charge per unit length on the inner and outer surfaces of the shell, respectively are:
A. λ□ and λc
B. −λ□ and λc + λ□
C. −λ□ and λc − λc
D. λ□ + λc and λc − λ□
E. λ□ − λc and λc + λ□
ans: B

32. Charge is distributed uniformly on the surface of a large flat plate. The electric field 2 cm from the plate is 30N/C. The electric field 4 cm from the plate is:
A. 120N/C
B. 80N/C
C. 30N/C
D. 15N/C
E. 7.5N/C
ans: C

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33. A particle with charge Q is placed outside a large neutral conducting sheet. At any point in the interior of the sheet the electric field produced by charges on the surface is directed:
A. toward the surface
B. away from the surface
C. toward Q
D. away from Q
E. none of the above
ans: C

34. A hollow conductor is positively charged. A small uncharged metal ball is lowered by a silk thread through a small opening in the top of the conductor and allowed to touch its inner surface. After the ball is removed, it will have:
A. a positive charge
B. a negative charge
C. no appreciable charge
D. a charge whose sign depends on what part of the inner surface it touched
E. a charge whose sign depends on where the small hole is located in the conductor
ans: C

35. A spherical conducting shell has charge Q. A particle with charge q is placed at the center of the cavity. The charge on the inner surface of the shell and the charge on the outer surface of the shell, respectively, are:
A. 0, Q
B. q, Q − q
C. Q, 0

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D. −q, Q + q
E. −q, 0
ans: D

36. A particle with a charge of 5.5×10−8C is 3.5 cm from a particle with a charge of −2.3×10−8 C. The potential energy of this two-particle system, relative to the potential energy at infinite separation, is:
A. 3.2 × 10−4 J
B. −3.2 × 10−4 J
C. 9.3 × 10−3 J
D. −9.3 × 10−3 J
E. zero
ans: B

37. A particle with a charge of 5.5 × 10−8C is fixed at the origin. A particle with a charge of −2.3×10−8 C is moved from x = 3.5 cm on the x axis to y = 4.3 cm on the y axis. The change in potential energy of the two-particle system is:
A. 3.1 × 10−3 J
B. −3.1 × 10−3 J
C. 6.0 × 10−5 J
D. −6.0 × 10−5 J
E. 0
ans: C

38. A particle with a charge of 5.5 × 10−8 C charge is fixed at the origin. A particle with a charge of −2.3 × 10−8 C charge is moved from x = 3.5 cm on the x axis to y = 3.5 cm on the y axis. The change in the potential energy of the two-particle system is:
A. 3.2 × 10−4 J
B. −3.2 × 10−4 J
C. 9.3 × 10−3 J
D. −9.3 × 10−3 J
E. 0
ans: E

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39. Three particles lie on the x axis: particle 1, with a charge of 1×10−8 C is at x = 1 cm, particle 2, with a charge of 2 × 10−8 C, is at x = 2 cm, and particle 3, with a charge of −3 × 10−8 C, is at x = 3 cm. The potential energy of this arrangement, relative to the potential energy for infinite separation, is:
A. +4.9 × 10−4 J
B. −4.9 × 10−4 J
C. +8.5 × 10−4 J
D. −8.5 × 10−4 J
E. zero
ans: B

40. Two identical particles, each with charge q, are placed on the x axis, one at the origin and the other at x = 5 cm. A third particle, with charge −q, is placed on the x axis so the potential energy of the three-particle system is the same as the potential energy at infinite separation. Its x coordinate is:
A. 13 cm
B. 2.5 cm
C. 7.5 cm
D. 10 cm
E. −5 cm
ans: A

41. Choose the correct statement:
A. A proton tends to go from a region of low potential to a region of high potential
B. The potential of a negatively charged conductor must be negative
C. If □ E = 0 at a point P then V must be zero at P
D. If V = 0 at a point P then □ E must be zero at P
E. None of the above are correct
ans: E

42. If 500 J of work are required to carry a charged particle between two points with a potential difference of 20V, the magnitude of the charge on the particle is:
A. 0.040C

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B. 12.5C
C. 20C
D. cannot be computed unless the path is given
E. none of these
ans: B

43. The potential difference between two points is 100V. If a particle with a charge of 2C is transported from one of these points to the other, the magnitude of the work done is:
A. 200 J
B. 100 J
C. 50 J
D. 100 J
E. 2 J
ans: A

44. During a lightning discharge, 30C of charge move through a potential difference of 1.0×108 V in 2.0 × 10−2 s. The energy released by this lightning bolt is:
A. 1.5 × 1011 J
B. 3.0 × 109 J
C. 6.0 × 107 J
D. 3.3 × 106 J
E. 1500 J
ans: B

45. An electron is accelerated from rest through a potential difference V . Its final speed is proportional to:
A. V
B. V 2
C. √V
D. 1/V
E. 1/√V
ans: C

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46. Two large parallel conducting plates are separated by a distance d, placed in a vacuum, and connected to a source of potential difference V . An oxygen ion, with charge 2e, starts from rest on the surface of one plate and accelerates to the other. If e denotes the magnitude of the electron charge, the final kinetic energy of this ion is:
A. eV/2
B. eV/d
C. eV d
D. V d/e
E. 2eV
ans: E

47. An electron volt is :
A. the force acting on an electron in a field of 1N/C
B. the force required to move an electron 1 meter
C. the energy gained by an electron in moving through a potential difference of 1 volt
D. the energy needed to move an electron through 1 meter in any electric field
E. the work done when 1 coulomb of charge is moved through a potential difference of 1 volt.
ans: C

48. An electron has charge −e and mass me. A proton has charge e and mass 1840me. A "proton volt" is equal to:
A. 1 eV
B. 1840 eV
C. (1/1840) eV
D. √1840 eV
E. (1/√1840) eV
ans: A

49. Two conducting spheres are far apart. The smaller sphere carries a total charge Q. The larger sphere has a radius that is twice that of the smaller and is neutral. After the two spheres are connected by a conducting wire, the charges on the smaller and larger spheres, respectively, are:
A. Q/2 and Q/2
B. Q/3 and 2Q/3

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C. 2Q/3 and Q/3
D. zero and Q
E. 2Q and −Q
ans: B

50. A conducting sphere with radius R is charged until the magnitude of the electric field just outside its surface is E. The electric potential of the sphere, relative to the potential far away, is:
A. zero
B. E/R
C. E/R2
D. ER
E. ER2
ans: D

51. A 5-cm radius conducting sphere has a surface charge density of 2 ×10−6 C/m2 on its surface. Its electric potential, relative to the potential far away, is:
A. 1.1 × 104 V
B. 2.2 × 104 V
C. 2.3 × 105 V
D. 3.6 × 105 V
E. 7.2 × 106 V
ans: A

52. A hollow metal sphere is charged to a potential V . The potential at its center is:
A. V
B. 0
C. −V
D. 2V
E. πV
ans: A

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53. Positive charge is distributed uniformly throughout a non-conducting sphere. The highest electric potential occurs:
A. at the center
B. at the surface
C. halfway between the center and surface
D. just outside the surface
E. far from the sphere
ans: A

54. A total charge of 7×10−8 C is uniformly distributed throughout a non-conducting sphere with a radius of 5 cm. The electric potential at the surface, relative to the potential far away, is about:
A. −1.3 × 104 V
B. 1.3 × 104 V
C. 7.0 × 105 V
D. −6.3 × 104 V
E. 0
ans: B

55. Eight identical spherical raindrops are each at a potential V , relative to the potential far away. They coalesce to make one spherical raindrop whose potential is:
A. V/8
B. V/2
C. 2V
D. 4V
E. 8V
ans: D

56. A metal sphere carries a charge of 5 × 10−9 C and is at a potential of 400V, relative to the potential far away. The potential at the center of the sphere is:
A. 400V
B. −400V
C. 2 × 10−6 V
D. 0
E. none of these
ans: A

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57. A 5-cm radius isolated conducting sphere is charged so its potential is +100V, relative to the potential far away. The charge density on its surface is:
A. +2.2 × 10−7 C/m2
B. −2.2 × 10−7 C/m2
C. +3.5 × 10−7 C/m2
D. −3.5 × 10−7 C/m2
E. +1.8 × 10−8 C/m2
ans: E

57. A conducting sphere has charge Q and its electric potential is V , relative to the potential far away. If the charge is doubled to 2Q, the potential is:
A. V
B. 2V
C. 4V
D. V/2
E. V/4
ans: B

58. The potential difference between the ends of a 2-meter stick that is parallel to a uniform electric field is 400V. The magnitude of the electric field is:
A. zero
B. 100V/m
C. 200V/m
D. 400V/m
E. 800V/m
ans: E

59. In a certain region of space the electric potential increases uniformly from east to west and does not vary in any other direction. The electric field:
A. points east and varies with position
B. points east and does not vary with position
C. points west and varies with position
D. points west and does not vary with position
E. points north and does not vary with position
ans: B



60. If the electric field is in the positive x direction and has a magnitude given by E = Cx2, where C is a constant, then the electric potential is given by V =:
A. 2Cx
B. −2Cx
C. Cx3/3
D. −Cx3/3
E. −3Cx3
ans: D

61. The work required to carry a particle with a charge of 6.0C from a 5.0-V equipotential surface to a 6.0-V equipotential surface and back again to the 5.0-V surface is:
A. 0
B. 1.2 × 10−5 J
C. 3.0 × 10−5 J
D. 6.0 × 10−5 J
E. 6.0 × 10−6 J
ans: A

62. The equipotential surfaces associated with a charged point particles are:
A. radially outward from the particle
B. vertical planes
C. horizontal planes
D. concentric spheres centered at the particle
E. concentric cylinders with the particle on the axis.
ans: D

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63. The electric field in a region around the origin is given by E = C(xˆi + yˆj), where C is a constant. The equipotential surfaces in that region are:
A. concentric cylinders with axes along the z axis
B. concentric cylinders with axes along the x axis
C. concentric spheres centered at the origin
D. planes parallel to the xy plane
E. planes parallel to the yz plane
ans: A

64.Room temperature is about 20 degrees on the:
Select correct option:

Kelvin scale
Celsius scale
Fahrenheit scale
Absolute scale

65.A particle with zero mass and energy E carries momentum:
Select correct option:

Ec
Ec2
vEc
E/c

66.The quantum number ms is most closely associated with what property of the electron in an atom?
Select correct option:

Magnitude of the orbital angular momentum
Energy
z component of the spin angular momentum
z component of the orbital angular momentum

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67.J.J.Thompson's measurement of e/m for electrons provides evidence of the:
Select correct option:

Wave nature of matter
Particle nature of matter
Wave nature of radiation
Particle nature of radiation

68.During a slow adiabatic expansion of a gas:
Select correct option:

The pressure remains constant
Energy is added as heat
Work is done on the gas
No energy enters or leaves as heat

69. The electric potential in a certain region of space is given by V = −7.5x2 + 3x, where V is in volts and x is in meters. In this region the equipotential surfaces are:
A. planes parallel to the x axis
B. planes parallel to the yz plane
C. concentric spheres centered at the origin
D. concentric cylinders with the x axis as the cylinder axis
E. unknown unless the charge is given
ans: B

70. A particle with charge q is to be brought from far away to a point near an electric dipole. No work is done if the final position of the particle is on:
A. the line through the charges of the dipole
B. a line that is perpendicular to the dipole moment

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C. a line that makes an angle of 45◦ with the dipole moment
D. a line that makes an angle of 30◦ with the dipole moment
E. none of the above
ans: B

71. Equipotential surfaces associated with an electric dipole are:
A. spheres centered on the dipole
B. cylinders with axes along the dipole moment
C. planes perpendicular to the dipole moment
D. planes parallel to the dipole moment
E. none of the above
ans: E

72. The units of capacitance are equivalent to:
A. J/C
B. V/C
C. J2/C
D. C/J
E. C2/J
ans: E

73. A farad is the same as a:
A. J/V
B. V/J
C. C/V
D. V/C
E. N/C
ans: C

74. A capacitor C "has a charge Q". The actual charges on its plates are:
A. Q, Q
B. Q/2, Q/2
C. Q, −Q
D. Q/2, −Q/2
E. Q, 0
ans: C





75. Each plate of a capacitor stores a charge of magnitude 1mC when a 100-V potential difference is applied. The capacitance is:
A. 5 μF
B. 10 μF
C. 50 μF
D. 100 μF
E. none of these
ans: B

76. To charge a 1-F capacitor with 2C requires a potential difference of:
A. 2V
B. 0.2V
C. 5V
D. 0.5V
E. none of these
ans: A

77. The capacitance of a parallel-plate capacitor with plate area A and plate separation d is given by:
A. □0d/A
B. □0d/2A
C. □0A/d
D. □0A/2d
E. Ad/□0
ans: C

78. The capacitance of a parallel-plate capacitor is:
A. proportional to the plate area
B. proportional to the charge stored
C. independent of any material inserted between the plates
D. proportional to the potential difference of the plates

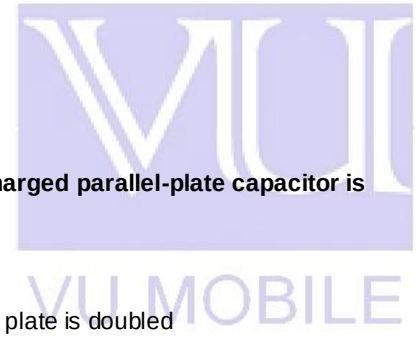
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E. proportional to the plate separation
ans: A

79. The plate areas and plate separations of five parallel plate capacitors are
capacitor 1: area A0, separation d0
capacitor 2: area 2A0, separation 2d0
capacitor 3: area 2A0, separation d0/2
capacitor 4: area A0/2, separation 2d0
capacitor 5: area A0, separation d0/2
Rank these according to their capacitances, least to greatest.
A. 1, 2, 3, 4, 5
B. 5, 4, 3, 2, 1
C. 5, 3 and 4 tie, then 1, 2
D. 4, 1 and 2 tie, then 5, 3
E. 3, 5, 1 and 2 tie, 1, 4
ans: D

80. The capacitance of a parallel-plate capacitor can be increased by:
A. increasing the charge
B. decreasing the charge
C. increasing the plate separation
D. decreasing the plate separation
E. decreasing the plate area
ans: D

81. If both the plate area and the plate separation of a parallel-plate capacitor are doubled, the capacitance is:
A. doubled
B. halved
C. unchanged
D. tripled
E. quadrupled
ans: C





82. If the plate area of an isolated charged parallel-plate capacitor is doubled:
A. the electric field is doubled
B. the potential difference is halved
C. the charge on each plate is halved
D. the surface charge density on each plate is doubled
E. none of the above
ans: B

83. If the plate separation of an isolated charged parallel-plate capacitor is doubled:
A. the electric field is doubled
B. the potential difference is halved
C. the charge on each plate is halved
D. the surface charge density on each plate is doubled
E. none of the above
ans: E

84. Pulling the plates of an isolated charged capacitor apart:
A. increases the capacitance
B. increases the potential difference
C. does not affect the potential difference
D. decreases the potential difference
E. does not affect the capacitance
ans: B

85. If the charge on a parallel-plate capacitor is doubled:
A. the capacitance is halved
B. the capacitance is doubled
C. the electric field is halved
D. the electric field is doubled
E. the surface charge density is not changed on either plate
ans: D

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86. A parallel-plate capacitor has a plate area of 0.2m2 and a plate separation of 0.1mm. To obtain an electric field of 2.0 × 106 V/m between the plates, the magnitude of the charge on each plate should be:
A. 8.9 × 10−7 C
B. 1.8 × 10−6 C
C. 3.5 × 10−6 C
D. 7.1 × 10−6 C
E. 1.4 × 10−5 C
ans: D

87. A parallel-plate capacitor has a plate area of 0.2m2 and a plate separation of 0.1 mm. If the charge on each plate has a magnitude of 4 × 10−6 C the potential difference across the plates is approximately:

A. 0
B. 4 × 10−2 V
C. 1 × 102 V
D. 2 × 102 V
E. 4 × 108 V
ans: D

88. The capacitance of a spherical capacitor with inner radius a and outer radius b is proportional to:
A. a/b
B. b − a
C. b2 − a2
D. ab/(b − a)
E. ab/(b2 − a2)
ans: D

89. The capacitance of a single isolated spherical conductor with radius R is proportional to:
A. R
B. R2
C. 1/R
D. 1/R2
E. none of these
ans: A



90. The capacitance of a cylindrical capacitor can be increased by:
A. decreasing both the radius of the inner cylinder and the length
B. increasing both the radius of the inner cylinder and the length
C. increasing the radius of the outer cylindrical shell and decreasing the length
D. decreasing the radius of the inner cylinder and increasing the radius of the outer cylindrical shell
E. only by decreasing the length
ans: B

91. A battery is used to charge a series combination of two identical capacitors If the potential difference across the battery terminals is V and total charge Q flows through the battery during the charging process then the charge on the positive plate of each capacitor and the potential difference across each capacitor are:
A. Q/2 and V/2, respectively
B. Q and V , respectively
C. Q/2 and V , respectively
D. Q and V/2, respectively
E. Q and 2V , respectively
ans: D

92. A battery is used to charge a parallel combination of two identical capacitors. If the potential difference across the battery terminals is V and total charge Q flows through the battery during the charging process then the charge on the positive plate of each capacitor and the potential difference across each capacitor are:
A. Q/2 and V/2, respectively
B. Q and V , respectively
C. Q/2 and V , respectively
D. Q and V/2, respectively
E. Q and 2V , respectively
ans: C

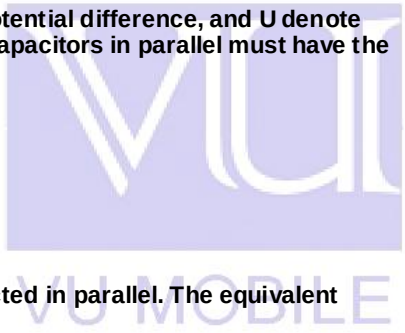
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93. A 2-μF and a 1-μF capacitor are connected in series and a potential difference is applied across the combination. The 2-μF capacitor has:
A. twice the charge of the 1-μF capacitor
B. half the charge of the 1-μF capacitor
C. twice the potential difference of the 1-μF capacitor
D. half the potential difference of the 1-μF capacitor
E. none of the above
ans: D

94. A 2-μF and a 1-μF capacitor are connected in parallel and a potential difference is applied across the combination. The 2-μF capacitor has:
A. twice the charge of the 1-μF capacitor
B. half the charge of the 1-μF capacitor
C. twice the potential difference of the 1-μF capacitor
D. half the potential difference of the 1-μF capacitor
E. none of the above
ans: A

95. Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in series must have the same:
A. Q only
B. V only
C. U only
D. Q and U only
E. V and U only
ans: A

96. Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in parallel must have the same:
A. Q only
B. V only
C. U only
D. Q and U only
E. V and U only
ans: B





97. Capacitors C1 and C2 are connected in parallel. The equivalent capacitance is given by:
A. C1C2/(C1 + C2)
B. (C1 + C2)/C1C2
C. 1/(C1 + C2)
D. C1/C2
E. C1 + C2
ans: E

98. Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by:
A. C1C2/(C1 + C2)
B. (C1 + C2)/C1C2
C. 1/(C1 + C2)
D. C1/C2
E. C1 + C2
ans: A

99. Capacitors C1 and C2 are connected in series and a potential difference is applied to the combination. If the capacitor that is equivalent to the combination has the same potential difference, then the charge on the equivalent capacitor is the same as:
A. the charge on C1
B. the sum of the charges on C1 and C2
C. the difference of the charges on C1 and C2
D. the product of the charges on C1 and C2
E. none of the above
ans: A

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100. Capacitors C1 and C2 are connected in parallel and a potential difference is applied to the combination. If the capacitor that is equivalent to the combination has the same potential difference, then the charge on the equivalent capacitor is the same as:
A. the charge on C1
B. the sum of the charges on C1 and C2
C. the difference of the charges on C1 and C2
D. the product of the charges on C1 and C2
E. none of the above
ans: B

101. Two identical capacitors are connected in series and two, each identical to the first, are connected in parallel. The equivalent capacitance of the series connection is the equivalent capacitance of parallel connection.

A. twice
B. four times
C. half
D. one-fourth
E. the same as
ans: D

102. Two identical capacitors, each with capacitance C, are connected in parallel and the combination is connected in series to a third identical capacitor. The equivalent capacitance of this arrangement is:
A. 2C/3
B. C
C. 3C/2
D. 2C
E. 3C
ans: A

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103. A 2-μF and a 1-μF capacitor are connected in series and charged from a battery. They store charges P and Q, respectively. When disconnected and charged separately using the same battery, they have charges R and S, respectively. Then:
A. R > S > Q = P
B. P > Q > R = S
C. R > P = Q > S
D. R =P >S = Q
E. R > P > S = Q
ans: A

104. A 20-F capacitor is charged to 200V. Its stored energy is:
A. 4000 J
B. 4 J
C. 0.4J
D. 2000 J
E. 0.1J
ans: C

105. A charged capacitor stores 10C at 40V. Its stored energy is:
A. 400 J
B. 4 J
C. 0.2J
D. 2.5J
E. 200 J
ans: E

106. A 2-μF and a 1-μF capacitor are connected in series and charged by a battery. They store energies P and Q, respectively. When disconnected and charged separately using the same battery, they store energies R and S, respectively. Then:
A. R > P > S > Q
B. P > Q > R > S
C. R > P > Q > S
D. P >R>S >Q
E. R > S > Q > P
ans: E

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107. The quantity (1/2)0E2 has the significance of:
A. energy/farad
B. energy/coulomb
C. energy
D. energy/volume
E. energy/volt
ans: D

108. Capacitors A and B are identical. Capacitor A is charged so it stores 4 J of energy and capacitor B is uncharged. The capacitors are then connected in parallel. The total stored energy in the capacitors is now:
A. 16 J
B. 8 J
C. 4 J
D. 2 J
E. 1 J
ans: D

109. To store a total of 0.040 J of energy in the two identical capacitors shown, each should have a capacitance of: 200V
A. 0.10 µF
B. 0.50 µF0.10 µF
C. 1.0 µJ
D. 1.5 µF
E. 2.0 µF
ans: C

110. A battery is used to charge a parallel-plate capacitor, after which it is disconnected. Then the plates are pulled apart to twice their original separation. This process will double the:
A. capacitance
B. surface charge density on each plate
C. stored energy
D. electric field between the two places
E. charge on each plate
ans: C



111. A parallel-plate capacitor has a plate area of 0.3m2 and a plate separation of 0.1 mm. If the charge on each plate has a magnitude of 5×10−6 C then the force exerted by one plate on the other has a magnitude of about:
A. 0
B. 5N
C. 9N
D. 1 × 104 N
E. 9 × 105 N
ans: B

112. A certain capacitor has a capacitance of 5.0 µF. After it is charged to 5.0 µC and isolated, the plates are brought closer together so its capacitance becomes 10 µF. The work done by the agent is about:
A. zero
B. 1.25 × 10−6 J
C. −1.25 × 10−6 J
D. 8.3 × 10−7 J
E. −8.3 × 10−7 J
ans: C

113. A dielectric slab is slowly inserted between the plates of a parallel plate capacitor, while the potential difference between the plates is held constant by a battery. As it is being inserted:
A. the capacitance, the potential difference between the plates, and the charge on the positive plate all increase
B. the capacitance, the potential difference between the plates, and the charge on the positive plate all decrease
C. the potential difference between the plates increases, the charge on the positive plate decreases, and the capacitance remains the same
D. the capacitance and the charge on the positive plate decrease but the potential difference between the plates remains the same

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E. the capacitance and the charge on the positive plate increase but the potential difference between the plates remains the same
ans: E

114. An air-filled parallel-plate capacitor has a capacitance of 1 pF. The plate separation is then doubled and a wax dielectric is inserted, completely filling the space between the plates. As a result, the capacitance becomes 2 pF. The dielectric constant of the wax is:
A. 0.25
B. 0.5
C. 2.0
D. 4.0
E. 8.0
ans: D

115. One of materials listed below is to be placed between two identical metal sheets, with no, air gap, to form a parallel-plate capacitor. Which produces the greatest capacitance?

A. material of thickness 0.1mm and dielectric constant 2
B. material of thickness 0.2mm and dielectric constant 3
C. material of thickness 0.3mm and dielectric constant 2
D. material of thickness 0.4mm and dielectric constant 8
E. material of thickness 0.5mm and dielectric constant 11
ans: E

116. Two capacitors are identical except that one is filled with air and the other with oil. Both capacitors carry the same charge. The ratio of the electric fields Eair/Eoil is:
A. between 0 and 1
B. 0
C. 1
D. between 1 and infinity
E. infinite
ans: D



117. A parallel-plate capacitor, with air dielectric, is charged by a battery, after which the battery is disconnected. A slab of glass dielectric is then slowly inserted between the plates. As it is being inserted:
A. a force repels the glass out of the capacitor
B. a force attracts the glass into the capacitor
C. no force acts on the glass
D. a net charge appears on the glass
E. the glass makes the plates repel each other
ans: B

118. Two parallel-plate capacitors with the same plate separation but different capacitance are connected in parallel to a battery. Both capacitors are filled with air. The quantity that is NOT the same for both capacitors when they are fully charged is:
A. potential difference
B. energy density
C. electric field between the plates
D. charge on the positive plate
E. dielectric constant
ans: D

119. Two parallel-plate capacitors with the same plate area but different capacitance are connected in parallel to a battery. Both capacitors are filled with air. The quantity that is the same for both capacitors when they are fully charged is:
A. potential difference
B. energy density
C. electric field between the plates
D. charge on the positive plate
E. plate separation
ans: A

120. Two parallel-plate capacitors with different plate separation but the same capacitance are connected in series to a battery. Both capacitors are filled with air. The quantity that is NOT the same for both capacitors when they are fully charged is:

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A. potential difference
B. stored energy
C. electric field between the plates
D. charge on the positive plate
E. dielectric constant
ans: C

121. Two parallel-plate capacitors with different capacitance but the same plate separation are connected in series to a battery. Both capacitors are filled with air. The quantity that is the same for both capacitors when they are fully charged is:
A. potential difference
B. stored energy
C. energy density
D. electric field between the plates
E. charge on the positive plate
ans: E

122. A car battery is rated at 80A · h. An ampere-hour is a unit of:
A. power
B. energy
C. current
D. charge
E. force
ans: D

123. Current has units:
A. kilowatt·hour
B. coulomb/second
C. coulomb
D. volt
E. ohm
ans: B



124. Current has units:
A. kilowatt·hour
B. ampere
C. coulomb
D. volt
E. ohm
ans: B

125. The units of resistivity are:
A. ohm
B. ohm·meter
C. ohm/meter
D. ohm/meter2
E. none of these
ans: B

126. The rate at which electrical energy is used may be measured in:
A. watt/second
B. watt·second
C. watt
D. joule·second
E. kilowatt·hour
ans: C

127. Energy may be measured in:
A. kilowatt
B. joule·second
C. watt
D. watt·second
E. volt/ohm
ans: D

128. Which one of the following quantities is correctly matched to its unit?

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A. Power – kW·h
B. Energy – kW
C. Potential difference – J/C
D. Current – A/s
E. Resistance – V/C
ans: C

129. Current is a measure of:
A. force that moves a charge past a point
B. resistance to the movement of a charge past a point
C. energy used to move a charge past a point
D. amount of charge that moves past a point per unit time
E. speed with which a charge moves past a point
ans: D

130. A 60-watt light bulb carries a current of 0.5A. The total charge passing through it in one hour is:
A. 120C
B. 3600C
C. 3000C
D. 2400C
E. 1800C
ans: E

The number of significant figures in 0.00150 is:

► 5
► 4
► 3
► 2

One revolution is the same as:

► 1 rad
► 57 rad
► π/2 rad
► π rad
► 2π rad

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A body to be in equilibrium under the combined action of several forces:

► All the forces must be applied at the same point
all the forces must be applied at the same point
► all of the forces form pairs of equal and opposite forces
► any two of these forces must be balanced by a third force
► the sum of the torques about any point must equal zero

A bucket of water is pushed from left to right with increasing speed across a horizontal surface. Consider the pressure at two points at the same level in the water.

► It is the same
► It is higher at the point on the left
► It is higher at the point on the right
► At first it is higher at the point on the left but as the bucket speeds up it is lower there

An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

► A. 200 Hz
► B. 300 Hz
► C. 400 Hz
► D. 600 Hz

Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by

► C1C2/(C1 + C2)
► (C1 + C2)/C1C2

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► 1/(C1 + C2)
► C1/C2

If the potential difference across a resistor is doubled:

► only the current is doubled
► only the current is halved
► only the resistance is doubled
► only the resistance is halved

By using only two resistors, R1 and R2, a student is able to obtain resistances of 3 Ω, 4Ω, 12Ω, and 16 Ω. The values of R1 and R2 (in ohms) are:

► 3, 4
► 2, 12
► 3, 16
► 4, 12

Faraday's law states that an induced emf is proportional to:

► the rate of change of the electric field
► the rate of change of the magnetic flux
► the rate of change of the electric flux
► the rate of change of the magnetic field

A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

► 1000V
► 500V
► 250V
► 100V

The wavelength of red light is 700 nm. Its frequency is _____.

4.29 10^14

4.29 10^14 SEE PGE NO.92

4.29 10^14

► 4.30 * 104 Hertz
► 4.30 * 103 Hertz
► 4.30 * 105 Hertz
► 4.30 * 102 Hertz

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A laser in a compact disc player generates light that has a wavelength of 780 nm in air. The light then enters into the plastic of a CD. If the index of refraction of plastic is 1.55, the speed of this light once enter the plastic is _____.

► 3.00 * 108 m/s
► 1.94 * 108 m/s
► 4.29 * 108 km/h
► 3.00 * 108 km/h

Which of the following electromagnetic radiations has photons with the greatest energy?

► blue light
► yellow light
► x rays
► radio waves

A virtual image is one:

► toward which light rays converge but do not pass through
► from which light rays diverge as they pass through
► toward which light rays converge and pass through
► from which light rays diverge but do not pass through

What is the unit of magnification factor?

► meter.Kelvin

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► radian.Kelvin
► degree.Kelvin
► no units

During an adiabatic process an object does 100 J of work and its temperature decreases by 5K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is.

► 20 J/K
► 100 J/K
► 15 J/K
► 5 J/K

An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

► a change in entropy
► a decrease of internal energy
► an increase of pressure
► a change in temperature

The Stern-Gerlach experiment makes use of:

► a strong uniform magnetic field
► a strong non-uniform magnetic field
► a strong uniform electric field
► a strong non-uniform electric field

A large collection of nuclei are undergoing alpha decay. The rate of decay at any instant is proportional to:



► the number of undecayed nuclei present at that instant
► the time since the decays started
► the time remaining before all have decayed
► the half-life of the decay

As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, a distance of 0.75m from the origin. The z component of the angular momentum around the origin is:

► 6.0kg · m2/s
► 9.0kg · m2/s
► 11 kg · m2/s
► 14 kg · m2/s

A net torque applied to a rigid object always tends to produce:

► linear acceleration
► rotational equilibrium
► angular acceleration
► rotational inertia

An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

► 1.57 rad/s
► 2.0 rad/s
► 6.3 rad/s
► 12.6 rad/s

In simple harmonic motion, the restoring force must be proportional to the:

► amplitude
► frequency
► velocity
► displacement

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Mercury is a convenient liquid to use in a barometer because:

► it is a metal
► it has a high boiling point
► it expands little with temperature
► it has a high density

The units of the electric field are:

► J/m
► J/(C·m)
► N/C
► J·C

A farad is the same as a

► J/V
► V/J
► C/V
► V/C

We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5H. We also need a capacitance of:

► 1 F
► 1mF
► 1 µF
► 100 µF

Which of the following statements is NOT TRUE about electromagnetic waves?

► Electromagnetic waves satisfy the Maswell's Equation.
► Electromagnetic waves can not travel through space.
► The receptions of electromagnetic waves require an antenna.
► The electromagnetic radiation from a burning candle is unpolarized.

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Radio waves and light waves are _____.

► Longitudinal waves
► Transverse waves
► Electromagnetic and transverse both
► Electromagnetic and longitudinal both

Wien's Law states that, *lmax* = _____ K.

► 2.90 * 10-3 Hertz
► 2.90 * 10-3 s
► 2.90 * 10-3 kg
► 2.90 * 10-3 m

Interference of light is evidence that:

► the speed of light is very large
► light is a transverse wave
► light is a wave phenomenon
► light is electromagnetic in character

Fahrenheit and Kelvin scales agree numerically at a reading of:

► -40
► 0
► 273
► 574

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According to the theory of relativity:

► moving clocks run fast
► energy is not conserved in high speed collisions
► the speed of light must be measured relative to the ether
► none of the above are true

Light from a stationary spaceship is observed, and then the spaceship moves directly away from the observer at high speed while still emitting the light. As a result, the light seen by the observer has:

► higher frequency and a longer wavelength than before
► lower frequency and a shorter wavelength than before **(Not Sure)**
► higher frequency and a shorter wavelength than before
► lower frequency and a longer wavelength than before

How fast should you move away from a 6.0 × 1014 Hz light source to observe waves with a frequency of 4.0 × 1014 Hz?

► 0.20c
► 0.38c
► 0.45c
► 0.51c

The quantum number n is most closely associated with what property of the electron in a hydrogen atom?

► Energy
► Orbital angular momentum
► Spin angular momentum
► Magnetic moment



The quantum number ms is most closely associated with what property of the electron in an atom?

► Magnitude of the orbital angular momentum
► Energy
► z component of the spin angular momentum
► z component of the orbital angular momentum

As the wavelength of a wave in a uniform medium increases, its speed will _____.

► Decrease
► Increase
► Remain the same
► None of these

Question # 1 of 5 (Start time: 05:02:08 PM) Total Marks: 1

A force of 5000N is applied outwardly to each end of a 5.0-m long rod with a radius of 34.0 cm and a Young's modulus of 125 × 108 N/m2. The elongation of the rod is:
Select correct option:

0.0040mm
0.14mm
0.55mm (Correct)
1.42mm

An object attached to one end of a spring makes 20 complete oscillations in 10 s. Its period is:
Select correct option:

2Hz
0.5 s (Correct)
0.5Hz
2 s

Question # 3 of 5 (Start time: 05:04:59 PM) Total Marks: 1

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back and forth in a leftward and rightward direction. This type of wave is known as a ____.
Select correct option:

Mechanical
Electromagnetic
Transverse
Longitudinal (Correct)

Question # 4 of 5 (Start time: 05:06:27 PM) Total Marks: 1

Which of the following statements about completely submerged objects resting on the ocean bottom is correct?
Select correct option:

The buoyant force acting on the object is equal to the object's weight.
The apparent weight of the object depends on the object's density. (Correct)
The displaced volume of fluid is greater than the volume of the object.
The weight of the object and the buoyant force are equal and opposite.

Question # 5 of 5 (Start time: 05:08:03 PM) Total Marks: 1

The flow rate of blood through the average human aorta is about 9.0 ? 101 cm3/s. If the aorta has a radius of 1.0 cm, what is the velocity of the blood flow?
Select correct option:

14 cm/s
32 cm/s
37 cm/s
29 cm/s (Correct)

Question # 1 of 10 (Start time: 09:42:03 PM) Total Marks: 1

A laser in a compact disc player generates light that has a wavelength of 780 nm in air. The light then enters into the plastic of a CD. If the index of refraction of plastic is 1.55, the speed of this light once enter the plastic is _____.
Select correct option:

3.00 x10^8 m/s
1.94 x 10^8 m/s (Correct)
4.29 x 10^8 km/h
3.00 x 10^8 km/h

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Question # 2 of 10 (Start time: 09:42:41 PM) Total Marks: 1

The quantum number ms is most closely associated with what property of the electron in an atom?
Select correct option:

Magnitude of the orbital angular momentum
Energy
z component of the spin angular momentum (Correct)
z component of the orbital angular momentum

Question # 3 of 10 (Start time: 09:43:13 PM) Total Marks: 1

Metal pipes, used to carry water, sometimes burst in the winter because:
Select correct option:

Metal contractst more han water
Outside of the pipe contracts more than the inside
Metal becomes brittle when cold
Water expands when it freezes (Correct)

Question # 4 of 10 (Start time: 09:44:19 PM) Total Marks: 1

An erect object is in front of a convex mirror a distance greater than the focal length. The image is:
Select correct option:

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Real, inverted, and smaller than the object
Virtual, inverted, and larger than the object
Real, inverted, and larger than the object
Virtual, erect, and smaller than the object (Correct)

Question # 5 of 10 (Start time: 09:45:48 PM) Total Marks: 1

Select the correct statement:
Select correct option:

Ultraviolet light has a longer wavelength than infrared (Correct)
Blue light has a higher frequency than x rays
Radio waves have higher frequency than gamma rays
Gamma rays have higher frequency than infrared waves

Question # 6 of 10 (Start time: 09:46:50 PM) Total Marks: 1

A virtual image is one:
Select correct option:

toward which light rays converge but do not pass through
from which light rays diverge but do not pass through (Correct)
from which light rays diverge as they pass through
toward which light rays converge and pass through

Question # 7 of 10 (Start time: 09:47:39 PM) Total Marks: 1

The wavelength of light beam A is twice the wavelength of light beam B. The energy of a photon in beam A is:
Select correct option:

half the energy of a photon in beam B (Correct)
one-fourth the energy of a photon in beam B
equal to the energy of a photon in beam B
twice the energy of a photon in beam B

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Question # 8 of 10 (Start time: 09:49:09 PM) Total Marks: 1

Evidence for the wave nature of matter is:
Select correct option:

Electron di?raction experiments of Davisson and Germer (Correct)
Thompson's measurement of e/m
Young's double slit experiment
The Compton s' e?ect

Question # 9 of 10 (Start time: 09:50:36 PM) Total Marks: 1

Evidence that a gas consists mostly of empty space is the fact that:
Select correct option:

The density of a gas becomes much greater when it is liquefied (Correct)
Gases exert pressure on the walls of their containers
Gases are transparent
Heating a gas increases the molecular motion

Question # 10 of 10 (Start time: 09:51:53 PM) Total Marks: 1

In a photoelectric effect experiment at a frequency above cut off, the stopping potential is proportional to:
Select correct option:

the energy of the least energetic electron before it is ejected
the energy of the least energetic electron after it is ejected
the energy of the most energetic electron before it is ejected

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the energy of the most energetic electron after it is ejected (Correct)

X

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PHY101 MCQS 22 TO 45 LECTURE WISE BY NAJAM AHMED

LECTURE 22

the force of attraction or repulsion acting along a straight line between two electric charges is directly proportional to the product of the charges and inversely to the square of the distance between them is called:

Ohm law

Columb's law check this

Ainstain law

None of the given

The constant of proportionality will take different values depending upon the:

Law

Mass

quantity

units pg#61

MKS system, charge is measured in:

Ohm law

Columb pg#61

Ainstain law

None of the given

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Charge is quantized. This means that charge comes in..... unit/s:

Certain pg#61

One

Two

Three

charge is never:

created or destroyed

created

destroyed

created or destroyed both pg#61

the total charge at an earlier time is equal to the charge at a

later time pg#61

The electric field is also an example of a

vector field pg#62

Scalar field

Megnatic field

None

It is easy to find the torque experienced by an electric dipole that is placed in a uniform electric field

True pg#63

false

LECTURE 23

when charges are continuously distributed over some region of space, For this, we need to break up the region into little pieces to calculate:

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magnetic field

both

electric field pg#64

none of the given

The..... of any vector field is a particularly important concept.

Flux pg#65

Point

Direction

Magnitude

.....is the measure of the "flow" or penetration of the field vectors through an imaginary fixed surface

Flux pg#65

Point

Direction

Magnitude

LECTURE 24

The electric potential (or simply potential) is the energy of a unit charge in an:

electric field pg#68

megnatic field

both

none

Potentials are more positive in regions which have more positive charge.

True pg#68

False

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The electric potential is a scalar quantity

True pg#68

False

The electric potential exists only because the electrostatic force is conservative.

True pg#68

False

LECTURE 25

The electric field comes from the charges that have been pushed to the plates by the battery

True pg#69

false

The amount of charge pushed on to the conductors is proportional to the potential difference between the battery terminals.

True pg#69

false

When a battery is connected to a capacitor, positive and negative charges appear on the opposite plates

True pg#71

False

LECTURE 26

Current flows because something forces it around a circuit.

True pg#73

False

EMF stands for: electromotive force.

True pg#73

False

EMF is not a force.

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True pg#73

False

…… is the difference in electric potentials between two parts of a circuit.

EMF pg#73

Electric force

Megnatic field

None of the given

K The sum of the potential differences encountered in moving around a closed circuit is zero. Is known as irchoff's Law

True pg#75

False

We can apply Kirchoff's Law to a circuit that consists of a resistor and capacitor in order to see how current flows through it

True pg#75

False

LECTURE 27

The magnetic field exerts a force upon any charge that moves in the

Circle

Line

medium

field pg#77

The unit of magnetic field that is used most commonly is the tesla.

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True pg#77

False

electric force and magnetic force are very different

True pg#77

False

The electric force is non-zero even if the charge is stationary

True pg#77

False

The Lorentz Force can be used to select charged particles of whichever velocity we want.

True pg#77

False

A magnetic field can be strong enough to lift an elephant

True pg#78

False

magnetic field can never increase or decrease the energy of a particle.

True pg#78

False

The fact that a magnetic field bends charged particles is responsible for shielding the earth from harmful effects of the "solar wind".

True pg#78

False

A large number of charged particles are released from the sun and reach the earth. These can destroy

Earth

Moon

Planents

Life pg#78

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A current that goes around a loop (any shape) produces a

Pulse

Electric field

None of the given

magnetic field pg#79

Magnetic fields are produced by currents.

True pg#79

False

LECTURE 28

A fundamental law of magnetism states that the net flux through a closed surface is always

one

zero pg#81

three

two

when the magnetic flux changes in a circuit, an electro motive force is induced which is proportional to the rate of change of flux is known as Faraday's Law for Induced EMF.

True pg#82

False

LECTURE 29

Alternating current (AC) is current that flows first in one direction along a wire, and then in the reverse direction.

True pg#85

False

square of any AC wave is always.....Therefore its average is not zero.

negative

equal

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none of the given

positive pg#85

AC is generated by a coil rotating in a magnetic field

True pg#86

False

3. AC is particular useful because make it possible to.....voltages.

step up or step down pg#86

step up

step down

none

The basic transformer consists of .....coil/s

One

Four

Two pg#86

three

The inductance (called the self-inductance if there is only one coil)

True pg#86

False

The unit of inductance is called Henry.

True pg#86

False

y, 1 Henry= 1 Tesla meter square/Ampare

True pg#87

False

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Inductance, like capacitance, is purely geometrical and depends only upon the shape and sizes of wires. It does not depend on the current.

True pg#87

False

When we pass current through an inductor a ..... magnetic field is produced.

Constant

Changing pg#88

Equal

None of the given

energy can be stored in a capacitor as well as in an inductor

True pg#88

False

LECTURE 30

But Maxwell realized that the above 4 (amperes,faraday's,gauses law) were not consistent with the conservation of charge,

True pg#90

False

the electromagnetic waves inside a micro wave oven have wavelength of 6 cm

True pg#91

False

radio waves are a few metres long

True pg#91

False

The production of electromagnetic waves is done by forcing current to vary rapidly in a small piece of wire

True pg#92

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False

The reception of electromagnetic waves requires an

Recever

Device

Tv

Antenna pg#93

The incoming wave has an electric field that forces the electrons to run up and down the

antenna wire pg#93

Recever

Device

None

electric field of a wave is perpendicular to the direction of its motion

true pg#93

false

LECTURE 31

Light travels very fast but its speed is not

Finit

Constant

Vary

Infinite pg#94

https://welcometogameslab.blogspot.com/

Light is electromagnetic wave

True pg#94

False

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different wavelength correspond to different colours

True pg#94

False

If light contained all frequencies with equal strength, it would appear as ..... to us.

Black

Green

White pg#94

Yellow

The sky appears blue to us on a clear day because tiny dust particles high above in the atmosphere reflect a lot of the blue light coming from the sun

True pg#94

False

The speed of light in vacuum is a fixed constant of nature which we usually call

C pg#96

B

D

A

a medium light can travel slower or faster than:

C pg#96

B

D

A

Light coming from air into water bends toward the normal. Conversely, light from a source in the water will bend away from the normal. What if you keep increasing the angle with respect to the normal so that the light bends and begins to just follow the surface? This phenomenon is called total internal reflection and is called the criti θc cal the criticl angle.

Fibre optic cables, which are now common everywhere

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True pg#97

False

if the cable is bent, the light will continue to travel along it.

True pg#97

False

Optical fibres now carry thousands of telephone calls in a cable whose diameter is only a little bigger than a human hair!

True pg#97

False

LECTURE 32

When an object is heate o o o d, it emits light. BELOW 800C

True pg#98

False

When the temperature is around 800 C, object is red hot.

True pg#98

False

Around 2500 C object is yellowish-white

True pg#98

False

At temperatures lower than 800 C, infrared (IR) light is emitted but our eyes cannot see this.

True pg#98

False

Each chemical element produces a very distinct pattern of colors called an

emission

emission and spectrum both pg#99

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emission spectrum

spectrum

The atmosphere contains various gases which absorb light at many different

Frequincies

Wavelengths pg#100

Both above.

None

Birefringence can occur in any material that possesses some asymmetry in its structure

True pg#101

False

LECTURE 33

Coherent means that both coherent waves should have a fixed phase relative to each other

True pg#102

False

it is very.....to produce coherent light from two separate sources

difficult pg#102

simpile

common

none

When light falls upon a thin film of soapy water, oil, etc. it is reflected from

two surfaces. Pg#104

Three surfaces.

Four surfaces.

None

Whatis diffraction?

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The bending of light around objects is known as diffraction.

LECTURE 34

Light is actually electric and magnetic waves that can travel through empty space

True pg#106

False

The electric and magnetic waves are perpendicular to each other and to the direction of travel

True pg#106

False

Explanation of the photoelectric puzzle came from Einstein

True pg#107

False

Einstein got Nobel Prize in 1905

True pg#107

False

Microwaves, radio and TV waves, X-rays, -rays, etc. are all photons

True pg#108

False

Fluorescence and phosphorence are two phenomena observed in some materials

True pg#108

False

LASER stands for: Light mplification by the timulated mission of adiation

Lasers are important because they emit a very large number of .....all with one single frequency

Photons pg#109

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Electrons

Both above

None

LECTURE 35

When light falls on a flat surface, the angle of incidence equals the angle of

incidence

reflection pg# 110

both above

none

virtual image is actually

not a real image pg#112

real image

can be real or can't be real

none

flat surface has .....radius of curvature

finite

infinite pg#114

bothe above

none

The "strength" of a lens is measured in diopters

True pg#114

False

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LECTURE 36

heat is the flow of internal energy due to a

Positional difference

Electric flux

none

temperature difference pg#115

joule is the unit of work

True pg#117

False

K = thermal conductivity

A = cross-sectional area

L = Length

Q = Heat transferred

t = duration of heat transfer

delta T = temperature difference

n the heat transferred in time is $Q = KA\left(\frac{\Delta T}{L}\right)t$, This formula allows us to measure if all the other quantities in it are measured

true pg#120

false

Conduction is one possible way by which heat is transferred from one portion of a system to another

true pg#120

false

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LECTURE 37

Work and heat are called path variables

true pg#121

false

LECTURE38

More heat (such as inside the core of a star) will break up atomic nuclei into protons and

Electron

Neutrons pg#126

Both above

None

In statistical mechanics, we interpret entropy as the degree of

Orderd

Disorder pg#128

Both above

None

LECTURE39

The Special Theory of Relativity was created in (1995) by

Faraday's

OHM

Albert Einstein pg#129

None of the above

we measure time by looking at some phenomenon that repeats itself:

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true pg#129

false

LECTURE40

Einstein realized that the photoelectric effect could be explained in:

1903

1904

1905 pg#140

1906

The word "quantum" means:

packet or bundle

packet

bundle

both packet or bundle pg#147

Atoms or molecules can only exist in certain energy states. These are also called

"allowed levels" or quantum states.

allowed levels

quantum states

both "allowed levels" or quantum states. Pg#147

Quantum mechanics is all about probabilities

True pg#150

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false

Atoms are building blocks of the

Elements pg#153

Both above

None

All atoms of the same element have the same

Weight

Lenth

With

Mass pg#153

Atoms of different elements are:

Same

Normal

None of the given

Different pg#153

The Bohr model gave wonderful results when compared against the hydrogen spectrum

True pg#157

False

Electrons can be thought of as little spinning balls of charge.

True pg#157

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False

The diameter of the nucleus is about 10 million times smaller that the overall diameter of the atom.

True pg#161

False

BEST OF LUCK

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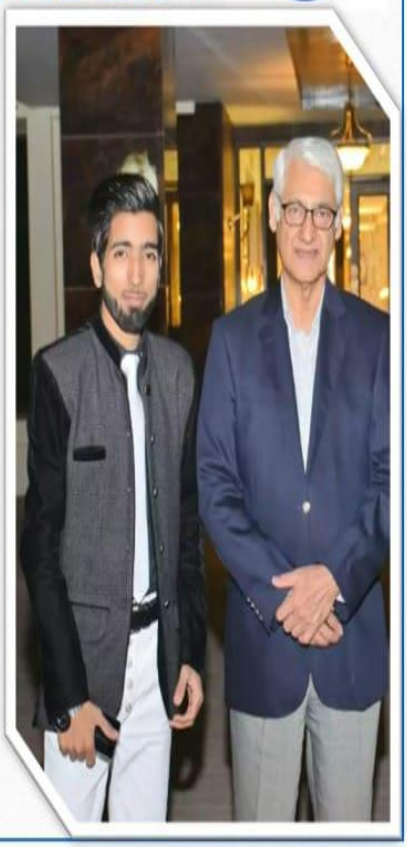
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Question No:1 **(Marks:1)** **Vu-Topper RM**

The image produced by a convex mirror of an erect object in front of the mirror is always:

A. Real, erect, and larger than the object
B. Real, erect, and smaller than the object
C. Virtual, erect, and larger than the object
D. Virtual, erect, and smaller than the object

Question No:2 **(Marks:1)** **Vu-Topper RM**

In some movies, you sometimes see an actor looking in a mirror and you can see his face in the mirror. During the filming of this scene, what does the actor see in the mirror?

A. His face
B. Your face
C. The director's face
D. The movie camera

Question No:3 **(Marks:1)** **Vu-Topper RM**

A femtometer is:

A. 10-9 m
B. 10-12 m
C. 10-15 m
D. 10-18 m

Question No:4 **(Marks:1)** **Vu-Topper RM**

The "strength" of a lens is measured in _____.

A. Meters
B. Diopters
C. Provatans
D. Lense meter

Vu Topper RM

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Question No:5 **(Marks:1)** **Vu-Topper RM**

When an object (iron/tungsten) is heated, it emits light. It emits infrared (IR) light when it is heated at a temperature _____.

A. Below 800 degree C
B. Above 2500 degree C
C. Between 800 degree C to 1600 degree C
D. Between 1600 degree C to 2500 degree C

Question No:6 **(Marks:1)** **Vu-Topper RM**

The reason there are two slits, rather than one, in a Young's experiment is:

A. To increase the intensity
B. To create a path length difference
C. One slit is for frequency, the other for wavelength
D. One slit is for Electric fields , the other is for Magnetic fields

Question No:7 **(Marks:1)** **Vu-Topper RM**

An electron is moving at 0.6c.If we calculate its kinetic energy using (1/2) mv2, we get a result that is:

A. Just right
B. Just half enough
C. About 28% too low
D. Twice the correct value

Question No:8 **(Marks:1)** **Vu-Topper RM**

The theoretical upper limit for the frequency of electromagnetic waves is:

A. There is no upper limit
B. Both
C. None of these
D. Just slightly greater than that of red light

Vu TopperRM

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Question No:9 **(Marks:1)** **Vu-Topper RM**

One revolution is the same as:

A. 1 rad
B. 57 rad
C. π/2 rad
D. 2π rad

Question No:10 **(Marks:1)** **Vu-Topper RM**

For a body to be in equilibrium under the combined action of several forces:

A. All the forces must be applied at the same point
B. All the forces must be applied at the same point
C. All of the forces form pairs of equal and opposite forces
D. Any two of these forces must be balanced by a third force

Question No:11 **(Marks:1)** **Vu-Topper RM**

Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by

A. C1/C2
B. 1/(C1 + C2)
C. C1C2/(C1 + C2)
D. (C1 + C2)/C1C2

Question No:12 **(Marks:1)** **Vu-Topper RM**

If the potential difference across a resistor is doubled:

A. Only the current is halved
B. Only the resistance is halved
C. Only the current is doubled
D. Only the resistance is doubled

Vu Topper RM

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Question No:13 **(Marks:1)** **Vu-Topper RM**

By using only two resistors, R1 and R2, a student is able to obtain resistances of 3Ω, 4Ω, 12Ω, and 16 Ω. The values of R1 and R2 (in ohms) are:

A. 3, 4
B. 2, 12
C. 3, 16
D. 4, 12

Question No:14 **(Marks:1)** **Vu-Topper RM**

Faraday's law states that an induced emf is proportional to:

A. The rate of change of the electric flux
B. The rate of change of the electric field
C. The rate of change of the magnetic field
D. The rate of change of the magnetic flux

Question No:15 **(Marks:1)** **Vu-Topper RM**

A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

A. 500V
B. 250V
C. 100V
D. 1000V

Question No:16 **(Marks:1)** **Vu-Topper RM**

Which of the following electromagnetic radiations has photons with the greatest energy?

A. X rays
B. Blue light
C. Radio waves
D. Yellow light

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Question No:17 **(Marks:1)** **Vu-Topper RM**

We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5H. We also need a capacitance of:

A. 1 F
B. 1mF
C. 1 µF
D. 100 µF

Question No:18 **(Marks:1)** **Vu-Topper RM**

The wavelength of red light is 700 nm. Its frequency is 4.29 10^14

A. 4.30 * 104 Hertz
B. 4.30 * 103 Hertz
C. 4.30 * 105 Hertz
D. 4.30 * 102 Hertz

4.29 10^14 SEE PGE NO.92

4.29 10^14

Question No:19 **(Marks:1)** **Vu-Topper RM**

Which of the following statements is NOT TRUE about electromagnetic waves?

A. Electromagnetic waves cannot travel through space.
B. Electromagnetic waves satisfy the Maswell's Equation.
C. The receptions of electromagnetic waves require an antenna.
D. The electromagnetic radiation from a burning candle is unpolarized.

Question No:20 **(Marks:1)** **Vu-Topper RM**

Radio waves and light waves are

A. Transverse waves
B. None of the these
C. Longitudinal waves
D. Electromagnetic and transverse both

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Question No:21 **(Marks:1)** **Vu-Topper RM**

An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

A. A change in entropy
B. An increase of pressure
C. A change in temperature
D. A decrease of internal energy

Question No:22 **(Marks:1)** **Vu-Topper RM**

The Stern-Gerlach experiment makes use of:

A. A strong uniform electric field
B. A strong uniform magnetic field
C. A strong non-uniform electric field
D. A strong non-uniform magnetic field

Question No:23 **(Marks:1)** **Vu-Topper RM**

A net torque applied to a rigid object always tends to produce:

A. Rotational inertia
B. Linear acceleration
C. Rotational equilibrium
D. Angular acceleration

Question No:24 **(Marks:1)** **Vu-Topper RM**

As the wavelength of a wave in a uniform medium increases, its speed will _____.

A. Increase
B. Decrease
C. None of these
D. Remain the same

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Question No:25 **(Marks:1)** **Vu-Topper RM**

An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

A. 200 Hz
B. 300 Hz
C. 400 Hz
D. 600 Hz

Question No:26 **(Marks:1)** **Vu-Topper RM**

Room temperature is about 20 degrees on the:

A. Kelvin scale
B. Celsius scale
C. Absolute scale
D. Fahrenheit scale

Question No:27 **(Marks:1)** **Vu-Topper RM**

A particle with zero mass and energy E carries momentum:

A. Ec
B. E/c
C. Ec2
D. vEc

Question No:28 **(Marks:1)** **Vu-Topper RM**

J.J.Thompson's measurement of e/m for electrons provides evidence of the:

A. Wave nature of matter
B. Wave nature of radiation
C. Particle nature of matter
D. Particle nature of radiation

Vu Topper RM

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Question No:29 **(Marks:1)** **Vu-Topper RM**

During a slow adiabatic expansion of a gas:

A. Energy is added as heat
B. Work is done on the gas
C. The pressure remains constant
D. No energy enters or leaves as heat

Question No:30 **(Marks:1)** **Vu-Topper RM**

A force of 5000N is applied outwardly to each end of a 5.0-m long rod with a radius of 34.0 cm and a Young's modulus of 125 × 108 N/m2. The elongation of the rod is:

A. 0.55mm
B. 1.42mm
C. 2.24mn
D. 0.56nm

Question No:31 **(Marks:1)** **Vu-Topper RM**

An object attached to one end of a spring makes 20 complete oscillations in 10 s. Its period is: Select correct option:

A. 2 s
B. 2Hz
C. 0.5 s
D. 0.5Hz

Question No:32 **(Marks:1)** **Vu-Topper RM**

back and forth in a leftward and rightward direction. This type of wave is known as a

A. Transverse
B. Mechanical
C. Longitudinal
D. Electromagnetic

Vu Topper RM

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Question No:33 **(Marks:1)** **Vu-Topper RM**

The flow rate of blood through the average human aorta is about 9.0 ? 101 cm3/s. If the aorta has a radius of 1.0 cm, what is the velocity of the blood flow?

A. 14 cm/s
B. 32 cm/s
C. 37 cm/s
D. 29 cm/s

Question No:34 **(Marks:1)** **Vu-Topper RM**

Metal pipes, used to carry water, sometimes burst in the winter because:

A. Metal contractst more han water
B. Metal becomes brittle when cold
C. Water expands when it freezes
D. Outside of the pipe contracts more than the inside

Question No:35 **(Marks:1)** **Vu-Topper RM**

Which one of is in the order of decreasing frequency?

A. Ultraviolet rays
B. Visible light
C. Radio waves
D. All

Question No:36 **(Marks:1)** **Vu-Topper RM**

If velocity of a conductor moving through a magnetic field B is made zero then motional emf is:

-VBL ZERO

Question No:37 **(Marks:1)** **Vu-Topper RM**

The lenz's law refers to:

Induced current

Vu Topper RM

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Question No:38 **(Marks:1)** **Vu-Topper RM**
The increase in capacitance of a capacitor due to presence of dielectric is due to:
Electric polarization

Question No:39 **(Marks:1)** **Vu-Topper RM**
inductance L and a capacitance C has an oscillation frequency f. Two inductors, each with inductance L, and two capacitors, each with capacitance C, are all wired in series and the circuit is completed. The oscillation
F

Question No:40 **(Marks:1)** **Vu-Topper RM**
Which of the following is equivalent to one coulomb?
One ampere second

Question No:41 **(Marks:1)** **Vu-Topper RM**
Tesla is a unit of
Magnetic flux FLUX DENSITY

Question No:42 **(Marks:1)** **Vu-Topper RM**
The current in a car headlamp is 2.0 A. The headlamp is switched on for 3.0 minutes. How much charge passes through the headlamp?
360c

Question No:43 **(Marks:1)** **Vu-Topper RM**
In constructing a thermometer it is NECESSARY to use a substance that:
Undergoes some change when heated or cooled

Question No:44 **(Marks:1)** **Vu-Topper RM**
A gas which strictly obeys the gas laws under all conditions of temperature and pressure is called:
Perfect gases

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Question No:45 **(Marks:1)** **Vu-Topper RM**
In RL series circuit phase angle is given by:
Tan-1(wl/R)

Question No:46 **(Marks:1)** **Vu-Topper RM**
Which of the following have the greatest wavelength?
Radio waves

Question No:47 **(Marks:1)** **Vu-Topper RM**
When a charge is projected perpendicular to a uniform magnetic field, its path is:
A circular path

Question No:48 **(Marks:1)** **Vu-Topper RM**
When two objects come to common temperature, the body is said to be in:
Thermal equilibrium

Question No:49 **(Marks:1)** **Vu-Topper RM**
Which method(s) of heat transfer can take place in vacuum?
Electromagnetic waves

Question No:50 **(Marks:1)** **Vu-Topper RM**
A step-down transformer is used to:
Decrease the voltage

Question No:51 **(Marks:1)** **Vu-Topper RM**
In an adiabatic process:
The work done by the environment on the system equals the change in internal energy.

Vu Topper RM

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Question No:52 **(Marks:1)** **Vu-Topper RM**

The focal length of a spherical mirror is N times its radius of curvature where N is:

F=R/2

Question No:53 **(Marks:1)** **Vu-Topper RM**

Capacitive reactance Xc is:

1/2 π fc

Question No:54 **(Marks:1)** **Vu-Topper RM**

Force per unit charge is:

Electric field.

Question No:55 **(Marks:1)** **Vu-Topper RM**

If the kinetic energy of a free particle is much less than its rest energy then its kinetic energy is proportional to:

The square of the magnitude of its momentum

Question No:56 **(Marks:1)** **Vu-Topper RM**

Which physical quantity is produced by a calculation where a charge is multiplied by a potential difference (p.d)?

Energy

Question No:57 **(Marks:1)** **Vu-Topper RM**

Work done on a charged particle moving in uniform magnetic field is:

Zero

Question No:58 **(Marks:1)** **Vu-Topper RM**

Which of the following is equivalent to 1 V?

1 Joule/1 coulomb = 1 J/C

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Question No:59 **(Marks:1)** **Vu-Topper RM**
If Io is the peak value of an AC supply, then its rms value is given as Irms:?
IO/square root2

Question No:60 **(Marks:1)** **Vu-Topper RM**
Heat has the same units as:
Work

Question No:61 **(Marks:1)** **Vu-Topper RM**
Which of the following types of electromagnetic radiation at the vacuum?
(Radio, gamma, x)rays All of these travel at the same speed

Question No:62 **(Marks:1)** **Vu-Topper RM**
Inductive reactance XL of an inductor is pi
2(pi)Fl

Question No:63 **(Marks:1)** **Vu-Topper RM**
The sum of electric and magnetic force is called
Lorentz force

Question No:64 **(Marks:1)** **Vu-Topper RM**
Kirchhoff's 2nd rule is the manifestation of the law of conservation of
Energy

Question No:65 **(Marks:1)** **Vu-Topper RM**
The total energy in an circuit is 5.0 * 10-6 J. if L=25mH the maximum current is:
20mA

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Question No:66 **(Marks:1)** **Vu-Topper RM**
The potential difference between the ends of a conductor is 12V. How much electrical energy is converted to other forms of energy in the conductor when 100C of charge flows through it?
1200J

Question No:67 **(Marks:1)** **Vu-Topper RM**
The Focal length of convex lens is:
Positive

Question No:68 **(Marks:1)** **Vu-Topper RM**
The AC circuit in which current and voltage are in phase the power factor is
1 0.5

Question No:69 **(Marks:1)** **Vu-Topper RM**
The known laws of electromagnetism before James Clerk Maxwell were following
EXCEPT
Faraday's Law deborglie law

Question No:70 **(Marks:1)** **Vu-Topper RM**
When some dielectric is inserted between the plates of a capacitor, then capacitance;
Increased

Question No:71 **(Marks:1)** **Vu-Topper RM**
Which of the following is NOT the characteristics of electromagnetic waves?
They can travel with different speeds in vacuum depending on0.4m

Question No:72 **(Marks:1)** **Vu-Topper RM**
The quantity of deltaV/deltar is called
Potential gradient

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Vu Topper RM

Question No:73 **(Marks:1)** **Vu-Topper RM**
Which electromagnetic radiation transmits the highest photon energy?
Gamma rays

Question No:74 **(Marks:1)** **Vu-Topper RM**
The reaction of an particle at 50 Hz is 10.0 its reactance at 100Hz becomes
5.0

Question No:75 **(Marks:1)** **Vu-Topper RM**
Select the correct statement
Gamma Rays have higher frequency than infrared waves

Question No:76 **(Marks:1)** **Vu-Topper RM**
Potentiometer can be used as
Potential divider

Question No:77 **(Marks:1)** **Vu-Topper RM**
All of the followings are the different types of mental images; people generally experience them while close their eyes Except:
Solving an algebra equations

Question No:78 **(Marks:1)** **Vu-Topper RM**
The nation that motivational state of the organism is understood and explained in terms of positive and negative environmental stimuli BEST connects to;
Incentive approaches

Question No:79 **(Marks:1)** **Vu-Topper RM**
If one wants to increase the capacity of short-term memory, through which process more items can be held?
Chunking

Vu Topper RM

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Question No:80 **(Marks:1)** **Vu-Topper RM**
If potential difference across two plates of a capacitor is doubled, then
Four times

Question No:81 **(Marks:1)** **Vu-Topper RM**
How could the unit of potential difference the volt also be written?
J/C

Question No:82 **(Marks:1)** **Vu-Topper RM**
If magnetic field is doubled then magnetic energy density becomes
Four times

Question No:83 **(Marks:1)** **Vu-Topper RM**
The primary of a 3.1 step-up transformer is connected to a source and the secondary this situation is P. If R is connected directly to the source it will dissipate a power of:
P/9

Question No:84 **(Marks:1)** **Vu-Topper RM**
A particle having 2e charge falls through a potential difference of 5.V energy acquired by it is
10 eV

Question No:85 **(Marks:1)** **Vu-Topper RM**
The combined effect of resistance and reactance is known as
Impedance

Question No:86 **(Marks:1)** **Vu-Topper RM**
A transparent refracting medium bounded by the two curved surfaces called
Prism LENS

Vu Topper RM

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Question No:87 **(Marks:1)** **Vu-Topper RM**

A student rubs a rod held in his hand. Which action causes the rod to gain a large electrostatic charge

Rubbing a polythene rod with a woolen duster

Question No:88 **(Marks:1)** **Vu-Topper RM**

A charge of 30 C flows through an electric appliance in 2.0 minutes. What is the average current in the appliance?

0.25A

Question No:89 **(Marks:1)** **Vu-Topper RM**

Total flux through a closed surface depends on

Charge and -N CHARGE ENCLOSED

Question No:90 **(Marks:1)** **Vu-Topper RM**

Electromagnetic waves emitted from antenna are

Transverse

Question No:91 **(Marks:1)** **Vu-Topper RM**

Two parallel straight wires carrying current in same direction will.

Attract each other

Question No:92 **(Marks:1)** **Vu-Topper RM**

Charge carriers in electrolysis are:

Positive and negative ions

Question No:93 **(Marks:1)** **Vu-Topper RM**

A detector is used to detect the speed at a particular wave if the frequency of an electromagnetic signal is 400 MHz, the speed define the detector is

300000 km/s

Vu Topper RM

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Question No:94 **(Marks:1)** **Vu-Topper RM**
The product of resistance and capacitance is called
Time

Question No:95 **(Marks:1)** **Vu-Topper RM**
Greenhouse gases absorb ____ radiation
Infrared

Question No:96 **(Marks:1)** **Vu-Topper RM**
Electromagnetic waves do not transport
Charge

Question No:97 **(Marks:1)** **Vu-Topper RM**
Electric energy is measured :
Watt

Question No:98 **(Marks:1)** **Vu-Topper RM**
Of the following which might NOT vanish over one cycle of a cyclic process?
The work done by the substance

Question No:99 **(Marks:1)** **Vu-Topper RM**
The most refined form of matter is:
Light

Question No:100 **(Marks:1)** **Vu-Topper RM**
In Compton scattering from stationary particles the maximum change in wavelength can be made smaller by using.
More massive particles

Question No:101 **(Marks:1)** **Vu-Topper RM**
Hotness or coldness of an object is represented in terms of:
Temperature

Vu Topper RM

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Question No:102 **(Marks:1)** **Vu-Topper RM**
The unit of thermal conductivity might be:
Cal/(cm*s*C?)

Question No:103 **(Marks:1)** **Vu-Topper RM**
It is more di? Cult to measure the coe? Cient of volume expansion of a liquid than that of a solid because?
The containing evessel aslo xpands

Question No:104 **(Marks:1)** **Vu-Topper RM**
When we accelerate the charge, which types of waves are produced?
Electromagnetic waves

Question No:105 **(Marks:1)** **Vu-Topper RM**
Due to relative motion of observer and the frames of reference of events, time always:
Statics itself

Question No:106 **(Marks:1)** **Vu-Topper RM**
The allowed energy states of atoms or molecules can be described by a set of _____.
Quantum numbers

Question No:107 **(Marks:1)** **Vu-Topper RM**
Probability values are assigned on a scale of
0 to 1

Question No:108 **(Marks:1)** **Vu-Topper RM**
The magnitude of the momentum of a particle can never exceed:
E/c, where E is its total energy

Question No:109 **(Marks:1)** **Vu-Topper RM**
The ratio of the size of image to the size of object.
Angular magnification

Vu Topper RM

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Question No:110 **(Marks:1)** **Vu-Topper RM**
When white light passes through a prism, it splits into _____ colors.
7

Question No:111 **(Marks:1)** **Vu-Topper RM**
A heat of transformation of a substance is:
The energy per unit mass absorbed as heat during a phase transformation

Question No:112 **(Marks:1)** **Vu-Topper RM**
If electric and gravitational force on an electron in a uniform electric field balance each other, then the intensity of electric field will be:
Mg/q

Question No:113 **(Marks:1)** **Vu-Topper RM**
During a slow adiabatic expansion of a gas No energy enters or leaves as heat
52

Question No:114 **(Marks:1)** **Vu-Topper RM**
The electron states that constitute a single shell for an atom all have:
The same value of n

Question No:115 **(Marks:1)** **Vu-Topper RM**
An isothermal process for an ideal gas is represented on a p-V diagram by:
A portion of a hyperbola

Question No:116 **(Marks:1)** **Vu-Topper RM**
The concept of an electric field lines is introduced by:
Faraday

Vu Topper RM

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Question No:117 **(Marks:1)** **Vu-Topper RM**
The pauli exculusion principle is obeyed by:
All particles with spin quantum numbers of ½

Question No:118 **(Marks:1)** **Vu-Topper RM**
An electron (m=9.11 * 10-31kg) has a speed of 0.95C. its kinetic energy is;
1.8*10-31J

Question No:119 **(Marks:1)** **Vu-Topper RM**
An electron (m=9.11 x 10-31kg) has a momentum of 4.0x10-22kg*m/s. Its kinetic energy:
6.3*10-14J

Question No:120 **(Marks:1)** **Vu-Topper RM**
Force on a charged particle is zero when projected at angle with the magnetic field
0 degree

Question No:121 **(Marks:1)** **Vu-Topper RM**
In case of metallic conductors, the charge carriers are.
Electrons

Question No:122 **(Marks:1)** **Vu-Topper RM**
As compared to distance measured by an observer on Earth, the distance from earth star measured by an observer in a moving spaceship would seem:
Same

Question No:123 **(Marks:1)** **Vu-Topper RM**
Evidence that a gas consists mostly of empty space is the fact that
The density of a gas becomes much greater when it is
Liquefied

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Question No:124 **(Marks:1)** **Vu-Topper RM**
For an electron, the rest mass energy is:
0.511 MeV

Question No:125 **(Marks:1)** **Vu-Topper RM**
Antiparticle of electron is:
Position

Question No:126 **(Marks:1)** **Vu-Topper RM**
According to the first law of thermodynamics, applied to a gas, the increase in the internal energy during any process
Equals the heat input plus the work done on the gas

Question No:127 **(Marks:1)** **Vu-Topper RM**
The approach to personality stresses that people possess a basic goodness, and have a natural tendency to grow to higher levels of functioning is.
Humanistic

Question No:128 **(Marks:1)** **Vu-Topper RM**
In a photoelectric experiment at a frequency above cut off, the stopping potential is proportional to :
The energy of the most energetic electron after it is ejected

Question No:129 **(Marks:1)** **Vu-Topper RM**
There is a temperature at which the reading on the kelvin scale is numerically:
Equal to that on the Celsius scale

Question No:130 **(Marks:1)** **Vu-Topper RM**
When you stand in front of a plane mirror, your image is:
Virtual, erect and the same size as you

Vu Topper RM

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Question No:131 **(Marks:1)** **Vu-Topper RM**
The process for which entropy remains constant is:
A reversible process

Question No:132 **(Marks:1)** **Vu-Topper RM**
The appearance of colors in thin film is due to:
Interference

Question No:133 **(Marks:1)** **Vu-Topper RM**
The theoretical uppr limit for the frequency of electromagnetic waves is:
There is no upper limit

Question No:134 **(Marks:1)** **Vu-Topper RM**
Capacitance of a capacitor in vaccum is given by:
A eo /d

Question No:135 **(Marks:1)** **Vu-Topper RM**
AN electron in a K shell of an atom has the principal quantum number:
N =1

Question No:136 **(Marks:1)** **Vu-Topper RM**
Bending of light around the edges of an obstacle called:
Diffraction

Question No:137 **(Marks:1)** **Vu-Topper RM**
Interface of light is evidence that:
Light is a wave phenomenon

Question No:138 **(Marks:1)** **Vu-Topper RM**
In a photoelectric e? ect experiment at a frequency above cut o? the number of electrons ejected is proportional to:
The number of photons that hit the sample

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Question No:139 **(Marks:1)** **Vu-Topper RM**
In a photoelectric e? ect ecperiment the stopping potential is:
The electric potential that causes the electron current to vanish

Question No:140 **(Marks:1)** **Vu-Topper RM**
A free electron and a free proton have the same kinetic energy . This means that, compared to the matter wave associated with the proton, the matter wave associated with the electron has:
A longer wavelength and the same frequency

Question No:141 **(Marks:1)** **Vu-Topper RM**
Polarization experiments provide evidence that light is:
A transverse wave

Question No:142 **(Marks:1)** **Vu-Topper RM**
If light of low wavelength is used in young's double slit experiment, then width of the fringe will:
Decreases

Question No:143 **(Marks:1)** **Vu-Topper RM**
Lenz's law is in accordance with law of conservation of
Energy

Question No:144 **(Marks:1)** **Vu-Topper RM**
According to relatively theory a particle of mass m with a momentum of 2mc has a speed of:
c/2

Question No:145 **(Marks:1)** **Vu-Topper RM**
The two metallic strips that constitute some thermostats must di? Er in:
Coefficient of linear expansion

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Question No:146 **(Marks:1)** **Vu-Topper RM**

Phase difference between V and I of an AC through resistor is:

0 degree

Question No:147 **(Marks:1)** **Vu-Topper RM**

Average value of current and voltage over complete cycle is:

Zero

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Phy101 Searching file

by Orange Monkey Team

1. **According to the equation of work, when a boy sits in one place and studies for the whole night, he does**

   No work

2. **When the velocity of an aeroplane is doubled, the momentum_____**

   Increase uniformaly

3. **The momentum of an object at a given instant is independent of its.**

   Acceleration

4. **What sort of energy does flying bird possess?**

   Both potential and kinetic energy

5. **The acceleration of projectile in x-direction:**

   $0ms^{-2}$

6. **The projectile path is also known as its**

   Trajectory

7. **If force of friction is negligible, then acceleration of two freely falling objects of different masses is:**

   Smaller acceleration for smaller mass

8. **Which of the following statement/s is/ are true?**
   - **The weight of a man on the moon is smaller than that on the earth**
   - **The mass of a man is the same on both the moon and the earth**
   - **We cannot determine our own mass in the outer space because there is no gravity**

1 only

9. **When the momentum of a body is doubled, its kinetic energy_____**

Becomes four times its initial energy

10. **Which of the following statement is true**

Gravity is necessary to measure both weight and mass

11. **It is easier to walk on the concrete road than on ice because**

_the friction between the ice and the feet is less than that between the concrete and feet

12. **Pre-second refers to the dimensions of physical quantity**

_____Velocity

13. **If the mass of moving objects is doubled then its K.E becomes**

4 times

14. The slope of a velocity-time the graph at any point may be identified with

Average acceleration

15. K.E can be defined as the dot product of

Mass and velocity

16. A wheel of radius 50cm having the angular speed of 5 rad/s will have linear speed in m/s

2.5

19. Lighter objects always reach the ground later than heaver objects

2 and 3 only

17. In _____ object returns to its original position if displaced slightly

Stable equilibrium

18. Newton first law is also called_____

Law of inertia

19. A mosquito's buzz is often rated with a decibel rating of 40 db _

20

20. The product of force and time is called change in _____

Momentum or impulse…. Both are correct

21.People try to keep the _____ over their feet n order to feel stable

Center of gravity

22. If the magnitude of force applied is incorrect the work done will____ be increased unit of distance is

Light year

23. Power is equal to the dot product of for4ce and

Velocity

24. Product of vector a with itself is equal to

A^2

25. For a body to be in equilibrium under the combined action of several forces

The sum of the torques about any point must equal zero

26.The ratio of circumference of a circle to its diameter is equal to

Pai

27.A net torque applied to a rigid object always tends to produce

Angular acceleration

28. The work done by gravity during the decent of a projectile

Is positive

29. A man with his arms at his sides is spinning on a light friction less

His angular momentum remains the same

30. Example of physical quantity is

Length

31.Swimming becomes possible because of

Third law of motion

32. The scalar product of two vectors is maximum when they are

Parallel

33. Which of the following statements are true of sound waves_____ identify all that apply

A sound wave is a mechanical wave

34. The angular momentum vector of earth about its rotation axis. Due to its daily rotation is directed

north

35. For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about

the same point

36. People try to keep the _____ over their feet in order to feel stable

Centre of gravity

37. Add two vectors of length 4m & 5m but their orientation is not known___the length after addition

Between 9m and 1 m

38. Which statement is not true for acceleration?
riding your bike straight down the street at a constant speed

39. Acceleration of an object must be zero at a point where

The instantaneous velocity is constant

40. Which of the following statements are true of sound waves____identify that apply

A sound wave is mechanical wave

41. If the distance between all pairs of particles of the body do not change by applying a force then the body is sad to be

Rigid

42. A couple produces

Purely rotation motion

43. To determine if a rigid body is equilibrium the vector sum of the gravitational forces acting on the particles of the body can be replaced by a single force acting at

The center of gravity

44. Work may be done by

Only living organisms

45. The dimension of joule is

ML^2T^{-2}

46.The unit kg m^2/s can be used for

Angular momentum

47. Unit of distance is

Meter

48. The law of inertia was firstly formulated by

Galileo

49. What is the angle of projection of projectile for which its maximum height and horizontal range are equal

90

50. A body will be in translational equilibrium if

Ef=0

51. Suitable units for the gravitational constant G are

$m^3/kg\ s^2$

54. As the wavelength of a wave in a uniform medium increases its frequency will

Remains the same

52.A sound wave is a mechanical wave not an electromagnetic wave this means that

A medium is required in order for sound waves to transport energy

53. The gravitational constant G has the derived units

Nm^2/kg^2

54.If the formula F=Gm $11m^2/r2$ the quantity G

Is a universal constant if nature

55.An object living in a circle at constant speed

Has an acceleration of constant magnitude

56. The fundamental dimension of angular momentum are

ML^2T^{-2} $Kgm^{2/}s$

57. The area under the velocity time graph is

Distance

58. A particle oscillating in simple harmonic motion is

In equilibrium at the ends of its path because the acceleration is zero there

59. An elastic collision is one in which

K.E and momentum are both conserved

65. When brakes are applied to a fact moving car the passenger will be thrown

Forward

60. Work has dimension like power

Power

61. A 1kg block is lifted vertically 1m by a boy. The work done by the boy is about.

1J

62. 0.0003711 in scientific notation is

3.7 * 10^-4

68. A bullet shoot straight up returns to its starting point in 10 sec. the initial speed was

49m/s

63. The rate of change of momentum of a body is equal to

Applied force

64. If you travelled for 2.5 hours with an average speed 48 miles/hours. The distance travelled is

120 miles

65. A stone is thrown up from the surface of earth, then it reaches a maximum height, kinetic energy is equal to:

Zero

66. A force of 100N acts upon a body for 5 seconds. What will be the change in momentum?

500Ns

67. The relative speed of approach is always equal to relative speed of separation in?

Perfectly elastic collision

68. If the magnitude of force applied is increased, the work done will ?

Be increased

69. 6 joule of work done in 3 second then power is?

2 watt

70. An object moving in the circle at constant speed.

Must have only one force acting on it

71. A vector a is added to vector b. the resultant vector(a+b) have greatest magnitude when:

When vector A and B are per pendicular

72. A force of 120 newton is exerted on a 40 kg container which site on a floor. If the friction force between floor and container is 80 Newton. What is the magnitude of acceleration of container?

2ms^-2

73. The angle between rectangular components of a vector is

45 degree

74. Newton second law of motion is also called ?

Law of acceleration

75. The goal of all scientific enquiry or scientific method is?

Predicting natural events based on known patterns

76. If the total momentum of a system is changing

A next external force must be acting on the system

77.Change in momentum in one second is called?

Impulse

78. A/An _____ is the basic reason to change the motion of an object according to Newton's second law of motion.

Net force

79. An object moves in a circle at constant speed. The work done by centripetal force is zero because_?

The centripetal force is perpendicular to the velocity

80. The law of conservation of momentum applies to a system of colliding object only if

The net external impulse is zero

81.Force is a

Universal quantity

82. When the mass of the colliding body is much larger than the mass of the body at rest, its velocity after collision

Become double

83. The slope of a velocity – time the graph at any point may be identified with

Average acceleration

84.If the force acting on a body is doubled then the acceleration becomes

Double

85.Force:

Equal the time rate of chnge of momentum

86. When a spring is compressed or stretched, the potential energy of the spring

Increases

87. Which of the following statement is true

Weight is a force, mass is a measure of inertia

88. A ball is allowed to fall freely from certain height. It covers a distance is 1st second equal to

2g

89. Acceleration of an object must be zero at a point where:

The average velocity is zero

90. A body is changing its direction but its speed is constant, the object is:

Accelerating

91. Per-second refers to the dimension of physical quantity

Velocity

92. The scalar product of two vectors is maximum when they are

Parallel

93. Physics is one of the branches of

Physical sciences

94. Suppose no reaction force exists then which of the following case\s will occur?

- **We cannot swim**
- **We cannot jump**
- **We cannot walk**

1 2 and 3 all

95. Newton-Second is SI unit of:

Impulse

96. Work done in gravitational field

Negative

97. When body moves with constant acceleration the velocity time graph is:

Straight line

1. A sound wave has a wavelength of 3.0m. The distance from a compression center to the adjacent rarefaction center is:

    1. 0.75m
    2. 1.5m
    3. 3.0m
    4. need to know wave speed

2. The speed of sound in medium depends upon

    1. amplitude
    2. frequency
    3. wavelength
    4. properties of the medium

3. Which of the following will remain unchanged when a sound wave travels in air or in water?

    1. Amplitude
    2. Wavelength
    3. Frequency
    4. Speed

4. The vibrations or the pressure variations inside the inner air are converted into electrical signals by the _____.

    1. cochlea
    2. tympanic membrane
    3. pinna
    4. anvil

5. Sound and light waves both

    1. have similar wavelength
    2. obey the laws of reflection
    3. travel as longitudinal waves
    4. travel through vacuum

6. A mosquito's buzz is often rated with a decibel rating of 40 dB. Normal conversation is often rated at 60 dB. How many times more intense is normal conversation compared to a mosquito's buzz?

    1. 2
    2. 20
    3. 100
    4. 400

7. Point where force causes system to move without rotation is

    1. 2 times beyond centre of mass
    2. edge of the body
    3. mid point of body
    4. center of mass

8. The frequency which is not audible to the human ear is

    1. 50 Hz
    2. 500 Hz
    3. 5000 Hz
    4. 50000 Hz

9. If the speed of sound is 340 m/s a plane flying at 400m/s creates a conical shock wave with an apex half angle of:

    1. 0 (no shock wave)
    2. 32?
    3. 40?
    4. 58?

10. A municipal water supply is provided by a tall water tower. Water from this tower flows to a building. How does the water flow out of a faucet on the ground floor of a building compare with the water flow out of an identical faucet on the second floor of the building?

    1. Water flows more rapidly out of the ground-floor faucet.
    2. Water flows more rapidly out of the second-floor faucet.
    3. Water flows at the same speed out of both faucets.
    4. The speed of water flow cannot be determined unless the height of the water tower is known.

11. A sound wave has a wavelength of 3.0m. The distance from a compression center to the adjacent to the rarefaction center is:

    1. 0.75m
    2. 1.5m
    3. 3.0m
    4. need to know frequency

12. A plane produces a sonic boom only when:

    1. it emits sound waves of very long wavelength
    2. it emits sound waves of high frequencyit flys at high altitudes
    3. it flys faster than the speed of sound
    4. it flys faster than the speed of sound

13. acceleration of an object must be zero at a point where:

    1. the instantaneous velocity is zero but changing
    2. the instantaneous velocity is constant
    3. the average velocity is zero
    4. the instantaneous velocity is not zero but changing

14. Coulomb's law is only true for point charges whose sizes are

    1. large
    2. very small
    3. very large
    4. medium

15. A vector A is added to a vector B. The resultant vector A + B have greatest magnitude when:

    1. The magnitude of vector A + B does not depend on the directions of A and B
    2. When vectors A and B are parallel and in the opposite direction
    3. When vectors A and B are parallel and in the same direction
    4. When vectors A and B are perpendicular

16. Earth exerts a gravitational force on the Moon, keeping it in its orbit. The reaction to this force, in the sense of Newton's third law, is:

1. the centripetal force on the Moon
2. the nearly circular orbit on the Moon
3. the gravitational force on Earth by the Moon
4. the tides due to the Moon

17. The angular momentum vector of Earth about its rotation axis, due to its daily rotation, is directed:

1. south
2. north
3. tangent to the equator toward the west
4. tangent to the equator toward the east

18. Work may be done by _____.

1. only vehicles
2. only non-living objects
3. only living organisms
4. both living organisms and non-living objects

19. In simple harmonic motion, the restoring force must be proportional to the:

1. amplitude
2. frequency
3. velocity
4. displacement

20. The turning effect of a force is called the force's

1. momentum
2. torque/moment
3. distance
4. inertia

21. How much pressure is exerted on a submarine at a depth of 8.50 km in the Pacific Ocean? (The density of sea water = 1.025 * 103 kg/m3, and the atmospheric pressure at sea level = 1.01 * 105 Pa.)

1. 8.6 * 105 Pa
2. 8.7 * 106 Pa
3. 9.5 * 106 Pa

4. 8.6 * 106 Pa

22. A 4.0-m long steel beam with a cross-sectional area of 1.0 × 10-2 m2 and a Young's modulus of 2.0 × 1011 N/m2 is wedged horizontally between two vertical walls. In order to wedge the beam, it is compressed by 0.020mm. If the coefficient of static friction between the beam and the walls is 0.70 the maximum mass (including its own) it can bear without slipping is:

1. 3.6 kg
2. 36 kg
3. 71 kg
4. 710 kg

23. The torque about any two points has the same value when the body is in

1. translational equilibrium
2. neutral equilibrium
3. dynamic equilibrium
4. static equilibrium

24. The work done by gravity during the descent of a projectile:

1. depends for its sign on the direction of the y axis
2. is positive
3. is negative
4. is zero

25. Take the speed of sound to be 340m/s. A thunder clap is heard about 3 s after the lightning is seen. The source of both light and sound is:

1. moving overhead faster than the speed of sound
2. emitting a much higher frequency than is heard
3. emitting a much lower frequency than is heard
4. about 1000m away

26. The magnitude of the acceleration of a planet in orbit around the Sun is proportional to:

1. the mass of the planet
2. the mass of the Sun
3. the distance between the planet and the Sun

4. **the reciprocal of the distance between the planet and the Sun**

27. **The unit kg-m2/s can be used for:**

    1. **angular momentum**
    2. **rational kinetic energy**
    3. **rational inertia**
    4. **torque**

28. **Which of the following statements about floating object is correct?**

    1. **The object's density is greater than the density of the fluid on which it floats.**
    2. **The object's density is equal to the density of the fluid on which it floats.**
    3. **The displaced volume of fluid is greater than the volume of the object.**
    4. **The buoyant force equals the object's weight.**

29. **When a spring is compressed or stretched, the potential energy of the spring**

    1. **Decreases**
    2. **stays constant**
    3. **Increases**
    4. **Becomes zero**

30. **An ultrasonic wave is sent from a ship towards the bottom of the sea. It is found that the time interval between the sending and receiving of the wave is 1.6 s. What is the depth of the sea, if the velocity of sound in the seawater is 1400 m/s?**

    1. **1120 m**
    2. **560 m**
    3. **1400 m**
    4. **112 m**

31. **To determine if a rigid body is in equilibrium the vector sum of the gravitational forces acting on the particles of the body can be replaced by a single force acting at:**

    1. **a point on the boundary**

2. **the center of gravity**
3. **the geometrical center**
4. **the center of mass**

32. **A couple produces:**

1. **Purely linear motion**
2. **Purely rotational motion**
3. **Linear and rotational motion**
4. **No motion**

33. **In _____ object returns to its original position if displaced slightly**

1. **dynamic equilibrium**
2. **stable equilibrium**
3. **unstable equilibrium**
4. **rotational equilibrium**

34. **The momentum of an object at a given instant is independent of its:**

1. **acceleration**
2. **velocity**
3. **speed**
4. **mass**

35. **A vector is obtained by dividing the vector by its magnitude:**

1. **unit**
2. **position**
3. **normal**
4. **negative**

36. **According to the equation of work, when a boy sits in one place and studies for the whole night , he does**

1. **lot of work**
2. **no work**
3. **too much work**
4. **very little work**

37. **The application/s of dimensional analysis is/are:**
i. To convert a physical quantity from one system of units to an other.

ii. To check the dimensional correctness of a given equation.
iii. Establish a relationship between different physical quantities in an equation.

1. i only
2. ii & iii only
3. i & iii only
4. I, ii & iii

38. The angle between rectangular components of a vector is:

1. 0*
2. 45*
3. 60*
4. 90*

39. A qualitative definition of force is given by which law?

1. law of inertia

40. Which pair will always have the same magnitude to the rate of change of position?

1. Average speed and average velocity

41. Add two vectors of length 4 m & 5 m but their orientation is not known, The length after addition of these two vectors will be:

1. Less than 1 m
2. between 9 m and 1 m
3. between 9 m and 5 m
4. 9m

42. What is the angle of projection of projectile, for which its maximum height and horizontal range are equal?

1. 30*

2. 36*
3. 76*
4. 90*

43. A _____ vector is obtained by dividing the vector by its magnitude:

1. unit
2. position
3. normal
4. negative

44. When the momentum of a body is doubled, its kinetic energy _____ .

1. Becomes ten times its kinetic energy
2. Becomes four times its initial kinetic energy

45. A ball is thrown upward into air with a speed that is greater than terminal speed. It lands at the place where it was thrown. During its flight the force of air resistance is the greatest:

1. just after it is thrown
2. half way up
3. at the top of its trajectory
4. halfway down

46. An object moves in a circle at constant speed. The work done by the centripetal force is zero because:

1. the displacement for each revolution is zero
2. the average force for each revolution is zero
3. there is no firiction
4. the centripetal force is perpendicular to the velocity

47. It is easier to walk on concrete road than on ice because:

1. the amount of firection is same for both (ice & concrete road)
2. there is more friction on the ice than on concrete
3. ice is soft and spongy where as concrete is hard
4. the friction between the ice and the feet is less than that between the concrete and feet

48. An inellastic collidion is one in which:

1. momentum is not conserved but kinetic energy is conserved
2. total mass is not conserved but momentum is conserved
3. neither kinetic energy nor momentum is conserved
4. momentum is conserved but kinetic energy is not conserved

49. When you step on the accelerator to increase the speed of your car, the force that accelerates the car is

1. the force of friction of the road on the tires

50. Light year is a unit of:

1. accelration
2. velocity
3. time
4. distance

51. A body is changing its direction but speed is constant, the object is:

1. accelerating
2. slowing down
3. speeding
4. Displacing

52. The ratio of circumference of a circle to its diameter is equal to:

1. 2 π
2. π
3. π /2
4. one steradian

53. **Which of the following statements are TRUE of sound waves? Identify all that apply.**

1. **Sound can travel through a vacuum**
2. **A sound wave is a transverse wave**
3. **A sound wave is a mechanical wave**
4. **To hear the sound of a tuning fork, the tines of the fork must move air from the fork to one's ear**

54. **A force of 120 N is exerted on a 40 kg container which sits on a floor. If the frictional force between floor and container is 80 N. What is the magnitude of the acceleration of the container?**

1. **2 ms^-2**
2. **5 ms^-2**
3. **3 ms^-2**
4. **1 ms^-2**

55. **If a simple pendulum oscillates with an amplitude 50 mm and time period 2s, then its maximum velocity is**

1. **0.1 m/s**
2. **0.15 m/s**
3. **0.8 m/s**
4. **0.16 m/s**

56. **Acceleration in a body is always produced in the direction of:**

1. **Velocity**
2. **Weight**
3. **Force**
4. **Acceleration**

57. **The center of mass of a uniform disk of radius R is located:**

1. **at the center**
2. **a distance R/3 from the center**
3. **a distance R/2 from the center**

4. on the rim

58. For a body to be in equilibrium under the combined action of several forces:

1. any two of these forces must be balanced by a third force
2. the sum of components of all the forces in any direction must equal zero
3. all of the forces form pairs of equal and opposite forces
4. all the forces must be applied at the same point

59. A source of frequency of 500 Hz emits waves of wavelength 0.4 m, how long does the wave take to travel 600 m?

1. 3 s

60. Momentum may be expressed in:

1. kg/(m s)
2. N s
3. gram s
4. kg/m

61. A force of 5000N is applied outwardly to each end of a 5.0-m long rod with a radius of 34.0 cm and a Young's modulus of 125 x 108 N/m^2. The elongation of the rod is:

1. 0.0020mm
2. 0.0040mm
3. 0.14mm
4. 0.55mm

62. The speed of a sound wave is determined by:

1. its amplitude

2. its intensity
3. the transmitting medium
4. number of harmonics present

63. Which of the following statement is TRUE of sound intensity and decibel levels? Identify all that apply.

1. The intensity of sound which corresponds to the threshold of pain is one trillion times more intense than the sound which corresonds to the threshold of hearing.
2. Two sounds which have a ratio of decible ratings equal to 2.0. This means that the second sound is twice as intense as the first sound.
3. Sound A is 20 times more intense than the sound B. So if sound B is rated at 30 dB, then sound A is rated at 50 dB.
4. A machine produces a sound which is rated at 60 dB. If two of the machines were used at the same time, the decibel rating would be 120 dB.

64. Sound waves are

1. longitudinal

65. For an ideal fluid flowing through a horizontal pipe, Bernoulli's equation states that the sum of the pressure and energy per unit volume along the pipe does which of the following? (Assume measurements are taken along the pipe in the direction of fluid flow.)

1. increases as the pipe diameter increases
2. decreases as the pipe diameter increases
3. remains constant as the pipe diameter increases
4. increases, then decreases as the pipe diameter increases

66. A 2.0-kg block travels around a 0.5-m radius circle with an angular velocity of 12 rad/s. Its angular momentum about the center of the circle is:

1. 6.0 $kg.m^2/s$
2. 12 $kg.m^2/s$
3. 48 $kg/m^2.s$
4. 72 $kg.m^2/s^2$

67. The fundamental dimensions of angular momentum are

1. $ML^{-2}T^{-2}$
2. MLT^{-1}
3. $ML^{2}T^{-1}$
4. $ML^{2}T^{-2}$

12. Power is equal to the dot product of force and:

c) Velocity

2) The frequency which is not audible to the human ear is

Select the correct option

a) 50000 Hz

b) 500 Hz

c) 5000 Hz

d) 50 Hz

3) A wheel of radius 50 cm having the angular speed of 5 rad/s will have linear speed in m/s?

Select the correct option

a) 1.5

b) 2.5

c) 3.5

d) 0.5

4) When breaks are applied to a fast-moving car, the passenger will be thrown:

Select the correct option

a) Backward

b) Forward

c) Upward

d) Downward

5) Stress can be measured in:

Select the correct option

a) N.m

b) N/m2

c) N.m2

d) N/m

6) A heavy particle moving with 5 m/s suffers an elastic collision with a light particle at rest. After collision, velocity of light particle will be:

Select the correct option

a) 10 m/s

b) 2.5 m/s

c) 20 m/s

d) 5 m/s

6) Acceleration in a body is always produced on the direction of:

Select the correct option

a) Velocity r

b) Weight

c) Acceleration

d) Force

7) The fundamental dimensions of angular momentum are

Select the correct option

a) ML-2T-2

b) ML2T-1

c) ML2T-2

d) MLT-1

8) Ali wants to lift a mass of 7.5 kg with constant velocity by a rope that passes through a frictionless pulley which is attached to the ceiling of room. Calculate the tension in the rope by neglecting the mass of the rope.

Select the correct option

a) 75 N

b) 7.5 Kg

c) 75 Kg

d) 0.75 N

9) The first condition of equilibrium implies that:

Select the correct option

a) ∑Fy = 0

b) ∑Fx = ∑Fy

c) ∑F = 0

d) ∑Fx = 0

10) Final take off velocity of an airplane is 67 m/s. The length of runway is 2 km, the constant acceleration is:

Select the correct option

a) 3.24 ms^-1

b) 3.24 ms^-2

c) 2.24 ms^-1

d) 2.24 ms^-2

11) The ultimate strength of a sample in the stress at which the simple:

Select the correct option

a) remain underwater

b) breaks

c) bends 180?

d) returns to its original shape when the stress is moved

12) The area under the velocity-time graphs is:

Select the correct option

a) Acceleration

b) Torque

c) Distance

d) Force

13) if you traveled for 2.5 hours with an average speed 48 miles/hours, the distance traveled is:

Select the correct option

a) 19.2 miles

b) 120 miles

c) 300 miles

d) 48 miles

14) The law of inertia was firstly formulated by:

Select the correct option

a) Newton

b) Einstein

c) Galileo

d) Aristotle

15) When the velocity of an aeroplance is doubled, the momentum ___.

Select the correct option

a) Is conserved

b) Becomes zero

c) Increase uniformly

d) Remains unchanged

16) As we move above, the body above the surface of the earth, the change in potential energy will be:

Select the correct option

a) Infinity

b) Positive

c) Zero

d) Negative

17) The slope of a velocity – time the graph at any point may be identified with:

Select the correct option

a) average acceleration

18) A body of mass 2 kg moving with velocity 4 m/s has K.E equals to:

Select the correct option

a) 16 J

b) 8 J

c) 32 J

d) 2 J

19) A ___ velocity is obtained by dividing the vector by its magnitude:

Select the correct option

a) negative

b) unit

c) position

d) normal

20) A particle oscillating in simple harmonic motion is:

Select the correct option

a) in equilibrium at the center of its path because the acceleration is zero there

b) never in equilibrium because it is in motion

c) in equilibrium at the end of its path because the acceleration is zero there

d) never in equilibrium because there is always a force

22) Young's modulus is a proportionality constant that relates the force per unit area applied perpendicularly at the surface of an object to:

Select the correct option

a) the fractional change in volume

b) the fractional change in length

c) the shear

d) the pressure

23) A bullet shoot straight up returns to its sharing point in 10 sec. the initial speed was:

Select the correct option

a) 49 m/s

b) 9.8 m/s

c) 98 m/s

d) 24.5 m/s

24) An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

Select the correct option

a) 1.57 rad/s

b) 12.6 rad/s

c) 2.0 rad/s

d) 6.3 rad/s

25) The dimensional units of ratio of work and power is:

Select the correct option

a) F

b) J

c) T

d) L

26) The newton-second is a unit of:

Select the correct option

a) power

b) angular momentum

c) linear momentum

d) work

27) A sound wave has a wavelength of 3.0m. The distance from a compression center to the adjacent rarefaction center is:

Select the correct option

a) 3.0m

b) 1.5m

c) 0.75m

d) need to know wave speed

28) For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

Select the correct option

a) the center of gravity

b) the same point

c) the geometrical center

d) the center of mass

29) If a simple pendulum oscillates with an amplitude 50 mm and time period 2s, then its maximum velocity is

Select the correct option

a) 0.8 m/s

b) 0.16 m/s

c) 0.15 m/s

d) 0.1 m/s

30) bank and forth is a leftward and rightward direction. This type of wave is known as a ___.

Select the correct option

a) Mechanical

b) Electromagnetic

c) Transverse

d) Longitudinal

18) A/an ___ is the basic reason to change in the motion of an object according to Newton's second law of motion

Select the correct option

a) acceleration

b) change in velocity

c) **net force**

d) decrease in inertia

19) If the distance between all pairs of particles of the body do not change by applying a force, then a body is said to be

Select the correct option

a) **rigid**

b) small

c) large

d) flexible

20) The center of mass of Earth's atmosphere is:

Select the correct option

a) near the outer boundary of the atmosphere

b) **near the center of earth**

c) a little less than halfways between earth's surface and the outer boundary of the atmosphere

d) near the surface of earth

21) An object attached to one end of a spring males 20 complete oscillations in 10 s. Its period is:

Select the correct option

a) 0.5 s

b) 0.5Hz

c) **2 s**

d) 2Hz

22) Whenever an object strikes a stationary object of equal mass:

Select the correct option

a) the first object must stop

b) the two objects cannot stick together

c) **none of these**

d) the collision must be elastic

23) If a wheel turns with constant angular speed then:

Select the correct option

a) each point on its moves with constant velocity

b) each point on its rim moves with constant acceleration

c) the wheel turns through equal angles in equal times

d) **the angle through which the wheel turns in each second increases as time goes on**

24) A stone is thrown up from the surface of earth, the it reaches

a maximum height, kinetic energy is equal to:

Select the correct option

a) ½ mv

b) mgh

c) 2mgh

d) **zero**

25) In Simple harmonic motion, the magnitude of the acceleration is

Select the correct option

a) greatest when the velocity is greatest

b) **proportional to the displacement**

c) inversely proportional to the displacement

d) constant

The angle between rectangular components of a vector is:

Select the correct option

a) 0^0

b) 45^0

c) 60^0

d) **90^0**

10) A ___ vector is obtained by dividing the vector by its magnitude

Select the correct option

a) **unit**

b) position

c) normal

d) negative

11) According to the equation of work, when a boy sits in one place and studies for the whole night, he does

Select the correct option

a) lot of work

b) **no work**

c) too much work

d) very little work

12) the projectile path is known as its:

Select the correct option

a) range

b) **trajectory**

c) curve

d) time of action

13) If force of friction is negligible, then acceleration of two freely falling objects of different masses is:

Select the correct option

a) None of these

b) **Smaller acceleration for smaller mass**

c) Variable

d) The same

14) Add two vectors of length 4 m & 5m but their orientation is not known. The length after addition of these two vectors will be:

Select the correct option

a) **between 9 m and 1 m**

b) 9 m

c) between 9 m and 5

d) Less than 1 m

15) The dimensional units of ratio of work and power is:

Select the correct option

a) L

b) F

c) J

d) **T**

16) A sphere of moment on inertia I rolls down an inclined plane without slipping. The ratio of the rotational kinetic energy to the transitional kinetic energy is nearly:

Select the correct option

a) **5/7**

b) 7/5

c) 2/5

d) 5/2

17) Swimming becomes possible because of:

Select the correct option

a) First law of motion

b) Second law of motion

c) **Third law of motion**

d) Law of torque

An object moves in a circle at constant speed. The work done by the centripetal force is zero because:

d) **the centripetal force is perpendicular to the velocity**

3) When the velocity of an aeroplane is doubled, the momentum ___.

Select the correct option

a) **Increase uniformly**

b) Becomes zero

c) Is conserved

d) Remains unchanged

4) The momentum of an object at a given instant is independent of its:

Select the correct option

a) **acceleration**

b) velocity

c) speed

d) mass

5) What sort of energy does flying bird possess?

Select the correct option

a) Potential energy

b) Kinetic energy

c) Elastic energy

d) **Both potential and kinetic energy**

6) The acceleration of projectile in x-direction:

Select the correct option

a) depends on initial velocity

b) depends on how long it is in the air

c) depends on y-acceleration

d) **0 m/s^2**

7) Light year is a unit of:

Select the correct option

a) acceleration

b) velocity

c) time

d) distance

8) The goal of all scientific inquiry (or scientific method) is:

Select the correct option

a) world peace

b) **predicting natural events based**

c) to make everyone rich and happy

As the wavelength of a wave in a uniform medium increases, its frequency will _____.	Decrease	
9	A bullet shot straight up returns to its starting point in 10 sec. the initial speed was:	49 m/s
11	Young's modulus is a proportionality constant that relates the force per unit area applied perpendicularly at the surface of an object to:	the fractional change in length
12	A ball is thrown upward into the air with a speed that is greater than terminal speed. It lands at the place where it was thrown. During its flight the force of air resistance is the greatest:	at the top of its trajectory
13	A sphere of the moment on inertia I roll down an inclined plane without slipping. The ratio of the rotational kinetic energy to the transitional kinetic energy is near:	5/7
14	The persistence of audible sound due to the successive	Reverberation

	reflections from the surrounding objects even after the source has stopped to produce that sound is called	
15	A municipal water supply is provided by a tall water tower. Water from this tower flows to a building. How does the water flow out of a faucet on the ground floor of a building compare with the water flow out of an identical faucet on the second floor of the building?	Water flows more rapidly out of the ground-floor faucet.
16	An elastic collision is one in which:	kinetic energy and momentum are both conserved
17	The ultimate strength of a sample is the stress at which the sample	Breaks/ returns to its original shape when the stress is removed
18	The dot product of vector A with itself is equal to:	A^2
19	A heavy particle moving with 5 m/s suffers an elastic collision with a light particle at rest. After the collision, the velocity of light particle will be:	5 m/s
20	Young's modulus can be correctly given in:	N/m^2
21	A net torque applied to a rigid object always tends to produce:	angular acceleration
23	The velocity-time graph is parallel to the time axis the acceleration of moving body is:	zero
24	What is the angle of projection	76

	of projectile, for which its maximum height and horizontal range are equal?	
25	A particle	
26	The dimensional units of the ratio of work and power is:	T
27	The magnitude of the acceleration on the planet in orbit around the Sun is proportional to.	the mass of the Sun
28	A Prosonic wave I gent from @ ship towards the bottom of the sea. N is found that the time interval between the sending and receiving of the wave is 1.6 5. What is the depth of the sea. if the velocity of sound in the seawater is	1120m
29	Water flows into © house at e velocity of 15 m/s through a pipe that has 9 radii of 0.40 m. The water then flows through a smaller pipe at a velocity of 45 m/s. What is the radius of the smuttier pipe?	0.23m
30	A mosquito's buzz is often rated with a decibel rating of 40 dB. Normal conversation is often rated at 60 dB. How many times more intense is normal conversation compared to 6 mosquito's buzz?	20
31	People try to keep the... over their feet. in order to feel stable.	center of gravity
32	If force is applied at the center of mass then torque is.	Zero

33	As the wavelength of a wave in a uniform medium increases. its speed will	Decrease
34	For an ideal fluid flowing through a horizontal pipe, Bernoulli's equation state that the sum of the pressure and energy per unit volume along the pipe does which of the following?	Remains constant as the pipe diameter increases
35	Which of the following statement is true?	Weight is a force, mass is a measure of inertia
36	A/an ---------- is the basic reason to change in the motion of an object according to Newton's second law of motion.	Net force
37	When spring is compressed or stretched, the potential energy of spring	Increase
38	The ultimate strength of a sample is the stress in which the sample:	Breaks
39	The value of k in coulomb's law depends upon:	Medium between two charges
40	The law of inertia was firstly formulated by	Newton
41	The ratio of the circumference of a circle to its diameter is equal to:	Pi
42	A water bed that is 1.5 m wide and 2.5 m long weighs 1055 N. assuming the entire lower surface of the bed in contact	280 pa

	with the floor, what is the pressure the bed exerts on the floor?	
43	The center of gravity is the average location of the …….. of an object	Weight
44	Acceleration in a body is always produced in the direction of	Force
45	As per coulomb's law, the force of attraction or repulsion between two-point charge directly proportional to the	Product os the magnitude of charges
46	One revolution is the same as:	2pi rad
47	A ---------- vector is obtained by dividing the vector by its magnitude:	Unit
48	Take the speed of the sound to be 340m/s. A thunderclap is heard about 3 s after the lightning is seen. The source of both light and sound is:	About 1000m away
49	The fundamental dimensions of angular momentum are :	ML2T-1
50	The approximate value of get an altitude above Earth equal to one Earth diameter is:	4.9m/s2
51	If the speed of sound is 340m/s a plane flying at 400rm/s creates @ conical shock wave with an apex half-angle of:	Zero
52	If the distance between all pairs of particulars of the body do not change by applying a force that a body is said to be	Rigid

PHY101 quiz 3(2022)

1. Lenz's law is in accordance with law of conservation of

   Energy…confirm

2. Which of the following types of electromagnetic radiation at the greatest speed in vacuum?

   All of these travels at the same speed…..confirm

3. Inductive reactance XL of an inductor is
   a. 2(pi)fL….confirm
4. Force per unit charge is:

   Electric field intensity…confirm

5. The sum of electric and magnetic force is called

   Lorentz force…confirm

6. Radio waves and light waves are _____.

   Electromagnetic and transverse both…confirm

7. Kirchhoff's 1st rule is the manifestation of the law of conservation of

   Charge…confirm

8. Kirchhoff's 2nd rule is the manifestation of the law of conservation of

   Energy…confirm

9. The total energy in an circuit is 5.0 * 10-6 J. if L=25mH the maximum current is:

   20mA…confirm

10. If Io is the peak value of an AC supply, then its rms value is given as firms

    Io/square root2….confirm

11. The potential difference between the ends of a conductor is 12V. How much electrical energy is converted to other forms of energy in the conductor when 100C of charge flows through it?

    1200J….confirm

12. A particle carrying a charge of 2e falls through potential difference of 3V the energy acquired by it will be:

6 eV…confirm

13. An inductor may store energy in its:

Magnetic field…confirm

14. The graphical representation of Ohm's law is:

Straight line…confirm

15. Radio waves and light waves are _____.

Longitudinal waves…confirm

16. The increase in capacitance of a capacitor due to presence of dielectric is due to:

Electric polarization

17. The Focal length of convex lens is:

Positive…confirm

18. The AC circuit in which current and voltage are in phase the power factor is

1…confirm 0.5 60 DEGREE

19. The wavelength of red light is 700 nm. Its frequency is _____.

4.29*10^14 hertz…confirm

20. The known laws of electromagnetism before James Clerk Maxwell were following EXCEPT

Faraday's Law…confirm deborglie law

21. When some dielectric is inserted between the plates of a capacitor, then capacitance;

Increased…..confirm

22. Which of the following is NOT the characteristic of electromagnetic waves?

i. They can travel with different speeds in vacuum depending on frequency…..confirm

23. If electric and gravitational force on an electron in uniform electric field balance each other, then the intensity of electric field will be:

Mg/q

23. The increase in capacitance of

Larger plates provide greater capacity to store electric charge. Therefore, as **the area of the plates increase**, capacitance increases. Capacitance is directly proportional to the electrostatic force field between the plates. ... Therefore, as the dielectric constant increases, capacitance increases. 07-Jun-2017

a

Electric energy is measured

Watt…confirm

When we accelerate the charge, which types of waves are produced?

Electromagnetic waves….confirm

A particle device transmitted a signal in the form of electromagnetic waves of frequency 7.5 * 108Hz. What is the wavelength of the signal?

0.4m

A step-down transformer is used to:

Decrease the voltage

27. The quantity of delta V /delta r is called

Potential gradient

28. Which electromagnetic radiation transmits the highest photon energy?

Gamma rays….confirm

29. The reaction of a particle at 50 Hz is 10.0 its reactance at 100Hz becomes

5.0

30. Work done on a charged particle moving in uniform magnetic field is

Zero….confirm

31. Select the correct statement

Gamma Rays have higher frequency than infrared waves…confirm

32. Which of the following have the greatest wavelength?

Radio waves

33. Potentiometer can be used as

Potential divider….confirm

34. All of the followings are the different types of mental images; people generally experience them while close their eyes except:

Solving an algebra equations

35. The nation that motivational state of the organism is understood and explained in terms of positive and negative environmental stimuli BEST connects to;

Incentive approaches

36. If one wants to increase the capacity of short-term memory, through which process more items can be held?

Chunking….confirm

37. Which physical quantity is produced by a calculation where a charge is

Energy…confirm

38. The current in a car headlamp is 2.0 A. The Headlamp is switched on for 3.0

360 C

39. If potential difference across two plates of a capacitor is doubled, then

Four times….confirm

40. How could the unit of potential difference the volt also be written?

J/C

41. If magnetic field is doubled then magnetic energy density becomes

Four times….confirm

42. The primary of a 3.1 step-up transformer is connected to a source and the secondary this situation is P. If R is connected directly to the source it will dissipate a power of:

P/9….confirm

43. A particle having 2e charge falls through a potential difference of 5.V energy acquired by it is

10 eV

44. A generator supplies 100V to the primary call of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is

1000V

45. The combined effect of resistance and reactance is known as

Impedance…confirm

46. A transparent refracting medium bounded by the two curved surfaces called

LENS **Prism….confirm**

47. A student rubs a rod held in his hand. Which action causes the rod to gain a large electrostatic charge

Rubbing a polythene rod with a woolen duster….confirm

48. A charge of 30 C flows through an electric appliance in 2.0 minutes. What is the average current in the appliance?

0.25A….confirm

Total flux through a closed surface depends on

CHARGE ENCLOSED**Charge and -N...confirm**

49. Electromagnetic waves emitted from antenna are

Transverse….confirm

50. Two parallel straight wires carrying current in same direction will.

Attract each other…confirm

51. Charge carriers in electrolysis are:

Positive and negative ions…confirm

52. Capacitive reactance XC is

$1/2\pi fC$

53. The number of turns becomes double, but length remain same then magnetic field in the solenoid become

Double…confirm

54. If velocity of a conductor moving through Magnetic field B is made zero then motional emf is

-vBL ZERO

The amplitude modulation frequency ranges from

540 kHz to 1600 kHz…confirm from net

55. Average value of current and voltage over complete cycle is:

Zero…. Confirm

56. A detector is used to detect the speed at a particular wave if the frequency of an electromagnetic signal is 400 MHz, the speed define the detector is

300000 km/s

57. The product of resistance and capacitance is called

Time….confirm

58. Greenhouse gases absorb ____ radiation

Infrared…confirm

59. In RL series circuit phase angle is given by:

Tan $^{-1}$(w_L/ R)…confirm

60. One tesla is equal to

N/Am

61. An LC series circuit an inductance L and a capacitance C has an oscillation frequency f. Two inductors each with inductance L, and two capacitors, each with capacitance C, are all wired in series and the circuit is completed. The oscillation frequency is:

f…..confirm

62. Capacitance of a capacitor in vacuum is given by

Aεo/d….confirm

63. Force on a charged particle is zero when projected at angle with the magnetic field

0 degree….confirm

64. If an electron of charge is accelerated through a potential difference V, it will acquire energy.

QV **Ve….confirm**

65. In case of metallic conductors, the charge carriers are:

Electrons…confirm

66. Tesla is a unit of FLUX DENSITY

Magnetic flux….confirm

67. When a charge is projected perpendicular to a uniform magnetic field, its path is:

Circular…..confirm

68. Electromagnetic waves do not transport

Charge….confirm

69. The concept of an electric field lines is introduced by

Faraday….confirm

70. A capacitor in an LC oscillator has a maximum potential difference of 15V and a maximum energy of 360 uJ. At a certain instant the energy in the capacitor is 40 uJ. At that instant what is the emf induced in the inductor

5V….confirm

71. The wave form of alternating voltage is a:

Sine wave….confirm

72. If velocity of a conductor moving through a magnetic field B is made zero then motional emf is:

ZERO **-vBL…confirm**

72. Which one of is in the order of decreasing frequency?

Ultraviolet rays, visible light, radio waves…confirm

73. The lenz's law refers to:

Induced current….confirm

74. Which of the following statements is Not true about electromagnetic waves?

The electromagnetic radiation from a burning candle is polarized…confirm

75. Phase different between V and I of on A.C through resistor is:

0 degree

76. Which of the following is equivalent to 1 V?

1 J/C…..confirm

77. Which of the following is equivalent to one coulomb?

One ampere second….confirm

PHY101 QUIZ 4

ORANGE MONKEY TEAM

You are being informed that the 4th online quiz of Physics (PHY101) is going to held on 07th March, 2022 to 08th March, 2022. The quiz will consist of Multiple Choice Questions (MCQs) and cover Lessons 23 to 43.

NOTE

"QUIZ ATTEMPT KRNY SEY PHELY GOOGLE ZAROOR SATH MAIN OPEN RKH LYNA BCUZ MCQS NEW ARAHY HEIN ABHE BHE THAT"S WHY WAHN PER SEARHC KRLYNA JOO YAHN SEY NA MILEIN >>>JAZAKALLAH"

1. **The reason there are two slits, rather than one, in a young's experiment is:**

   To create a path length difference….confirm from net

2. **According to the kinetic theory of gasses, the pressure of a gas is due to:**

   Change of momentum of molecules as the strike the well….confirm from net

3. **What is the name of the relationship between the peak wavelength for a blackbody radiator and its temperature.**

   Wien's law

4. **What is the unit of magnification factor?**

   No units

5. **The special theory of relativity is applicable to the object moving with maximum velocity equal to**

   More than speed of light

6. **A free electron and a free proton have the same kinetic energy . this means that, compared to the matter wave associated with the proton, the matter wave associated with the electron has:**

   A longer wavelength and the same frequency

7. **No entropy change is associated with:**

   Adiabatic process

8. **Electromagnetic waves emitted from antenna are:**

   Transverse

9. **A virtual image is one**

   From which light rays diverge but do not pass through

10. **A detector is used to detect the speed at a particular wave if the frequency of an electromagnetic signal is 400 MHz, the speed define the detector is**

    300000 km/s

11. **Evidence that molecules of a gas are in constant motion is:**

    Two gases interdi? Use quickly

12. **Of the following which is the best evidence for the wave nature of matter?**

    The photoelectric e? ect THE REFLECTION OF ELECTRON BY CRYSTAL

13. **For holography we use a beam of:**

    Laser

14. **Interference of light is evidence that:**

    Light is a wave phenomenon

15. **Which of the following types of electromagnetic radiation travels at the greatest speed in vaccum?**

    (Radio, gamma, x)rays .. all options….confirm

16. **The plants look green. It is because they have chorophyll in them. Chorophyll makes leaves green to absorb_____.**

    Green only

17. **The most refined form of matter is_____.**

    Light

18. **The known laws of electromagnetism before james clerk Maxwell were . Except**

    Faraday's law deborglie law

19. **Which of the following electromagnetic radiations has photons with the greatest energy?**

    Radio waves

20. **The primary of a 3:1 step-up transformer is connected to a source and the secondary is connected to a register R. The power dissipated by R in this situation is P. if R is connected directly to the source it will dissipate a power of:**

    P/9

21. **The quantum number ms is most closely associated with whaat property of the electron in an atom?**

    Energy Z COMPONENT OF SPIN ANGULAR MOMENTUM

22. **The special theory of relatively is based on:**

    Two postulates

23. **The center of Newton ring is dark due to:**

    Destructive interference

24. **Which electromagnetic radiation transmits the highest photon energy/**

    Gamma rays

25. **Which methods of heat transfer can take place in vacuum?**

    Radiation

26. **An electron is moving at 0.6c. If we calculate its kinetic energy using (1/2) mV2. We get a result that is:**

    About 28% too low

27. **The wavelength of light beam A is twice the wavelength beam B. the energy of a photon in beam A is:**

    Half the energy of a photon in beam b

28. **The lanz's law refers to :**

    Induced current

29.

30. **Electric energy is measured :**

    Watt….confirm from ent

31. **The lenz's law refers to :**

    Induced current

32. **The films formed due to detergents(soap/shampoo) are due to**

    interference

33. **A free electron and a free proton have the same kinetic energy . this means that, compared to the matter wave associated with the proton, the matter wave associated with the electron has:**

    A longer wavelength and a smaller frequency …..confirm from net

34. **Of the following. Compton scattering from electrons is most easily observed for:**

    X rays

35. **Of the following which might NOT vanish over one cycle of a cyclic process?**

    The work done by the substance….confirm from net

36. **The most refined form of matter is:**

    Light

37. **How could the unit of potential difference, the volt, also be written?**

    J/C

38. **In Compton scattering from stationary particles the maximum change in wavelength can be made smaller by using.**

    More massive particles

39. **Hotness or coldness of an object is represented in terms of:**

Temperature

40. **The unit of thermal conductivity might be:**

Cal/(cm*s*C?)

41. **During the time that latent heat is involved in a change of state.**

The temperature does not change

42. **A laser in a compact disc player generates light that has a wavelength of 780 nm in air. The light then enters into the plastic of a CD. If the index of refraction of plastic is 1.55, the speed of this light once enter the plastic is _____.**

1.94 x 10^8 m/s

43. **Antiparticle of electron is**

Positron…..confirm from net

44. **When some dielectric is inserted between the plates of a capacitor then capacitance**

Increases….confirm from net

45. **The centre of Newton ring is dark due to:**

Destrictive interference

46. **If two objects are in thermal equilibrium with each other:**

They cannot be at different temperatures….confirm from net

47. **The graphical representation of Ohm's law is**

Straight line…..confirm from net

48. **A particular device transmitted a signal in the form of electromagnetic waves of frequency 7.5 * 108 Hz. What is the wavelength of the signal?**

0.4m

41. **The resolving power of a telescope can be increased by**

Increasing the lens diameter

49. **Which physical quantity is produced by a calculation where a charge is multiplied by a potential difference (p.d)?**

Energy….confirm form net

50. A transparent refracting medium bounded by the two curved surfaces called:

LENS

Prism….confirm from net

51. It is more di? Cult to measure the coe? Cient of volume expansion of a liquid than that of a solid because?

The containing evessel aslo xpands….confirm form net

52. The product of resistance and capacitance is called

Time….confirm from net

53. The potential difference between the ends of a conductor is 12 V. How much electrical energy is converted to other forms of energy in the conductor when 100C of charge flows through it?

1200….confirm form net

54. When we accelerate the charge, which types of waves are produced?

Electromagnetic waves

55. Due to relative motion of observer and the frames of reference of events, time always:

statics itself

56. The allowed energy states of atoms or molecules can be described by a set of _____.

Quantum numbers……confirm

57. Probability values are assigned on a scale of

0 to 1….confirm

58. The magnitude of the momentum of a particle can never exceed:

E/c, where E is its total energy

59.

60. The ratio of the size of image to the size of object.

Angular magnification

61. **When white light passes through a prism, it splits into _____ colors.**

7…….confirm from net

62. **A thermometer indicates 98.6?C. it may be:**

IN a cup of hot tea

63. **Tesla is a unit of**

FLUX DENSITY

Magnetic flux

64. **A light entering into glass prism from air does not give change in its:**

Frequency

65. **The total energy in an LC circuit is 5.0 x 10- 6 J. If L = 25mH the maximum current is:**

20mA

66. **Electromagnetic waves do not transport:**

Charge

67. **A heat of transformation of a substance is:**

The energy per unit mass absorbed as heat during a phase transformation

68. **Which of the following is true of all virtual images?**

None of these

69. **2dSin0=mA is called**

bragg equation

70. **No lens is perfect because _____.**

They suffer from aberration

71. **A basic postulate of Einstein's theory of relativity is**

The laws of physics must be the same of obersvers moving with uniform velocity relative to each other.

72. **Evidence that a gas consists mostly of empty space is the fact that**

Heating a gas increases the molecular motion

DENSITY OF GAS BECOME MUCH GREATER WHEN IT IS LIQUIFIED

73. **The number of turns becomes double, but length remain same, then magnetic field in the solenoid become.**

Double

74. **Which of the following electromagnetic radiations has photons with the greatest energy?**

X rays

75. **When two objects come to common temperature, the body is said to be in:**

Thermal equilibirium….confirm from net

76. **If electric and gravitational force on an electron in a uniform electric field balance each other, then the intensity of electric field will be:**

Mg/q

77. **Electromagnetic waves do not transport:**

Charge

78. **During a slow adiabatic expansion of a gas**

No energy enters or leaves as heat

79.

80. **A free electron and a free proton have the same kinetic energy. This means that, compared to the matter wave associated with the proton, the matter wave associated with the electron has.**

A longer wavelength and the same frequency

81. **The quantity delta(V)/delta(r) is called**

Potential gradient

82. **If two objects are in thermal equilibrium with each other.**

They cannot be at different temperatures

83. **A consequence of Einstein's theory of relatively is:**

Moving clocks run more slowly than when they are at rest

84. **In order that a signal process is both isothermal and isobaric**

A change of phase is essential

85. **The specific heat of a substance is:**

The amount of heat energy per unit mass to rales the temperature of the substance by 1?C

86. **The electron states that constitute a single shell for an atom all have:**

The same value of n

87. **An isothermal process for an ideal gas is represented on a p-V diagram by:**

A portion of a hyperbola

88. **The concept of an electric field lines is introduced by:**

Faraday

89. **The current in a car headlamp is 2.0A. The headlamp is switched on for 30 minutes. How much charge passes through the headlamp?**

360 C

90. **In some movies, you sometimes see an actor looking in a mirror and you can see his face in the mirror. During the filming of this scene, what does the actor see in the mirror?**

The movie camera

91. **The pauli exculusion principle is obeyed by:**

All particles with spin quantum numbers of ½

92. **The focal length of convex lens is:**

Positive

93. **An electron (m=9.11 * 10-31kg) has a speed of 0.95C. its kinetic energy is;**

1.8*10-31J

94. **An electron (m=9.11 x 10-31kg) has a speed of 4.0.95C. Its kinetic energy is:**

1.8x10-13J

95. **An electron (m=9.11 x 10-31kg) has a momentum of 4.0x10-22kg*m/s. Its kinetic energy:**

6.3*10-14J

96. Force on a charged particle is zero when projected at angle with the magnetic field

0 degree

97. Heat is:

Energy transferred by virtue of a temperature di?erence

98. When a charge is projected perpendicular to a uniform magnetic field, its path is:

Circular

99. If potential difference across two plates of a capacitor is doubled, then energy stored in it will be:

Four times

100. In case of metallic conductors, the charge carriers are.

Electrons

101. Room temperature is about 20 to 22 degrees on the:

Celcius scale

102. Room temperature is about 68 to 72 degrees on the

Fahreheit scale

103. As compared to distance measured by an observer on Earth, the distance from earth star measured by an observer in a moving spaceship would seem:

Same

104. Evidence that a gas consists mostly of empty space is the fact that

The density of a gas becomes much greater when it is liquefied….confrim

105.

106. For an electron, the rest mass energy is:

0.511 MeV……confirm from net

107. **Antiparticle of electron is:**

Position

108. **According to the first law of thermodynamics, applied to a gas, the increase in the internal energy during any process**

Equals the heat input plus the work done on the gas….confirm from net

109. **The approach to personality stresses that people possess a basic goodness, and have a natural tendency to grow to higher levels of functioning is.**

humanistic….confirm from net

110. **In a photoelectric experiment at a frequency above cut off, the stopping potential is proportional to :**

The energy of the most energetic electron after it is ejected….confirm from net

111. **There is a temperature at which the reading on the kelvin scale is numerically:**

Equal to that on the Celsius scale

112. **When you stand in front of a plane mirror, your image is:**

Virtual, erect and the same size as you

113. **When a ray of light travels from a lesser denser medium to denser medium then _____.**

It bends towards the normal

114. **Entropy measure the:**

Energy of a system

115. **The theoretical uppr limit for the frequency of electromagnetic waves is:**

There is no upper limit

116. The special theory of relativity is applicable to the object moving with maximum velocity equal to:

Speed of light

117. According to the special theory of relativity:

All forms of energy have mass-like properties

118. In constructing a thermometer it is necessary to use a substance that:

Undergoes some change when heated or cooled

119. A particle with rest mass m moves with speed 0.6c. its kinetic energy is:

0.25 mC2

120. J.J Thompson's measurement of e/m for electrons provides evidence of the:

Particle nature of matter

121. Polarization means _____.

To alter the wavelength of the light

122. Emission of electron by metals on heating is called

Thermonic emission

123. The specific heat of a substance is:

the amount of heat energy per unit mass to raise the temperature of the substance by 1?C

124. An erect object is in front of a convex mirror a distance greater than the focal length. The image is:

Virtual, erect, and smaller than the object…..cconfirm from net

125. The apperarance of colors in thin film is due to:

Interface…..confirm from net

126. A student rubs a rod held in his hand. Which action causes the rod to gain a large electrostatic charge?

Rubbing a polythene rod with a woolen duster

127. Two parallel straight wires carrying current in same direction

Attract each other

128.

129. One tesla is equal to

N/Am…..confirm from net

130. The process for which entropy remains constant is:

A reversible process

131. The appearance of colors in thin film is due to:

interference

132. Greenhouse gases absorb _____ radiation

Infrared

133. The theoretical uppr limit for the frequency of electromagnetic waves is:

There is no upper limit

134. Heat has the same units as:

Work

135. Capacitance of a capacitor in vaccum is given by:

A eo /d

136. AN electron in a K shell of an atom has the principal quantum number:

N =1

137. When an electromagnetic waves meets a reflecting surface, the direction taken by the reflected wave is dertermined by:

The angle of incidence

138. Capacitive reactance Xc is

1/2(pi)fc

139. Bending of light around the edges of an obstacle called:

Diffraction

140. In RL series circuit phase angle is given by:

Tan-1(wl/R)

141. Interface of light is evidence that:

Light is a wave phenomenon

142. In a photoelectric e? ect experiment at a frequency above cut o? the number of electrons ejected is proportional to:

The number of photons that hit the sample…..confirm from net

143. In a photoelectric e? ect ecperiment the stopping potential is:

The electric potential that causes the electron current to vanish…confirm from net

144. In an adiabatic process:

The work done by the environment on the system equals the change in internal energy….confirm from net

145. A free electron and a free proton have the same kinetic energy . This means that, compared to the matter wave associated with the proton, the matter wave associated with the electron has:

A longer wavelength and the same frequency…confirm from net

146. Polarization experiments provide evidence that light is:

A transverse wave….confirm from net

147. If light of low wavelength is used in young's double slit experiment, then width of the fringe will:

Decreases….confirm from net

148. Lenz's law is in accordance with law of conservation of

Energy….confirm from net

149. AN LC series circuit with an inductance L and a capacitance C has an oscillation frequency f. Two inductors, each with inductance L, and two capacitors, each with capacitance C, are all wired in series and the circuit is completed. The oscillation frequency is:

F

150. The known laws of electromagnetism before James Clerk Maxwell were following EXCEPT:

Faraday's Law deborglie law

151. A particle having 2e charge falls through a potential difference of 5V energy acquired by it is.

10eV

152. According to relatively theory a particle of mass m with a momentum of 2mc has a speed of:

c/2

153. If the kinetic energy of a free particle is much less than its rest energy then its kinetic energy is proportional to:

The square of the magnitude of its momentum

154. The two metallic strips that constitute some thermostats must di? Er in:

Coefficient of linear expansion

155. Phase difference between V and I of an AC through resistor is:

0 degree

156. Average value of current and voltage over complete cycle is:

zero

157. A detector is used to detect the speed of a particular wave. If the frequency of an electromagnetic signal is 400 MHz, the speed detected by detector is _____.

300000 km/s

158. Inductive reactance XL of an inductor is:

2π f L…..confirm from net

159. An inductor may store energy in its:

Magnetic field….confirm from net

160. If velocity of a conductor moving through a magnetic field B is made zero then motional emf is:

ZERO

-vBL

VIRTUAL UNIVERSTY
ORANGE MONKEY

PHY101-Physics

(LAST TIME PAPER)

(MCQ's)

Junaidfazal08@gmail.com
Bc190202640@vu.edu.pk

FOR MORE VISIT
VULMSHELP.COME

JUNAID MALIK

0304-1659294

AL-JUNAID TECH INSTITUTE



For CS619 Project Feel Free To Contact With Me

Ph# 0304-1659294
Email: junaidfazal08@gmail.com

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1. The units of the Planck constant h are those of:

A. energy

B. power

C. momentum

D. angular momentum

E. frequency

2. Which of the following is true of all virtual images?

A. They can be seen but not photographed

B. They are ephemeral

C. They are smaller than the objects

D. They are larger than the objects

E. None of the above

ans: E

3. In constructing a thermometer it is NECESSARY to use a substance that:

A. expands with rising temperature

B. expands linearly with rising temperature

C. will not freeze

D. will not boil

E. undergoes some change when heated or cooled

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4. A 60-watt light bulb carries a current of 0.5 A. The total charge passing through it in one hour is:

A. 120 C

B. 3600 C

C. 3000 C

D. 2400 C

E. 1800 C

5. If the charge on a parallel-plate capacitor is doubled:

A. the capacitance is halved

B. the capacitance is doubled

C. the electric field is halved

D. the electric field is doubled

6. the wavelength of red light is 700 nm. Its frequency is:

A. $4.30*10^4$ Hertz 4.29 10^14 SEE PGE NO.92

B. $4.30*10^5$ Hertz NOT SURE

C. $4.30*10^3$ Hertz

D. $4.30*10^2$ Hertz

7. A capacitor C "has a charge Q". The actual charges on its plates are:

A. Q, Q

B. Q/2, Q/2

C. Q , −Q

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D. Q/2, −Q/2

8. How fast should you move away from a 6.0×10^{14} Hz light source to observe waves with a frequency of 4.0×10^{14} Hz?

A. 0.20c

B. 0.38c

C. 0.45c

D. 0.51c

E. 0.76c

9. The units of the electric field are:

A. J/(C·m)

B. J/C

C. J·C

D. J/m

10. If two light waves are coherent:

A. their amplitudes are the same
B. their frequencies are the same
C. their wavelengths are the same
D. their phase difference is constant

11. In a photoelectric effect experiment at a frequency above cut off, the stopping potential is proportional to:

A. the energy of the least energetic electron before it is ejected

B. the energy of the least energetic electron after it is ejected

C. the energy of the most energetic electron before it is ejected

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D. the energy of the most energetic electron after it is ejected

12. The total energy in an LC circuit is 5.0 × 10−6 J. If L = 25 mH the maximum current is:

A. 10 mA

B. 14 mA

C. 20 mA

D. 28 mA

13. The quantum number ms is most closely associated with what property of the electron in an atom?

A. Magnitude of the orbital angular momentum

B. Energy

C. z component of the spin angular momentum

D. z component of the orbital angular momentum

14. If 500 J of work are required to carry a charged particle between two points with a potential difference of 20 V, the magnitude of the charge on the particle is:

A. 0.040 C

B. 12.5 C

C. 20 C

D. cannot be computed unless the path is given

15. The smallest particle of any chemical element that can exist by itself and yet retain the qualities that distinguish it as that element is

: A. an electron

B. a proton

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C. a neutron

D. an atom

E. a molecule

16. A particle with a charge of 5.5×10−8 C is 3.5 cm from a particle with a charge of −2.3×10−8 C. The potential energy of this two-particle system, relative to the potential energy at infinite separation, is:

A. 3.2 × 10−4 J

B. −3.2 × 10−4 J

C. 9.3 × 10−3 J

D. −9.3 × 10−3 J

E. zero

17. A farad is the same as a:

A. J/V

B. V/J

C. C/V

D. V/C

E. N/C

18. 1 weber is the same as:

A. 1 V/s

B. 1 T/s

C. 1 T/m

D. 1 T · m2

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19. coherent sources of light mean

A. They must have same speeds

B. They must have same frequency

C. They must have same amplitude

D. They must have fixed phase relative to each other
not sure

20. Room temperature is about 20 degrees on the:

A. Kelvin scale

B. Celsius scale

C. Fahrenheit scale

D. absolute scale

E. C major scale

21. The appearance of color in the thin film is due to:

- Interference

22. When you stand in front of a plane mirror, your image is:

A. real, erect, and smaller than you

B. real, erect, and the same size as you

C. virtual, erect, and smaller than you

D. virtual, erect, and the same size as you

23. In a photoelectric effect experiment at a frequency above cut off, the number of electrons ejected is proportional to:

A. their kinetic energy

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B. their potential energy

C. the work function

D. the frequency of the incident light

E. the number of photons that hit the sample

24. In a photoelectric effect experiment at a frequency above cut off, the stopping potential is proportional to:

A. the energy of the least energetic electron before it is ejected

B. the energy of the least energetic electron after it is ejected

C. the energy of the most energetic electron before it is ejected

D. the energy of the most energetic electron after it is ejected

E. the electron potential energy at the surface of the sample

25. The units of magnetic dipole moment are:

A. ampere

B. ampere·meter

C. ampere·meter2

D. ampere/meter

26. The primary of a 3 : 1 step-up transformer is connected to a source and the secondary is connected to a resistor R. The power dissipated by R in this situation is P. If R is connected directly to the source it will dissipate a power of:

A. P/9

B. P/3

C. P

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D. 3P

E. 9P

27. We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5 H. We also need a capacitance of:

A. 1 F

B. 1 mF

C. 1 μF

D. 100 μF

28. An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

A. a change in entropy

B. an increase of pressure

C. a change in temperature

D. a decrease of internal energy

30. A step-down transformer is used to:

A. increase the power

B. decrease the power

C. increase the voltage

D. decrease the voltage

31. Which of the following electromagnetic radiations has photons with the greatest energy?

A. blue light

B. yellow light

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C. x rays

D. radio waves

33. The Stern-Gerlach experiment makes use of:

A. a strong uniform magnetic field

B. a strong non-uniform magnetic field

C. a strong uniform electric field

D. a strong non-uniform electric field

34. Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in series must have the same:

A. Q only

B. V only

C. U only

D. Q and U only

35. Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in parallel must have the same:

A. Q only

B. V only

C. U only

D. Q and U only

36. . “The sum of the currents into a junction equals the sum of the currents out of the junction” is a consequence of:

A. Newton’s third law

B. Ohm’s law

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C. Newton's second law

D. conservation of energy

E. conservation of charge

ans: E

37. "The sum of the emf's and potential differences around a closed loop equals zero" is a consequence of:

A. Newton's third law

B. Ohm's law

C. Newton's second law

D. conservation of energy

E. conservation of charge

38. Evidence for the wave nature of matter is:

A. electron diffraction experiments of Davisson and Germer

B. Thompson's measurement of e/m

C. Young's double slit experiment

D. the Compton effect

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39. A conducting sphere with radius R is charged until the magnitude of the electric field just outside its surface is E. The electric potential of the sphere, relative to the potential far away, is:

A. zero

B. E/R

C. E/R2

D. ER

40. radio waves and light waves are ;

- Electromagnetic and transverse

41. An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

A. a change in entropy

B. an increase of pressure

C. a change in temperature

D. a decrease of internal energy

42. Capacitors A and B are identical. Capacitor A is charged so it stores 4 J of energy and capacitor B is uncharged. The capacitors are then connected in parallel. The total stored energy in the capacitors is now:

A. 16 J

B. 8 J

C. 4 J

D. 2 J

43. Capacitors C1 and C2 are connected in parallel. The equivalent capacitance is given by:

A. C1C2/(C1 + C2)

B. (C1 + C2)/C1C2

C. 1/(C1 + C2)

D. C1/C2

E. C1 + C2

ans: E

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44. Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by:

A. C1C2/(C1 + C2)

B. (C1 + C2)/C1C2

C. 1/(C1 + C2)

D. C1/C2

E. C1 + C2

45. During an adiabatic process an object does 100 J of work and its temperature decreases by 5 K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is:

A. 20 J/K

B. 24 J/K

C. 5 J/K

D. 15 J/K

46. A virtual image is one:

A. toward which light rays converge but do not pass through

B. from which light rays diverge but do not pass through

C. from which light rays diverge as they pass through

D. toward which light rays converge and pass through

47. A generator supplies 100 V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

A. 1000 V

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B. 500 V

C. 250 V

D. 100 V

48. The rate at which electrical energy is used may be measured in:

A. watt/second

B. watt·second

C. watt

D. joule·second

QUESTION: A vessel is filled with gas at some equilibrium pressure and temperature. Can all gas molecules in the vessel have same speed?

Answer: They can, as this possibility is not contradicted by any of our descriptions of the motion of gases. If the vessel contains more than a few molecules, it is highly improbable that all will have the same speed. Collisions will make their speeds scatter according to the Boltzmann distribution law."

QUESTION: Electric potential energy:

Answer:

The electric potential is the energy of a unit charge in an electric field. So in our MKS units the unit of potential is 1Joule/Coulombs=1Volt.Another useful unit is 1Electron Volt.

QUESTION: All objects radiate energy then why are we not able to see all the objects in the dark? Our eyes are not sensitive to infrared radiation so we do not see it. Cold bodies radiate mostly in the infrared zone (invisible to the human eye), but as the temperature increases the body will emmit higher frequencies with more intensity. So room temperature objects will not be seen due to black body radiation.

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QUESTION: In a cool room, metal and marble is cool but wood is not col when we touch. Why?

Answer: Because metal and stone conduct heat better than wood. Skin temperature is higher than room temperature, so when we touch metal, we give heat more quickly to the metal than wood. Another reason is that metal and stone store more energy than a like sized object of wood

QUESTION: what causes electric shock or voltage?

ANS: one clock is upstairs of the bedroom and second clock is in downstairs of the kitchen. Then which clock runs slowly?
Really, neither will run appreciably slower than the other. In the strictest sense, the clock in the upstairs bedroom is moving a few centimeters per year faster than the clock downstairs in the kitchen, because it is a few feet further from the Earth's center than the clock in the upstairs.

QUESTION: Difference between reflection and refraction?
ANS:

Reflection involves a change in direction of waves when they bounce off a barrier. Refraction of waves involves a change in the direction of waves as they pass from one medium to another. Refraction, or the bending of the path of the waves, is accompanied by a change in speed and wavelength of the waves.

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Question:

If an electron and proton have same De Broglie's wavelength, which particle has greater speed

The de Broglie wavelength of a particle is inversely proportional to its momentum p = m v; since a proton is about 1800 times more massive than an electron, its momentum at the same speed is 1800 times that of an electron, and therefore its wavelength 1800 times smaller. The electron has the longer wavelength.

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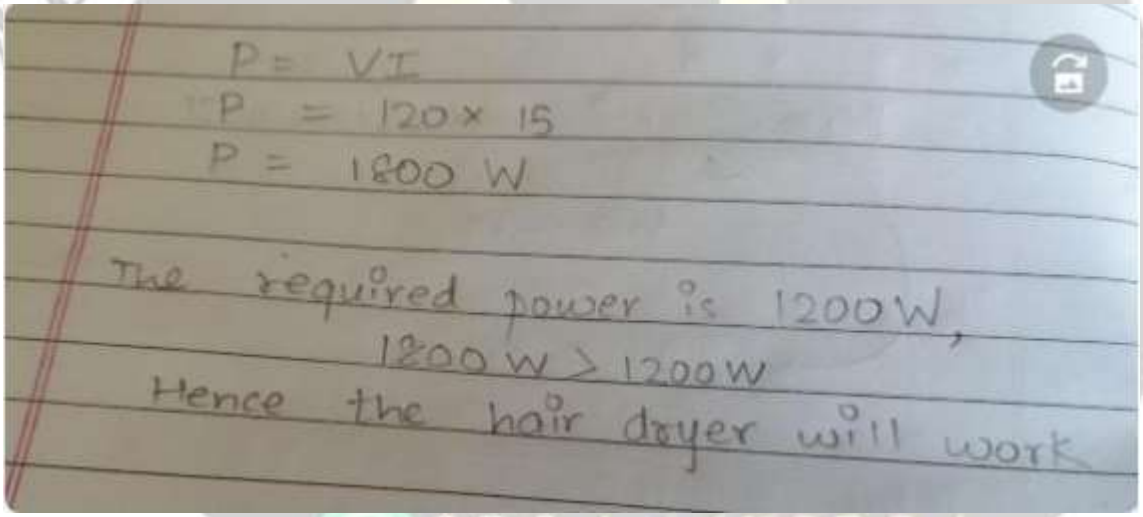
QUESTION: black and black body radiation?

ANS:

An object that absorbs all radiation falling on it, at all wavelengths, is called a black body. When a black body is at a uniform temperature, its emission has a characteristic frequency distribution that depends on the temperature. Its emission is called black-body radiation.

For examples include the heating element of a toaster and the filament of a light bulb.

QUESTION: If a 120 line to shocked is limited 15a by a safety fuse will it operate a 1200w hair dryer?



QUESTION: Define statically mechanics

Ans: - The study of heat, considered as arising from the random motion of the basic constituents of matter, is an area of physics called statistical mechanics.

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QUESTION: is it true to say that Centre of mass and Centre of graving are the same? Either yes or no explain in each case.

Answer: Many people assume that the terms "Centre of mass" and "Centre of gravity" are synonymous, but this is not the case. The center of mass and the Centre of gravity of an object are in the same position if the gravitational field in which the object exists is uniform.

QUESTION: What is the work done? If Force is 500N, height of downwards cliff is 30m Email and angle with land is 30 degrees.?

QUESTION: Magnetic Flux and Magnetic Flux density and when Flux is maximum and when it is zero?

ANS:

Because the induced EMF is proportional to the time rate of change of the magnetic flux, not to the magnetic flux itself, as shown in Faraday's law. In a sinusoidal wave, the maximum rate of change occurs at the zero crossing; in that point, the slope of the line tangent to the wave is the greatest, so the rate of change is the greatest (recall the derivative of a invariable function gives the slope of such function at each point). At the maximum or minimum value, the slope of the tangential line is zero, since it's a horizontal line, so the rate of change is zero.

The magnetic flux as a function of time is

$\Phi(t)=B\cdot A(t)=BA\cos(wt)$

$\Phi(t)=B\cdot A(t)=BA\cos(\omega t)\Phi(t)=B\cdot A(t)=BA\cos\ (\omega t)$
BB =the magnetic field and
A(t)A(t) =the area vector as a function of time
ωtωt =the angle between the field and the area vector as a function of time.
Cos(90)=0

But

Then the rate of change of the flux as a function of time is

$\Phi'(t) = -BA\omega\sin(\omega t)$

$\Phi'(t) = -BA\omega\sin(\omega t)\Phi'(t) = -BA\omega\sin(\omega t)$
the angle between(the area vector vector and the magnetic field) =zero.

flux =nonzero and equal to AB, its maximum,but the rate of change of the flux vanishes because

Sin(0)=0,sin(90)=1.

QUESTION: How Isotopes change Atomic Mass, Magnetic spin moment and Chemical behavior of Elements

ANS:

Isotopes are atoms that have the same number of protons and electrons, but a different number of neutrons. Changing the number of neutrons in an atom does not change the element. Atoms of elements with different numbers of neutrons are called "isotopes" of that element. Isotopes behave similarly to other isotopes of the same element in chemical reactions because such reactions don't depend on the properties of atomic nuclei. Rather they depend on the number and arrangement of an atom's electrons.

QUESTION: Similarity and difference of gland meter and electric motor

ANS:

The motor requires Electricity, but the generator produces electricity. The shaft of the motor is driven by the magnetic force developed between the armature and the field winding, while in the case of the generator the shaft is connected to the rotor and driven by mechanical force

QUESTION: Fundamental source of Electromagnetic radiation

ANS:

A moving charge particle is the fundamental source of electromagnetic radiation.

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QUESTION: A photon of green light has a wavelength of 530nm. Find the photos frequency, magnitude of momentum and electron volts. Express the energy in joles and electron volts.

ANS:

TAKE IDEA FROM THIS LINK

https://socratic.org/questions/calculate-the-energy-in-joules-of-a-photon- of-green-light-having-a-wavelength-of

https://www.numerade.com/questions/a-photon-of-green-light-has-a-wavelength-of-520-nm-find-the-photons-frequency-magnitude-of-momentum-/

QUESTION: Find the difference between two waves which interfere constructively and destructively.

ANS:

There are two types of interference, constructive and destructive. In constructive interference, the amplitudes of the two waves add together resulting in a higher wave at the point they meet. In destructive interference, the two waves cancel out resulting in a lower amplitude at the point they meet.

QUESTION: Similarity and difference of glvano meter and electric motor

ANS:

Galvanometers and electric motors are both based on the same basic principle of **an external magnetic field exerting** a force on a current carrying wire. The difference is in how far the device is allowed to move. Motors must be able to move freely and rotate through complete 360o turns.

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QUESTION:

A 2000kg car is moving with a velocity of 20 m/s collides and locks with a 1500 kg car at rest a stop sign. Show the momentum is conserved.

A 2000-kg car moving with a speed of 20 m/s collides with and sticks to a 1500-kg car at rest at a stop sign. Show that because momentum is conserved in the rest frame, momentum is also conserved in a reference frame moving with a speed of 10 m/s in the direction of the moving car

QUESTION:

If you hold water in a paper cup over flame, can you bring the water to boil, if then how?

Just fill a **paper cup** with **water** and start trying to burn it from the outside. The **water will boil** vigorously inside the **cup**, but the **paper** itself **wills** not combust. ... The **boiling water can**'t get **above** 212 degrees, and the **paper** is thin enough that this **water** keeps its temperature from climbing high enough to combust.

Question: What is Corona?

Ans:

A corona discharge is **an electrical discharge caused by the ionization of a fluid** such as air surrounding a conductor carrying a high voltage.A corona occurs at locations where the strength of the electric field (potential gradient) around a conductor exceeds the dielectric strength of the air.

Question: Greenhouse effect explain.

Ans:

The greenhouse effect is a natural process that warms the Earth's surface. When the Sun's energy reaches the Earth's atmosphere, some of it is

reflected back to space and the rest is absorbed and re-radiated by greenhouse gases. The absorbed energy warms the atmosphere and the surface of the Earth

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Phy101 Quiz 3

1. Lenz's law is in accordance with law of conservation of

   Energy…confirm

2. Which of the following types of electromagnetic radiation at the greatest speed in vacuum?

   All of these travels at the same speed….confirm

3. Inductive reactance XL of an inductor is

   2(pi)fL…. confirm

4. Force per unit charge is:

   Electric field intensity…confirm

5. The sum of electric and magnetic force is called

   Lorentz force…confirm

6. Radio waves and light waves are_____.

   Electromagnetic and transverse both…confirm

7. Kirchhoff's 1st rule is the manifestation of the law of conservation of

   Charge…confirm

8. Kirchhoff's 2nd rule is the manifestation of the law of conservation of

   Energy…confirm

9. The total energy in an circuit is 5.0 * 10-6 J. if L=25mH the maximum current is:

   20mA…confirm

10. If Io is the peak value of an AC supply, then its rms value is given as firms

    Io/square root2….confirm

11. The potential difference between the ends of a conductor is 12V. How much electrical energy is converted to other forms of energy in the conductor when 100C of charge flows through it?

1200J….confirm

12. A particle carrying a charge of 2e falls through potential difference of 3V the energy acquired by it will be:

6 eV…confirm

13. An inductor may store energy in its:

Magnetic field…confirm

14. The graphical representation of Ohm's law is:

Straight line…confirm

15. Radio waves and light waves are_____.

Longitudinal waves…confirm

16. The increase in capacitance of a capacitor due to presence of dielectric is due to:

Electric polarization

17. The Focal length of convex lens is:

Positive…confirm

18. The AC circuit in which current and voltage are in phase the power factor is

1…confirm 0.5 60 DEGREE

19. The wavelength of red light is 700 nm. Its frequency is_____.

4.29*10^14 hertz…confirm

21. The known laws of electromagnetism before James Clerk Maxwell were following EXCEPT

Faraday's Law…confirm deborglie law

20. When some dielectric is inserted between the plates of a capacitor, then capacitance;

Increased…..confirm

21. Which of the following is NOT the characteristics of electromagnetic waves?

They can travel with different speeds in vacuum depending on frequency…..confirm

22. If electric and gravitational force on an electron in uniform electric field balance each other, then the intensity of electric field will be:

Mg/q

23. The increase in capacitance of a Electric polarization

24. Electric energy is measured

WaTT…confirm

25. When we accelerate the charge, which types of waves are produced

Electromagnetic waves….confirm

26. A particle device transmitted a signal in the form of electromagnetic waves of frequency 7.5 * 108Hz. What is the wavelength of the signal.

0.4m

27. A step-down transformer is used to:

Decrease the voltage

27. The quantity of deltaV/deltar is called

Potential gradient

28. Which electromagnetic radiation transmits the highest photon energy?

Gamma rays….confirm

29. The reaction of an particle at 50 Hz is 10.0 its reactance at 100Hz becomes

5.0

30. Work done on a charged particle moving in uniform magnetic field is

Zero….confirm

31. Select the correct statement

Gamma Rays have higher frequency than infrared waves…confirm

32. Which of the following have the greatest wavelength

Radio waves

33. Potentiometer can be used as

Potential divider….confirm

34. All of the followings are the different types of mental images; people generally experience them while close their eyes Except:

Solving an algebra equations

35. The nation that motivational state of the organism is understood and explained in terms of positive and negative environmental stimuli BEST connects to;

Incentive approaches

36. If one wants to increase the capacity of short-term memory, through which process more items can be held?

Chunking….confirm

37. Which physical quantity is produced by a calculation where a charge is

Energy…confirm

38. The current in a car headlamp is 2.0 A. The Headlamp is switched an for 3.0

360 C

39. If potential difference across two plates of a capacitor is doubled, then

Four times….comfirm

40. How could the unit of potential difference the volt also be written?

J/C

41. If magnetic field is doubled then magnetic energy density becomes

Four times….confirm

42. The primary of a 3.1 step-up transformer is connected to a source and the secondary this situation is P. If R is connected directly to the source it will dissipate a power of:

P/9….confirm

43. A particle having 2e charge falls through a potential difference of 5.V energy acquired by it is

10 eV

44. A generator supplies 100V to the primary call of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is

1000V

45. The combined effect of resistance and reactance is known as

Impedance…confirm

46. A transparent refracting medium bounded by the two curved surfaces called
LENS Prism….confirm
47. A student rubs a rod held in his hand. Which action causes the rod to gain a large electrostatic charge
Rubbing a polythene rod with a woolen duster….confirm
48. A charge of 30 C flows through an electric appliance in 2.0 minutes. What is the average current in the appliance?
0.25A….confirm
51. Total flux through a closed surface depends on
Charge and -N...confirm CHARGE ENCLOSED
49. Electromagnetic waves emitted from antenna are
Transverse….confirm
50. Two parallel straight wires carrying current in same direction will.
Attract each other…confirm
51. Charge carriers in electrolysis are:
Positive and negative ions…confirm
52. Capacitive reactance XC is
$1/2\pi fC$
53. The number of turns becomes double, but length remain same then magnetic field in the solenoid become
Double…confirm
54. If velocity of a conductor moving through Magnetic field B is made zero then motional emf is
-vBL ZERO
58. The amplitude modulation frequency ranges from
540 kHz to 1600 kHz…confirm from net
55. Average value of current and voltage over complete cycle is:
Zero..confirm
56. A detector is used to detect the speed at a particular wave if the frequency of an electromagnetic signal is 400 MHz, the speed define the detector is
300000 km/s
57. The product of resistance and capacitance is called
Time….confirm
58. Greenhouse gases absorb _____radiation
Infrared…confirm
59. In RL series circuit phase angle is given by:
$\tan^{-1}(w_L/R)$…confirm
60. One tesla is equal to
N/Am
61. An LC series circuit an inductance L and a capacitance C /has an oscillation frequency f. Two inductors each with inductance L, and two capacitors, each with capacitance C, are all wired in series and the circuit is completed. The oscillation frequency is:
f…..confirm
62. Capacitance of a capacitor in vacuum is given by
Aεo/d….confirm
63. Force on a charged particle is zero when projected at angle with the magnetic field

0 degree….confirm

64. If an electron of charge is accelerated through a potential difference V, it will acquire energy.

QV Ve….confirm

65. In case of metallic conductors, the charge carriers are:

Electrons…confirm

66. Tesla is a unit of

Magnetic flux….confirm FLUX DENSITY

67. When a charge is projected perpendicular to a uniform magnetic field, its path is:

Circular…..confirm

68. Electromagnetic waves do not transport

Charge….confirm

69. The concept of an electric field lines is introduced by

Faraday….confirm

70. A capacitor in an LC oscillator has a maximum potential difference of 15V and a maximum energy of 360 uJ. At a certain instant the energy in the capacitor is 40 uJ. At that instant what is the emf induced in the inductor

5V….confirm

71. The wave form of alternating voltage is a:

Sine wave….confirm

72. If velocity of a conductor moving through a magnetic field B is made zero then motional emf is:

-vBL…confirm ZERO

72. Which one of is in the order of decreasing frequency?

Ultraviolet rays, visible light, radio waves…confirm

73. The lenz's law refers to:

Induced current….confirm

74. Which of the following statements is Not true about electromagnetic waves?

The electromagnetic radiation from a burning candle is polarized…confirm

75. Phase different between V and I of on A.C through resistor is:

0 degree

76. Which of the following is equivalent to 1 V

1 J/C…..confirm

77. Which of the following is equivalent to one coulomb

One ampere second….confirm

PHY101 (Physics)

Recent Papers with their answers

Q1. A photon of green light has a wavelength of 530nm. Find the photos frequency, magnitude of through 1m width.

Answer: To find the frequency of a photon with a wavelength of 530nm, we can use the formula:

c = λv

Where c is the speed of light, λ is the wavelength, and v is the frequency.

Rearranging the formula, we get:

v = c/λ

Plugging in the values, we get:

v = (3 x 10^8 m/s)/(530 x 10^-9 m)

v = 5.66 x 10^14 Hz

Therefore, the frequency of the photon is 5.66 x 10^14 Hz.

To find the magnitude of the photon's energy through a 1m width, we need to know the photon's energy. The energy of a photon can be calculated using the formula:

E = hf

Where E is the energy of the photon, h is Planck's constant, and f is the frequency of the photon.

Plugging in the values, we get:

E = (6.626 x 10^-34 J s) x (5.66 x 10^14 Hz)

E = 3.75 x 10^-19 J

To find the magnitude of the photon's energy through a 1m width, we need to know the photon flux density. The photon flux density is the number of photons passing through a unit area in a unit time. We can calculate the photon flux density using the formula:

Photon flux density = Power / (Area x c)

Where Power is the energy per second, Area is the cross-sectional area of the beam, and c is the speed of light.

Plugging in the values, we get:

Photon flux density = (3.75 x 10^-19 J/s) / (π x (0.5 x 1)^2 m^2 x (3 x 10^8 m/s))

Photon flux density = 1.50 x 10^9 photons/s/m^2

Therefore, the magnitude of the photon's energy through a 1m width is 1.50 x 10^9 photons/s/m^2.

Q3. Find the difference between two waves which interfere constructively and destructively.

Answer: When two waves interfere constructively, the amplitude of the resulting wave is increased. The difference between the two waves is such that their peaks and troughs line up, resulting in an increase in amplitude.

When two waves interfere destructively, the amplitude of the resulting wave is decreased. The difference between the two waves is such that their peaks and troughs are out of phase, resulting in a cancellation of the amplitude.

Q4. Similarity and difference of Galvano meter and electric motor.

Answer: The Galvanometer and Electric Motor are two different devices that have some similarities and differences.

Similarities:

Both devices have a coil of wire and a magnet that interact with each other to produce motion.

Both devices are based on the principle of electromagnetism.

Both devices can be used to measure electrical currents.

Differences:

The Galvanometer is a measuring instrument that is used to detect small currents, while the Electric Motor is a device that converts electrical energy into mechanical energy to produce motion.

The Galvanometer typically has a very low resistance and is used in circuits where a small current needs to be detected, while the Electric Motor has a higher resistance and is used in circuits where electrical energy needs to be converted into mechanical energy.

The Galvanometer typically has a sensitive pointer or meter to indicate the current, while the Electric Motor typically has a rotating shaft that produces mechanical work.

Q5. Which law used in blackbody radiation?

The law used in blackbody radiation is known as Planck's law. This law describes the spectral energy density of electromagnetic radiation emitted by a blackbody at a given temperature. It states that the energy density of radiation emitted by a blackbody at a given wavelength is proportional to the frequency of the radiation and the temperature of the blackbody. Planck's law was first introduced by Max Planck in 1900 as a theoretical explanation for the observed spectral distribution of radiation emitted by heated objects, such as a blackbody. It played a crucial role in the development of quantum mechanics and was later extended to include other phenomena, such as the photoelectric effect.

Q6. Difference between magnetic flux and magnetic flux density, when flux is max and when it is zero?

Magnetic flux and magnetic flux density are related but distinct concepts in magnetism.

Magnetic flux refers to the total magnetic field passing through a given area. It is represented by the symbol Φ and is measured in units of Weber (Wb). Magnetic flux is a scalar quantity and is calculated by multiplying the magnetic field

strength (B) by the area (A) through which it passes and the cosine of the angle between the field and the normal to the surface.

Magnetic flux density, on the other hand, refers to the strength of the magnetic field in a given region. It is represented by the symbol B and is measured in units of Tesla (T). Magnetic flux density is a vector quantity and is calculated by dividing the magnetic flux by the area through which it passes and the cosine of the angle between the field and the normal to the surface.

When magnetic flux is at its maximum, the magnetic field passing through a given area is at its highest level. This could occur when a magnetic field is applied to a material, creating a strong magnetic field that is passing through a surface. When magnetic flux is zero, there is no magnetic field passing through the given area. This could occur when a magnetic field is removed from a material or when a magnetic field cancels out another magnetic field passing through the same area, resulting in no net magnetic field.

Q7. How isotopes change atomic mass?

Isotopes are different forms of the same element that have the same number of protons but a different number of neutrons in their atomic nucleus. Since the number of protons determines the atomic number of an element, isotopes of the same element have the same atomic number. However, the different number of neutrons means that isotopes have different atomic masses.

When calculating the atomic mass of an element, the atomic masses of all the isotopes of that element are taken into account. The atomic mass of an element is the weighted average of the atomic masses of its isotopes, where the weighting is based on the abundance of each isotope in nature.

For example, the element carbon has two naturally occurring isotopes, carbon-12 and carbon-13. Carbon-12 has an atomic mass of 12 amu (atomic mass units) and makes up about 98.9% of all carbon atoms, while carbon-13 has an atomic mass of 13 amu and makes up about 1.1% of all carbon atoms. The atomic mass of carbon is therefore calculated as:

(0.989 x 12 amu) + (0.011 x 13 amu) = 12.01 amu

As can be seen in this example, the presence of isotopes with different atomic masses can cause the calculated atomic mass of an element to differ slightly from the atomic mass of its most abundant isotope. This effect can be more pronounced for elements with a wider range of isotopes or with isotopes that differ significantly in their relative abundance.

Q8. What is magnetic spin moment?

Magnetic spin moment, also known as spin magnetic moment or simply spin moment, refers to the intrinsic magnetic moment associated with the spin of an elementary particle, such as an electron or a proton. This magnetic moment arises from the spin of the particle, which can be thought of as a rotation of the particle about its own axis.

The spin moment of a particle can interact with an external magnetic field, causing the particle to experience a torque that tends to align its spin axis with the direction of the field. This is the basis of many magnetic phenomena, such as ferromagnetism and Para magnetism.

The spin moment of an electron is particularly important in magnetism, as it is the source of the magnetic moment in many materials. In ferromagnetic materials, the spin moments of electrons in neighboring atoms align with each other, producing a net magnetic moment that gives rise to a macroscopic magnetic field. In paramagnetic materials, the spin moments of electrons are randomly oriented, but can be aligned in the presence of an external magnetic field.

The spin moment of a particle is often represented by the symbol μs, and is typically measured in units of Bohr magnetons (μB), where one Bohr magneton is equal to the magnetic moment associated with the electron's orbital motion in the hydrogen atom. The magnitude of the spin moment depends on the type of particle and its intrinsic properties, such as its mass and spin.

Q9. Write down the Chemical behavior of elements.

The chemical behavior of elements refers to how they interact with other elements and compounds in chemical reactions. This behavior is largely determined by the electronic configuration of the elements, which dictates their chemical properties.

Elements in the same group of the periodic table tend to exhibit similar chemical behavior due to their similar electronic configurations. For example, the alkali metals (Group 1) are highly reactive with water and oxygen, while the noble gases (Group 18) are generally unreactive due to their full valence electron shells.

Chemical behavior can also be affected by the presence of other elements or compounds. For example, halogens (Group 17) are highly reactive with alkali metals to form ionic compounds, while nonmetals like carbon and nitrogen can form covalent bonds with each other to form molecules like carbon dioxide and nitrogen gas.

The chemical behavior of an element can also vary depending on its oxidation state, or the number of electrons it has gained or lost in a chemical reaction. For example, iron can exhibit both ferrous (Fe2+) and ferric (Fe3+) oxidation states, which can affect its reactivity with other compounds.

Overall, the chemical behavior of elements is a complex and varied field, but is essential to understanding how different elements can interact with each other to form the wide range of compounds and materials that we observe in the world around us.

Q10. For observing interference, why we use thin slit only? why don't we use thick slit?

Interference is a phenomenon that occurs when two or more waves overlap and combine to form a resultant wave. In order to observe interference, it is necessary to use a device that can create two or more coherent sources of waves that will interfere with each other.

One common device used for this purpose is a double-slit apparatus, which consists of two narrow slits separated by a small distance. When light waves pass through these slits, they diffract and interfere with each other, producing an interference pattern on a screen placed behind the slits.

The reason that thin slits are used instead of thick slits is that thin slits produce a more pronounced diffraction effect, which is essential for creating a clear interference pattern. Thick slits, on the other hand, do not produce as much diffraction, which can result in a weaker interference pattern or no interference pattern at all.

In addition, thin slits allow for a greater degree of control over the interference pattern, as the distance between the slits can be easily adjusted to produce different patterns. Thick slits, by contrast, are less flexible in terms of producing different interference patterns.

Overall, thin slits are preferred for observing interference because they produce a stronger diffraction effect and allow for greater control over the interference pattern.

Q11. Write about plum pudding model that (most important)

a) Who gave give

b) What was its basic features.

c) Why was it wrong?

The plum pudding model was a model of the atom proposed by J.J. Thomson in 1904. The model was based on the idea that atoms were made up of a diffuse cloud of positive charge with electrons embedded within it, much like the plums in a plum pudding.

The basic features of the plum pudding model were that atoms consisted of a positively charged sphere of matter with electrons embedded within it. The positive charge was assumed to be distributed uniformly throughout the sphere, with the electrons occupying the spaces between the positive charges.

Thomson proposed this model based on his discovery of the electron in 1897, which showed that atoms were not indivisible, as had previously been thought. Instead, he believed that atoms were made up of smaller subatomic particles, with the negative electrons held within a positively charged matrix.

However, the plum pudding model was ultimately found to be incorrect. In 1911, Ernest Rutherford and his colleagues performed the famous gold foil experiment, in which they fired alpha particles at a thin sheet of gold foil. They expected the alpha particles to pass through the foil with only minor deflections, in accordance with the plum pudding model. However, they observed that a small fraction of the alpha particles was deflected at large angles, and some even bounced back in the direction they had come from.

This result could not be explained by the plum pudding model, as it predicted that the positive charge was distributed uniformly throughout the atom. Rutherford realized that the positive charge must be concentrated in a small, dense nucleus at the center of the atom, with the electrons orbiting around it. This led to the development of the nuclear model of the atom, which is still the basis of our understanding of atomic structure today.

Q12. composition of sun.

The sun is a massive, luminous ball of gas that is at the center of our solar system. It is composed mostly of hydrogen and helium, with trace amounts of other elements.

About 74% of the sun's mass is hydrogen, while helium makes up about 24%. The remaining 2% is made up of trace amounts of other elements, such as oxygen, carbon, neon, and iron.

The sun's core, where nuclear fusion takes place, is the hottest part of the sun, with temperatures of up to 15 million degrees Celsius. At these temperatures, hydrogen atoms are fused together to form helium, releasing vast amounts of energy in the process. This energy radiates outwards through the sun's layers, eventually reaching the surface and escaping into space as light and heat.

In addition to the elements that make up the sun, it also contains various layers with different characteristics. The outermost layer, known as the corona, is composed of highly ionized gas that extends far into space. Beneath the corona is the chromosphere, which is a layer of cooler, dense gas that is visible during solar eclipses. The layer below the chromosphere is the photosphere, which is the visible surface of the sun and the layer where most of the sun's light and heat is emitted.

Overall, the sun is a complex and dynamic object, and its composition and structure play a crucial role in shaping the behavior of the solar system as a whole.

Mix Shorts

Question. 1: Do you think clock speed is slower than stationery time.

Question. 2: wave speed is function of frequency are you familiar with this, if we increase frequency than what happened to the speed of wave.

Question. 3: Magnetic field exert force or not while in the presence of charge particles?

Question. 4: what is luckawa force and what is range of it.

Question. 5: Numerical tha Neutron, Deuteron electron ki charges maloom krni thi.

Question. 6: Define capacitors resisters etc.

Long Questions:

Question 1. Define Quantum Revolution.

Question 2. Numerical tha magnetic field maloom krni thi.

Question 3. Capacitor parral me connected hen unka formula batane the ur charge maloom krni thi etc.

Question 4. Relativity ka numerical tha planks constant etc.

Question 2: Yes, wave speed is typically proportional to frequency, with higher frequencies corresponding to higher wave speeds. This relationship is expressed by the equation v = fλ, where v is the wave speed, f is the frequency, and λ is the wavelength.

Question 3: Yes, magnetic fields can exert a force on charged particles that are moving through them. This is the basis of many important phenomena, such as the operation of electric motors and the behavior of charged particles in particle accelerators.

Question 6: Capacitors and resistors are both electrical components that are used in electronic circuits. A capacitor stores electrical charge and is characterized by its capacitance, which is measured in farads (F). A resistor limits the flow of electrical current and is characterized by its resistance, which is measured in ohms (Ω).

Long Questions:

Question 1: The quantum revolution refers to a period of scientific discovery in the early 20th century that fundamentally changed our understanding of the nature of matter and energy. It was characterized by the development of quantum mechanics, a new branch of physics that describes the behavior of particles at the atomic and subatomic level. Key figures in the quantum revolution include Max Planck, Albert Einstein, Niels Bohr, Werner Heisenberg, and Erwin Schrödinger.

Question 3: To calculate the total charge on capacitors connected in parallel, you can use the equation Q = CV, where Q is the total charge, C is the capacitance of each capacitor, and V is the voltage applied to the capacitors. The charge on each capacitor is equal to the product of its capacitance and the voltage applied to it. Resistors, on the other hand, limit the flow of electrical current and are characterized by their resistance, which is measured in ohms (Ω).

MIX

1. Does your bathroom mirror show your older and younger than your actually.

2. What is the advantage of transmitting power at high voltage, will the transformer operate, if a battery is used for input voltage primary.

3. If the field charge inside a closed surface is known both the distribution of the charge is unsatisfied, can you use Gauss law to find electric filed.

4. Many traffic lights charge when a car roll up intersection. How dose light sense the presence of car.

ANSWERS

No, your bathroom mirror does not show you as older or younger than you actually are. However, mirrors do reflect an image that is flipped horizontally, so your left side appears on the right and vice versa.

Transmitting power at high voltage has the advantage of reducing energy losses due to resistance in the transmission lines. This is because power loss is proportional to the square of the current, and higher voltages allow for lower currents to achieve the same power transmission. A transformer will only operate if there is a changing magnetic field, so a battery would not be able to drive the primary coil of a transformer.

Yes, Gauss's law can be used to find the electric field inside a closed surface if the charge distribution is known. Gauss's law states that the total electric flux through a closed surface is proportional to the charge enclosed by the surface. By applying Gauss's law, it is possible to find the electric field at any point inside the closed surface.

Many traffic lights use sensors to detect the presence of vehicles at an intersection. These sensors can be inductive loops embedded in the roadway, which detect changes in magnetic fields as vehicles pass over them. They can also be infrared sensors that detect the presence of vehicles by measuring changes in the infrared radiation emitted by the vehicles. Once the presence of a vehicle is detected, the traffic light can change accordingly to allow for safe and efficient traffic flow.

1. What becomes of isotopes that undergo alpha decay? Which produces a higher counting rate or radiation dectr a gram of radioactive having a short half-life r gram having a long half-life.
2. Explain solar zone. What is its other name?
3. Distinguish between electric potential and electric potential energy. A negative charge move in the direction of uniform.
4. What is Yukawa Force and its range
5. Define thermometer, internal energy, Heat
6. What is the pressure of sun?

Answers:

1. Isotopes that undergo alpha decay emit alpha particles, which are helium nuclei consisting of two protons and two neutrons. As a result, the atomic number of the parent nucleus decreases by two, while the atomic mass decreases by four. The daughter nucleus produced by alpha decay has a lower energy state than the parent nucleus. A gram of radioactive material with a short half-life will produce a higher counting rate of radiation compared to a gram of radioactive material with a long half-life because it undergoes radioactive decay more rapidly.

2. The solar zone is the region of the sun's interior where energy is transported by radiation. This region extends from the center of the sun to about 0.7 solar radii. It is also known as the radiative zone because the energy is transported by radiation rather than convection.

3. Electric potential is the electric potential energy per unit charge at a point in an electric field. It is a scalar quantity and is measured in volts. Electric potential energy, on the other hand, is the energy that a charged particle possesses due to its position in an electric field. It is a scalar quantity and is measured in joules. If a negative charge moves in the direction of a uniform electric field, it will experience a force in the opposite direction to the field, and its potential energy will decrease.

4. The Yukawa force, also known as the nuclear force, is a strong attractive force between nucleons (protons and neutrons) that holds the atomic nucleus together. It is mediated by the exchange of mesons, which are subatomic particles with integer spin. The range of the Yukawa force is very short, on the order of 10^-15 meters, because the exchange particles have a relatively large mass.

5. A thermometer is a device used to measure temperature. Internal energy is the total energy of a system due to the motion and interactions of its particles. Heat is the energy that is transferred between two bodies at different temperatures when they are brought into thermal contact with each other.

6. The pressure at the center of the sun is estimated to be about 250 billion atmospheres. However, the pressure decreases rapidly as you move outwards from the center, reaching about 1 atmosphere at the visible surface of the sun, known as the photosphere.

PHYSICS 101

AN INTRODUCTION TO PHYSICS

This course of 45 video lectures, as well as accompanying notes, have been developed and presented by Dr. Pervez Amirali Hoodbhoy, professor of physics at Quaid-e-Azam University, Islamabad, for the Virtual University of Pakistan, Lahore.

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GENERAL INFORMATION

Purpose: This course aims at providing the student a good understanding of physics at the elementary level. Physics is essential for understanding the modern world, and is a definite part of its culture.

Background: It will be assumed that the student has taken physics and mathematics at the F.Sc level, i.e. the 12th year of schooling. However, B.Sc students are also likely to find the course useful. Calculus is not assumed and some essential concepts will be developed as the course progresses. Algebra and trigonometry are essential. However, for physics, the more mathematics one knows the better.

Scope and Duration: The course has 45 lectures, each of somewhat less than one hour duration. All main fields of physics will be covered, together with several applications in each.

Language: For ease of communication, all lectures are in Urdu. However, English or Latin technical terms have been used where necessary. The student must remember that further study and research in science is possible only if he or she has an adequate grasp of English.

Textbook: There is no prescribed textbook. However, you are strongly recommended to read a book at the level of "College Physics" by Halliday and Resnick (any edition). There are many other such books too, such as "University Physics" by Young and Freedman. Study any book that you are comfortable with, preferably by a well-established foreign author. Avoid local authors because they usually copy. After listening to a lecture, go read the relevant chapter. Please remember that these notes cover only some things that you should know and are not meant to be complete.

Assignments: There will be total Eight Assignment in this course and its schedules will be announced from time to time. The book you choose to consult will have many more. Those students who are seriously interested in the subject are advised to work out several of the questions posed there. In physics you cannot hope to gain mastery of the subject without extensive problem solving.

Examinations: Their schedules will be announced from time to time.

Tutors: Their duty is to help you, and they will respond to all genuine questions. However, please do not overload them as they have to deal with a large number of students. Happy studying!

Acknowledgements: I thank the Virtual University team and administration for excellent cooperation, as well as Mansoor Noori and Naeem Shahid, for valuable help.

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Summary of Lecture 1 – INTRODUCTION TO PHYSICS

1. Physics is a science. Science works according to the *scientific method*. The *scientific method* accepts only reason, logic, and experimental evidence to tell between what is scientifically correct and what is not. Scientists do not simply believe – they test, and keep testing until satisfied. Just because some "big scientist" says something is right, that thing does not become a fact of science. Unless a discovery is repeatedly established in different laboratories at different times by different people, or the same theoretical result is derived by clear use of established rules, we do not accept it as a scientific discovery. The real strength of science lies in the fact that it continually keeps challenging itself.

2. It is thought that the laws of physics do not change from place to place. This is why experiments carried out in different countries by different scientists – of any religion or race – have always led to the same results if the experiments have been done honestly and correctly. We also think that the laws of physics today are the same as they were in the past. Evidence, contained in the light that left distant stars billions of years ago, strongly indicates that the laws operating at that time were no different than those today. The spectra of different elements then and now are impossible to tell apart, even though physicists have looked very carefully.

3. This course will cover the following broad categories:
   a) *Classical Mechanics*, which deals with the motion of bodies under the action of forces. This is often called Newtonian mechanics as well.
   b) *Electromagnetism*, whose objective is to study how charges behave under the influence of electric and magnetic fields as well as understand how charges can create these fields.
   c) *Thermal Physics*, in which one studies the nature of heat and the changes that the addition of heat brings about in matter.
   d) *Quantum Mechanics*, which primarily deals with the physics of small objects such as atoms, nuclei, quarks, etc. However, Quantum Mechanics will be treated only briefly for lack of time.

4. Every physical quantity can be expressed in terms of three fundamental dimensions: Mass (M), Length (L), Time (T). Some examples:

Speed	LT^{-1}
Acceleration	LT^{-2}
Force	MLT^{-2}
Energy	ML^2T^{-2}
Pressure	$ML^{-1}T^{-2}$

You cannot add quantities that have different dimensions. So force can be added to force, but force can never be added to energy, etc. A formula is definitely wrong if the dimensions on the left and right sides of the equal sign are different.

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5. Remember that any function $f(x)$ takes as input a dimensionless *number x* and outputs a quantity f (which may, or may not have a dimension). Take, for example, the function $f(\theta) = \sin\theta$. You know the expansion: $\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots$ If θ had a dimension then you would be adding up quantities of different dimensions, and that is not allowed.

6. Do not confuse units and dimensions. We can use different units to measure the same physical quantity. So, for example, you can measure the mass in units of kilograms, pounds, or even in *sair* and *chatak!* In this course we shall always use the MKS or **M**etre-**K**ilogram-**S**econd system. When you want to convert from one system to another, be methodical as in the example below:

$$1\frac{mi}{hr} = 1\frac{mi}{hr} \times 5280\frac{ft}{mi} \times \frac{1}{3.28}\frac{m}{ft} \times \frac{1}{3600}\frac{hr}{s} = 0.447\frac{m}{s}$$

When you write it out in this manner, note that various quantities cancel out cleanly in the numerator and denominator. So you never make a mistake!

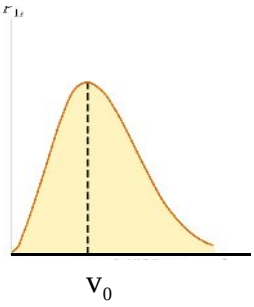
7. A good scientist first thinks of the larger picture and then of the finer details. So, estimating *orders of magnitude* is extremely important. Students often make the mistake of trying to get the decimal points right instead of the first digit – which obviously matters the most! So if you are asked to calculate the height of some building using some data and you come up with 0.301219 metres or 4.01219×10^6 metres, then the answer is plain nonsense even though you may have miraculously got the last six digits right. Physics is commonsense first, so use your intelligence before submitting any answer.

8. Always check your equations to see if they have the same dimensions on the left side as on the right. So, for example, from this principle we can see the equation $v^2 = u^2 + 2at$ is clearly wrong, whereas $v^2 = u^2 + 13a^2t^2$ could possibly be a correct relation. (Here v and u are velocities, a is acceleration, and t is time.) Note here that I use the word *possibly* because the dimensions on both sides match up in this case.

9. Whenever you derive an equation that is a little complicated, see if you can find a special limit where it becomes simple and transparent. So, sometimes it is helpful to imagine that some quantity in it is very large or very small. Where possible, make a "mental graph" so that you can picture an equation. So, for example, a formula for the distribution of molecular speeds in a gas could look like $f(v) = ve^{-(v-v_0)^2/a^2}$. Even without knowing the value of a you can immediately see that

a) $f(v)$ goes to zero for large values of v, and $v = 0$.

b) The maximum value of $f(v)$ occurs at v_0 and the function decreases on both side of this value.



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Summary of Lecture 2 – KINEMATICS I

1. *x(t)* is called displacement and it denotes the position of a body at time. If the displacement is positive then that body is to the right of the chosen origin and if negative, then it is to the left.

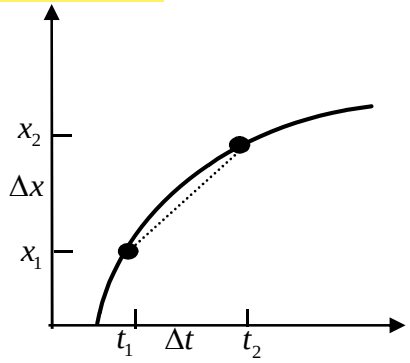
2. If a body is moving with average speed v then in time *t* it will cover a distance $d=vt$. But, in fact, the speed of a car changes from time to time and so one should limit the use of this formula to small time differences only. So, more accurately, one defines an average speed over the small time interval Δt as:

$$\text{average speed} = \frac{\text{distance travelled in time } \Delta t}{\Delta t}$$

3. We define *instantaneous velocity* at any time t as:

$$v = \frac{x(t_2)-x(t_1)}{t_2-t_1} \equiv \frac{\Delta x}{\Delta t} \quad .$$

Here Δx and Δt are both very small quantities that tend to zero but their ratio v does not.



4. Just as we have defined velocity as the rate of change of distance, similarly we can define *instantaneous acceleration* at any time t as:

$$a = \frac{v(t_2)-v(t_1)}{t_2-t_1} \equiv \frac{\Delta v}{\Delta t} \quad .$$

Here Δv and Δt are both very small quantities that tend to zero but their ratio *a* is not zero, in general. Negative acceleration is called deceleration. The speed of a decelerating body decreases with time.

5. Some students are puzzled by the fact that a body can have a very large acceleration but can be standing still at a given time. In fact, it can be moving in the opposite direction to its acceleration. There is actually nothing strange here because position,

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6. For constant acceleration and a body that starts from rest at $t = 0$, v increases linearly with time, $v \propto t$ (or $v = at$). If the body has speed v_0 at $t = 0$, then at time t, $v = at + v_0$.
7. We know in (6) above how far a body moving at constant speed moves in time t. But what if the body is changing its speed? If the speed is increasing linearly (i.e. constant acceleration), then the answer is particularly simple: just use the same formula as in (6) but use the average speed: $(v_0 + v_0 + at)/2$. So we get $x = x_0 + (v_0 + v_0 + at)t/2 = x_0 + v_0 t + \frac{1}{2}at^2$. This formula tells you how far a body moves in time t if it moves with constant acceleration a, and if started at position x_0 at $t=0$ with speed v_0.

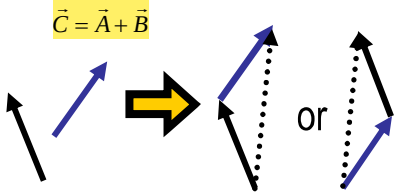
8. We can eliminate the time using (7), and arrive at another useful formula that tells us what the final speed will be after the body has traveled a distance equal to $x - x_0$ after time t, $v^2 = v_0^2 + 2a(x - x_0)$.

9. Vectors: a quantity that has a size as well as direction is called a *vector*. So, for example, the wind blows with some speed and in some direction. So the *wind velocity* is a vector.

10. If we choose axes, then a vector is fixed by its components along those axes. In one dimension, a vector has only one component (call it the x-component). In two dimensions, a vector has both x and y components. In three dimensions, the components are along the x,y,z axes.

11. If we denote a vector $\vec{r} = (x, y)$ then, $r_x = x = r\cos\theta$, and $r_y = y = r\sin\theta$. Note that $x^2 + y^2 = r^2$. Also, that $\tan\theta = y/x$.

13. Two vectors can be added together geometrically. We take any one vector, move it without changing its direction so that both vectors start from the same point, and then make a parallelogram. The diagonal of the parallelogram is the resultant.



14. Two vectors can also be added algebraically. In this case, we simply add the components of the two vectors along each axis separately. So, for example, Two vectors can be put together as $(1.5, 2.4) + (1, -1) = (2.5, 1.4)$.

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Summary of Lecture 3 – KINEMATICS II

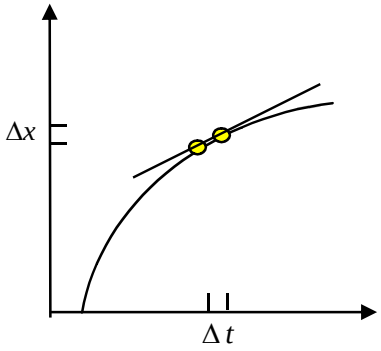
1. The concept of the derivative of a function is exceedingly important. The derivative shows how fast a function changes when its argument is changed. (Remember that for $f(x)$ we say that f is a function that depends upon the argument x. You should think of f as a machine that gives you the value f when you input x.)

2. Functions do not always have to be written as $f(x)$. $x(t)$ is also a function. It tells us where a body is at different times t.

3. The derivative of $x(t)$ at time t is defined as:

$$\frac{dx}{dt} \equiv \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{x(t+\Delta t) - x(t)}{\Delta t}.$$



4. Let's see how to calculate the derivative of a simple function like $x(t) = t^2$. We must first calculate the difference in x at two slightly different values, t and $t + \Delta t$, while remembering that we choose Δt to be extremely small:

$$\Delta x = (t + \Delta t)^2 - t^2$$

$$= t^2 + (\Delta t)^2 + 2t\Delta t - t^2$$

$$\frac{\Delta x}{\Delta t} = \Delta t + 2t \Rightarrow \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} = 2$$

5. In exactly the same way you can show that if $x(t) = t^n$ then:

$$\frac{dx}{dt} \equiv \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = nt^{n-1}$$

This is an extremely useful result.

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6. Let us apply the above to the function $x(t)$ which represents the distance moved by a body with constant acceleration (see lecture 2):

$$x(t) = x_0 + \mathrm{v}_0 t + \frac{1}{2}at^2$$

$$\frac{dx}{dt} = 0 + \mathrm{v}_0 + \frac{1}{2}a(2t) = \mathrm{v}_0 + a\,t$$

This clearly shows that $\frac{d\mathrm{v}}{dt} = 0 + a = a$ (acceleration is constant)

7. A stone dropped from rest increases its speed in the downward direction according to $\frac{d\mathrm{v}}{dt} = g \approx 9.8$ m/sec. This is true provided we are fairly close to the earth, otherwise the value of g decreases as we go further away from the earth. Also, note that if we measured distances from the ground up, then the acceleration would be negative.

8. A useful notation: write $\frac{d\mathrm{v}}{dt} = \frac{d}{dt}\left(\frac{dx}{dt}\right) = \frac{d^2x}{dt^2}$. We call $\frac{d^2x}{dt^2}$ the second derivative of x with respect to t, or the rate of rate of change of x with respect to t.

9. It is easy to extend these ideas to a body moving in both the x and y directions. The position and velocity in 2 dimensions are:

$$\vec{r} = x(t)\hat{i} + y(t)\,\hat{j}$$

$$\vec{\mathrm{v}} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\,\hat{j}$$

$$= \mathrm{v}_x\hat{i} + \mathrm{v}_y\,\hat{j}$$

Here the unit vectors $\hat{i}$ and $\hat{j}$ are fixed, meaning that they do not depend upon time.

10. The scalar product of two vectors $\vec{A}$ and $\vec{B}$ is defined as:

$$\vec{A}\cdot\vec{B} = AB\cos\theta$$

You can think of:

$$\vec{A}\cdot\vec{B} = (A)(B\cos\theta)$$

$$= (\text{length of } \vec{A}) \times (\text{projection of } \vec{B} \text{ on } \vec{A})$$

OR,

$$\vec{A}\cdot\vec{B} = (B)(A\cos\theta)$$

$$= (\text{length of } \vec{B}) \times (\text{projection of } \vec{A} \text{ on } \vec{B}).$$

Remember that for unit vectors $\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = 1$ and $\hat{i}\cdot\hat{j} = 0$.

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Summary of Lecture 4 – FORCE AND NEWTON'S LAWS

1. Ancient view: objects tend to stop if they are in motion; force is required to keep something moving. This was a natural thing to believe in because we see objects stop moving after some time; frictionless motion is possible to see only in rather special circumstances.

2. Modern view: objects tend to remain in their initial state; force is required to *change* motion. Resistance to changes in motion is called *inertia*. More inertia means it is harder to make a body accelerate or decelerate.

3. Newton's First Law: An object will remain at rest or move with constant velocity unless acted upon by a net external force. (A non-accelerating reference frame is called an inertial frame; Newton's First Law holds only in inertial frames.)
4. More force leads to more acceleration: $\Rightarrow \quad a \propto F$

5. The greater the mass of a body, the harder it is to change its state of motion. More mass means more inertia. In other words, more mass leads to less acceleration:

$$\Rightarrow \quad a \propto \frac{1}{m}$$

Combine both the above observations to conclude that:

$$a \propto \frac{F}{m}$$

6. Newton's Second Law: $a = \frac{F}{m}$ (or, if you prefer, write as $F = ma$).

7. $F = ma$ is one relation between three independent quantities (m, a, F). For it to be useful, we must have separate ways of measuring mass, acceleration, and force. Acceleration is measured from observing the rate of change of velocity; mass is a measure of the amount of matter in a body (e.g. two identical cars have twice the mass of a single one). Forces (due to gravity, a stretched spring, repulsion of two like charges, etc) will be discussed later.

8. Force has dimensions of [mass] $\times$ [acceleration] $= M L T^{-2}$. In the MKS system the unit of force is the Newton. It has the symbol N where:
1 Newton = 1 kilogram.metre/second^2.

9. Forces can be internal or external. For example the mutual attraction of atoms within a block of wood are called internal forces. Something pushing the wood

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is an external force. In the application of $F = ma$, remember that F stands for the total external force upon the body.

10. Forces are vectors, and so they must be added vectorially:

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \cdots$$

This means that the components in the $\hat{x}$ direction must be added separately, those in the $\hat{y}$ direction separately, etc.

11. Gravity acts directly on the mass of a body - this is a very important experimental observation due to Newton and does not follow from $F = ma$. So a body of mass m_1 experiences a force $F_1 = m_1 g$ while a body of mass m_2 experiences a force $F_2 = m_2 g$, where g is the acceleration with which any body (big or small) falls under the influence of gravity. (Galileo had established this important fact when he dropped different masses from the famous leaning tower of Pisa!)

12. The weight of a body W is the force which gravity exerts upon it, $W = mg$. Mass and weight are two completely different quantities. So, for example, if you used a spring balance to weigh a kilo of grapes on earth, the same grapes would weigh only 1/7 kilo on the moon.

13. Newton's Third Law: for every action there is an equal and opposite reaction. More precisely, $F_{AB} = -F_{BA}$, where F_{AB} is the force exerted by body B upon A whereas F_{BA} is the force exerted by body A upon B. Ask yourself what would happen if this was not true. In that case, a system of two bodies, even if it is completely isolated from the surroundings, would have a net force acting upon it because the net force acting upon both bodies would be $F_{AB} + F_{BA} \neq 0$.

14. If action and reaction are always equal, then why does a body accelerate at all? Students are often confused by this. The answer: in considering the acceleration of a body you must consider only the (net) force acting upon that body. So, for example, the earth pulls a stone towards it and causes it to accelerate because there is a net force acting upon the stone. On the other hand, by the Third Law, the stone also pulls the earth towards it and this causes the earth to accelerate towards the stone. However, because the mass of the earth is so large, we are only able to see the acceleration of the stone and not that of the earth.

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Summary of Lecture 5 – APPLICATIONS OF NEWTON'S LAWS – I

1. An obvious conclusion from $F = ma$ is that if $F = 0$ then $a = 0$! How simple, yet how powerful ! This says that for any body that is not accelerating the *sum of all the forces acting upon it* must vanish.

2. Examples of systems in equilibrium: a stone resting on the ground; a pencil balanced on your finger; a ladder placed against the wall, an aircraft flying at a constant speed and constant height.

3. Examples of systems out of equilibrium: a stone thrown upwards that is at its highest point; a plane diving downwards; a car at rest whose driver has just stepped on the car's accelerator.

4. If you know the acceleration of a body, it is easy to find the force that causes it to accelerate. Example: An aircraft of mass m has position vector,

$$\vec{r} = (at + bt^3)\hat{i} + (ct^2 + dt^4)\,\hat{j}$$

What force is acting upon it?

SOLUTION:
$$\vec{F} = m\frac{d^2x}{dt^2}\hat{i} + m\frac{d^2y}{dt^2}\hat{j}$$
$$= 6\,b\,m\,t\,\hat{i} + m(2c + 12d\,t^2)\,\hat{j}$$

5. The other way around is not so simple: suppose that you know F and you want to find x. For this you must solve the equation,

$$\frac{d^2x}{dt^2} = \frac{F}{m}$$

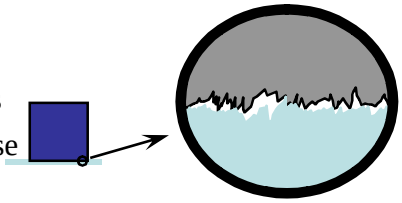
This may or may not be easy, depending upon F (which may depend upon both x as well as t if the force is not constant).

6. Ropes are useful because you can pull from a distance to change the direction of a force. The tension, often denoted by T, is the force you would feel if you cut the rope and grabbed the ends. For a massless rope (which may be a very good approximation in many situations) the tension is the same at every point along the rope. Why? Because if you take any small slice of the rope it weighs nothing (or very little). So if the force on one side of the slice was any different from the force force on the other side, it would be accelerating hugely. All this was for the "ideal rope" which has no mass and never breaks. But this idealization if often good enough.

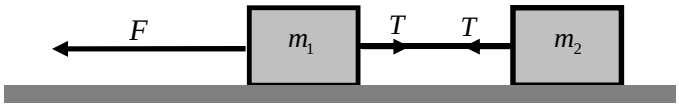
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7. We are all familiar with frictional force. When two bodies rub against each other, the frictional force acts upon each body separately opposite to its direction of motion (i.e it acts to slow down the motion). The harder you press two bodies against each other, the greater the friction. Mathematically, $\vec{F} = \mu\vec{N}$, where $\vec{N}$ is the force with which you press the two bodies against each other (normal force). The quantity μ is called the coefficient of friction (obviously!). It is large for rough surfaces, and small for smooth ones. Remember that $\vec{F} = \mu\vec{N}$ is an empirical relation and holds only approximately. This is obviously true: if you put a large enough mass on a table, the table will start to bend and will eventually break.

8. Friction is caused by roughness at a microscopic level - if you look at any surface with a powerful microscope you will see unevenness and jaggedness. If these big bumps are levelled somehow, friction will still not disappear because there will still be little bumps due to atoms. More precisely, atoms from the two bodies will interact each other because of the electrostatic interaction between their charges. Even if an atom is neutral, it can still exchange electrons and there will be a force because of surrounding atoms.



9. Consider the two blocks below on a frictionless surface:



We want to find the tension and acceleration: The total force on the first mass is $F - T$ and so $F - T = m_1 a$. The force on the second mass is simply T and so $T = m_2 a$. Solving the above, we get: $T = \dfrac{m_2 F}{m_1 + m_2}$ and $a = \dfrac{F}{m_1 + m_2}$.

10. There is a general principle by which you solve equilibrium problems. For equilibrium, the sum of forces in every direction must vanish. So $F_x = F_y = F_z = 0$. You may always choose the x, y, z directions according to your convenience. So, for example, as in the lecture problem dealing with a body sliding down an inclined plane, you can choose the directions to be along and perpendicular to the surface of the plane.

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Summary of Lecture 6 – APPLICATIONS OF NEWTON'S LAWS – II

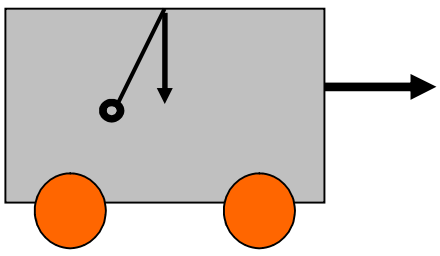
1. As a body moves through a body it displaces the fluid. it has to exert a force on the fluid to push it out of the way. By Newton's third law, the fluid pushes back on the body with an equal and opposite force. The direction of the fluid resistance force on a body is always opposite to the direction of the body's velocity relative to the fluid.

2. The magnitude of the fluid resistance force usually increases with the speed of the body through the fluid. Typically, $f = kv$ (an empirical law!). Imagine that you drop a ball bearing into a deep container filled with oil. After a while the ball bearing will approach its maximum (terminal) speed when the forces of gravity and friction balance each other: $mg = kv$ from which $v_{final} = mg / k$.

3. The above was a simple example of equilibrium under two forces. In general, while solving problems you should a)draw a diagram, b)define an origin for a system of coordinates, c)identify all forces (tension, normal, friction, weight, etc) and their x and y components, d)Apply Newton's law separately along the x and y axes. e) find the accelerations, then velocities, then displacements. This sounds very cook-book, and in fact it will occur to you naturally how to do this while solving actual problems.

4. Your weight in a lift: suppose you are in a lift that is at rest or moving at constant velocity. In either case $a=0$ and the normal force N and the force due to gravity are exactly equal, $N - Mg = 0 \Rightarrow N = Mg$. But if the lift is accelerating downwards then $Mg - N = Ma$ or $N = M(g - a)$. So now the normal force (i.e. the force with which the floor of the lift is pushing on you) is decreased. Note that if the lift is accelerating downwards with acceleration a (which it will if the cable breaks!) then $N=0$ and you will experience weightlessness just like astronauts in space do. Finally, if the lift is accelerating upwards then a is negative and you will feel heavier.

5. Imagine that you are in a railway wagon and want to know how much you are accelerating. You are not able to look out of the windows. A mass is hung from the roof. Find the acceleration of the car from the angle made by the mass.



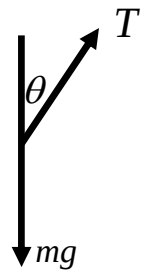
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We first balance the forces vertically: $T\cos\theta = mg$

And then horizontally: $T\sin\theta = ma$

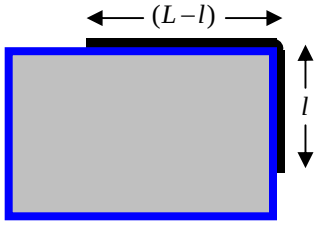
From these two equations we find that: $\tan\theta = \frac{a}{g}$

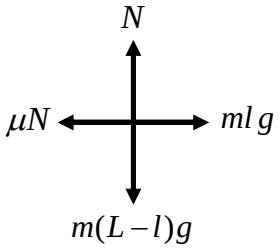
Note that the mass m doesn't matter - it cancels out!



6. Friction is a funny kind of force. It doe not make up its mind which way to act until some other force compels it to decide. Imagine a block lying on the floor. If you push it forward, friction will act backward. And if you push it to the left, friction will act to the right. In other words, the direction of the frictional force is always in the opposite direction to the applied force.

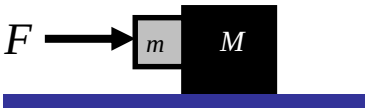
7. Let us solve the following problem: a rope of total length L and mass per unit length m is put on a table with a length l hanging from one edge. What should be l such that the rope just begins to slip?





To solve this, look at the balance of forces in the diagram below: in the vertical direction, the normal force balances the weight of that part of the rope that lies on the table: $N = m(L-l)g$. In the horizontal direction, the rope exerts a force mlg to the right, which is counteracted by the friction that acts to the left. Therefore $\mu N = mlg$. Substituting N from the first equation we find that $l = \frac{\mu L}{\mu + 1}$. Note that if μ isvery small then even a small piece of string that hangs over the edge will cause the entire string to slip down.

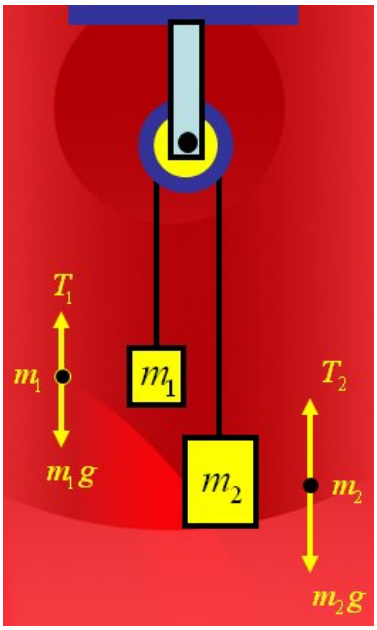
9. In this problem, we would like to calculate the minimum force F such that the small block does not slip downwards. Clearly, since the 2 bodies move together, $F = (m+M)a$. This gives $a = \frac{F}{(m+M)}$. We want the friction μN to be at least as large as the downwards force, mg.



So, we put $N = ma = m\left(\frac{F}{(m+M)}\right)$ from which the minimum horizontal force needed to prevent slippage is $F = \frac{(m+M)g}{\mu}$.

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There is an error in video lecture # 6 (video recording at 41:00) please consider the following problem with correct solution.
Consider two bodies of unequal masses m_1 and m_2 connected by the ends of a string, which passes over a frictionless pulley as shown in the diagram.



$m_2 > m_1$

$$T - m_1 g = m_1 a$$

$$m_2 g - T = m_2 a$$

$$m_2 g - m_1 g = (m_1 + m_2) a$$

$$a = \frac{(m_2 - m_1)}{(m_1 + m_2)} g$$

$m_2 > m_1$

$$T - m_1 g = m_1 a$$

$$a = \frac{T - m_1 g}{m_1} \quad \text{(i}$$

$$m_2 g - T = m_2 a$$

$$a = \frac{m_2 g - T}{m_2} \quad \text{(ii}$$

equating (i and (ii

$$\frac{T - m_1 g}{m_1} = \frac{m_2 g - T}{m_2}$$

$$m_2 T - m_1 m_2 g = m_1 m_2 g - m_1 T$$

$$m_2 T + m_1 T = m_1 m_2 g + m_1 m_2 g$$

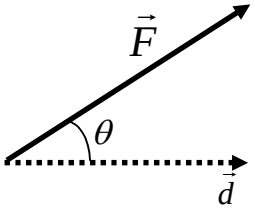
$$T(m_2 + m_1) = 2 m_1 m_2 g$$

$$T = \frac{2 m_1 m_2 g}{(m_2 + m_1)}$$

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Summary of Lecture 7 – WORK AND ENERGY

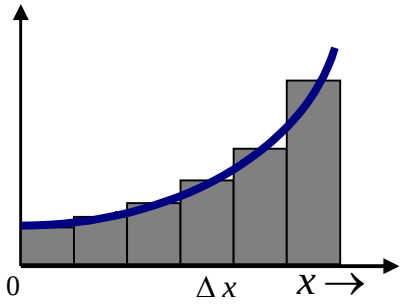
1. Definition of work: force applied in direction of displacement × displacement. This means that if the force F acts at an angle θ with respect to the direction of motion, then $W = \vec{F} \cdot \vec{d} = Fd\cos\theta$



2. a) Work is a scalar - it has magnitude but no direction.
   b) Work has dimensions: $M \times (L\ T^{-2}) \times L\ =\ M\ L^2T^{-2}$
   c) Work has units: 1 Newton × 1 Metre ≡ 1 Joule (J)

3. Suppose you lift a mass of 20 kg through a distance of 2 metres. Then the work you do is 20 kg × 9.8 Newtons × 2 metres = 392 Joules. On the other hand, the force of gravity is directed opposite to the force you exert and the work done by gravity is -392 Joules.

4. What if the force varies with distance (say, a spring pulls harder as it becomes longer). In that case, we should break up the distance over which the force acts into small pieces so that the force is approximately constant over each bit. As we make the pieces smaller and smaller, we will approach the exact result:



Now add up all the little pieces of work:

$$W = \Delta W_1 + \Delta W_2 + \cdots + \Delta W_N = F_1\Delta x + F_2\Delta x + \cdots + F_N\Delta x \equiv \sum_{n=1}^{N} F_n \Delta x$$

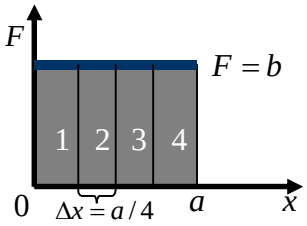
To get the exact result let $\Delta x \to 0$ and the number of intervals N $\to \infty$: $W = \lim_{\Delta x \to 0} \sum_{n=1}^{\infty} F_n \Delta x$

Definition: $W = \lim_{\Delta x \to 0} \sum_{n=1}^{\infty} F_n\ \Delta x \equiv \int_{x_i}^{x_f} F(x)dx$ is called the integral of F with respect to x from

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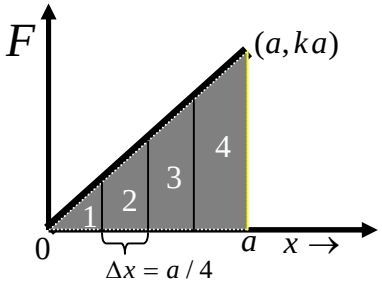
x_i to x_f. This quantity is the work done by a force, constant or non-constant. So if the force is known as a function of position, we can always find the work done by calculating the definite integral.

5. Just to check what our result looks like for a constant force, let us calculate W if $F = b$,



$$\frac{1}{4}a(b)+\frac{1}{4}a(b)+\frac{1}{4}a(b)+\frac{1}{4}a(b)=ab \qquad \therefore \int_0^a F dx = ab$$

6. Now for a less trivial case: suppose that $F=kx$, i.e. the force increases linearly with x.



Area of shaded region $= \frac{1}{2}(a)(ka) = k\frac{a^2}{2}$ $\qquad \therefore \quad \int_0^a F dx = k\frac{a^2}{2}$

7. Energy is the capacity of a physical system to do work:
   - it comes in many forms – mechanical, electrical, chemical, nuclear, etc
   - it can be stored
   - it can be converted into different forms
   - it can never be *created* or *destroyed*

8. Accepting the fact that energy is conserved, let us derive an expression for the kinetic energy of a body. Suppose a *constant force* accelerates a mass *m* from speed *0* to speed v over a *distance d*. What is the work done by the force? Obviously the answer is:

$W = Fd$. But $F = ma$ and $v^2 = 2ad$. This gives $W = (ma)d = \frac{mv^2}{2d}d = \frac{1}{2}mv^2$. So, we conclude that the work done by the force has gone into creating kinetic energy. and that the amount of kinetic energy possessed by a body moving with speed v is $\frac{1}{2}mv^2$.

9. The work done by a force is just the force multiplied by the distance – it does not depend upon time. But suppose that the same amount of work is done in half the time. We then say that the *power* is twice as much.

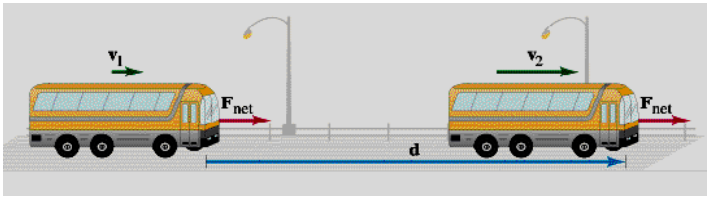
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We define:

$$\text{Power} = \frac{\text{Work done}}{\text{Time taken}}$$

If the force does not depend on time: $\frac{\text{Work}}{\text{Time}} = \frac{F\,\Delta x}{\Delta t} = F\,\text{v}$. Therefore, Power = F v.

10. Let's work out an example. A constant force accelerates a bus (mass m) from speed v_1 to speed v_2 over a distance d. What work is done by the engine?



Recall that for constant acceleration, $\text{v}_2^2 - \text{v}_1^2 = 2a\,(x_2 - x_1)$ where: v_2 = final velocity, x_2 = final position, v_1 = initial velocity, x_1 = initial position. Hence, $a = \frac{\text{v}_2^2 - \text{v}_1^2}{2d}$. Now calculate the work done: $W = Fd = mad = m\frac{\text{v}_2^2 - \text{v}_1^2}{2d}\,d = \frac{1}{2}m\text{v}_2{}^2 - \frac{1}{2}m\text{v}_1{}^2$. So the work done has resulted in an increase in the quantity $\frac{1}{2}m\text{v}^2$, which is kinetic energy.

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Summary of Lecture 8 – CONSERVATION OF ENERGY

1. Potential energy is, as the word suggests, the energy "locked up" up somewhere and which can do work. An object can store energy as the result of its position. Stored energy of position is referred to as potential energy. **Potential energy** is the stored energy of position possessed by an object. It has the ability or capacity to do work like other forms of energies.Potential energy can be converted into kinetic energy, $\frac{1}{2}mv^2$. As I showed you earlier, this follows directly from Newton's Laws.

2. If you lift a stone of mass m from the ground up a distance x, you have to do work against gravity. The (constant) force is mg, and so $W = mgx$. By conservation of energy, the work done by you was transformed into gravitational potential energy whose values is exactly equal to mgx. Where is the energy stored? Answer: it is stored neither in the mass or in the earth - it is stored in the gravitational field of the combined system of stone+earth.

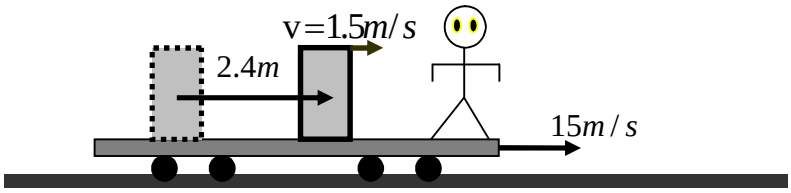
3. Suppose you pull on a spring and stretch it by an amount x away from its normal (equilibrium) position. How much energy is stored in the spring? Obviously, the spring gets harder and harder to pull as it becomes longer. When it is extended by length x and you pull it a further distance dx, the small amount of work done is $dW = Fdx = kxdx$. Adding up all the small bits of work gives the total work:

$$W = \int_0^x F dx = \int_0^x k x dx = \frac{1}{2} k x^2$$

This is the work you did. Maybe you got tired working so hard. What was the result of your working so hard? Answer: this work was transformed into energy stored in the spring. The spring contains energy exactly equal to $\frac{1}{2}kx^2$.

4. Kinetic energy obviously depends on the frame you choose to measure it in. If you are running with a ball, it has zero kinetic energy with respect to you. But someone who is standing will see that it has kinetic energy! Now consider the following situation: a box of mass 12kg is pushed with a constant force so that so that its speed goes from zero to 1.5m/sec (as measured by the person at rest on the cart) and it covers a distance of 2.4m. Assume there is no friction.

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mass of box = 12 *kg*

Let's first calculate the change in kinetic energy:

$$\Delta K = K_f - K_i = \frac{1}{2}(12kg)(1.5m/s)^2 - 0 = 13.5J$$

And then the (constant) acceleration:

$$a = \frac{v_f^2 - v_i^2}{2(x_f - x_i)} = \frac{(1.5m/s)^2 - 0}{2(2 \cdot 4m)} = 0.469m/s^2$$

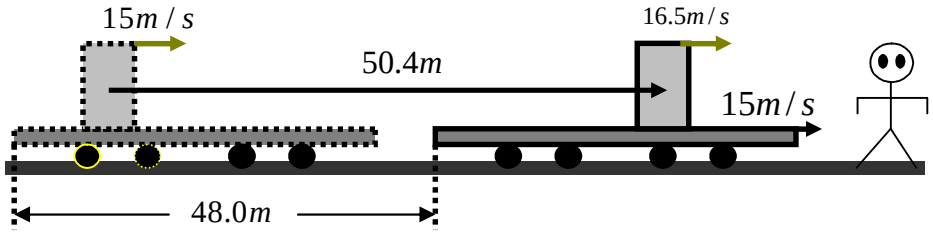
This acceleration results from a constant net force given by:

$$F = ma = (12kg)(0.469\,m/s^2) = 5.63N$$

From this, the work done on the crate is:

$$W = F\Delta x = (5.63N)(2.4m) = 13.5J \quad \text{(same as } \Delta K = 13.5J \text{ !)}$$

5. Now suppose there is somebody standing on the ground, and that the trolley moves at 15 m/sec relative to the ground:



Let us repeat the same calculation:

$$\Delta K' = K'_f - K'_i = \frac{1}{2}mv_f^{\,2} - \frac{1}{2}mv_i^{\,2}$$

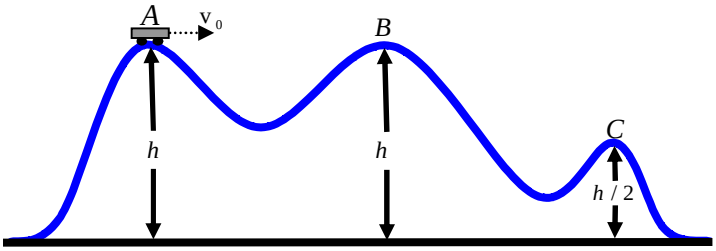
$$= \frac{1}{2}(12kg)(16.5m/s)^2 - \frac{1}{2}(12kg)(15.0m/s)^2 = 284J$$

This example clearly shows that work and energy have different values in different frames.

6. The total mechanical energy is: $E_{mech} = KE + PE$. If there is no friction then E_{mech} is conserved. This means that the sum does not change with time. For example: a ball is thrown upwards at speed v_0. How high will it go before it stops? The loss of potential energy is equal to the gain of potential energy. Hence, $\frac{1}{2}mv_0^2 = mgh \Rightarrow h = \frac{v_0^2}{2g}$.

Now look at the smooth, frictionless motion of a car over the hills below:

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Even though the motion is complicated, we can use the fact that the total energy is a constant to get the speeds at the points B,C,D:

At point A: $\frac{1}{2}mv_A^2 + mgh = \frac{1}{2}mv_B^2 + mgh \quad \Rightarrow \quad v_B = v_C$

At point C: $\frac{1}{2}mv_A^2 + mgh = \frac{1}{2}mv_C^2 + mg\frac{h}{2} \quad \Rightarrow \quad v_C = \sqrt{v_A^2 + gh}$

At point D: $\frac{1}{2}mv_A^2 + mgh = \frac{1}{2}mv_D^2 \quad \Rightarrow \quad v_D = \sqrt{v_A^2 + 2gh}$

7. Remember that potential energy has meaning only for a force that is conservative. A conservative force is that for which the work done in going from point A to point B is independent of the path chosen. Friction is an example of a non-conservative force and a potential energy cannot be defined. For a conservative force, $F = -\frac{dV}{dx}$. So, for a spring, $V = \frac{1}{2}kx^2$ and so $F = -kx$.

8. Derivation of $F = -\frac{dV}{dx}$: If the particle moves distance Δx in a potential V, then change in PE is ΔV where, $\Delta V = -F\,\Delta x$. From this, $F = -\frac{\Delta V}{\Delta x}$. Now let $\Delta x \to 0$. Hence, $F = -\lim_{\Delta x \to 0}\frac{\Delta V}{\Delta x} = -\frac{dV}{dx}$.

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Summary of Lecture 9 – MOMENTUM

1. Momentum is the "quantity of motion" possessed by a body. More precisely, it is defined as:

Mass of the body × Velocity of the body.

The dimensions of momentum are MLT^{-1} and the units of momentum are kg-m/s.

2. Momentum is a vector quantity and has both magnitude and direction, $p = m\vec{v}$. We can easily see that Newton's Second Law can be reexpressed in terms of momentum. When I wrote it down originally, it was in the form $m\vec{a} = \vec{\mathbf{F}}$. But since $\frac{d\vec{v}}{dt} = \vec{a}$, this can also be written as $\frac{d\vec{\mathbf{p}}}{dt} = \vec{\mathbf{F}}$ (new form). In words, the rate of change of momentum of a body equals the total force acting upon it. Of course, the old and new are exactly the same,

$$\frac{d\vec{\mathbf{p}}}{dt} = \frac{d(m\vec{\mathbf{v}})}{dt} = m\frac{d\vec{\mathbf{v}}}{dt} = m\vec{\mathbf{a}} = \vec{\mathbf{F}}.$$

3. When there are many particles, then the total momentum $\vec{\mathbf{P}}$ is,

$$\vec{\mathbf{P}} = \vec{\mathbf{p}}_1 + \vec{\mathbf{p}}_2 + \cdots \vec{\mathbf{p}}_N$$

$$\frac{d}{dt}\vec{\mathbf{P}} = \frac{d\vec{\mathbf{p}}_1}{dt} + \frac{d\vec{\mathbf{p}}_2}{dt} + \cdots \frac{d\vec{\mathbf{p}}_N}{dt}$$

$$= \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2 + \cdots \vec{\mathbf{F}}_N = \vec{\mathbf{F}}$$

This shows that when there are several particles, the rate at which the total momentum changes is equal to the total force. It makes sense!

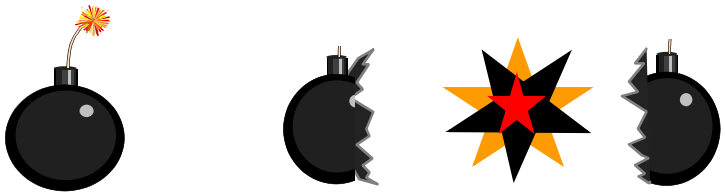
4. A very important conclusion of the above is that if the sum of the total external forces vanishes, then the total momentum is conserved, $\sum \vec{F}_{ext} = 0 \Rightarrow \frac{d}{dt}\vec{\mathbf{P}} = 0$. This is quite independent of what sort of forces act between the bodies - electric, gravitational, etc. - or how complicated these are. We shall see why this is so important from the following examples.

5. Two balls, which can only move along a straight line, collide with each other. The initial momentum is $P_i = m_1u_1 + m_2u_2$ and the final momentum is $P_f = m_1v_1 + m_2v_2$. Obviously one ball exerts a force on the other when they collide, so its momentum changes. But, from the fact that there is no external force acting on the balls, $P_i = P_f$, or $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.

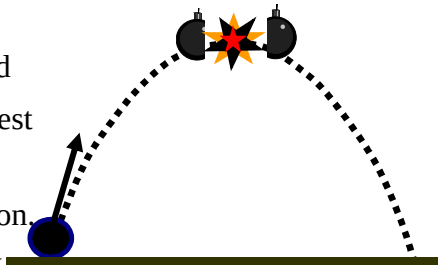
6. A bomb at rest explodes into two fragments. Before the explosion the total momentum is

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zero. So obviously it is zero after the explosion as well, $\mathbf{P}_f = \mathbf{0}$. During the time that the explosion happens, the forces acting upon the pieces are very complicated and changing rapidly with time. But when all is said and done, there are two pieces flying away with a total zero final momentum $\mathbf{P}_f = m_1\mathrm{v}_1 + m_2\mathrm{v}_2$. Hence $m_1\mathrm{v}_1 = -m_2\mathrm{v}_2$. In other words, the fragments fly apart with equal momentum but in opposite directions. The centre-of-mass stays at rest. So, knowing the velocity of one fragment permits knowing the velocity of the other fragment.



7. If air resistance can be ignored, then we can do some interesting calculations with what we have learned. So, suppose a shell is fired from a cannon with a speed 10 m/s at an angle 60^0 with the horizontal. At the highest point in its path it explodes into two pieces of equal masses. One of the pieces retraces its path to the cannon. Let us find the velocity of the other piece immediately after the explosion.



Solution: After the explosion: $P_{1x} = -5\dfrac{M}{2}$ (why?). But $P_{1x} + P_{2x} = P_x = M \times 10\cos 60$

$$\Rightarrow P_{2x} = 5M + 5\frac{M}{2}. \text{ Now use: } P_{2x} = \frac{M}{2}\mathrm{v}_{2x} \Rightarrow \mathrm{v}_{2x} = 15\, m/s.$$

8. When you hit your thumb with a hammer it hurts, doesn't it? Why? Because a large amount of momentum has been destroyed in a short amount of time. If you wrap your thumb with foam, it will hurt less. To understand this better, remember that force is the rate of change of momentum: $F = \dfrac{dp}{dt} \Rightarrow dp = Fdt$. Now define the **impulse** I as:

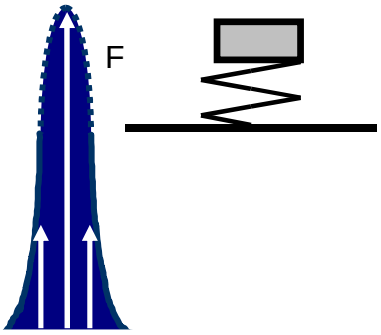
force × time over which the force acts.

If the force changes with time between the limits , then one should define I as,

$I = \int_{t_1}^{t_2} Fdt$. Since $\int_{t_1}^{t_2} Fdt = \int_{p_i}^{p_f} dp$, therefore $I = p_f - p_i$. In words, the change of momentum equals the impulse, which is equal to the area under the curve of force versus time. Even if you wrap your thumb in foam, the impulse is the same. But the force is definitely not!

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9. Sometimes we only know the force numerically (i.e. there is no expression like F=something). But we still know what the integral means: it is the area under the curve of force versus time. The curve here is that of a hammer striking a table. Before the hammer strikes, the force is zero, reaches a peak, and goes back to zero.



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Summary of Lecture 10 – COLLISIONS

1. Collisions are extremely important to understand because they happen all the time - electrons collide with atoms, a bat with a ball, cars with trucks, star galaxies with other galaxies,...In every case, the sum of the initial momenta equals the sum of the final momenta. This follows directly from Newton's Second Law, as we have already seen.

2. Take the simplest collision: two bodies of mass m_1 and m_2 moving with velocities u_1 and u_2. After the collision they are moving with velocities v_1 and v_2. From momentum conservation,

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 \Rightarrow m_1(u_1 - v_1) = m_2(v_2 - u_2)$$

This is as far as we can go. There are two unknowns but only one equation. However, if the collision is elastic then,

$$\tfrac{1}{2}m_1u_1^2 + \tfrac{1}{2}m_2u_2^2 = \tfrac{1}{2}m_1v_1^2 + \tfrac{1}{2}m_2v_2^2 \Rightarrow \tfrac{1}{2}m_1(u_1^2 - v_1^2) = \tfrac{1}{2}m_2(v_2^2 - u_2^2).$$

Combine the two equations above,

$$u_1 + v_1 = v_2 + u_2 \Rightarrow u_1 - u_2 = v_2 - v_1.$$

In words, this says that in an elastic collision the relative speed of the incoming particles equals the relative speed of the outgoing particles.

3. One can solve for v_1 and v_2 (please do it!) easily and find that:

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \left(\frac{2m_2}{m_1 + m_2}\right)u_2$$

$$v_2 = \left(\frac{2m_1}{m_1 + m_2}\right)u_1 + \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2$$

Notice that if $m_1 = m_2$, then $v_1 = u_2$ and $v_2 = u_1$. So this says that after the collision, the bodies will just reverse their velocities and move on as before.

4. What if one of the bodies is much heavier than the other body, and the heavier body is at rest? In this case, $m_2 >> m_1$ and $u_2 = 0$. e can immediately see that $v_1 = -u_1$ and $v_2 = 0$. This makes a lot of sense: the heavy body continues to stay at rest and the light body just bounces back with the same speed. In the lecture, you saw a demonstration of this!

5. And what if the lighter body (rickshaw) is at rest and is hit by the heavier body (truck)? In this case, $m_2 << m_1$ and $u_2 = 0$. From the above equation we see that $v_1 = u_1$ and $v_2 = 2u_1$. So the truck's speed is unaffected, but the poor rickshaw is thrust in the direction of the truck at twice the truck's speed!

6. Sometimes we wish to slow down particles by making them collide with other particles.

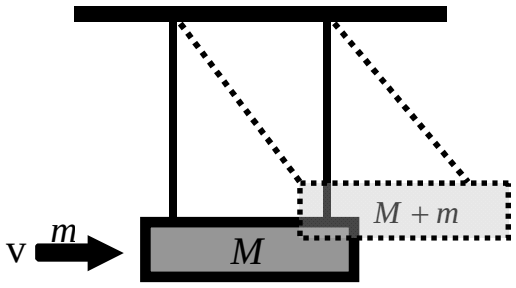
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In a nuclear reactor, neutrons can be slowed down in this way. let's calculate the fraction by which the kinetic energy of a neutron of mass m_1 decreases in a head-on collision with an atomic nucleus of mass m_2 that is initially at rest:

Solution: $\frac{K_i - K_f}{K_i} = 1 - \frac{K_f}{K_i} = 1 - \frac{v_f^2}{v_i^2}$

For a target at rest: $v_f = (\frac{m_1 - m_2}{m_1 + m_2})v_i$ $\quad\quad$ $\therefore \frac{K_i - K_f}{K_i} = \frac{4m_1m_2}{(m_1 + m_2)^2}.$

7. A bullet with mass m, is fired into a block of wood with mass M, suspended like a pendulum and makes a completely inelastic collision with it. After the impact, the block swings up to a maximum height y. What is the initial speed of the bullet?



Solution:

By conservation of momentum in the direction of the bullet,

$mv = (m + M)V, \Rightarrow v = \frac{(m + M)}{m}V$

The block goes up by distance y, and so gains potential energy. Now we can use the conservation of energy to give, $\frac{1}{2}(m + M)V^2 = (m + M)gy,$ where V is the velocity acquired by the block+bullet in the upward direction just after the bullet strikes. Now use $V = \sqrt{2gy}$. So finally, the speed of the bullet is: $v = \frac{(m + M)}{m}\sqrt{2gy}.$

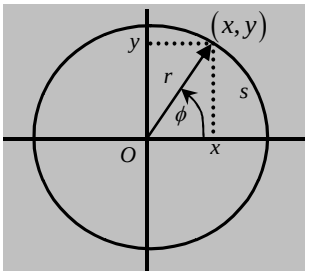
8. In 2 or 3 dimensions, you must apply conservation of momentum in each direction separately. The equation $\vec{P}_i = \vec{P}_f$ looks as if it is one equation, but it is actually 3 separate equations: $p_{ix} = p_{fx}, p_{iy} = p_{fy}, p_{iz} = p_{fz}$. On the other hand, suppose you had an elastic collision. In that case you would have only one extra equation coming from energy conservation, not three.

9. What happens to energy in an inelastic collision? Let's say that one body smashes into another body and breaks it into 20 pieces. To create 20 pieces requires doing work against the intermolecular forces, and the initial kinetic energy is used up for this.

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Summary of Lecture 11 – ROTATIONAL KINEMATICS

1. Any rotation is specified by giving two pieces of information:
   a) The point about which the rotation occurs, i.e. the origin.
   b) The angle of rotation is denoted by ϕ in the diagram.



2. The arc length = radius × angular displacement, or $s = r\phi$. Here ϕ is measured in radians. The maximum value of ϕ is 2π radians, which corresponds to 360 degrees or one full revolution. From this it follows that 1 radian = 57.3^0 or 1 radian = 0.159 revolution. Obviously, if $\phi = 2\pi$, then $s = 2\pi r$, which is the total circumference.

3. Suppose that there is a particle located at the tip of the radius vector. Now we wish to describe the rotational *kinematics* of this particle, i.e. describe its motion as goes around the circle. So, suppose that the particle moves from angle ϕ_1 to ϕ_2 in time $t_2 - t_1$. Then, the *average angular speed* $\bar{\omega}$ is defined as, $\bar{\omega} = \dfrac{\phi_2 - \phi_1}{t_2 - t_1} = \dfrac{\Delta\phi}{\Delta t}$. Suppose that we look at $\bar{\omega}$ over a very short time. Then, $\omega = \lim\limits_{\Delta t \to 0} \dfrac{\Delta\phi}{\Delta t} = \dfrac{d\phi}{dt}$ is called the instantaneous angular speed.

4. To familarize ourselves with the notion of angular speed, let us compute ω for a clock second, minute and hour hands:

$$\omega_{\text{second}} = \frac{2\pi}{60} = 0.105\ rad/s,$$

$$\omega_{\text{minute}} = \frac{2\pi}{60 \times 60} = 1.75 \times 10^{-3}\ rad/s,$$

$$\omega_{\text{hour}} = \frac{2\pi}{60 \times 60 \times 12} = 1.45 \times 10^{-4}\ rad/s.$$

5. Just as we defined accelaration for linear motion, we also define acceleration for circular motion:

$$\bar{\alpha} \equiv \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\Delta\omega}{\Delta t} \qquad \text{(average angular speed)}$$

Hence, $\alpha = \lim\limits_{\Delta t \to 0} \dfrac{\Delta\omega}{\Delta t}$ becomes $\alpha = \dfrac{d\omega}{dt} = \dfrac{d}{dt}\dfrac{d\phi}{dt} = \dfrac{d^2\phi}{dt^2}$ (angular acceleration). Let us see what this means for the speed with which a particle goes around. Now use $s = r\phi$. Differentiate with respect to time t: $\dfrac{ds}{dt} = r\dfrac{d\phi}{dt}$. The rate of change of arc length s is clearly

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what we should call the circular speed, v. So $v = r\omega$. Since r is held fixed, it follows that $\frac{dv}{dt} = r\frac{d\omega}{dt}$. Now define $a_T = \frac{dv}{dt}$. Obviously, $a_T = r\alpha$. Here T stands for tangential, i.e. in the direction of increasing s.

6. Compare the formulae for constant linear and angular accelerations:

LINEAR	ANGULAR
$v = v_0 + at$	$\omega = \omega_0 + \alpha t$
$x = x_0 + v_0 t + \frac{1}{2}at^2$	$\phi = \phi_0 + \omega_0 t + \frac{1}{2}\alpha t^2$
$v^2 = v_0^2 + 2a(x - x_0)$	$\omega^2 = \omega_0^2 + 2\alpha(\phi - \phi_0)$

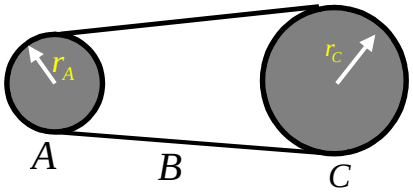
Why are they almost identical even though they describe two totally different physical situations. Answer: because the mathematics is identical!

7. The angular speed of a car engine is increased from 1170 rev/min to 2880 rev/min in 12.6 s. a)Find the average angular acceleration in rev/min^2. (b) How many revolutions does the engine make during this time?

SOLUTION: this is a straightforward application of the formulae in point 5 above.

$$\alpha = \frac{\omega_f - \omega_i}{t} = 8140\,\text{rev/min}^2, \qquad \phi = \omega_i t + \frac{1}{2}\alpha t^2 = 425\,\text{rev}.$$

8. Wheel A of radius r_A = 10.0 cm is coupled by a chain B to wheel C of radius r_C = 25.0 cm. Wheel A increases its angular speed from rest at a uniform rate of 1.60 rad/s2. Determine the time for wheel C to reach a rotational speed of 100 rev/min.

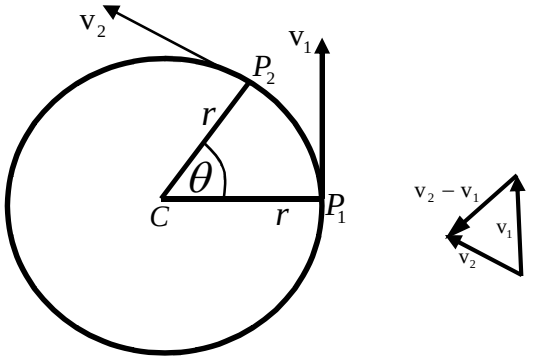


SOLUTION: Obviously every part of the chain moves with the same speed and so $v_A = v_C$. Hence $r_A\omega_A = r_C\omega_C \Rightarrow \omega_A = \frac{r_C\omega_C}{r_A}$. From the definition of acceleration, $\alpha = \frac{\omega_A - 0}{t}$. From this, $t = \frac{\omega_A}{\alpha} = \frac{r_C\omega_C}{r_A\alpha} = 16.4\ s.$

9. Imagine a disc going around. All particles on the disc will have same 'ω' and 'α' but different 'v' and 'a'. Clearly 'ω' and 'α' are simpler choices !!

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10. Now consider a particle going around a circle at constant speed. You might think that constant speed means no acceleration. Bu this is wrong! It is changing its direction and accelerating. This is called "centripetal acceleration", meaning acceleration directed towards the centre of the circle. Look at the figure below:



Note that the distance between points P_1 and P_2 is $\Delta r = \mathrm{v}\Delta t \approx r\theta$. Similarly, $\Delta \mathrm{v} \approx \mathrm{v}\theta \Rightarrow \bar{a} = \frac{\Delta \mathrm{v}}{\Delta t} \approx \frac{\mathrm{v}\theta}{r\theta/\mathrm{v}} = \frac{\mathrm{v}^2}{r}$. More generally, $a = \lim_{\Delta t \to 0} \frac{\Delta \mathrm{v}}{\Delta t} = \frac{\mathrm{v}^2}{r}$. In vector form, $\vec{\mathrm{a}}_R = -\frac{\mathrm{v}^2}{r}\hat{r}$. The negative sign indicates that the acceleration is towards the centre.

12. Vector Cross Products: The vector crossproduct of two vectors is defined as: $\vec{A} \times \vec{B} = AB\sin\theta\, \hat{n}$ where $\hat{n}$ is a unit vector that is perpendicular to both $\vec{A}$ and $\vec{B}$. Apply this definition to unit vectors in 3-dimensions: $\hat{i} \times \hat{j} = \hat{k},\ \hat{k} \times \hat{i} = \hat{j},\ \hat{j} \times \hat{k} = \hat{i}$.

13. Some key properties of the crossproduct:

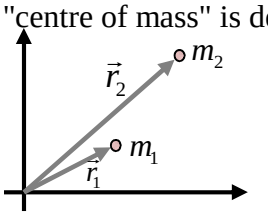
- $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$
- $\vec{A} \times \vec{A} = 0$
- $\left(\vec{A} + \vec{B}\right) \times \vec{C} = \left(\vec{A} \times \vec{B}\right) + \left(\vec{A} \times \vec{C}\right)$
- $\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$

14. The cross product is only definable in 3 dimensions and has no meaning in 2-d. This is unlike the dot product which as a meaning in any number of dimensions.

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Summary of Lecture 12 – PHYSICS OF MANY PARTICLES

1. A body is made of a collection of particles. We would like to think of this body having a "centre". For two masses the "centre of mass" is defined as: $\vec{r}_{cm} \equiv \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$.



In 2 dimensions (i.e. a plane) this is actually two equations:

$x_{cm} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$ and $y_{cm} = \frac{m_1y_1 + m_2y_2}{m_1 + m_2}$. These give the coordinates of the centre of mass of the two-particle system.

2. Example: one mass is placed at $x = 2cm$ and a second mass, equal to the first, is placed at $x = 6cm$. The cm position lies halfway between the two as you can see from:

$$x_{cm} = \frac{mx_1 + mx_2}{m + m} = \frac{2m + 6m}{2m} = 4cm.$$

Note that there is no physical body that is actually located at $x_{cm} = 4cm$! So the centre of mass can actually be a point where there is no matter. Now suppose that the first mass is three times bigger than the first:

$$x_{cm} = \frac{(3m)x_1 + mx_2}{3m + m} = \frac{2(3m) + 6m}{4m} = 3cm$$

This shows that the cm lies closer to the heavier body. This is always true.

3. For N masses the obvious generalization of the centre of mass position is the following:

$$\vec{r}_{cm} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + \cdots + m_N\vec{r}_N}{m_1 + m_2 + \cdots + m_N} = \frac{1}{M}\left(\sum_{n=1}^{n=N} m_n\vec{r}_n\right).$$

In words, this says that the following: choose any origin and draw vectors $\vec{r}_1, \vec{r}_2, \cdots \vec{r}_N$ that connect to the masses $m_1, m_2, \cdots m_N$. Heavier masses get more importance in the sum. So suppose that m_2 is much larger than any of the others. If so, $\vec{r}_{cm} \approx \frac{m_2\vec{r}_2}{m_2} = \vec{r}_2$. Hence, the cm is very close to the position vector of m_2.

4. For symmetrical objects, it is easy to see where the cm position lies: for a sphere or circle it lies at the centre; for a cylinder it is on the axis halfway between the two faces, etc.

5. Our definition of the cm allows Newton's Second Law to be written for entire collection

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of particles:

$$\vec{v}_{cm} = \frac{d\vec{r}_{cm}}{dt} = \frac{1}{M}\left(\sum m_n \vec{v}_n\right)$$

$$\vec{a}_{cm} = \frac{d\vec{v}_{cm}}{dt} = \frac{1}{M}\left(\sum m_n \vec{a}_n\right)$$

$$\therefore \quad M\,\vec{a}_{cm} = \sum \vec{F}_n = \sum\left(\vec{F}_{ext} + \vec{F}_{int}\right) \qquad \text{use } \sum \vec{F}_{int} = 0$$

$$\Rightarrow \quad \sum \vec{F}_{ext} = M\,\vec{a}_{cm} \text{ (the sum of external forces is what causes acceleration)}$$

In the above we have used Newton's Third Law as well: $\vec{F}_{12} + \vec{F}_{21} = 0$ etc.

6. Consider rotational motion now for a rigid system of N particles. Rigid means that all particles have a fixed distance from the origin. The kinetic energy is,

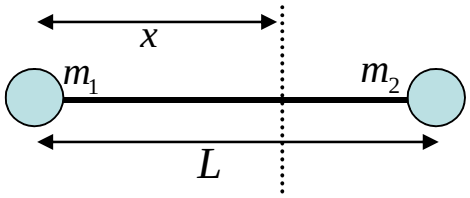
$$K = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 + \frac{1}{2}m_3 v_3^2 + \cdots$$

$$= \frac{1}{2}m_1 r_1^2\omega^2 + \frac{1}{2}m_2 r_2^2\omega^2 + \frac{1}{2}m_3 r_3^2\omega^2 + \cdots$$

$$= \frac{1}{2}\left(\sum m_i r_i^2\right)\omega^2$$

Now suppose that we define the "moment of inertia" $I \equiv \sum m_i r_i^2$. Then clearly the kinetic energy is $K = \frac{1}{2} I\omega^2$. How similar this is to $K = \frac{1}{2}Mv^2$!

7. To familiarize ourselves with I, let us consider the following: Two particles m_1 and m_2 are connected by a light rigid rod of length L. Neglecting the mass of the rod, find the rotational inertia I of this system about an axis perpendicular to the rod and at a distance x from m_1.



Answer: $I = m_1 x^2 + m_2\left(L - x\right)^2$. Of course, this was quite trivial. Now we can ask a more interesting question: For what x is I the largest? Now, near a maximum, the slope of a function is zero. So calculate $\frac{dI}{dx}$ and then put it equal to zero:

$$\therefore \quad \frac{dI}{dx} = 2m_1 x - 2m_2\left(L - x\right) = 0 \quad \Rightarrow \quad x_{\max} = \frac{m_2 L}{m_1 + m_2}.$$

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8. Although matter is made up of discrete atoms, even if one takes small pieces of any body, there are billions of atoms within it. So it is useful to think of matter as being continuously distributed. Since a sum $\sum$ becomes an integral $\int$, it is obvious that the new definitions of I and $\vec{\mathrm{R}}_{\mathrm{cm}}$ becom:

$$I \equiv \int r^2 dm \quad \text{and} \quad \vec{\mathrm{R}}_{\mathrm{cm}} \equiv \frac{1}{M}\int \vec{r} dm.$$

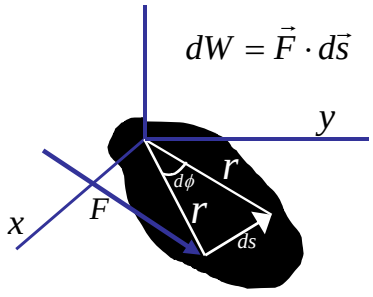
9. A simple application: suppose there is a hoop with mass distributed uniformly over it. The moment of inertia is: $I = \int r^2 dm = R^2 \int dm = MR^2$.

10. A less trivial application: instead of a hoop as above, now consider a solid plate:

$$I = \int r^2 dm \quad (dm = 2\pi r dr \rho_0)$$
$$= \int_0^R 2\pi r^3 dr \rho_0 = \frac{1}{2}(\pi R^2 \rho_0) R^2 = \frac{1}{2} M R^2$$

11. You have seen that it is easier to turn things (e.g. a nut, when changing a car's tyre after a puncture) when the applied force acts at a greater distance. This is because the *torque* τ is greater. We define $\vec{\tau} = \vec{r} \times \vec{F}$ from the magnitude is $\tau = rF\sin\theta$. Here θ is the angle between the radius vector and the force.

12. Remember that when a force $\vec{F}$ acts through a distance $d\vec{r}$ it does an amount of work equal to $\vec{F} \cdot d\vec{r}$. Now let us ask how much work is done when a torque acts through a certain angle as in the diagram below:



The small amount of work done is:

$$dW = \vec{F} \cdot d\vec{s} = F\cos\theta\, ds = \left(F\cos\theta\right)\left(rd\phi\right) = \tau d\phi$$

Add the contributions coming from from all particles,

$$dW_{net} = \left(F_1\cos\theta_1\right) r_1 d\phi + \left(F_2\cos\theta_2\right) r_2 d\phi + \cdots + \left(F_n\cos\theta_n\right) r_n d\phi$$

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$$= (\tau_1 + \tau_2 + \cdots + \tau_n)\, d\phi$$

$$\therefore dW_{net} = \left(\sum \tau_{ext}\right) d\varphi = \left(\sum \tau_{ext}\right) \omega dt \qquad .....(1)$$

Now consider the change in the kinetic energy K,

$$dK = d\left(\frac{1}{2} I\omega^2\right) = I\omega d\omega = (I\alpha)\omega dt \qquad .....(2)$$

By conservation of energy, the change in K must equal the work done, and so:

$$dW_{net} = dK \quad \Rightarrow \quad \sum \tau_{ext} = I\alpha$$

In words this says that the total torque equals the moment of inertia times the angular acceleration. This is just like Newton's second law, but for rotational motion !

13.A comparison between linear and rotational motion quantities and formulae:

LINEAR	ROTATIONAL
x, M	ϕ, I
$v = \frac{dx}{dt}$	$\omega = \frac{d\phi}{dt}$
$a = \frac{dv}{dt}$	$\alpha = \frac{d\omega}{dt}$
$F = Ma$	$\tau = I\alpha$
$K = \frac{1}{2} Mv^2$	$K = \frac{1}{2} I\omega^2$
$W = \int F dx$	$W = \int \tau d\phi$

14. Rotational and translational motion can occur simultaneously. For example a car's wheel rotates and translates. In this case the total kinetic energy is clearly the sum of the energies of the two motions: $K = \frac{1}{2} M\, v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$.

15. It will take a little work to prove the following fact that I simply stated above: for a system of N particles, the total kinetic energy divides up neatly into the kinetic energy of rotation and translation. Start with the expression for kinetic energy and write $\vec{v}_{cm} + \vec{v}'_i$ where $\vec{v}'_i$ is the velocity of a particle with respect to the cm frame,

$$K = \sum \frac{1}{2} m_i v_i^{\,2} = \sum \frac{1}{2} m_i \vec{v}_i \cdot \vec{v}_i$$

$$= \sum \frac{1}{2} m_i \left(\vec{v}_{cm} + \vec{v}'_i\right) \cdot \left(\vec{v}_{cm} + \vec{v}'_i\right)$$

$$= \sum \frac{1}{2} m_i \left(v_{cm}^2 + 2\vec{v}_{cm} . \vec{v}'_i + v'^2_i\right)$$

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Now, $\sum m_i \vec{v}_{cm} \cdot \vec{v}'_i = \vec{v}_{cm} \cdot \sum m_i \vec{v}'_i = \vec{v}_{cm} \cdot \sum \vec{p}'_i = 0$. Why? because the total momentum is xero in the cm frame! So this brings us to our result that,

$$K = \sum \frac{1}{2} m_i v_{cm}^2 + \sum \frac{1}{2} m_i v_i'^2$$
$$= \frac{1}{2} M v_{cm}^2 + \sum \frac{1}{2} m_i r_i'^2 \omega^2.$$

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Summary of Lecture 13 – ANGULAR MOMENTUM

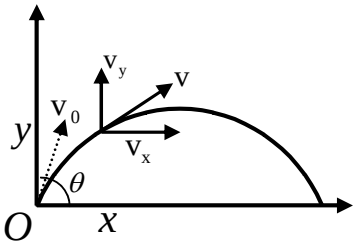
1. Recall the definition of angular momentum: $\vec{L} = \vec{r} \times \vec{p}$. The magnitude can be written in several different but equivalent ways,

   (a) $L = rp\sin\theta$

   (b) $L = (r\sin\theta)\, p = r_{\perp} p$

   (c) $L = r(p\sin\theta) = r\, p_{\perp}$

2. Let us use this definition to calculate the angular momentum of a projectile thrown from the ground at an angle θ. Obviously, initial angular momentum is zero (why?).



We know what the projectile's coordinates will be at time t after launch,

$$x = (\mathrm{v}_0 \cos\theta)t, \quad y = (\mathrm{v}_0 \sin\theta)t - \frac{1}{2}gt^2$$

as well as the velocity components,

$$\mathrm{v}_x = \mathrm{v}_0 \cos\theta,\ \mathrm{v}_y = \mathrm{v}_0 \sin\theta - gt.$$

Hence, $\vec{L} = \vec{r} \times \vec{p} = (x\hat{i} + y\,\hat{j}) \times (\mathrm{v}_x \hat{i} + \mathrm{v}_y\,\hat{j})m = m(x\,\mathrm{v}_y - y\,\mathrm{v}_x)\hat{k}$

$$= m\left(\frac{1}{2}gt^2 \mathrm{v}_0 \cos\theta - gt^2 \mathrm{v}_0 \cos\theta\right)\hat{k} = -\frac{m}{2}gt^2 \mathrm{v}_0 \cos\theta\, \hat{k}.$$

In the above, $\hat{k} = \hat{i} \times \hat{j}$ is a unit vector perpendicular to the paper. You can see here that the angular momentum increases as t^2.

2. Momentum changes because a force makes it change. What makes angular momentum change? Answer: torque. Here is the definition again: $\vec{\tau} = \vec{r} \times \vec{F}$. Now let us establish a very important relation between torque and rate of change of L.

Begin:

$\vec{L} = \vec{r} \times \vec{p}$. At a slightly later time, $\vec{L} + \Delta\vec{L} = (\vec{r} + \Delta\vec{r}) \times (\vec{p} + \Delta\vec{p})$

$$= \vec{r} \times \vec{p} + \vec{r} \times \Delta\vec{p} + \Delta\vec{r} \times \vec{p} + \Delta\vec{r} \times \Delta\vec{p}$$

By subtracting, $\Delta\vec{L} = \vec{r} \times \Delta\vec{p} + \Delta\vec{r} \times \vec{p}$ $\dfrac{\Delta\vec{L}}{\Delta t} = \dfrac{\vec{r} \times \Delta\vec{p} + \Delta\vec{r} \times \vec{p}}{\Delta t} = \vec{r} \times \dfrac{\Delta\vec{p}}{\Delta t} + \dfrac{\Delta\vec{r}}{\Delta t} \times \vec{p}.$

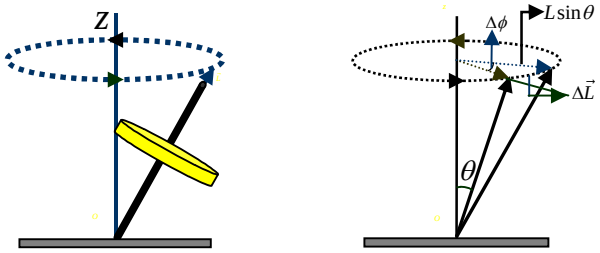
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Now divide by the time difference and then take limit as $\Delta t \to 0$:

$$\because \quad \lim_{\Delta t \to 0} \frac{\Delta \vec{L}}{\Delta t} = \frac{d\vec{L}}{dt} \quad \therefore \quad \frac{d\vec{L}}{dt} = \vec{r} \times \frac{d\vec{p}}{dt} + \frac{d\vec{r}}{dt} \times \vec{p}$$

But $\frac{d\vec{r}}{dt}$ is $\vec{\mathrm{v}}$ and $\vec{p} = m\vec{\mathrm{v}}$! Also, $\frac{d\vec{r}}{dt} \times \vec{p} = \vec{\mathrm{v}} \times m\vec{\mathrm{v}} = m\left(\vec{\mathrm{v}} \times \vec{\mathrm{v}}\right) = 0$. So we arrive at $\frac{d\vec{L}}{dt} = \vec{r} \times \frac{d\vec{p}}{dt}$. Now use Newton's Law, $\vec{F} = \frac{d\vec{p}}{dt}$. Hence we get the fundamental equation $\frac{d\vec{L}}{dt} = \vec{r} \times \vec{F}$, or $\frac{d\vec{L}}{dt} = \vec{\tau}$. So, just as a particle's momentum changes with time because of a force, a particle's *angular* momentum changes with time because of a torque.

3. As you saw in the lecture, the spinning top is an excelllent application of $\frac{d\vec{L}}{dt} = \vec{\tau}$.



Start from $\vec{\tau} = \vec{r} \times \vec{F}$ where $\vec{F} = m\vec{g}$ $\therefore$ $\tau = Mgr\sin\theta$. But $\vec{\tau}$ is perpendicular to $\vec{L}$ and so it cannot change the magnitude of $\vec{L}$. Only the direction changes. Since $\Delta\vec{L} = \vec{\tau}\ \Delta t$, you can see from the diagram that $\Delta\phi = \frac{\Delta L}{L\sin\theta} = \frac{\tau\ \Delta t}{L\sin\theta}$. So the precession speed ω_P is: $\omega_P = \frac{\Delta\phi}{\Delta t} = \frac{\tau}{L\sin\theta} = \frac{Mgr\sin\theta}{L\sin\theta} = \frac{Mgr}{L}$. As the top slows down due to friction and L decreases, the top precesses faster and faster.

4. Now consider the case of many particles. Choose any origin with particles moving with respect to it. We want to write down the total angular momentum,

$$\vec{L} = \vec{L}_1 + \vec{L}_2 + \cdots + \vec{L}_N = \sum_{n=1}^{N} \vec{L}_n$$

$$\frac{d\vec{L}}{dt} = \frac{d\vec{L}_1}{dt} + \frac{d\vec{L}_2}{dt} + \cdots + \frac{d\vec{L}_N}{dt} = \sum_{n=1}^{N} \frac{d\vec{L}_n}{dt}$$

Since $\frac{d\vec{L}_n}{dt} = \vec{\tau}_n$, it follows that $\frac{d\vec{L}}{dt} = \sum_{n=1}^{N} \vec{\tau}_n$. Thus the time rate of change of the total angular momentum of a system of particles equals the net torque acting on the system. I showed earlier that internal forces cancel. So also do internal torques, as we shall see.

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5. The torque on a system of particles can come both from external and internal forces. For example, there could be charged particles which attract/repel each other while they are all in an external gravitational field. Mathematically, $\sum \vec{\tau} = \sum \vec{\tau}_{\text{int}} + \sum \vec{\tau}_{ext}$. Now, if the forces between two particles not only are equal and opposite but are also directed along the line joining the two particles, then can easily show that the total internal torque, $\sum \vec{\tau}_{\text{int}} = 0$. Take the case of two particles,

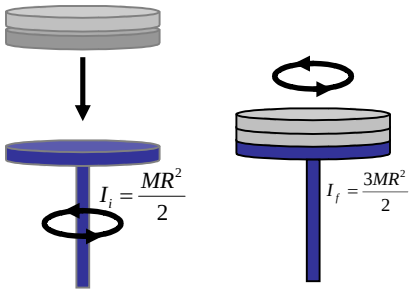
$$\sum \vec{\tau}_{\text{int}} = \vec{\tau}_1 + \vec{\tau}_2 = \vec{r}_1 \times \vec{F}_{12} + \vec{r}_2 \times \vec{F}_{21}$$

But $\vec{F}_{12} = -\vec{F}_{21} = F\hat{r}_{12}, \quad \therefore \sum \vec{\tau}_{\text{int}} = (\vec{r}_1 - \vec{r}_2) \times \vec{F}_{12} = \vec{r}_{12} \times (F\hat{r}_{12}) = F(\vec{r}_{12} \times \hat{r}_{12}) = 0$

Thus net external torque acting on a system of particles is equal to the time rate of change of the of the total angular momentum of the system.

6. It follows from $\frac{d\vec{L}}{dt} = \vec{\tau}$ that if no net external torque acts on the system, then the angular momentum of the system does not change with the time: $\frac{d\vec{L}}{dt} = 0 \quad \Rightarrow \quad \vec{L} =$ a constant.

This is simple but extremely important. Let us apply this to the system shown here. Two stationary discs, each with $I = \frac{1}{2}MR^2$, fall on top of a rotating disc. The total angular momentum is unchanged so, $I_i\omega_i = I_f\omega_f \;\Rightarrow\; \omega_f = \omega_i\left(\frac{I_i}{I_f}\right)$.



Hence, $\omega_f = \omega_i\left(\frac{MR^2}{2} \times \frac{2}{3MR^2}\right) = \frac{1}{3}\omega_i$.

7. You shoud be aware of the similarities and differences between the equations for linear and rotational motion: $\vec{F} = \frac{d\vec{p}}{dt} \Leftrightarrow \vec{\tau} = \frac{d\vec{L}}{dt}$. One big difference is that for momentum, $\vec{p} = m\vec{\text{v}}$, it does not matter where you pick the origin. But $\vec{L}$ definitely depends on the choice of the origin. So changing $\vec{r}$ to $\vec{c} + \vec{r}$ changes $\vec{L}$ to $\vec{L}'$:

$$\vec{L}' = \vec{r}' \times \vec{p} = (\vec{c} + \vec{r}) \times \vec{p} = \vec{c} \times \vec{p} + \vec{L}.$$

8. Linear and angular acceleration:

$$\vec{\text{a}} = \frac{d\vec{\text{v}}}{dt} = \frac{d}{dt}(\vec{\omega} \times \vec{r}) = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}$$
$$= \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{\text{v}}$$

So the acceleration has a tangential and radial part, $\vec{\text{a}} = \vec{\text{a}}_{\text{T}} + \vec{\text{a}}_{\text{R}}$.

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Summary of Lecture 14 – EQUILIBRIUM OF RIGID BODIES

1. A rigid body is one where all parts of the body are fixed relative to each other (for example, a pencil). Fluids and gases are non-rigid.

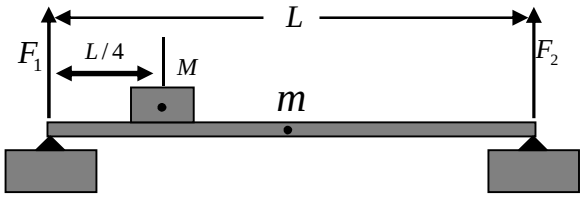
2. The translational motion of the centre of mass of a rigid body is governed by:

$$\frac{d\vec{P}}{dt} = \vec{F} \text{ where } \vec{F} = \sum \vec{F}_{ext} \text{ is the } \textit{net} \text{ external force.}$$

Similarly, for rotational motion, $\frac{d\vec{L}}{dt} = \vec{\tau}$ where $\vec{\tau} = \sum \vec{\tau}_{ext}$ is the net external torque.

3. A rigid body is in mechanical equilibrium if **both** the linear momentum $\vec{P}$ and angular momentum $\vec{L}$ have a constant value. i.e., $\frac{d\vec{P}}{dt} = 0$ **and** $\frac{d\vec{L}}{dt} = 0$. **Static equilibrium** refers to $\vec{P} = 0$ and $\vec{L} = 0$.

4. As an example of static equilibrium, consider a beam resting on supports:



We want to find the forces F_1 and F_2 with which the supports push on the rod in the upwards direction. First, balance forces in the vertical y direction:

$$\sum F_y = F_1 + F_2 - Mg - mg = 0$$

Now demand that the total torque vanishes:

$$\sum \tau_y = (F_1)(0) + (F_2)(L) - (Mg)(L/4) - (mg)(L/2) = 0$$

From these two conditions you can solve for F_1 and F_2,

$$F_1 = \frac{(3M + 2m)g}{4}, \quad \text{and} \quad F_2 = \frac{(M + 2m)g}{4}.$$

5. Angular momentum and torque depend on where you choose the origin of your coordinates. However, I shall now prove that for a body in equilibrium, the choice of origin does not matter. Let's start with the origin O and calculate the torque about O,

$$\vec{\tau}_O = \vec{\tau}_1 + \vec{\tau}_2 + \cdots + \vec{\tau}_N = \vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 + \cdots + \vec{r}_N \times \vec{F}_N$$

Now, if we take a second point P, then all distances will be measured from P and each

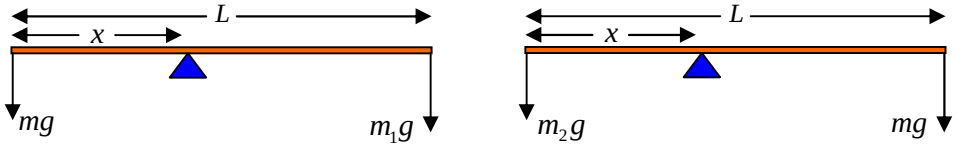
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vector will be shifted by an amount $\vec{r}_P$. Hence the torque about P is,

$$\begin{aligned}\vec{\tau}_P &= (\vec{r}_1 - \vec{r}_P)\times\vec{F}_1 + (\vec{r}_2 - \vec{r}_P)\times\vec{F}_2 + \cdots + (\vec{r}_N - \vec{r}_P)\times\vec{F}_N \\ &= \left[\vec{r}_1\times\vec{F}_1 + \vec{r}_2\times\vec{F}_2 + \cdots + \vec{r}_N\times\vec{F}_N\right] - \left[\vec{r}_P\times\vec{F}_1 + \vec{r}_P\times\vec{F}_2 + \cdots + \vec{r}_P\times\vec{F}_N\right] \\ &= \vec{\tau}_O - \left[\vec{r}_P\times\left(\vec{F}_1 + \vec{F}_2 + \cdots + \vec{F}_N\right)\right] \\ &= \vec{\tau}_O - \left[\vec{r}_P\times\left(\sum\vec{F}_{ext}\right)\right]\end{aligned}$$

but $\sum\vec{F}_{ext} = 0,$ for a body in translational equilibrium $\therefore \quad \vec{\tau}_P = \vec{\tau}_O.$

6. Let us use the equilibrium conditions to do something of definite practical importance. Consider the balance below which is in equilibrium when two known weights are hung as shown. We want to know m in terms of m_1 and m_2.



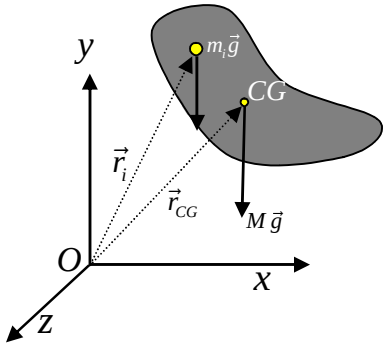
Taking the torques about the knife edge in the two cases, we have:

$$mgx = m_1 g(L - x) \text{ and } m_2 gx = mg(L - x)$$

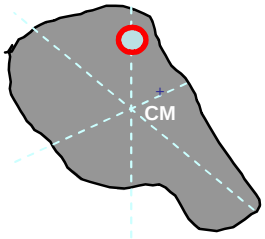
$$\Rightarrow \frac{m}{m_2} = \frac{m_1}{m} \text{ or } m = \sqrt{m_1 m_2}.$$

Remarkably, we do not need the values of x or L.

7. **Centre of Gravity.** The centre of gravity is the average location of the weight of an object. This is not quite the same as the centre of mass of a body (see lecture 12) but suppose the gravitational acceleration $\vec{g}$ has the same value at all points of a body. Then: 1) The weight is equal to $M\vec{g},$ and 2)the centre of gravity coincides with the centre of mass. Remember that weight is force, so the CG is really the centre of gravitational force acting on the body. The net force on the whole body = sum of forces over all individual particles, $\sum\vec{F} = \sum m_i\vec{g}$. If $\vec{g}$ has the same value at all points of the body, then $\sum\vec{F} = \vec{g}\sum m_i = M\vec{g}$. So the net torque about the origin O is $\sum\vec{\tau} = \sum(\vec{r}_i\times m_i\vec{g}) = \sum(m_i\vec{r}_i\times\vec{g})$ Hence, $\sum\vec{\tau} = M\vec{r}_{cm}\times\vec{g} = \vec{r}_{cm}\times M\vec{g}$. So the torque due to gravity about the centre of mass of a body (i.e. at $\vec{r}_{cm} = 0$) is zero !!



8. In the demonstration I showed, you saw how to find the CG of an irregular object by simply suspending it on a pivot,

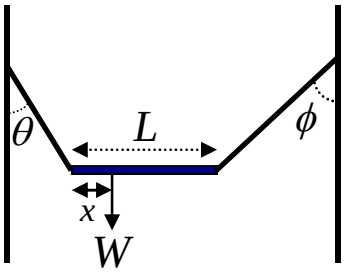


9. Let's solve a problem of static equilibrium: A non-uniform bar of weight W is suspended at rest in a horizontal position by two light cords. Find the distance *x* from the left-hand end to the center of gravity.

Solution : Call the tensions T_1 and T_2 . Put the forces in both directions equal to zero,

a) $T_2 \sin\phi - T_1 \sin\theta = 0$ (horizontal)

b) $T_2 \cos\phi + T_1 \cos\theta - W = 0$ (vertical) $\Rightarrow T_2 = \dfrac{W}{\sin(\theta+\phi)}$



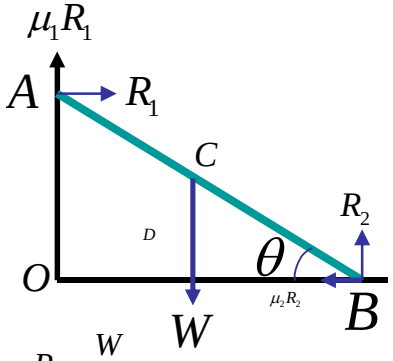
The torque about any point must vanish. Let us choose that point to be one end of the bar,

$$-Wx + (T_2 \cos\phi)L = 0 \Rightarrow x = \frac{(T_2 \cos\phi)L}{W} = \frac{L\cos\phi}{\sin(\theta+\phi)}$$

10.Here is another problem of the same kind: find the least angle θ at which the rod can lean to the horizontal without slipping.

Solution : Considering the translational equilibrium of the rod, $R_1 = \mu_2 R_2$ and $R_2 + \mu_1 R_1 = W$. This gives, $R_2 = \dfrac{W}{(1+\mu_1\mu_2)}$. Now consider rotational equilibrium about the point A: $R_2 \times OB = W \times OD + \mu_2 R_2 \times OA$

or, $R_2 \times AB\cos\theta = W \times \dfrac{AB\cos\theta}{2} + \mu_2 R_2 \times AB\sin\theta.$



This gives $\cos\theta\left(R_2 - \dfrac{W}{2}\right) = \mu_2 R_2 \sin\theta$ from which $\tan\theta = \dfrac{R_2 - \dfrac{W}{2}}{\mu_2 R_2}$ with $R_2 = \dfrac{W}{(1+\mu_1\mu_2)}$.

Using this value of R_2, we get $\tan\theta = \dfrac{1-\mu_1\mu_2}{2\mu_2}$.

10. **Types of Equilibrium.** In the lecture you heard about:

a)Stable equilibrium: object returns to its original position if displaced slightly.

b)Unstable equilibrium: object moves farther away from its original position if displaced slightly.

c)Neutral equilibrium: object stays in its new position if displaced slightly.

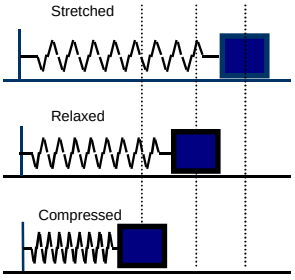
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Summary of Lecture 15 – OSCILLATIONS: I

1. An oscillation is any self-repeating motion. This motion is characterized by:
   a) The period T, which is the time for completing one full cycle.
   b) The frequency $f = 1/T$, which is the number of cycles per second. (Another frequently used symbol is ν).
   c) The amplitude A, which is the maximum displacement from equilibrium (or the size of the oscillation).

2. Why does a system oscillate? It does so because a force is always directed towards a central equilibrium position. In other words, the force always acts to return the object to its equilibrium position. So the object will oscillate around the equilibrium position. The restoring force depends on the displacement $F_{\text{restore}} = -k\,\Delta x,$ where Δx is the distance away from the equilibrium point, the negative sign shows that the force acts towards the equilibrium point, and k is a constant that gives the strength of the restoring force.

3. Let us understand these matters in the context of a spring tied to a mass that can move freely over a frictionless surface. The force $F(x) = -kx$ (or -$k\Delta x$ because the extension Δx will be called x for short). In the first diagram x is positive, x is negative in the second, and zero in the middle one. The energy stored in the spring, $U(x) = \frac{1}{2}kx^2,$ is positive in the first and third diagrams and zero in the middle one. Now we will use Newton's second law to derive a differential euation that describes the motion of the mass: From $F(x) = -kx$ and $ma = F$ it follows that $m\frac{d^2x}{dt^2} = -kx,$ or $\frac{d^2x}{dt^2} + \omega^2 x = 0$ where $\omega^2 \equiv \frac{k}{m}$. This is the equation of motion of a simple harmonic oscillator (SHO) and is seen widely in many different branches of physics. Although we have derived it for the case of a mass and spring, it occurs again and again. The only difference is that $\omega,$ which is called the oscillator frequency, is defined differently depending on the situation.



4. In order to solve the SHO equation, we shall first learn how to differentiate some elementary trignometric functions. So let us first learn how to calculate $\frac{\mathrm{d}}{\mathrm{dt}}\cos\omega t$ starting from the basic definition of a derivative:

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Start : $x(t) = \cos \omega t$ and $x(t + \Delta t) = \cos \omega (t + \Delta t)$. Take the difference:

$$x(t + \Delta t) - x(t) = \cos \omega (t + \Delta t) - \cos \omega t$$
$$= -\sin \omega \Delta t \sin(\omega t + \omega \Delta t / 2)$$
$$\approx -\omega \Delta t \sin \omega t \quad \text{as } \Delta t \text{ becomes very small.}$$
$$\therefore \frac{d}{dt} \cos \omega t = -\omega \sin \omega t.$$

(Here you should know that $\sin \theta \approx \theta$ for small θ, easily proved by drawing triangles.)

You should also derive and remember a second important result:

$$\frac{d}{dt} \sin \omega t = \omega \cos \omega t \ .$$

(Here you should know that $\cos \theta \approx 1$ for small θ.)

5. What happens if you differentiate twice?

$$\frac{d^2}{dt^2}(\sin \omega t) = \omega \frac{d}{dt} \cos \omega t = -\ \omega^2 \sin \omega t$$
$$\frac{d^2}{dt^2}(\cos \omega t) = -\omega \frac{d}{dt} \sin \omega t = -\ \omega^2 \cos \omega t.$$

So twice differentiating either $\sin \omega t$ or $\cos \omega t$ gives the same function back!

6. Having done all the work above, now you can easily see that any function of the form $x(t) = a \cos \omega t + b \sin \omega t$ satisfies $\frac{d^2 x}{dt^2} = -\omega^2 x$. But what do ω, a, b represent ?

a) The significance of ω becomes clear if your replace t by $t + \frac{2\pi}{\omega}$ in either $\sin \omega t$ or $\cos \omega t$. You can see that $\cos \omega \left(t + \frac{2\pi}{\omega} \right) = \cos(\omega t + 2\pi) = \cos \omega t$. That is, the function merely repeats itself after a time $2\pi / \omega$. So $2\pi / \omega$ is really the period of the motion T, $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$. The frequency ν of the oscillator is the number of complete vibrations per unit time: $\nu = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$ so $\omega = 2\pi\nu = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$.

Sometimes ω is also called the angular frequency. Note that $\dim[\omega] = T^{-1}$, from it is clear that the unit of ω is radian/second.

b) To understand what a and b mean let us note that from $x(t) = a \cos \omega t + b \sin \omega t$ it follows that $x(0) = a$ and that $\frac{\mathrm{d}}{\mathrm{dt}} x(t) = -\omega a \sin \omega t + \omega b \cos \omega t = \omega b$ (at $t = 0$). Thus, a is the initial position, and b is the initial velocity divided by ω.

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c) To understand what a and b mean let us note that from $x(t) = a\cos\omega t + b\sin\omega t$ it follows that $x(0) = a$ and that $\frac{\mathrm{d}}{\mathrm{dt}}x(t) = -\omega a\sin\omega t + \omega b\cos\omega t = \omega b$ (at $t = 0$). Thus, a is the initial position, and b is the initial velocity divided by ω.

d) The solution can also be written as: $x(t) = x_m\cos(\omega t + \phi)$. Since cos and sin never beome bigger than 1, or less than -1, it follows that $-x_m \le x \le +x_m$. For obvious reason x_m is called the amplitude of the motion. The frequency of the simple harmonic motion is independent of the amplitude of the motion.

e) The quantity $\theta = \omega t + \phi$ is called the phase of the motion. The constant ϕ is called the *phase constant.* A different value of ϕ just means that the origin of time has been chosen differently.

7. **Energy of simple harmonic motion**. Put $\phi = 0$ for convenience, and so imagine a mass whose position oscillates like $x = x_m\cos\omega t$. Let us first calculate the potential energy:

$$U = \frac{1}{2}kx^2 = \frac{1}{2}kx_m^2\cos^2\omega t.$$

Now calculate the kinetic energy:

$$K = \frac{1}{2}m\mathrm{v}^2 = \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 = \frac{1}{2}m\omega^2x_m^2\sin^2\omega t = \frac{1}{2}kx_m^2\sin^2\omega t$$

The sum of potential + kinetic is:

$$E = K + U = \frac{1}{2}kx_m^2\cos^2\omega t + \frac{1}{2}kx_m^2\sin^2\omega t$$
$$= \frac{1}{2}kx_m^2\left(\cos^2\omega t + \sin^2\omega t\right) = \frac{1}{2}kx_m^2.$$

Note that this is independent of time and energy goes from kinetic to potential, then back to kinetic etc.

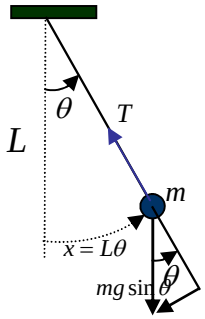
8. From the above, you can see that $\mathrm{v} = \frac{dx}{dt} = \pm\sqrt{\frac{k}{m}\left(x_m^2 - x^2\right)}$. From this it is clear that the speed is maximum at $x = 0$ and that the speed is zero at $x = \pm x_m$.

9. Putting two springs in parallel makes it harder to stretch them, and $k_{eff} = k_1 + k_2$. In series they are easier to stretch, and $k_{eff} = \left(\frac{k_1k_2}{k_1 + k_2}\right)$. So a mass will oscillate faster in the first case as compared to the second.

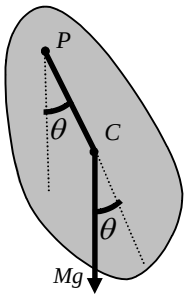
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Summary of Lecture 16 – OSCILLATIONS: II

1. In this chapter we shall continue with the concepts developed in the previous chapter that relate to simple harmonic motion and the simple harmonic oscillator (SHO). It is really very amazing that the SHO occurs again and again in physics, and in so many different branches.
2. As an example illustrating the above, consider a mass suspended a string. From the diagram, you can see that $F = -mg\sin\theta$. For small values of θ we know that $\sin\theta \approx \theta$. Using $x = L\theta$ (length of arc), we have $F = -mg\theta = -mg\dfrac{x}{L} = -\left(\dfrac{mg}{L}\right)x$. So now we have a restoring force that is proportional to the distance away from the equilibrium point. Hence we have a SHO with $\omega = \sqrt{g/L}$. What if we had not made the small θ approximation? We would still have an oscillator (i.e. the motion would be self repeating) but the solutions of the differential equation would be too complicated to discuss here.



3. If you take a common object (like a piece of cardboard) and pivot it at some point, it will oscillate when disturbed. But this is not the simple pendulum discussed above because all the mass is not concentrated at one point. So now let us use the ideas of torque and angular momentum discussed earlier for many particle systems. You can see that $\tau = -Mgd\sin\theta$. For small θ, $\sin\theta \approx \theta$ and so $\tau = -Mgd\theta$. But we also know that $\tau = I\alpha$ where I is the moment of inertia and α is the angular acceleration, $\alpha = \dfrac{d^2\theta}{dt^2}$. Hence, we have $I\dfrac{d^2\theta}{dt^2} = -Mgd\theta$, or, $\dfrac{d^2\theta}{dt^2} = -\left(\dfrac{Mgd}{I}\right)\theta$. From this we immediately see that the oscillation frequency is $\omega = \sqrt{\dfrac{Mgd}{I}}$. Of course, we have used the small angle approximation over here again. Since all variables except I are known, we can use this formula to tell us what I is about any point. Note that we can choose to put the pivot at any point on the body. However, if you put the pivot exactly at the centre of mass then it will not oscillate. Why? Because there is no restoring force and the torque vanishes at the cm position, as we saw earlier.



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4. Suppose you were to put the pivot at point P which is at a distance L from the centre of mass of the irregular object above. What should L be so that you get the same formula as for a simple pendulum?

   Answer: $T = 2\pi\sqrt{\frac{L}{g}} = 2\pi\sqrt{\frac{I}{Mgd}} \Rightarrow L = \frac{I}{Md}$

   P is then called the centre of gyration - when suspended from this point it appears as if all the mass is concentrated at the cm position.

5. Sum of two simple harmonic motions of the same period along the same line:

$$x_1 = A_1 \sin \omega t \text{ and } x_2 = A_2 \sin(\omega t + \phi)$$

   Let us look at the sum of x_1 and x_2,

$$\begin{aligned} x = x_1 + x_2 &= A_1 \sin \omega t + A_2 \sin(\omega t + \phi) \\ &= A_1 \sin \omega t + A_2 \sin \omega t \cos\phi + A_2 \sin\phi \cos \omega t \\ &= \sin \omega t (A_1 + A_2 \cos\phi) + \cos \omega t (A_2 \sin\phi) \end{aligned}$$

   Let $A_1 + A_2 \cos\phi = R\cos\theta$ and $A_2 \sin\phi = R\sin\theta$. Using some simple trigonometry, you can put x in the form, $x = R\sin(\omega t + \theta)$. It is easy to find R and θ:

   $R = \sqrt{A_1^2 + A_2^2 + A_1 A_2 \cos\phi}$ and $\tan\theta = \frac{A_2 \sin\phi}{A_1 + A_2 \cos\phi}$.

   Note that if $\phi = 0$ then $R = \sqrt{A_1^2 + A_2^2 + A_1 A_2} = \sqrt{(A_1 + A_2)^2} = A_1 + A_2$ and $\tan\theta = 0$ $\Rightarrow \theta = 0$. So we get $x = (A_1 + A_2)\sin \omega t$. This is an example of *constructive interference.* If $\phi = \pi$ then $R = \sqrt{A_1^2 + A_2^2 - A_1 A_2} = \sqrt{(A_1 - A_2)^2} = A_1 - A_2$ and $\tan\theta = 0$ $\Rightarrow \theta = 0$. Now we get $x = (A_1 - A_2)\sin \omega t$. This is *destructive interference.*

6. Composition of two simple harmonic motions of the same period but now at right angles to each other:

   Suppose $x = A\sin \omega t$ and $y = B\sin(\omega t + \phi)$. These are two independent motions. We can write $\sin \omega t = \frac{x}{A}$ and $\cos \omega t = \sqrt{1 - x^2/A^2}$.

   From this, $\frac{y}{B} = \sin \omega t \cos\phi + \sin\phi \cos \omega t = \frac{x}{A}\cos\phi + \sin\phi\sqrt{1 - x^2/A^2}$. Now square and rearrange terms to find:

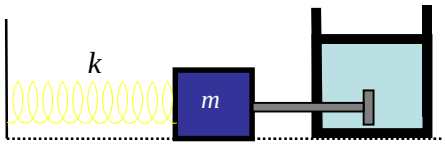
$$\frac{x^2}{A^2} + \frac{y^2}{B^2} - 2\frac{xy}{AB}\cos\phi = \sin^2\phi$$

   This is the equation for an ellipse (see questions at the end of this section).

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7. If two oscillations of different frequencies at right angles are combined, the resulting motion is more complicated. It is not even periodic unless the two frequencies are in the ratio of integers. This resulting curve are called Lissajous figures. Specifically, if $x = A\sin\omega_x t$ and $y = B\sin\left(\omega_y t + \phi\right),$ then periodic motion requires $\frac{\omega_x}{\omega_y} =$ integers. You should look up a book for more details.

8. **Damped harmonic motion**: Typically the frictional force due to air resistance, or in a liquid, is proportional to the speed. So suppose that the damping force $= -b\frac{dx}{dt}$ (why negative sign?). Now apply Newton's law to a SHO that is damped: $-kx - b\frac{dx}{dt} = m\frac{d^2x}{dt^2}$ Rearrange slightly to get the equation for a damped SHO: $m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0.$



Its solution for $\frac{k}{m} \geq \left(\frac{b}{2m}\right)^2$ is $x = x_m e^{-bt/2m}\cos\left(\omega' t + \phi\right)$. The frequency is now changed: $\omega' = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}$. The damping causes the amplitude to decrease with time and when $bt/2m = 1,$ the amplitude is $1/e \approx 1/2.7$ of its initial value.

9. **Forced oscillation and resonance.** There is a characteristic value of the driving frequency ω at which the amplitude of oscillation is a maximum. This condition is called resonance. For negligible damping resonance occurs at $\omega = \omega_0$. Here ω_0 is the natural frequency of the system and is given by $\omega_0 = \sqrt{\frac{k}{m}}$. The equation of motion is: $m\frac{d^2x}{dt^2} + kx = F_0\cos\omega t$. You should check that this is solved by putting $x = \frac{F_0}{m\left(\omega_0^2 - \omega^2\right)}\cos\omega t$ (just subsitute into the equation and see!). Note that the amplitude "blows up" when $\omega \to \omega_0$. This is because we have no damping term here. With damping, the amplitude is large when $\omega \to \omega_0$ but remains finite.

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Summary of Lecture 17 – PHYSICS OF MATERIALS

1. **Elasticity :** the property by virtue of which a body tends to regain its original shape and size when external forces are removed. If a body completely recovers its original shape and size , it is called perfectly elastic. Quartz, steel and glass are very nearly elastic.

2. **Plasticity :** if a body has no tendency to regain its original shape and size , it is called perfectly plastic. Common plastics, kneaded dough, solid honey, etc are plastics.

3. **Stress** characterizes the strength of the forces causing the stretch, squeeze, or twist. It is defined usually as force/unit area but may have different definitions to suit different situations. We distinguish between three types of stresses:
   a) If the deforming force is applied along some linear dimension of a body, the stress is called *longitudinal stress* or *tensile stress* or *compressive stress.*
   b) If the force acts normally and uniformly from all sides of a body, the stress is called *volume stress.*
   c) If the force is applied tangentially to one face of a rectangular body, keeping the other face fixed, the stress is called tangential or shearing stress.

4. Strain: When deforming forces are applied on a body, it undergoes a change in shape or size. The fractional (or relative) change in shape or size is called the strain.

$$\text{Strain} = \frac{\text{change in dimension}}{\text{original dimension}}$$

Strain is a ratio of similar quantities so it has no units. There are 3 different kinds of strain:

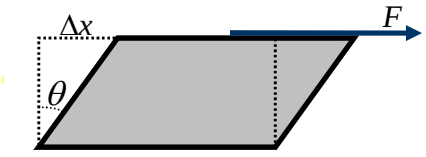
a) *Longitudinal (linear) strain* is the ratio of the change in length (ΔL) to original length (l), i.e., the linear strain $= \dfrac{\Delta l}{l}$.

b) *Volume strain* is the ratio of the change in volume (ΔV) to original volume (V)

Volume strain $= \dfrac{\Delta V}{V}$.

c) *Shearing strain :* The angular deformation (θ) in radians is called shearing stress.

For small θ the shearing strain $\equiv \theta \approx \tan\theta = \dfrac{\Delta x}{l}$.



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5. Hooke's Law: for small deformations, stress is proportional to strain.

$$\text{Stress} = \text{E} \times \text{Strain}$$

The constant E is called the modulus of elasticity. E has the same units as stress because strain is dimensionless. There are three moduli of elasticity.

(a) Young's modulus (Y) for linear strain:

$$\text{Y} \equiv \frac{\text{longitudinal stress}}{\text{longitudinal strain}} = \frac{F/A}{\Delta l/l}$$

(b) Bulk Modulus (B) for volume strain: Let a body of volume V be subjected to a uniform pressure ΔP on its entire surface and let ΔV be the corresponding decrease in its volume. Then,

$$\text{B} \equiv \frac{\text{Volume Stress}}{\text{Volume Strain}} = -\frac{\Delta P}{\Delta V/V}.$$

$1/B$ is called the compressibility. A material having a small value of B can be compressed easily.

(c) Shear Modulus (η) for shearing strain: Let a force F produce a strain θ as in the diagram in point 4 above. Then,

$$\eta \equiv \frac{\text{shearing stress}}{\text{shearing strain}} = \frac{F/A}{\theta} = \frac{F}{A\tan\theta} = \frac{Fl}{A\Delta x}.$$

6. When a wire is stretched, its length increases and radius decreases. The ratio of the lateral strain to the longitudinal strain is called Poisson's ratio, $\sigma = \frac{\Delta r/r}{\Delta l/l}$. Its value lies between 0 and 0.5.

7. We can calculate the work done in stretching a wire. Obviously, we must do work against a force. If x is the extension produced by the force F in a wire of length l, then $F = \frac{YA}{l}x$. The work done in extending the wire through Δl is given by,

$$W = \int_0^{\Delta l} F dx = \frac{YA}{l}\int_0^{\Delta l} x dx = \frac{YA}{l}\frac{(\Delta l)^2}{2}$$

$$= \frac{YA}{l}\frac{(\Delta l)^2}{2} = \frac{1}{2}(Al)\left(\frac{Y\Delta l}{l}\right)\left(\frac{\Delta l}{l}\right) = \frac{1}{2}\times \text{volume} \times \text{stress} \times \text{strain}$$

Hence, Work / unit volume $= \frac{1}{2}\times \text{stress} \times \text{strain}$. We can also write this as,

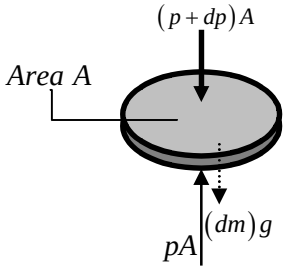
$$W = \frac{1}{2}\left(\frac{YA\Delta l}{l}\right)\Delta l = \frac{1}{2}\times \text{load} \times \text{extension}.$$

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8. A fluid is a substance that can flow and does not have a shape of its own. Thus all liquids and gases are fluids. Solids possess all the three moduli of elasticity whereas a fluid possess only the bulk modulus (B). A fluid at rest cannot sustain a tangential force. If such force is applied to a fluid, the different layers simply slide over one another. Therefore the forces acting on a fluid at rest have to be normal to the surface. This implies that the free surface of a liquid at rest, under gravity, in a container, is horizontal.

9. The normal force per unit area is called pressure, $P = \frac{\Delta F}{\Delta A}$. Pressure is a scalar quantity. Its unit is Newtons/metre^2, or Pascal (Pa). Another scalar is density, $\rho = \frac{\Delta m}{\Delta V}$, where Δm is the mass of a small piece of the material and ΔV is the volume it occupies.

10. Let us calculate how the pressure in a fluid changes with depth. So take a small element of fluid volume submerged within the body of the fluid: $dm = \rho dV = \rho A dy \quad \therefore \ (dm)g = \rho g A dy$



Now let us require that the sum of the forces on the fluid element is zero: $pA - (p+dp)A - \rho g A dy = 0 \quad \Rightarrow \frac{dp}{dy} = -\rho g.$

Note that we are taking the origin ($y = 0$) at the bottom of the liquid. Therefore as the elevation increases (dy positive), the pressure decreases (dp negative). The quantity ρg is the weight per unit volume of the fluid. For liquids, which are nearly incompressible, ρ is practically constant.

$\therefore \rho g = \text{constant} \Rightarrow \frac{dp}{dy} = \frac{\Delta p}{\Delta y} = \frac{p_2 - p_1}{y_2 - y_1} = -\rho g \Rightarrow \ p_2 - p_1 = -\rho g (y_2 - y_1).$

11. **Pascal's Principle**: Pressure applied to an enclosed fluid is transmitted to every portion of the fluid and to the walls of the containing vessel. This follows from the above: if h is the height below the liquid's surface, then $p = p_{ext} + \rho g h$. Here p_{ext} is the pressure at the surface of the liquid, and so the difference in pressure is $\Delta p = \Delta p_{ext} + \Delta(\rho g h)$. Therefore, $\Delta(\rho g h) = 0 \ \Rightarrow \Delta p = \Delta p_{ext}$. (I have used here the fact that liquids are incompressible). So the pressure is everywhere the same.

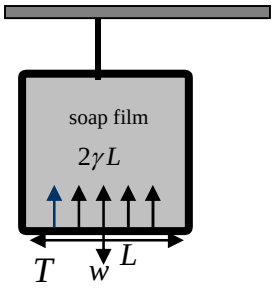
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Summary of Lecture 18 – PHYSICS OF FLUIDS

1. A fluid is matter that has no definite shape and adjusts to the container that it is placed in. Gases and liquids are both fluids. All fluids are made of molecules. Every molecules attracts other molecules around it.

2. Liquids exhibit surface tension. A liquid has the property that its free surface tends to contract to minimum possible area and is therefore in a state of tension. The molecules of the liquid exerts attractive forces on each other, which is called cohesive forces. Deep inside a liquid, a molecule is surrounded by other molecules in all directions. Therefore there is no net force on it. At the surface, a molecule is surrounded by only half as many molecules of the liquid, because there are no molecules above the surface.

3. The force of contraction is at right angles to an imaginary line of unit length, tangential to the surface of a liquid, is called its surface tension: $\gamma = \frac{F}{L}$. Here F is the force exerted by the "skin" of the liquid. The SI unit of the surface tension is N/m.

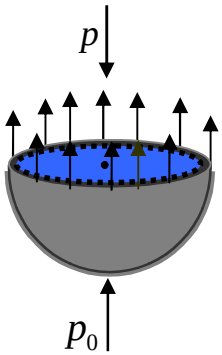


4. Quantitative measurement of surface tension: let w be the weight of the sliding wire, $T =$ force with which you pull the wire downward. Obviously, $T + w = F =$ net downward force. Since film has both front and back surfaces, the force F acts along a total length of $2L$. The surface tension in the film is defined as, $\gamma = \frac{F}{2L} \Rightarrow F = 2\gamma L.$ Hence, the surface tension is $\gamma = \frac{w+T}{2L}$.

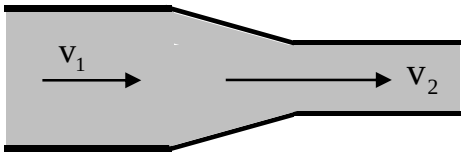
5. Let's ask how much work is done when we stretch the skin of a liquid. If we move the sliding wire through a displacement $\Delta x,$ the work done is $F\Delta x$. Now F is a conservative force, so there is potential energy $\Delta U = F\Delta x = \gamma L\Delta x$ where L is the length of the surface layer $L\Delta x = \Delta A =$ change in area of the surface. Thus $\gamma = \frac{\Delta U}{\Delta A}$. So we see that surface tension is the surface potential energy per unit area !

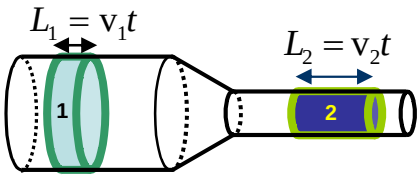
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7. The surface tension causes a pressure difference between the inside and outside of a soap bubble or a liquid drop. A soap bubble consists of two spherical surface films with a thin layer of liquid between them. Let p = pressure exerted by the upper half, and p_0 = external pressure. $\therefore$ force exerted due to surface tension is $2(2\pi r\gamma)$ (the "2" is for two surfaces). In equilibrium the net forces must be equal: $\pi r^2 p = 2(2\pi r\gamma) + \pi r^2 p_0$. So the excess pressure is $p - p_0 = \dfrac{4\gamma}{r}$. For a liquid drop, the difference is that there is only surface and so, excess pressure = $p - p_0 = \dfrac{2\gamma}{r}$.



8. From the fact that liquids are incompressible, equal volumes of liquid must flow in both sections in time t, i.e. $V_1/t = V_2/t \Rightarrow V_1 = V_2$. But you can see that $V_1 = A_1L_1 = A_1v_1t$ and similarly that $V_2 = A_2L_2 = A_2v_2t$. Hence $A_1v_1 = A_2v_2$. This means that liquid will flow faster when the tube is narrower, and slower where it is wider.





9. **Bernoulli's Equation :** When a fluid flows steadily, it obeys the equation:

$$p + \frac{1}{2}\rho v^2 + \rho g y = \text{ constant}$$

This famous equation, due to Daniel Bernoulli about 300 years ago, tells you how fast a fluid will flow when there is also a gravitational field acting upon. For a derivation, see any of the suggested references. In the following, I will only explain the meaning of the various terms in the formula and apply it to a couple of situations.

10. Let us apply this to water flowing in a pipe whose crossection decreases along its length. (A_1 is area of the wide part, etc).There is no change in the height so $y_1 = y_2$ and the gravitational potential cancels,

$$p_1 + \frac{1}{2}\rho v_1^2 + \cancel{\rho g y_1} = p_2 + \frac{1}{2}\rho v_2^2 + \cancel{\rho g y_2} \Rightarrow p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2.$$

Now, since the liquid is incompressible, it flows faster in the narrower part:

$$A_1v_1 = A_2v_2 \Rightarrow v_2 = \left(\frac{A_1}{A_2}\right)v_1 \Rightarrow p_2 = p_1 - \frac{1}{2}\rho\left(v_2^2 - v_1^2\right).$$

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This means that the pressure is smaller where the fluid is flowing faster! This is exactly why an aircraft flies: the wing shape is curved so that when the aircraft moves through the air, the air moves faster on the top part of the wing than on the lower part. Thus, the air pressure is lower on the top compared to the bottom and there is a net pressure upwards. This is called lift.

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Summary of Lecture 19 – PHYSICS OF SOUND

1. Sound waves correspond to longitudinal oscillations of density. So if sound waves are moving from left to right, as you look along this direction you will find the density of air greater in some places and less in others. Sound waves carry energy. The minimum energy that humans can hear is about 10^{-12} watts per cm^3 (This is called I_0, the threshold of hearing.)

2. To measure the intensity of sound, we use *decibels* as the unit. Decibels (db) are a relative measure to compare the intensity of different sounds with one another,

$$R \equiv \text{relative intensity of sound } I = \log_{10}\frac{I}{I_0} \text{ (decibels)}$$

Typically, on a street without traffic the sound level is about 30db, a pressure horn creates about 90db, and serious ear damage happens around 120db.

3. A sound wave moving in the x direction with speed v is described by

$$\rho(x,t) = \rho_m \sin\frac{2\pi}{\lambda}(x - \mathrm{v}t)$$

where $\rho(x,t)$ is the density of air at a point x at time t. Let us understand various aspects of this formula.

a) Suppose that as time t increases, we move in such a way as to keep $x - \mathrm{v}t$ constant. So if at $t = 0$ the value of x is 0.23 (say), then at $t = 1$ the value of x would be v+0.23, etc. In other words, to keep the density $\rho(x,t)$ constant, we would have to move with the speed of sound , i.e. v.

b) In the expression for $\rho(x,t)$, replace x by $x + \lambda$. What happens? Answer: nothing, because $\sin\frac{2\pi}{\lambda}(x + \lambda - \mathrm{v}t) = \sin\frac{2\pi}{\lambda}(x - \mathrm{v}t)$. This is why we call λ the "wavelength", meaning that length after which a wave repeats itself.

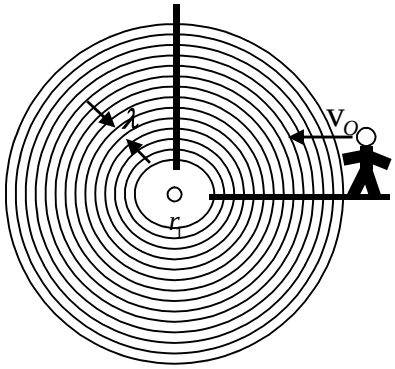
c) In the expression for $\rho(x,t)$, replace t by $t + T$ where $T = \lambda / \mathrm{v}$. What happens? Again the answer: nothing. T is called the time period of the sound wave, meaning that time after which it repeats itself. The frequency is the number of cycles per second and is obviously related to T through $\nu = 1/T$.

d) It is also common to introduce the *wavenumber* k the and *angular frequency* ω:

$$k = \frac{2\pi}{\lambda} \quad \text{and} \quad \omega = \frac{2\pi}{T} = 2\pi\nu$$

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4. **Doppler Effect**: The relative motion between source and observer causes the observer to receive a frequency that is different from that emitted by the source. One must distinguish between two cases:



Moving observer, source at rest. If the observer was at rest, the number of waves she would receive in time t would be t/T (or $\mathrm{v}t/\lambda$). But if she is moving towards the source with speed v_0 (as in the above figure), the additional number of waves received is obviously $\mathrm{v}_0 t/\lambda$. By definition,

$$\nu' = \text{frequency actually heard} = \frac{\text{number of waves received}}{\text{unit time}}$$

$$\therefore \nu' = \frac{\frac{\mathrm{v}t}{\lambda} + \frac{\mathrm{v}_0 t}{\lambda}}{t} = \frac{\mathrm{v}+\mathrm{v}_0}{\lambda} = \frac{\mathrm{v}+\mathrm{v}_0}{\mathrm{v}/\nu} = \nu\frac{\mathrm{v}+\mathrm{v}_0}{\mathrm{v}}$$

We finally conclude that the frequency actuall heard is $\nu' = \nu\left(1+\frac{\mathrm{v}_0}{\mathrm{v}}\right)$. So as the observer runs towards the source, she hears a higher frequency (higher pitch).

Moving source, observer at rest : As the source runs towards the observed, more waves will have to packed together. Each wavelength is reduced by $\frac{\mathrm{v}_s}{\nu}$. So the wavelength seen by the observer is $\lambda' = \frac{\mathrm{v}}{\nu} - \frac{\mathrm{v}_s}{\nu}$. From this, the frequency that she hears is $\nu' = \frac{\mathrm{v}}{\lambda} = \frac{\mathrm{v}}{(\mathrm{v}-\mathrm{v}_s)/\nu} = \nu\frac{\mathrm{v}}{(\mathrm{v}-\mathrm{v}_s)}$.

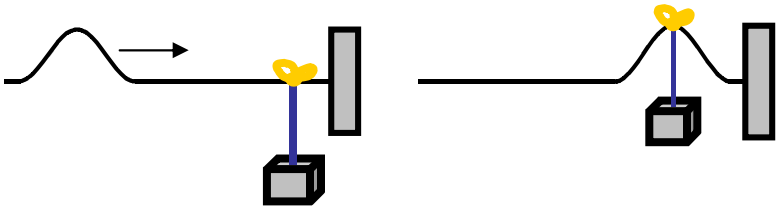
Moving source and moving observer : $\nu' = \nu\frac{\mathrm{v}+\mathrm{v}_o}{\mathrm{v}-\mathrm{v}_s}$. As you can easily see, the above two results are special cases of this.

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Summary of Lecture 20 – WAVE MOTION

1. Wave motion is any kind of self-repeating (periodic, or oscillatory) motion that transports energy from one point to another. Waves are of two basic kinds:
   (a) **Longitudinal Waves**: the oscillation is parallel to the direction of wave travel.
   Examples: sound, spring, "P-type" earthquake waves.
   (b) **Transverse Waves**: the oscillation is perpendicular to the direction of wave travel.
   Examples: radio or light waves, string, "S-type" earthquake waves.

2. Waves transport energy, not matter. Taking the vibration of a string as an example, each segment of the string stays in the same place, but the work done on the string at one end is transmitted to the other end. Work is done in lifting the mass at the other end below.

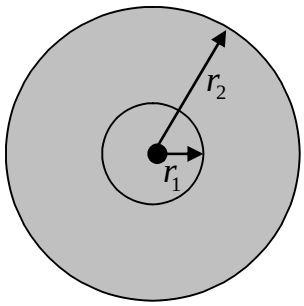


3. The height of a wave is called the amplitude. The average power (or intensity) in a wave is proportional to the square of the amplitude. So if $a(t) = a_0 \sin(\omega t - kx)$ is a wave of some kind, then a_0 is the amplitude and $I \propto a_0^2$.

4. A sound source placed at the origin will radiate sound waves in all directions equally. These are called spherical waves. For spherical waves the amplitude $\propto \frac{1}{r}$ and so the power $\propto \frac{1}{r^2}$.

We can easily see why this is so. Consider a source of sound and draw two spheres:
Let P_1 be the total radiated power and I_1 the intensity at r_1, etc. All the power (and energy) that crosses r_1 also crosses r_2 since none is lost in between the two. We have that,

$4\pi r_1^2 I_1 = P_1$ and $4\pi r_2^2 I_2 = P_2$. But $P_1 = P_2 = P$, and so $\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}$ or $I \propto \frac{1}{r^2}$.



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5. We have encountered waves of the kind $y(x,t) = y_0 \sin(kx - \omega t)$ in the previous lecture. Obviously $y(0,0) = 0$. But what if the wave is not zero at $x = 0$, $t = 0$? Then it could be represented by $y(x,t) = y_m \sin(kx - \omega t - \phi)$, where $kx - \omega t - \phi$ is called the phase and ϕ is called the phase constant. Note that you can rewrite $y(x,t)$ either as,

a) $y(x,t) = y_m \sin\left[k\left(x - \frac{\phi}{k}\right) - \omega t\right]$,

or as, b) $y(x,t) = y_m \sin\left[kx - \omega\left(t + \frac{\phi}{\omega}\right)\right]$.

The two different ways of writing the same expression can be interpreted differently. In (a) x has effectively been shifted to $x - \frac{\phi}{k}$ whereas in (b) t has been shifted to $t + \frac{\phi}{\omega}$. So the phase constant only moves the wave forward or backward in space or time.

6. When two sources are present the total amplitude at any point is the sum of the two separate amplitudes, $y(x,t) = y_1(x,t) + y_2(x,t)$. Now you remember that the power is proportional to the *square* of the amplitude, so $P \propto (y_1 + y_2)^2$. This is why *interference* happens. In the following we shall see why. Just to make things easier, suppose the two waves have equal amplitude. So lets take the two waves to be :

$$y_1(x,t) = y_m \sin(kx - \omega t - \phi_1) \quad \text{and} \quad y_2(x,t) = y_m \sin(kx - \omega t - \phi_2)$$

The total amplitudes is: $y(x,t) = y_1(x,t) + y_2(x,t)$

$$= y_m\left[\sin(kx - \omega t - \phi_1) + \sin(kx - \omega t - \phi_2)\right]$$

Now use the trigonometric formula, $\sin B + \sin C = 2\sin\frac{1}{2}(B + C) \times \cos(B - C)$ to get,

$$y(x,t) = y_m\left[\sin(kx - \omega t - \phi_1) + \sin(kx - \omega t - \phi_2)\right]$$

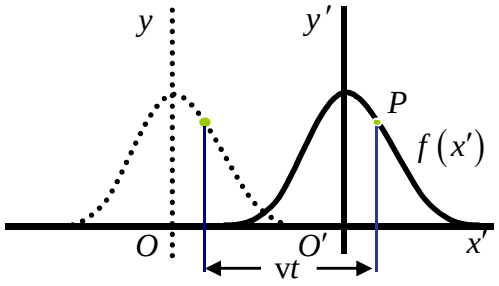
$$= \left[2y_m \cos\left(\frac{\Delta\phi}{2}\right)\right] \times \sin(kx - \omega t - \phi').$$

Here $\Delta\phi = \phi_2 - \phi_1$ is the difference of phases, and $\phi' = \frac{(\phi_1 + \phi_2)}{2}$ is the sum. So what do we learn from this? That if $\phi_2 = \phi_1$, then the two waves are in phase and the resultant amplitude is maximum (because $\cos 0 = 1$). And that if $\phi_2 = \phi + \pi$, then the two waves are out of phase and the resultant amplitude is minimum (because $\cos \pi/2 = 0$). The two waves have interfered with each other and have increased/decreased their amplitude in these two extreme cases. In general $\cos\left(\frac{\Delta\phi}{2}\right)$ will be some number that lies between -1 and +1.

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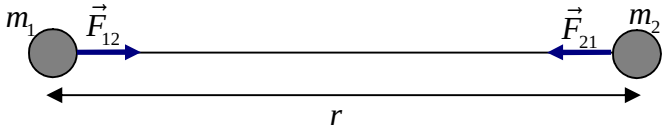
7. There was no time in the lecture to prove it, but you can look up any good book to find a proof for the formaula that the speed of sound in a medium is: $\mathrm{v} = \sqrt{\frac{B}{\rho}}$ where B is the bulk modulus and ρ is the density of the medium.

8. **The speed of a pulse.** A pulse is a burst of energy (sound, electromagnetic, heat,...) and could have any shape. Mathematically any pulse has the form $y(x,t) = f(x - \mathrm{v}t)$. Here f is any function (e.g. sin, cos, exp,....). Note that at time $t = 0$, $y(x,0) = f(x)$ and the shape would look as on the left in the diagram below. At a late time t, it will look just the same, but shifted to the right. In other words at time t, $y(x,t) = f(x')$ where $x' = x - \mathrm{v}t$. Fix your attention on any one point of the curve and follow it as the pulse moves to the right. From $x - \mathrm{v}t = \text{constant}$ it follows that $\frac{dx}{dt} - \mathrm{v} = 0$, or $\mathrm{v} = \frac{dx}{dt}$. This is called the *phase velocity* because we derived it using the constancy of phase.

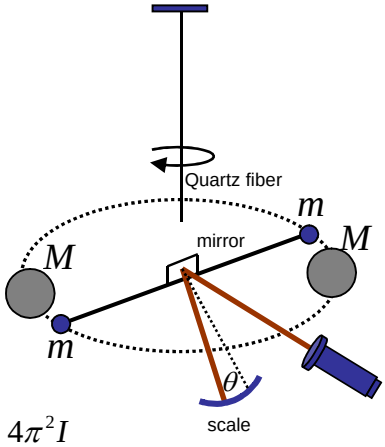


Summary of Lecture 21 – GRAVITY

1. Newton's law of universal gravitation states that the force of attraction between two masses m_1 and m_2 is $F \propto \frac{m_1 m_2}{r^2}$ and is directed along the line joining the two bodies. Putting in a constant of proportionality, $F = G\frac{m_1 m_2}{r^2}$. Now let's be a bit careful of the direction of the force. Looking at the diagram below, $\vec{F}_{21}$ = Force on m_2 by m_1, $\vec{F}_{12}$ = Force on m_1 by m_2 , $\left|\vec{F}_{12}\right| = \left|\vec{F}_{21}\right| = F$. By Newton's Third Law, $\vec{F}_{12} = -\vec{F}_{21}$.



2. The gravitational constant G is a very small quantity and needs a very sensitive experiment. An early experiment to find G involved suspending two masses and measuring the attractive force. From the figure you can see that the is gravitational torque is $2\left(\frac{GmM}{r^2}\right)\frac{L}{2}$. A thread provides the restoring torque $\kappa\theta$. The deflection θ can be measured by observing the beam of the light reflected from the small mirror. In equilibrium the torques balance, $\frac{GmML}{r^2} = \kappa\theta$. Hence $G = \frac{\kappa\theta r^2}{GmML}$. How to find κ? It can be found from observing the period of free oscillations, $T = 2\pi\sqrt{\frac{I}{\kappa}} \Rightarrow \kappa = \frac{4\pi^2 I}{T^2}$ with $I = \frac{mL^2}{2}$. The modern value is $G = 6.67259 \times 10^{-11} N.m^2/kg^2$



3. The magnitude of the force with which the Earth attracts a body of mass m towards its centre is $F = \frac{GmM_E}{R_E{}^2}$, where $R_E = 6400\,km$ is the radius of the Earth and M_E is the mass. The material does not matter - iron, wood, leather, etc. all feel the force in proportion to their masses. If the body can fall freely, then it will accelerate. So, $F = mg = \frac{GmM_E}{R_E{}^2}$.

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We measure g, the acceleration due to gravity, as 9.8m/s^2. From this we can immediately deduce the Earth's mass: $M_E = \frac{gR_E^{\,2}}{G} = 5.97\times10^{24}\,kg$. What a remarkable achievement!

We can do still more: the volume of the Earth $= V_{\rm E} = \frac{4}{3}\pi R_{\rm E}^{\,3} = 1.08\times10^{21}\,m^3$. Hence the] density of the Earth $= \rho_{\rm E} = \frac{V_{\rm E}}{M_{\rm E}} = 5462\,{\rm kg}\,m^{-3}$. So this is 5.462 times greater than the density of water and tells us that the earth must be quite dense inside.

4. The **gravitational potential** is an important quantity. It is the work done in moving a unit mass from infinity to a given point R, and equals $V(r) = -\frac{GM}{R}$.

   Proof: Conservation of energy says, $dV = -Fdr \Rightarrow \int_{V(R)}^{0} dV = -\int_{R}^{\infty} drF(r)$

   Integrate both sides: $0 - V(R) = GM\int_R^\infty \frac{dr}{r^2} = -GM\left[\frac{1}{r}\right]_R^\infty, \qquad \therefore V(R) = -\frac{GM}{R}$

5. Using the above formula, let us calculate the change in potential energy ΔU when we raise a body of mass *m* to a height *h* above the Earth's surface.

$$\Delta U = GMm\left(\frac{1}{R_E} - \frac{1}{R_E + h}\right) = GMm\left(1 - \frac{1}{1 + h/R_E}\right) = GMm\left(1 - \left(1 + h/R_E\right)^{-1}\right)$$

   Now suppose that the distance *h* is much smaller than the Earth's radius. So, for $h \ll R_E$, $\left(1 + h/R_E\right)^{-1} = 1 - h/R_E$. So we find $\Delta U = GMm\left(1 - \left(1 - h/R_E\right)\right) = m\left(\frac{GM}{R_E}\right)h = mgh.$

6. We can use the expression for potential energy and the law of conservation of energy to find the minimum velocity needed for a body to escape the Earth' gravity. Far away from the Earth, the potential energy is zero, and the smallest value for the kinetic energy is zero. Requiring that $\left(KE + PE\right)_{r=R} = \left(KE + PE\right)_{r=\infty}$ gives $\frac{1}{2}m{\rm v}_e^2 - \frac{GMm}{R_E} = 0 + 0$. From this, ${\rm v_e} = \sqrt{\frac{2GM}{R_E}} = \sqrt{2gR_E}$. Putting in some numbers we find that for the Earth ${\rm v_e} = 11.2\,{\rm km/s}$ and for the Sun ${\rm v_e} = 618\,{\rm km/s}$. For a Black Hole, the escape velocity is so high that nothing can escape, even if it could move with the speed of light! (Nevertheless, Black Holes can be observed because when matter falls into them, a certain kind of radiation is emitted.)

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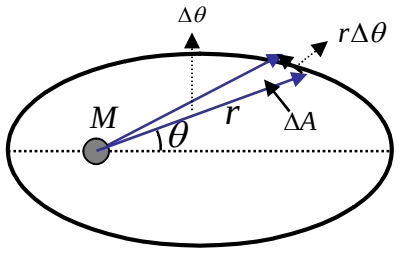
7. **Satellite problems**: A satellite is in circular orbit over the Earth's surface. The condition for equilibrium, $\frac{mv_o^2}{r} = \frac{GMm}{r^2} \Rightarrow v_o = \sqrt{\frac{GM}{r}}$. If R_E is the Earth's radius, and h is the height of the satellite above the ground, then $r = R_E + h$. Hence, $v_o = \sqrt{\frac{GM}{R_E + h}}$. For $h << R_E$, we can approximate $v_o = \sqrt{\frac{GM}{R_E}} = \sqrt{gR_E}$. We can easily calculate the time for one complete revolution, $T = \frac{2\pi}{\omega} = \frac{2\pi r}{v_o} = 2\pi r\sqrt{\frac{r}{GM}} = \frac{2\pi}{\sqrt{GM}} r^{3/2}$. This gives the important result, observed by Kepler nearly 3 centuries ago that $T^2 = \frac{4\pi^2}{GM} r^3$, or $T^2 \propto r^3$.

8. What is the total energy of a satellite moving in a circular orbit around the earth? Clearly, it has two parts, kinetic and potential. Remeber that the potential energy is negative. So, $E = KE + PE = \frac{1}{2}mv_o^2 - \frac{GM_E m}{r}$. But, $v_o^2 = \frac{GM}{r}$ as we saw earlier and therefore, $E = \frac{1}{2}\frac{GMm}{r} - \frac{GMm}{r} = -\frac{1}{2}\frac{GMm}{r}$. Note that the magnitude of the potential energy is larger than the kinetic energy. If it wasn't, the satellite would not be bound to the Earth!

9. A famous discovery of the astronomer Johann Kepler some 300 years ago says that the line joining a planet to the Sun sweeps out equal areas in equal intervals of time. We can easily see this from the conservation of angular momentum. Call ΔA the area swept out in time Δt. Then from the diagram below you can see that $\Delta A = \frac{1}{2} r\left(r\Delta\theta\right)$. Divide this by Δt and then take the limit where it becomes very small,

$$\frac{dA}{dt} = \lim_{\Delta t \to 0} \frac{\Delta A}{\Delta t} = \frac{1}{2} r^2 \left(\lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} \right) = \frac{1}{2} r^2 \omega = \frac{L}{2m}.$$

Since L is a constant, we have proved one of Kepler's laws (with so little effort)!



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Summary of Lecture 22 – ELECTROSTATICS I

1. Like charges repel, unlike charges attract. But by how much? Coulomb's Law says that this depends both upon the strength of the two charges and the distance between them. In mathematical terms, $F \propto \frac{q_1 q_2}{r^2}$ which can be converted into an equality, $F = k\frac{q_1 q_2}{r^2}$. The constant of proportionality will take different values depending upon the units we choose. In the MKS system, charge is measured in Coulombs (C) and $k = \frac{1}{4\pi\varepsilon_0}$ with $\varepsilon_0 = 8.85\times10^{-12} C^2/Nm^2$ and hence $k = 8.99\times10^9 Nm^2/C^2$.

2. The situation is quite similar to that of gravity, except that electric charges and not masses are the source of force. In vector form, $\vec{F}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}^2}\hat{r}_{12}$ is the force exerted by 2 on 1, where the unit vector is $\hat{r}_{12} = \frac{\vec{r}_{12}}{r_{12}}$. On the other hand, $\vec{F}_{21} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}^2}\hat{r}_{21}$ is the force exerted by 1 on 2. By Newton's Third Law, $\vec{F}_{12} = -\vec{F}_{21}$. For many charges, the force on charge 1 is given by, $\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \cdots$

3. Charge is quantized. This means that charge comes in certain units only. So the size of a charge can only be 0, $\pm e, \pm 2e, \pm 3e, \cdots$ where $e = 1.602\times10^{-19} C$ is the value of the charge present on a proton. By definition we call the charge on a proton positive. This makes the charge on an electron negative.

4. Charge is conserved. This means that charge is never created or destroyed. Equivalently, in any possible situation, the total charge at an earlier time is equal to the charge at a later time. For example, in any of the reactions below the initial charge = final charge:

$e^- + e^+ \rightarrow \gamma + \gamma$ (electron and positron annihilate into neutral photons)

$\pi^0 \rightarrow \gamma + \gamma$ (neutral pion annihilates into neutral photons)

$H^2 + H^2 \rightarrow H^3 + p$ (two deuterons turn into tritium and proton)

5. **Field :** this a quantity that has a definite value at any point in space and at any time. The simplest example is that of a scalar field, which is a *single number* for any value of x, y, z, t. Examples: temperature inside a room $T(x, y, z, t)$, density in a blowing wind $\rho(x, y, z, t)$, $\cdots$ There are also *vector fields,* which comprise of three numbers at each value of x, y, z, t. Examples: the velocity of wind, the pressure inside a fluid, or even a sugarcane field. In

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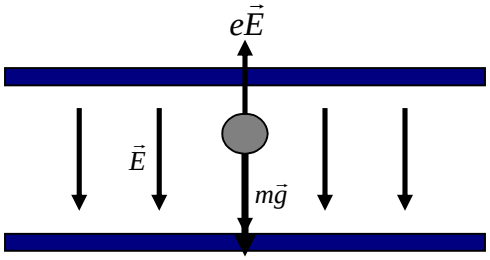
every case, there are 3 numbers: $\vec{V}(x,y,z,t) = \{V_1(x,y,z,t), V_2(x,y,z,t), V_3(x,y,z,t)\}$.

6. The electric field is also an example of a vector field, and will be the most important for our purpose. It is defined as the force on a unit charge. Or, since we don't want the charge to disturb the field it is placed in, we should properly define it as the force on a "test" charge q_0, $E \equiv \frac{F}{q_0}$. Here q_0 is very very small. The electric field due to a point charge can be calculated by considering two charges. The force between them is $F = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2}$ and so $E = \frac{F}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$. A way to visualize E fields is to think of lines starting on positive charges and ending on negative charges. The number of lines leaving/entering gives the amount of charge.

7. Typical values for the maginitude of the electric field E :

Inside an atom-	10^{11} N/C
Inside TV tube-	10^{5} N/C
In atmosphere-	10^{2} N/C
Inside a wire-	10^{-2} N/C

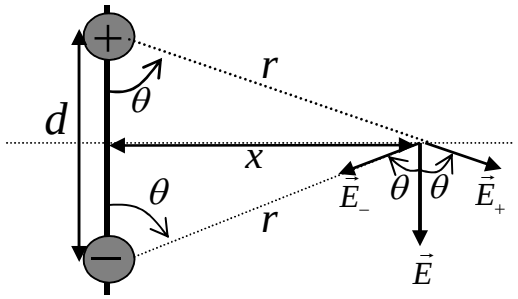
8. Measuring charge. One way to do this is to balance the gravitational force pulling a charged particle with mass m with the force exerted on it by a known electric field (see below). For equilibrium, the two forces must be equal and so $mg = qE$. The unknown charge q can then be found from $q = \frac{mg}{E}$.



9. Given several charges, one can find the total electric field at any point as the sum of the fields produced by the charges at that point individually, $\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \cdots\cdots$ or $\vec{E} = \sum_i \vec{E}_i = k\sum_i \frac{q_i}{r_i^2}\hat{r}_i \quad (i = 1,2,3,\cdots)$. Here $\hat{r}_i$ is the unit vector pointing from the charge to the point of observation.

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10. Let us apply the principle we have just learned to a system of two charges $+q$ and $-q$ which are separated by a distance d (see diagram). Then, $\vec{E} = \vec{E}_+ + \vec{E}_-$. Just to make things easier (not necessary; one can do it for any point) I have taken a point that lies on the x-axis. The magnitudes of the electric field due to the two charges are equal; $E_+ = E_- = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{q}{x^2 + (d/2)^2}$. The vertical components cancel out, and the net electric field is directed downwards with magnitude, $E = E_+ \cos\theta + E_- \cos\theta = 2E_+ \cos\theta.$ From the diagram below you can see that $\cos\theta = \frac{d/2}{\sqrt{x^2 + (d/2)^2}}$. Substituting this, we find: $E = 2\frac{1}{4\pi\varepsilon_0}\frac{q}{x^2 + (d/2)^2}\frac{d/2}{\sqrt{x^2 + (d/2)^2}} = \frac{1}{4\pi\varepsilon_0}\frac{qd}{\left[x^2 + (d/2)^2\right]^{3/2}}$.



11. The result above is so important that we need to discuss it further. In particular, what happens if we are very far away from the dipole, meaning $x >> d$? Let us first define the *dipole moment* as the product of the charge $\times$ the separation between them $p = qd$. Then, $E = \frac{1}{4\pi\varepsilon_0}\frac{p}{x^3}\frac{1}{\left[1 + (d/2x)^2\right]^{3/2}} = \frac{1}{4\pi\varepsilon_0}\frac{p}{x^3}\left[1 + \left(\frac{d}{2x}\right)^2\right]^{-3/2} = \frac{1}{4\pi\varepsilon_0}\frac{p}{x^3}$. In the above, $d/2x$ has been neglected in comparison to 1. So finally, we have found that for $x >> d$, $E = \frac{1}{4\pi\varepsilon_0}\frac{p}{x^3}$.

12. It is easy to find the torque experienced by an electric dipole that is placed in a uniform electric field: The magnitude is $\tau = F\frac{d}{2}\sin\theta + F\frac{d}{2}\sin\theta = Fd\sin\theta,$ and the direction is perpendicular and into the plane. Here θ is the angle between the dipole and the electric field. So $\tau = (qE)d\sin\theta = pE\sin\theta.$

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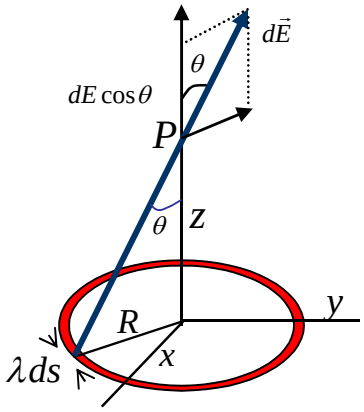
Summary of Lecture 23 – ELECTROSTATICS II

1. In the last lecture we learned how to calculate the electric field if there are any number of point charges. But how to calculate this when charges are continuously distributed over some region of space? For this, we need to break up the region into little pieces so that each piece is small enough to be like a point charge. So, $\vec{E} = \Delta\vec{E}_1 + \Delta\vec{E}_2 + \Delta\vec{E}_3 + \cdots$, or $\vec{E} = \sum \Delta\vec{E}_i$ is the total electric field. Remember that $\vec{E}$ is a vector that can be resolved into components, $\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$. In the limit where the pieces are small enough, we can write it as an integral, $\vec{E} = \int d\vec{E}$ (or $E_x = \int dE_x$, $E_y = \int dE_y$, $E_z = \int dE_z$)

2. Charge Density: when the charges are continuously distributed over a region - a line, the surface of a material, or inside a sphere - we must specify the *charge density*. Depending upon how many dimensions the region has, we define:
   (a) For linear charge distribution: $dq = \lambda ds$
   (b) For surface charge distribution: $dq = \sigma dA$
   (c) For volume charge distribution: $dq = \rho dV$

   The dimensions of λ, σ, ρ are determined from the above definitions.

3. As an example of how we work out the electric field coming from a continuous charge distribution, let us work out the electric field from a uniform ring of charge at the point P.



The small amount of charge λds gives rise to an electric field whose magnitude is

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{\lambda ds}{r^2} = \frac{\lambda ds}{4\pi\varepsilon_0\left(z^2 + R^2\right)}$$

The component in the z direction is $dE_z = dE\cos\theta$ with $\cos\theta = \frac{z}{r} = \frac{z}{\left(z^2 + R^2\right)^{1/2}}$.

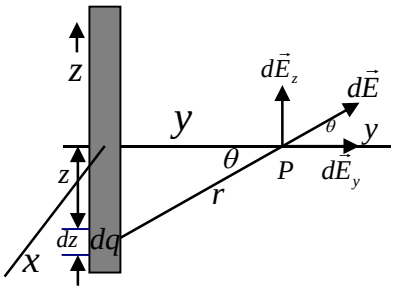
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So $dE_z = \dfrac{z\lambda ds}{4\pi\varepsilon_0\left(z^2+R^2\right)^{3/2}}$. Since *s*, which is the arc length, does not depend upon *z* or *R*,

$E_z = \dfrac{z\lambda}{4\pi\varepsilon_0\left(z^2+R^2\right)^{3/2}}\int ds = \dfrac{z\lambda\left(2\pi R\right)}{4\pi\varepsilon_0\left(z^2+R^2\right)^{3/2}} = \dfrac{qz}{4\pi\varepsilon_0\left(z^2+R^2\right)^{3/2}}$. Answer!!

Note that if you are very far away, the ring looks like a point: $E_z = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{z^2}, \quad \left(z >> R\right).$

4. As another example, consider a continuous distribution of charges along a wire that lies along the *z*-axis, as shown below. We want to know the electric field at a distance *x* from the wire. By symmetry, the only non-cancelling component lies along the *y*-axis.



Applying Coulomb's law to the small amount of charge λdz along the *z* axis gives,

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda dz}{y^2+z^2}$$

the component along the *y* direction is $dE_y = dE\cos\theta$. Integrating this gives,

$$E_y = \int dE = \int_{z=-\infty}^{z=\infty}\cos\theta dE = 2\int_{z=0}^{z=\infty}\cos\theta dE = \frac{\lambda}{2\pi\varepsilon_0}\int_{z=0}^{z=\infty}\cos\theta\frac{dz}{y^2+z^2}.$$

The rest is just technical: to solve the integral, put $z = y\tan\theta \Rightarrow dz = y\sec^2\theta d\theta$. And so,

$E = \dfrac{\lambda}{2\pi\varepsilon_0 y}\displaystyle\int_{\theta=0}^{\theta=\pi/2}\cos\theta d\theta = \frac{\lambda}{2\pi\varepsilon_0 y}$. Now, we could have equally well taken the x axis. The only thing that matters is the distance from the wire, and so the answer is better written as:

$$E = \frac{\lambda}{2\pi\varepsilon_0 r}.$$

5. The **flux** of any vector field is a particularly important concept. It is the measure of the "flow" or penetration of the field vectors through an imaginary fixed surface. So, if there is a uniform electric field that is normal to a surface of area *A*, the flux is $\Phi = EA$. More generally, for any surface, we divide the surface up into little pieces and take the

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component of the electric field normal to each little piece, $\Phi_E = \sum \vec{E}_i \cdot \Delta\vec{A}_i$. If the pieces are made small enough, then in this limit, $\Phi = \int \vec{E} \cdot d\vec{A}$.

5. Let us apply the above concept of flux to calculate the flux leaving a sphere which has a charge at its centre. The electric field at any point on the sphere has magnitude equal to $\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$ and it is directed radially outwards. Let us now divide up the surface of the sphere into small areas. Then $\Phi = \sum E\Delta A = E\sum \Delta A = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\left(4\pi r^2\right)$. So we end up with the important result that the flux leaving this closed surface is $\Phi = \frac{q}{\varepsilon_0}$.

6. **Gauss's Law**: the total electric flux leaving a closed surface is equal to the charge enclosed by the surface divided by ε_0. We can express this directly in terms of the mathematics we have learned, $\Phi \equiv \int \vec{E} \cdot d\vec{A} = \frac{q_{enclosed}}{\varepsilon_0}$. Actually, we have already seen why this law is equivalent to Coulomb's Law in point 5 above, but let's see it again. So, applying Gauss's Law to a sphere containing charge, $\varepsilon_0 \int \vec{E} \cdot d\vec{A} = \varepsilon_0 \int E\, dA = q_{enclosed}$. If the surface is a sphere, then E is constant on the surface and $\varepsilon_0 E \int dA = q$ and from this $\varepsilon_0 E\left(4\pi r^2\right) = q \Rightarrow E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$. This is Coulomb's Law again, but the power of Gauss's law is that it holds for any shape of the (closed) surface and for any distribution of charge.

7. Let us apply Gauss's Law to a hollow sphere that has charges only on the surface. At any distance r from the centre, Gauss's Law is $\varepsilon_0 E\left(4\pi r^2\right) = q_{enclosed}$. Now, if we are inside the sphere then $q_{enclosed} = 0$ and there is no electric field. But if we are outside, then the total charge is $q_{enclosed} = q$ and $E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$, which is as if all the charge was concentrated at the centre.

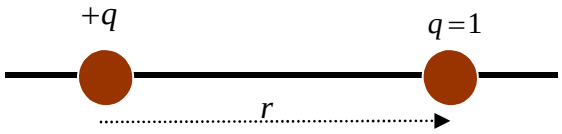
8. Unfortunately, it will not be possible for me to prove Gauss's Law in the short amount of time and space available but the general method can be outlined as follows: take any volume and divide it up into little cubes. Each little cube may contain some small amount of charge. Then show that for each little cube, Gauss's Law follows from Coulomb's Law. Finally, add up the results. For details, consult any good book on electromagnetism.

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Summary of Lecture 24 – ELECTRIC POTENTIAL ENERGY

1. You are already familiar with the concept of gravitational potential energy. When you lift a weight, you have to do work against the downwards pull of the Earth. That work is stored as potential energy. Suppose a force $\vec{F}$ acts on something and displaces it by $d\vec{s}$. Then the work done is $\vec{F} \cdot d\vec{s}$. The work done in going from point a to point b (call it W_{ab}) is then got by adding together the little bits of work, $W_{ab} = \int_a^b \vec{F} \cdot d\vec{s}$. The change in potential energy is defined as $\Delta U = U_b - U_a = -W_{ab}$. Remember always that we can only define the potential U at a point if the force is conservative.

2. The electrostatic force is conservative and can be represented by a potential. Let us see how to calculate the potential. So consider two charges separated by a distance as below.



Let us take the point a very far from the fixed charge q, and the unit charge at the point b to be at a distance R from q. Then the work you did in bringing the unit charge from infinity to R is, $W_{ab} = \int_{\infty}^{R} (-qE)dr = -\frac{1}{4\pi\varepsilon_0}\int_{\infty}^{R} dr\frac{q}{r^2} = -\frac{1}{4\pi\varepsilon_0} q\left(\frac{1}{R} - \frac{1}{\infty}\right) = -\frac{q}{4\pi\varepsilon_0}\frac{1}{R}$. Since the charges repel each other, it is clear that you had to do work in pushing the two charges closer together. So where did the negative sign come from? Answer: the force you exert on the unit charge is directed *towards* the charge q, i.e. is in the negative direction. This is why $\vec{F} \cdot d\vec{s} = (-qE)dr$. Now $\Delta U = U(R) - U(\infty) = \frac{q}{4\pi\varepsilon_0}\frac{1}{R}$. If we take the potential at ∞ to be zero, then the electric potential due to a charge q at the point r is $U(r) = \frac{q}{4\pi\varepsilon_0}\frac{1}{r}$.

Remember that we know how to calculate the force given the potential: $F = -\frac{dU}{dr}$. Apply this here and you see that $F = \frac{q}{4\pi\varepsilon_0}\frac{1}{r^2}$, (which also has the correct repulsive sign).

3. From the above, it is quite obvious that the potential energy of two charges q_1, q_2 is, $U(r) = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$. Compare this with the formula for gravitational energy, $U(r) = -G\frac{m_1 m_2}{r}$. What is the difference? From here, you can see that the gravitational force is always negative (which means attractive), whereas the electrostatic force can be both attractive

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or repulsive because we have both + and - charges in nature.

3. The electric potential (or simply potential) is the energy of a unit charge in an electric field. So, in our MKS units, the unit of potential is $1\ \frac{\text{Joule}}{\text{Coulomb}} = 1$ Volt. Another useful unit is "electron volt" or eV. The definition is:

One electron - volt = energy gained by moving one electron charge through one Volt

$$= \left(1.6\times10^{-19}C\right)\times 1V = 1.6\times10^{-19}J$$

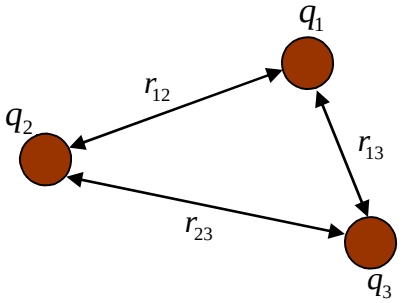
It is useful to note that 1 Kev $= 10^{3}eV$ (kilo-electron-volt)
1 Mev = $10^{6}eV$ (million-electron-volt)
1 Gev $= 10^{9}eV$ (giga-electron-volt)
1 Tev $= 10^{12}eV$ (tera-electron-volt)

4. Every system seeks to minimize its potential energy (that is why a stone falls down!). So, positive charges accelerate toward regions of lower potential, but negative charges accelerate toward regions of higher potential. Note that only the potential difference matters - even if a charge is placed in a region where there is a high potential, it will not want to move unless there is some other place where the potential is higher/lower.

5. Given a system of charges, we can always compute the force - and hence the potential - that arises from them. Here are some important general statements:
   a)Potentials are more positive in regions which have more positive charge.
   b)The electric potential is a scalar quantity (a scalar field, actually).
   c)The electric potential determines the force through $F = -\frac{dU}{dr}$, and hence the electric field because $F = qE$.
   d)The electric potential exists only because the electrostatic force is conservative.

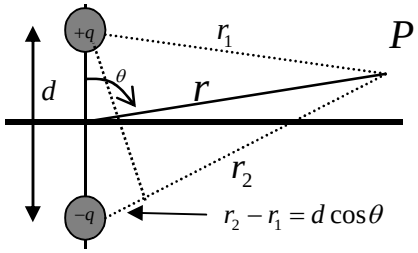
6. To compute the potential at a point, the potentials arising from charges 1, 2, $\cdots$ N must be added up: $V = V_1 + V_2 + \cdots V_N = \sum_{i=1}^{N} V_i = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{N}\frac{q_i}{r_i}$. Here r_i is the distance of the i'th charge from the point where the potential is being calculated or measured. As an example, the potential from the three charges is:

$$\mathrm{V}(r) = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r_{12}} + \frac{1}{4\pi\varepsilon_0}\frac{q_1q_3}{r_{13}} + \frac{1}{4\pi\varepsilon_0}\frac{q_2q_3}{r_{23}}.$$

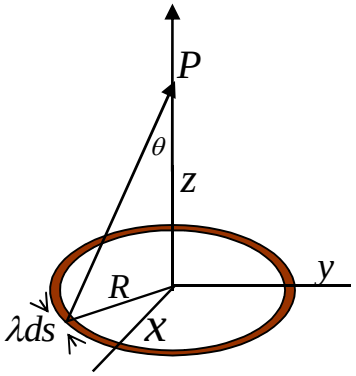


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7. Let us apply these concepts to the dipole system considered earlier. With two charges, $V_P = V_1 + V_2 = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r_1} + \frac{-q}{r_2}\right) = \frac{q}{4\pi\varepsilon_0}\frac{r_2 - r_1}{r_1 r_2}$. We are particularly interested in the situation where $r >> d$. from the diagram you can see that $r_2 - r_1 \approx d\cos\theta$ and that $r_1 r_2 \approx r^2$. Hence, $V \approx \frac{q}{4\pi\varepsilon_0}\frac{d\cos\theta}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}$. So we have calculated the potential at any θ with such little difficulty. Note that $V = 0$ at $\theta = \frac{\pi}{2}$.

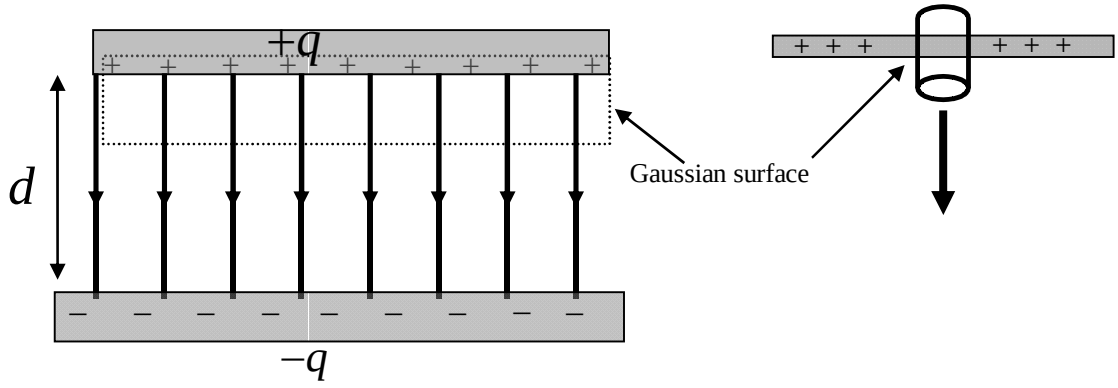


8. Now let us calculate the potential which comes from charges that are uniformly spread over a ring. This is the same problem as in the previous lecture, but simpler. Give the small amount of potential coming from the small amount of charge $dq = \lambda ds$ some name, $dV = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2}$. Then obviously $V = \int dV = \frac{1}{4\pi\varepsilon_0}\int\frac{dq}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{q}{\sqrt{R^2 + z^2}}$.



Summary of Lecture 25 – CAPACITORS AND CURRENTS

1. Two conductors isolated from one another and from their surroundings, form a *capacitor*. These conductors may be of any shape and size, and at any distance from each other. If a potential difference is created between the conductors (say, by connecting the terminals of a battery to them), then there is an electric field in the space between them. The electric field comes from the charges that have been pushed to the plates by the battery. The amount of charge pushed on to the conductors is proportional to the potential difference between the battery terminals (which is the same as between the capacitor plates). Hence, $Q \propto V$. To convert this into an equality, we write $Q = CV$. This provides the definition of capacitance, $C = \frac{Q}{V}$.
2. Using the above definition, let us calculate the capacitance of two parallel plates separated by a distance d as in the figure below.



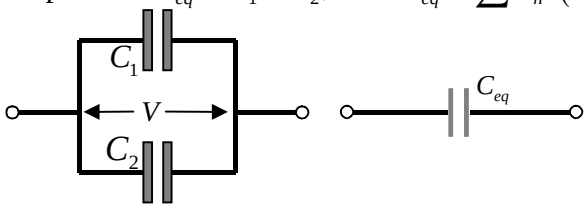
Recall Gauss's Law: $\Phi \equiv \int \vec{E} \cdot d\vec{A} = \frac{q_{enclosed}}{\varepsilon_0}$. Draw any Gaussian surface. Since the electric field is zero above the top plate, the flux through the area A of the plate is $\Phi = EA = \frac{Q}{\varepsilon_0}$, where Q is the total charge on the plate. Thus, $E = \frac{Q}{\varepsilon_0 A}$ is the electric field in the gap between the plates. The potential difference is $V = \frac{E}{d}$, and so $C = \frac{Q}{V} = \frac{\varepsilon_0 A}{d}$. You can see that the capacitance will be large if the plates are close to each other, and if the plates have a large area. We have simplified the calculation here by assuming that the electric field is strictly directed downwards. This is only true if the plates are infinitely long. But we can usually neglect the side effects. Note that any arrangement with two plates forms a capacitor: plane, cylindrical, spherical, etc. The capacitance depends upon the geometry, the size of plates and the gap between them.

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3. One can take two (or more) capacitors in various ways and thus change the amount of charge they can contain. Consider first two capacitors connected in parallel with each other. The same voltage exists across both. For each capacitor, $q_1 = C_1V,\ q_2 = C_2V$ where V is the potential between terminals *a* and *b.* The total charge is:

$$Q = q_1 + q_2 = C_1V + C_2V = (C_1 + C_2)V$$

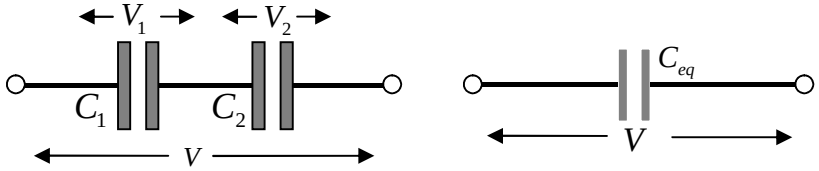
Now, let us define an "effective" or "equivalent" capacitance as $C_{eq} = \frac{Q}{V}$. Then we can immediately see that for 2 capacitors $C_{eq} = C_1 + C_2,$ and $C_{eq} = \sum C_n$ $\left(\text{for } n \text{ capacitors}\right)$.



4. We can repeat the analysis above when the capacitors are put in series. Here the difference is that now we must start with $V = V_1 + V_2,$ where V_1 and V_2 are the voltages across the two. Clearly the same charge had to cross both the capacitors. Hence,

$$V = V_1 + V_2 = \frac{Q}{C_1} + \frac{Q}{C_2} = Q(\frac{1}{C_1} + \frac{1}{C_2}).$$

From our definition, $C_{eq} = \frac{Q}{V}$, it follows that $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$. The total capacitance is now less than if they were in parallel. In general, $\frac{1}{C_{eq}} = \sum \frac{1}{C_n}$ $\left(\text{for } n \text{ capacitors}\right)$.



5. When a battery is connected to a capacitor, positive and negative charges appear on the opposite plates. Some energy has been transferred from the battery to the capacitor, and now been stored in it. When the capacitor is discharged, the energy is recovered. Now let us calculate the energy required to charge a capacitor from zero to V volts.

Begin: the amount of energy required to transfer a small charge dq to the plates is $dU = \text{v}dq,$ where v is the voltage at a time when the charge is $q = C\text{v}$. As time goes on, the total charge increases until it reaches the final charge Q (at which point the voltage becomes V). So,

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$$dU = \mathrm{v}dq = \frac{q}{C}dq \quad \Rightarrow \quad U = \int dU = \int_0^Q \frac{q}{C}dq = \frac{Q^2}{2C} = \frac{1}{2}CV^2.$$

But where in the capacitor is the energy stored? Answer, it is present in the electric field in the volume between the two plates. We can calculate the energy density:

$$u = \frac{\text{energy stored in capacitor}}{\text{volume of capacitor}} = \frac{U}{Ad} = \frac{\frac{1}{2}CV^2}{Ad} = \frac{\varepsilon_0}{2}\left(\frac{V}{d}\right)^2 = \frac{1}{2}\varepsilon_0 E^2.$$

In the above we have used $C = \frac{\varepsilon_0 A}{d}$, derived earlier. The important result here is that $u \propto E^2$. Turning it around, wherever there is an electric field, there is energy available.

6. **Dielectrics.** Consider a free charge $+Q$. Around it is an electric field, $E = \frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{Q}{r^2}$.

Now suppose this charge is placed among water molecules. These molecules will polarise, i.e. the centre of positive charge and centre of negative charge will be slightly displaced. The negative part of the water molecule will be attracted toward the positive charge $+Q$. So, in effect, the electric field is weakened by $\frac{1}{\varepsilon_r}$ and becomes, $\frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{Q}{r^2}$. Here I have introduced a new quantity ε_r called "dielectric constant". This is a number that is usually bigger than one and measures the strength of the polarization induced in the material. For air, $\varepsilon_r = 1.0003$ while $\varepsilon_r \approx 80$ for pure water. The effect of a dielectric is to increase the capacitance of a capacitor: if the air between the plates of a capacitor is replaced by a dielectric, $C = \frac{\varepsilon_0 A}{d} \rightarrow \varepsilon_r \frac{\varepsilon_0 A}{d}$.

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Summary of Lecture 26 – ELECTRIC POTENTIAL ENERGY

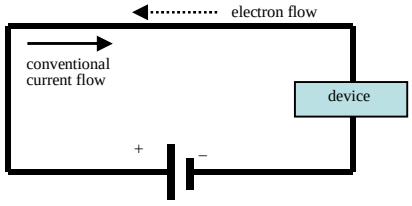
1. Electric current is the flow of electrical charge. If a small amount of charge dq flows in time dt, then the current is $i = \frac{dq}{dt}$. If the current is constant in time, then in time t, the current that flows is $q = i \times t$. The unit of charge is ampere, which is define as:

$$1 \text{ ampere} = \frac{1 \text{ coulomb}}{\text{second}}$$

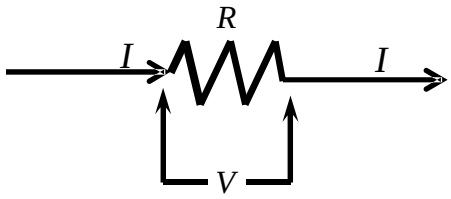
A car's battery supplies upto 50 amperes when starting the car, but often we need to deal with smaller values:

$$1 \text{ milliampere} = 1 \text{ ma} = 10^{-3} A$$
$$1 \text{ microampere} = 1 \mu A = 10^{-6} A$$
$$1 \text{ nanoampere} = 1 \text{ n}A = 10^{-9} A$$
$$1 \text{ picoampere} = 1 \text{ p}A = 10^{-12} A$$

2. The direction of current flow is the direction in which positive charges move. However, in a typical wire, the positive charges are fixed to the atoms and it is really the negative charges (electrons) that move. In that case the direction of current flow is reversed.

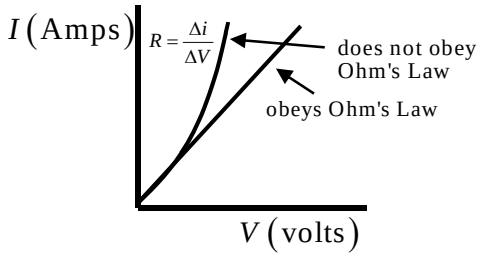


3. Current flows because something forces it around a circuit. That "something" is EMF, electromotive force. But remember that we are using bad terminology and that EMF is not a force - it is actually the difference in electric potentials between two parts of a circuit. So, in the figure below, $V = V_a - V_b$ is the EMF which causes current to flow in the resistor. How much current? Generally, the larger V is , the more current will flow and we expect $I \propto V$. In general this relation will not be completely accurate but when it holds, we say Ohm's Law applies: $I = \frac{V}{R}$. Here, $R = \frac{V}{I}$ is called the resistance.

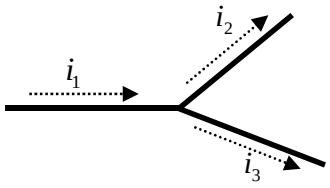


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4. Be careful in understanding Ohm's Law. In general the current may depend upon the applied voltage in a complicated way. Another way of saying this is that the resistance may depend upon the current. Example: when current passes through a resistor, it gets hot and its resistance increases. Only when the graph of current versus voltage is a straight line does Ohm's Law hold. Else, we can only define the "incremental resistance".



5. Charge is always conserved, and therefore current is conserved as well. This means that when a current splits into two currents the sum remains constant, $i_1 = i_2 + i_3$.



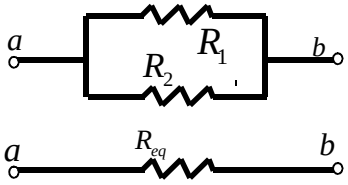
6. When resistors are put in series with each other, the same current flows through both. So, $V_1 = iR_1$ and $V_2 = iR_2$. The total potential drop across the pair is $V = V_1 + V_2 = i(R_1 + R_2)$.
$\Rightarrow R_{eq} = R_1 + R_2$. So resistors in series add up.



7. Resistors can also be put in parallel. This means that the same voltage V is across both. So the currents are $i_1 = \frac{V}{R_1},\ i_2 = \frac{V}{R_2}$.

Since $i = i_1 + i_2$ it follows that $i = \frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2}$

$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$ or $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$.



This makes sense: with two possible paths the current will find less resistance than if only one was present.

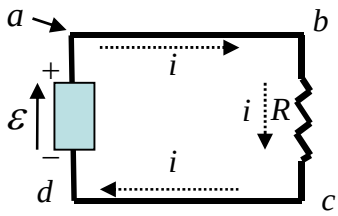
8. When current flows in a circuit work is done. Suppose a small amount of charge *dq* is moved through a potential difference V. Then the work done is $dW = Vdq = V\,idt$. Hence

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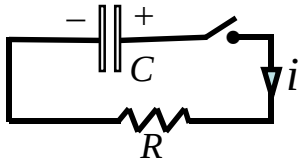
$Vidt = i^2 Rdt$ (because $i = \frac{V}{R}$). The rate of doing work, i.e. power, is $P = \frac{dW}{dt} = i^2 R$.

This is an important formula. It can also be written as $P = \frac{V^2}{R}$, or as $P = iV$. The unit of power is: $1 \text{ volt-ampere} = 1\frac{\text{joule}}{\text{coulomb}} \cdot \frac{\text{coulomb}}{\text{second}} = 1\frac{\text{joule}}{\text{second}} = 1 \text{ watt}.$

9. Kirchoff's Law: The sum of the potential differences encountered in moving around a closed circuit is zero. This law is easy to prove: since the electric field is conservative, therefore no work is done in taking a charge all around a circuit and putting it back where it was. However, it is very useful in solving problems. As a trivial example, consider the circuit below. The statement that, starting from any point a we get back to the same potential after going around is: $V_a - iR + \mathcal{E} = V_a$. This says $-iR + \mathcal{E} = 0$.



10. We can apply Kirchoff's Law to a circuit that consists of a resistor and capacitor in order to see how current flows through it. Since $q = VC$, we can see that $\frac{q}{C} + iR = 0$. Now differentiate with respect to time to find $\frac{1}{C}\frac{dq}{dt} + \frac{di}{dt}R = 0$, or $\frac{di}{dt} = -\frac{1}{RC}i$. This equation has solution: $i = i_0 e^{-\frac{t}{RC}}$. The product RC is called the time constant τ, and it gives the time by which the current has fallen to $1/e \approx 1/2.7$ of the initial value.



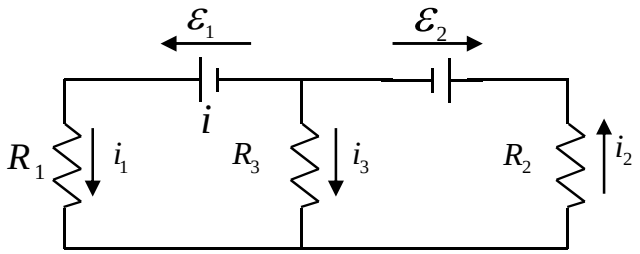
A reminder about the exponential function, $e^x \equiv 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$

From this, $\frac{\mathrm{d}}{\mathrm{dx}}e^x = 1 + \frac{2x}{2!} + \frac{3x^2}{3!} + \frac{4x^3}{4!} + \cdots = e^x$ Similarly, $\frac{\mathrm{d}}{\mathrm{dx}}e^{-x} = -e^{-x}$

11. Circuits often have two or more loops. To find the voltages and currents in such situations, it is best to apply Kirchoff's Law. In the figure below, you see that there are 3 loops and you can see that:

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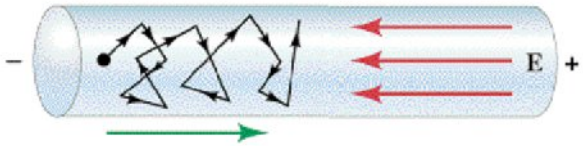
$$i_1 + i_3 = i_2$$
$$\mathcal{E}_1 - i_1R_1 + i_3R_3 = 0$$
$$-i_3R_3 - i_2R_2 + \mathcal{E}_2 = 0$$



Make sure that you understand each of these, and then check that the solution is:

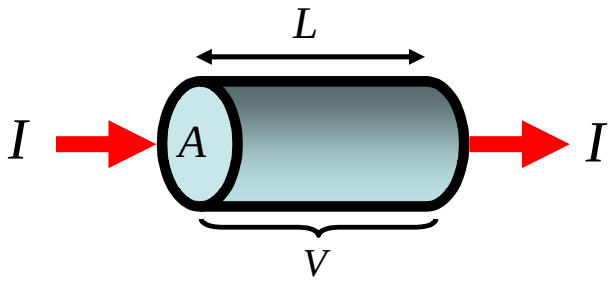
$$i_1 = \frac{\mathcal{E}_1(R_2 + R_3) - \mathcal{E}_2R_3}{R_1R_2 + R_2R_3 + R_1R_3} \qquad i_2 = \frac{\mathcal{E}_1R_3 - \mathcal{E}_2(R_1 + R_3)}{R_1R_2 + R_2R_3 + R_1R_3} \qquad i_3 = \frac{-\mathcal{E}_1R_2 - \mathcal{E}_2R_1}{R_1R_2 + R_2R_3 + R_1R_3}$$

12. A charge inside a wire moves under the influence of the applied electric field and suffers many collisions that cause it to move on a highly irregular, jagged path as shown below.



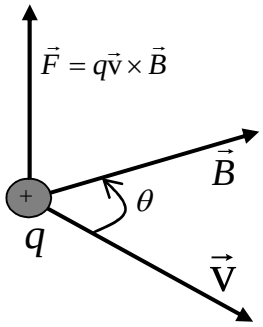
Nevertheless, it moves on the average to the right at the "drift velocity" (or speed).

13. Consider a wire through which charge is flowing. Suppose that the number of charges per unit volume is *n*. If we multiply *n* by the crossectional area of the wire *A* and the length *L*, then the charge in this section of the wire is $q = (nAL)e$. If the drift velocity of the charges is v_d, then the time taken for the charge to move through the wire is $t = \frac{L}{v_d}$. hence the current is $i = \frac{q}{t} = \frac{nALe}{L/v_d} = nAev_d$. From this we can calculate the drift velocity of the charges in terms of the measured current, $v_d = \frac{i}{nAe}$. The current density, which is the current per unit crossectional area is defined as $j = \frac{i}{A} = nev_d$. If j varies inside a volume, then we can easily generalize and write, $i = \int \vec{j} \cdot d\vec{A}$.



Summary of Lecture 27 – THE MAGNETIC FIELD

1. The magnetic field exerts a force upon any charge that moves in the field. The greater the size of the charge, and the faster it moves, the larger the force. The direction of the force is perpendicular to both the direction of motion and the magnetic field. If θ is the angle between $\vec{v}$ and $\vec{B}$, then $F = qvB\sin\theta$ is the magnitude of the force. This vanishes when $\vec{v}$ and $\vec{B}$ are parallel ($\theta = 0$), and is maximum when they are perpendicular.



2. The unit of magnetic field that is used most commonly is the *tesla.* A charge of one coulomb moving at 1 metre per second perpendicularly to a field of one tesla experiences a force of 1 newton. Equivalently,

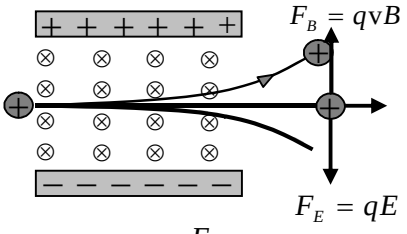
$$1 \text{ tesla} = 1\ \frac{\text{newton}}{\text{coulomb} \cdot \text{meter/second}} = 1\ \frac{\text{newton}}{\text{ampere} \cdot \text{meter}} = 10^4 \text{ gauss (CGS unit)}$$

In order to have an appreciation for how much a tesla is, here are some typical values of the magnetic field in these units:

Earth's surface	10^{-4} T
Bar magnet	10^{-2} T
Powerful electromagnet	1 T
Superconducting magnet	5 T

3. When both magnetic and electric fields are present at a point, the total force acting upon a charge is the vector sum of the electric and magnetic forces, $\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$. This is known as the *Lorentz Force.* Note that the electric force and magnetic force are very different. The electric force is non-zero even if the charge is stationary, and it is in the same direction as $\vec{E}$.

4. The Lorentz Force can be used to select charged particles of whichever velocity we want. In the diagram below, particles enter from the left with velocity v. They experience a force due to the perpendicular magnetic field, as well as force downwards because of an electric field. Only particles with speed $v = E / B$ are undeflected and keep going straight.



$qE = qvB \Rightarrow v = \frac{E}{B}$ velocity selector !!

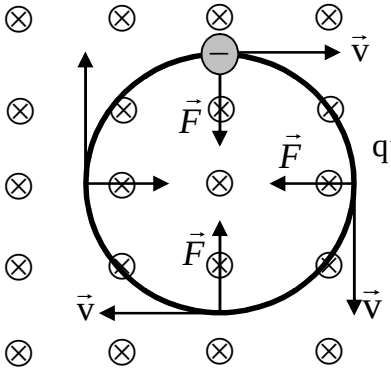
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5. A magnetic field can be strong enough to lift an elephant, but it can never increase or decrease the energy of a particle. Proof: suppose the magnetic force $\vec{F}$ moves a particle through a displacement $d\vec{r}$. Then the small amount of work done is,

$$\begin{aligned} dW = \vec{F}\cdot d\vec{r} &= q(\vec{\mathrm{v}}\times\vec{\mathrm{B}})\cdot d\vec{r} \\ &= q(\vec{\mathrm{v}}\times\vec{\mathrm{B}})\cdot\frac{d\vec{r}}{dt}dt \\ &= q(\vec{\mathrm{v}}\times\vec{\mathrm{B}})\cdot\vec{\mathrm{v}}\,dt = 0. \end{aligned}$$

Basically the force and direction of force are orthogonal, and hence there can be no work done on the particle or an increase in its energy.

6. A magnetic field bends a charged particle into a circular orbit because the particle feels a force that is directed perpendicular to the magnetic field. As we saw above, the particle cannot change its speed, but it certainly does change direction! So it keeps bending and bending until it makes a full circle. The radius of orbit can be easily calculated: the magnetic and centrifugal forces must balance each other for equilibrium. So, $\mathrm{qv}B = m\frac{\mathrm{v}^2}{r}$ and we find that $r = \frac{m\mathrm{v}}{qB}$. A strong B forces the particle into a tighter orbit, as you can see. We can also calculate the angular frequency, $\omega = \frac{\mathrm{v}}{r} = \frac{qB}{m}$. This shows that a strong B makes the particle go around many times in unit time. There are a very large number of applications of these facts.

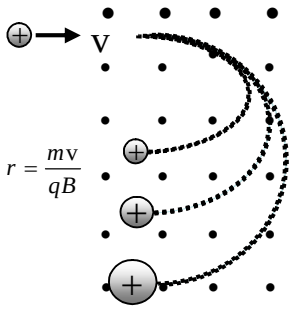


$$\mathrm{qv}B = m\frac{\mathrm{v}^2}{r} \;\Rightarrow\; r = \frac{m\mathrm{v}}{qB}.$$

7. The fact that a magnetic field bends charged particles is responsible for shielding the earth from harmful effects of the "solar wind". A large number of charged particles are released from the sun and reach the earth. These can destroy life. Fortunately the earth's magnetic field deflects these particles, which are then trapped in the "Van Allen" belt around the earth.

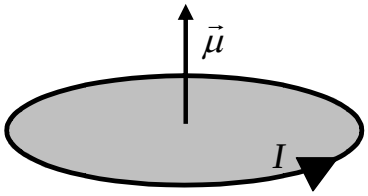
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8. The mass spectrometer is an extremely important equipment that works on the above principle. Ions are made from atoms by stripping away one electron. Then they pass through a velocity selector so that they all have the same speed. In a beam of many different ions, the heavier ones bend less, and lighter ones more, when they are passed through a *B* field.

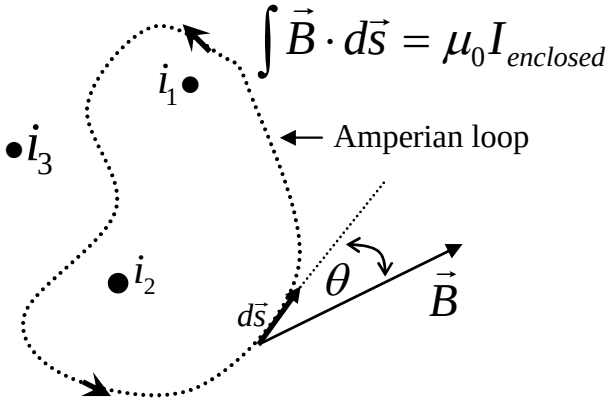


9. A wire carries current, and current is flowing charges. Since each charge experiences a force when placed in a magnetic field, you might expect the same for the current. Indeed, that is exactly the case, and we can easily calculate the force on a wire from the force on individual charges. Suppose N is the total number of charges and they are moving at the average (or drift) velocity $\vec{v}_d$. Then the total force is $\vec{F} = Ne\vec{v}_d \times \vec{B}$. Now suppose that the wire has length *L*, crossectional area *A*, and it has *n* charges per unit volume. Then clearly $N = nAL$, and so $\vec{F} = nALe\vec{v}_d \times \vec{B}$. Remember that the current is the charge that flows through the wire per unit time, and so $nAe\vec{v}_d = \vec{I}$. We get the important result that the force per unit length on the wire is $\vec{F} = \vec{I} \times \vec{B}$.

10. A current that goes around a loop (any shape) produces a magnetic field. We define the *magnetic moment* as the product of current and area, $\vec{\mu} = IA\hat{z}$. Here A is the area of the loop and I the current flowing around it. The direction is perpendicular to the plane of the loop, as shown.



11. Magnetic fields are produced by currents. Every small bit of current produces a small amount of the *B* field. Ampere's Law, illustrated below, says that if one goes around a loop (of any shape or size) then the integral of the B field around the loop is equal to the enclosed current. In the loop below $I = I_1 + I_2$. Here I_3 is excluded as it lies outside.



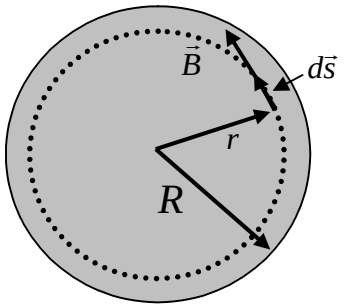
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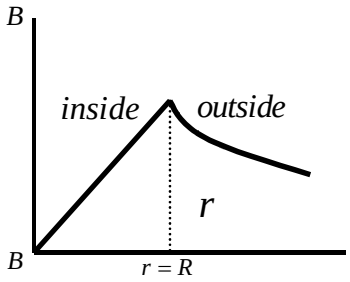
12. Let us apply Ampere's Law to a circular loop of radius r outside an infinitely long wire carrying current I through it. The magnetic field goes around in circles, and so $\vec{B}$ and $d\vec{s}$ are both in the same direction. Hence, $\int \vec{B} \cdot d\vec{s} = B\int ds = B(2\pi r) = \mu_0 I$. We get the important result that $B = \dfrac{\mu_0 I}{2\pi r}$.

13. Assuming that the current flows uniformly over the crossection, we can use Ampere's Law to calculate the magnetic field at distance r, where r now lies inside the wire.

$$B(2\pi r) = \mu_0 I\left(\frac{\pi r^2}{\pi R^2}\right) \Rightarrow B = \frac{\mu_0 I r}{2\pi R^2}$$

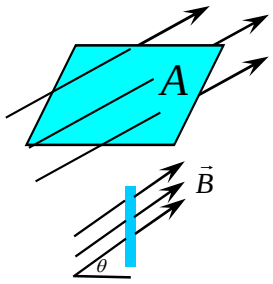
Here is a sketch of the B field inside and outside the wire as a function of distance r.



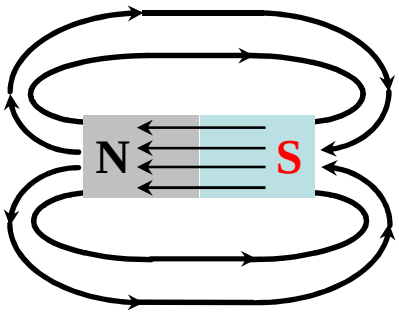


Summary of Lecture 28 – ELECTROMAGNETIC INDUCTION

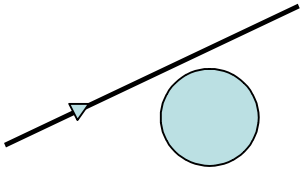
1. Earlier we had defined the flux of any vector field. For a magnetic field, this means that flux of a uniform magnetic field (see figure) is $\Phi = B_{\perp} A = BA\cos\theta$. If the field is not constant over the area then we must add up all the little pieces of flux: $\Phi_B = \int \vec{B} \cdot d\vec{A}$. The dimension of flux is magnetic field $\times$ area, and the unit is called weber, where $1 \text{ weber} = 1 \text{ tesla} \cdot \text{metre}^2$.



2. A fundamental law of magnetism states that the net flux through a closed surface is always zero, $\Phi_B = \int \vec{B} \cdot d\vec{A} = 0$. Note that this is very different from what you learned earlier in electrostatics where the flux is essentially the electric charge. There is no such thing as a magnetic charge! What we call the magnetic north (or south) pole of a magnet are actually due to the particular electronic currents, not magnetic charges. In the bar magnet below, no matter which closed surface you draw, the amount of flux leaving the surface is equal to that entering it.



Example : A sphere of radius *R* is placed near a long, straight wire that carries a steady current *I*. The magnetic field generated by the current is *B*. Find the total magnetic flux passing through the sphere.



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Answer: zero, of course!

3. Faraday's Law for Induced EMF: when the magnetic flux changes in a circuit, an electro-motive force is induced which is proportional to the rate of change of flux. Mathematically, $\mathcal{E} = -\frac{d\Phi_B}{dt}$ where $\mathcal{E}$ is the induced emf. If the coil consists of N turns, then $\mathcal{E} = -N\frac{d\Phi_B}{dt}$.

How does the flux through a coil change? Consider a coil and magnet. We can:

a) move the magnet,

b) change the size and shape of the coil by squeezing it,

c) move the coil.

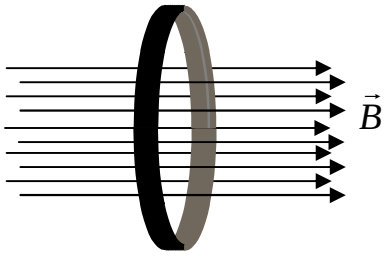
In all cases, the flux through the coil changes and $\frac{d\Phi_B}{dt}$ is non-zero leading to an induced emf.

Example: A flexible loop has a radius of 12cm and is in a magnetic field of strength 0.15T. The loop is grasped at points A and B and stretched until it closes. If it takes 0.20s to close the loop, find the magnitude of the average induced emf in it during this time.

Solution: Here the loop area changes, hence the flux. So the induced emf is:

$$\mathcal{E} = -\frac{d\Phi_B}{dt} \approx -\left[\frac{\text{final flux} - \text{initial flux}}{\text{time taken}}\right] = -\left[\frac{0 - \pi(0.12)^2 \times 0.15}{0.2}\right] = 0.034 \text{ Volts.}$$

Example : A wire loop of radius 0.30m lies so that an external magnetic field of +0.30T is perpendicular to the loop. The field changes to -0.20T in 1.5s. Find the magnitude of the average induced EMF in the loop during this time.



Solution: Again, we will find the initial and final fluxes first and then divide by the time taken for the change.

Use $\Phi = BA = B\pi r^2$ to calculate the flux.

$$\Phi_i = \quad 0.30 \times \pi \times (0.30)^2 = 0.085 \text{ Tm}^2$$

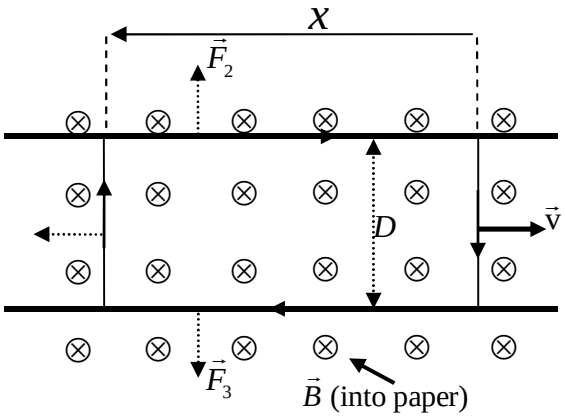
$$\Phi_f = -0.20 \times \pi \times (0.30)^2 = -0.057 \text{ Tm}^2$$

$$\varepsilon = \frac{\Delta\Phi}{\Delta t} = \frac{\Phi_f - \Phi_i}{\Delta t} = \frac{0.085 - 0.057}{1.5} = 0.095 \text{ V}$$

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4. Remember that the electromotive force is not really a force but the difference in electric potentials between two points. In going around a circular wire, where the electric field is constant as a function of angle, the emf is $\varepsilon = E(2\pi r)$. More generally, for any size or shape of a closed circuit, $\varepsilon = \int \vec{E} \cdot d\vec{s}$. So Faraday's Law reads: $\int \vec{E} \cdot d\vec{s} = -\dfrac{d\Phi_B}{dt}$.

Example : A conducting wire rests upon two parallel rails and is pulled towards the right with speed v. A magnetic field B is perpendicular to the plain of the rails as shown below. Find the current that flows in the circuit.



Solution: As the wire is pulled to the right, the area of the circuit increases and so the flux increases. Measure x as above so that $x = \text{v}t$, i.e. x keeps increasing as we pull. The flux at any value of x is $\Phi_B = BDx$, and so,

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d(BDx)}{dt} = -BD\frac{dx}{dt} = BD\text{v}.$$

To calculate the current, we simply use Ohm's Law: $I = \dfrac{\varepsilon}{R} = \dfrac{BD\,\text{v}}{R}$. Here R is the resistance of the circuit.

Example : Find the power dissipated in the above circuit using I^2R, and then by directly calculating the work you do by pulling the wire.

Solution: Clearly $I^2R = \dfrac{B^2D^2\,\text{v}^2}{R}$ is the power dissipated, as per usual formula. Now let us calculate the force acting upon the piece of wire (of length D) that you are pulling. From the formula for the force on a wire, $\vec{F} = I\,\vec{L} \times \vec{B}$, the magnitude is $F = IBD = \dfrac{B^2D^2\text{v}}{R}$. So, the power is $P = F\text{v} = \dfrac{B^2D^2\text{v}^2}{R}$. This is exactly the value calculated above!

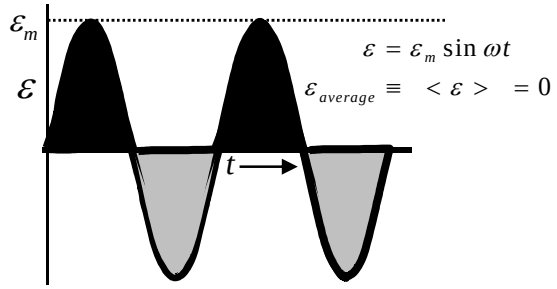
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5. **Lenz's Law :** The direction of any magnetic induction effect is such as to oppose the cause of the effect. Imagine a coil wound with wire of finite resistance. If the magnetic field decreases, the induced EMF is positive. This produces a positive current. The magnetic field produced by the current opposes the decrease in flux. Of course, because of finite resistance in loop, the induced current cannot completely oppose the change in flux.

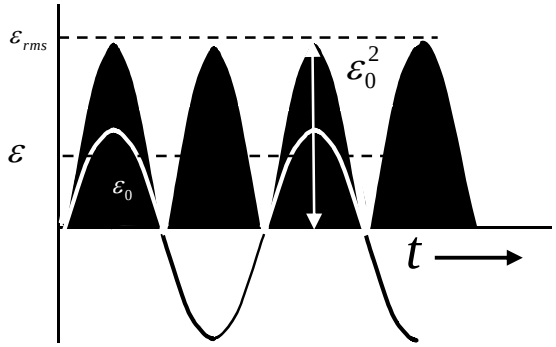
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Summary of Lecture 29 – ALTERNATING CURRENT

1. Alternating current (AC) is current that flows first in one direction along a wire, and then in the reverse direction. The most common AC is sinuisoidal in which the current (and voltage) follow a sine function, as in the graph below. The average value is zero because the current flows for the same time in one direction as in the other.



However, the square of any AC wave is always positive. Thus its average is not zero. As we shall see, upon averaging the square we get half the square of the peak value.



The calculation follows: if there is a sine wave of amplitude (height) equal to one and frequency ω, ($\omega = 2\pi / T$, T=time period), then the squared amplitude is $\sin^2 \omega t$ and its average is:

$$< \sin^2 \omega t > \equiv \frac{1}{T}\int_0^T dt \sin^2 \omega t = \frac{1}{T}\int_0^T dt(\frac{1-\cos 2\omega t}{2}) = \frac{1}{T}\int_0^T dt \frac{1}{2} = \frac{1}{2}$$

Taking the square root gives the root mean square value as $1/\sqrt{2}$ of the maximum value. Of course, it does not matter whether this is of the voltage or current:

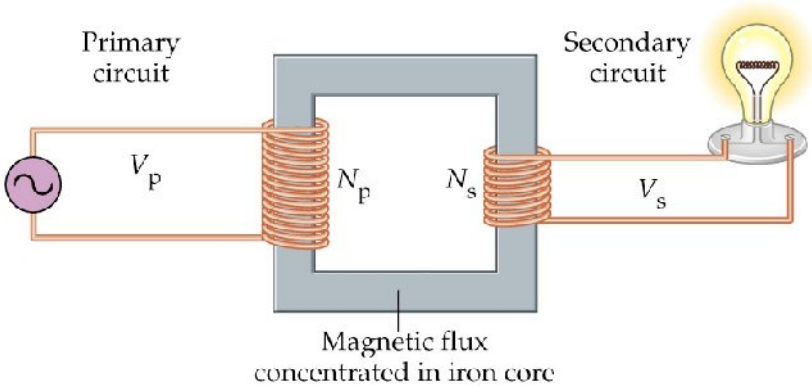
$$\varepsilon_{rms} = \sqrt{\frac{\varepsilon_m^2}{2}} = \frac{\varepsilon_m}{\sqrt{2}} = 0.707\varepsilon_m, \text{ and } I_{rms} = \sqrt{\frac{I_m^2}{2}} = \frac{I_m}{\sqrt{2}} = 0.707 I_m.$$

Exactly the same results are obtained for cosine waves. This is what one expects since the difference between sine and cosine is only that one starts earlier than the other.

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2. AC is generated by a coil rotating in a magnetic field. We know from Faraday's Law, $\mathcal{E} = -\frac{d\Phi_B}{dt}$, that a changing magnetic flux gives rise to an emf. Imagine a magnetic field and a coil of area *A*, rotating with frequency ω so that the flux through the coil at any instant of time is $\Phi_B = BA\cos\omega t$. Then the induced emf is $\varepsilon = BA\omega\sin\omega t$.

3. AC is particular useful because *transformers* make it possible to step up or step down voltages. The basic transformer consists of two coils - the primary and secondary - both wrapped around a core (typically iron) that enhances the magnetic field.



Suppose that the flux in the core is Φ and that its rate of change is $\frac{\partial\Phi}{\partial t}$. Call the number of turns in the primary and secondary N_p and N_s respectively. Then, from Faraday's law, the primary emf is $\varepsilon_p = -N_p\frac{\partial\Phi}{\partial t}$ and the secondary emf is $\varepsilon_s = -N_s\frac{\partial\Phi}{\partial t}$. The ratio is, $\frac{\varepsilon_p}{\varepsilon_s} = \frac{N_p}{N_s}$. So, the secondary emf is $\varepsilon_s = \frac{N_s}{N_p}\varepsilon_p$. If $\frac{N_s}{N_p}$ is less than one, then it is called a step-down transformer because the secondary voltage is less than the input voltage. Else, it is a step-up transformer. Both types are used.

4. If this is a lossless transformer (and good transformers are 99% lossless), then the input power must equal the output power, $\varepsilon_p I_p = \varepsilon_s I_s$. From above, this shows that the ratio of currents is $I_s = \frac{N_p}{N_s}I_p$.

5. Whatever the shape or size of a current carrying loop, the magnetic flux that passes through it is proportional to the current, $\Phi \propto I$. The inductance *L* (called the self-inductance if

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there is only one coil) is the constant of proportionality in the relation $\Phi = LI$. The unit of inductance is called Henry, 1 Henry= 1 Tesla metre^2 / Ampere. Note that inductance, like capacitance, is purely geometrical and depends only upon the shape and sizes of wires. It does not depend on the current.

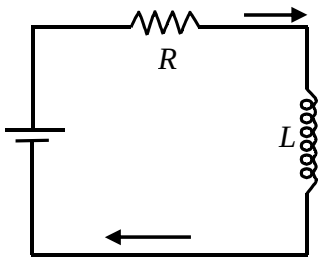
6. Let us calculate the inductance of a long coil wound with n turns per unit length. As calculated earlier, the B field is $B = \mu_0 nI$, and the flux passing through N turns of the coil of length l and area A is $N\Phi_B = (nl)(BA) = \mu_0 n^2 IAl$. From the definition, we find:

$L = \dfrac{N\Phi_B}{I} = \dfrac{\mu_0 n^2 lIA}{I} = \mu_0 n^2 lA$ (inductance of long solenoid).

Example: Find the inductance of a coil with 3500 turns, length 10 cm, and radius 5cm.

Solution: $L = 4\pi \cdot 10^{-7} \dfrac{T \cdot m}{A} \cdot 3500^2 \cdot \dfrac{\pi (0.05m)^2}{0.10m} = 1.21 \dfrac{T \cdot m^2}{A} = 1.21H.$

7. When the current changes through an inductor, by Faraday's Law it induces an emf equal to $-\dfrac{d\Phi_B}{dt} = -\dfrac{d(LI)}{dt} = -L\dfrac{dI}{dt}$. Let us use this to calculate the current through the circuit shown here, where a resistor is present. Then, by using Kirchoff's Law, $\varepsilon = IR + L\dfrac{dI}{dt}$ or $\varepsilon I = I^2 R + LI\dfrac{dI}{dt}$. In words: the power expended by the battery equals energy dissipated in the resistor + work done on inductor.



8. Before we go on to the AC case, suppose that we suddenly connect a battery to an inductor so that the voltage suddenly increases from zero to ε across the circuit. Then, $L\dfrac{dI}{dt} + IR = \varepsilon$ has solution: $I(t) = \dfrac{\varepsilon}{R}\left(1 - e^{-t/\tau}\right)$ where $\tau = \dfrac{L}{R}$. You can easily see that this true using $\dfrac{dI}{dt} = \dfrac{\varepsilon}{R}\dfrac{1}{\tau}e^{-t/\tau}$. The solution shows that the current increases from zero to a maximum of $\dfrac{\varepsilon}{R}$, i.e. that given by Ohm's Law. We need to pay a little attention to the "time constant" τ, which is the time after which the constant approaches 63% of its final value. Units: $[\tau] = \dfrac{[L]}{[R]} = \dfrac{\text{henry}}{\text{ohm}} = \dfrac{\text{volt.second / ampere}}{\text{ohm}} = \left(\dfrac{\text{volt}}{\text{ampere.ohm}}\right)\text{second} = \text{second}.$

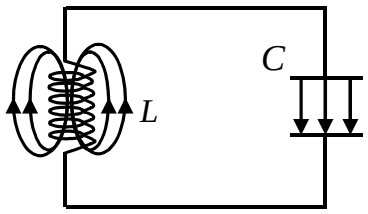
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Note that when t=τ, then $I = \frac{\varepsilon}{R}\left(1-e^{-1}\right) = \frac{\varepsilon}{R}\left(1-0.37\right) = \frac{\varepsilon}{R}0.63$

9. When we pass current through an inductor a changing magnetic field is produced. This, by Faraday's Law, induces an emf across the coil. So work has to be done to force the current through. How much work? The power, or rate of doing work, is emf $\times$ current. Let U_B be the work done in passing current I. Then, $\frac{dU_B}{dt} = \left(L\frac{dI}{dt}\right)I = LI\frac{dI}{dt}$. Let us integrate $dU_B = LI\,dI$. Then, $\int_0^{U_B} dU_B = \int_0^I LIdI \;\Rightarrow\; U_B = \frac{1}{2}LI^2$. This is an important result. It tells us that an inductor L carrying current I requires work $\frac{1}{2}LI^2$. By conservation of energy, this is also the energy stored in the inductor. Compare this result with the result $U_E = \frac{1}{2}CV^2$ for a capacitor. Notice that $C \leftrightarrow L$ and $V \leftrightarrow I$.

10. Let us use the result derived earlier for the inductance of a solenoid and the magnetic field in it, $L = \mu_0 n^2 lA$ and $B = \mu_0 nI$. Putting this into $U_B = \frac{1}{2}LI^2$ gives $U_B = \frac{1}{2}(\mu_0 n^2 lA)(B/\mu_0 n)^2$. Divide the energy by the volume of the solenoid, $\frac{U_B}{\text{volume}} = \frac{B^2}{2\mu_0}$. This directly gives the energy density (energy per unit volume) contained in a magnetic field.

11. **Electromagnetic oscillations in an LC circuit.**



We know that energy can be stored in a capacitor as well as in an inductor. What happens when we connect them up together and put some charge on the capacitor? As it discharges, it creates a current that transfers energy to the inductor. The total energy remains constant, of course.

This means that the sum $U = U_B + U_E = \frac{1}{2}LI^2 + \frac{1}{2}\frac{q^2}{C}$ is constant. Differentiate this:

$0 = \frac{dU}{dt} = \frac{d}{dt}\left(\frac{1}{2}LI^2 + \frac{1}{2}\frac{q^2}{C}\right)$. Since $I = \frac{dq}{dt}$, and $\frac{dI}{dt} = \frac{d^2q}{dt^2}$, $\therefore \quad \frac{d^2q}{dt^2} + \frac{1}{LC}q = 0$

To make this look nicer, put $\omega^2 \equiv \frac{1}{LC} \Rightarrow \frac{d^2q}{dt^2} + \omega^2 q = 0$. We have seen this equation many times earlier. The solution is: $q = q_m \cos\omega t$. The important result here is that the

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charge, current, and voltage will oscillate with frequency $\omega = \sqrt{\frac{1}{LC}}$. This oscillation will go on for forever if there is no resistance in the circuit.

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Summary of Lecture 30 – ELECTROMAGNETIC WAVES

1. Before the investigations of James Clerk Maxwell around 1865, the known laws of electromagnetism were:

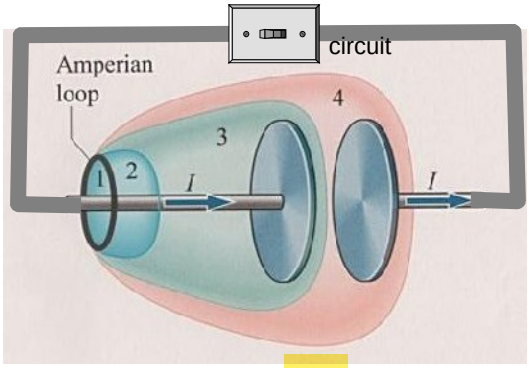
   a) Gauss' law of electricity: $\int \vec{E} \cdot d\vec{A} = \frac{q}{\varepsilon_0}$ (integral is over any closed surface)

   b) Gauss' law of magnetism: $\int \vec{B} \cdot d\vec{A} = 0$ (integral is over any closed surface)

   c) Faraday's law of induction: $\int \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$ (integral is over any closed loop)

   d) Ampere's law: $\int \vec{B} \cdot d\vec{s} = \mu_0 I$ (integral is over any closed loop)

2. But Maxwell realized that the above 4 laws were not consistent with the conservation of charge, which is a fundamental principle. He argued that if you take the space between two capacitors (see below) and take different surfaces 1,2,3,4 then applying Ampere's Law gives an inconsistency: $\left[\int \vec{B} \cdot d\vec{s}\right]_{(1,2,4)} \neq \left[\int \vec{B} \cdot d\vec{s}\right]_3$ because obviously charge cannot flow in the gap between plates. So Ampere's Law gives different results depending upon which surface is bounded by the loop shown!



Maxwell modified Ampere's law as follows: $\int \vec{B} \cdot d\vec{s} = \mu_0 \left(I + I_d\right)$ where the "displacement current" is $I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$. Let's look at the reasoning that led to Maxwell's discovery of the displacement current. The current that flows in the circuit is $I = \frac{dQ}{dt}$. But the charge on the capacitor plate is $Q = \varepsilon_0 EA$. Hence, $I = \frac{d}{dt}\left(\varepsilon_0 EA\right) = \varepsilon_0 \frac{d(EA)}{dt} = \varepsilon_0 \frac{d\Phi_E}{dt} \equiv I_D$. In words, the changing electric field in the gap acts as source of the magnetic field in just the same way as the current in the outside wires. This is really the most important point - a magnetic field may have two separate reasons for existence - flowing charges or changing electric fields.

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3. The famous Maxwell's equations are as follows:

a) $\int \vec{E} \cdot d\vec{S} = \frac{Q}{\varepsilon_0}$

b) $\int \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$

c) $\int \vec{B} \cdot d\vec{S} = 0$

d) $\int \vec{B} \cdot d\vec{\ell} = \mu_0 \left(I + \varepsilon_0 \frac{d\Phi_E}{dt} \right)$

Together with the Lorentz Force $\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right)$ they provide a complete description of all electromagnetic phenomena, including waves.

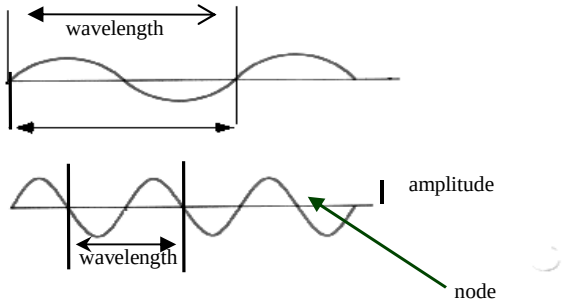
4. Electromagnetic waves were predicted by Maxwell and experimentally discovered many years later by Hertz. Note that for these waves:

a) Absolutely no medium is required - they travel through vacuum.

b) The speed of propagation is c for all waves in the vacuum.

c) There is no limit to the amplitude or frequency.

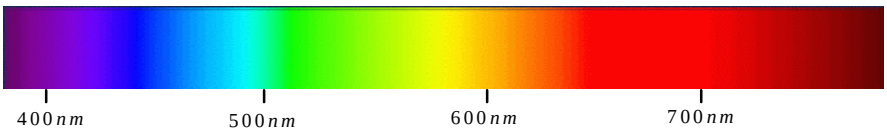
A wave is characterized by the amplitude and frequency, as illustrated below.



Example: Red light has λ = 700 nm. The frequency ν is calculated as follows:

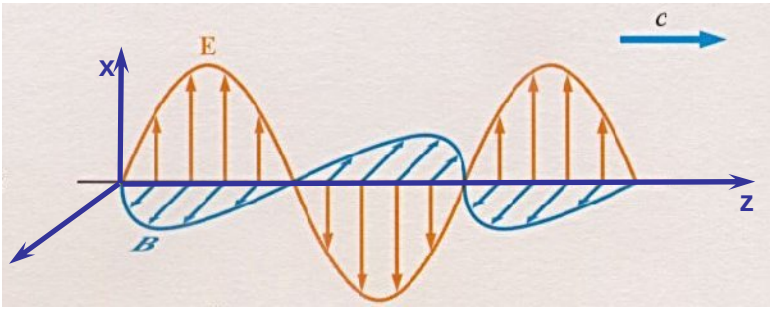
$$\nu = \frac{3.0 \times 10^8 m/\sec}{7 \times 10^{-7} m} = 4.29 \times 10^{14} Hertz$$

By comparison, the electromagnetic waves inside a microwave oven have wavelength of 6 cm, radio waves are a few metres long. For visible light, see below. On the other hand, X-rays and gamma-rays have wavelengths of the size of atoms and even much smaller.



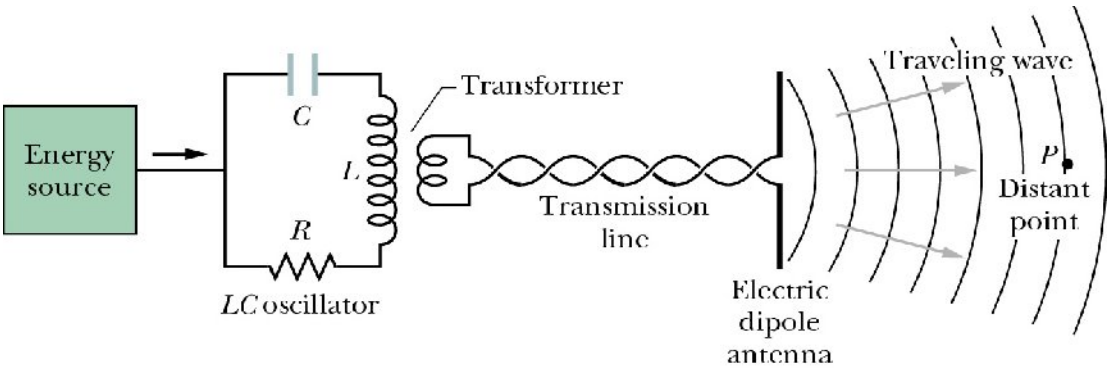
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5. We now consider how electromagnetic waves can travel through empty space. Suppose an electric field (due to distant charges in an antenna) has been created. If this changes then this creates a changing electric flux $\frac{d\Phi_E}{dt}$ which, through $\int \vec{B}\cdot d\vec{s} = \mu_0\varepsilon_0\frac{d\Phi_E}{dt}$, creates a changing magnetic flux $\frac{d\Phi_B}{dt}$. This, through $\int \vec{E}\cdot d\vec{s} = -\frac{d\Phi_B}{dt}$, creates a changing electric field. This chain of events in free space then allows a wave to propagate.



In the diagram above, an electromagnetic wave is moving in the z direction. The electric field is in the x direction, $E_x = E_0\sin(kz-\omega t)$, and the magnetic field is perpendicular to it, $B_y = B_0\sin(k\,z-\omega t)$. Here $\omega = kc$. From Maxwell's equations the amplitudes of the two fields are related by $E_0 = cB_0$. Note that the two fields are in phase with each other.

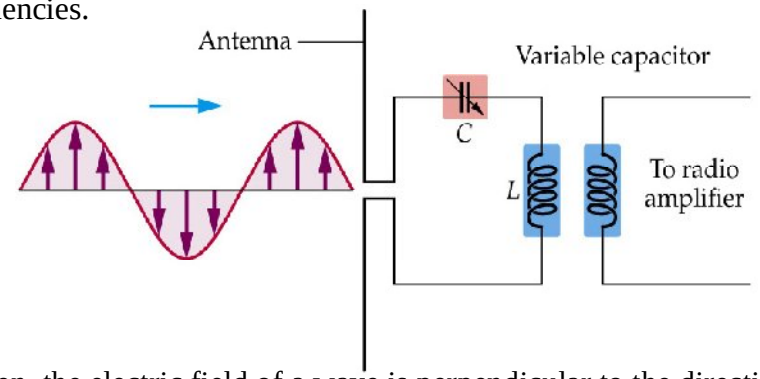
6. The production of electromagnetic waves is done by forcing current to vary rapidly in a small piece of wire. Consider your mobile phone, for example. Using the power from the battery, the circuits inside produce a high frequency current that goes into a "dipole antenna" made of two small pieces of conductor. The electric field between the two oppositely charged pieces is rapidly changing and so creates a magnetic field. Both fields propagate outwards, the amplitude falling as $1/r$.



The power, which is the square of the amplitude, falls off as $I(\theta) \propto \frac{\sin^2\theta}{r^2}$. The $\sin^2\theta$ dependence shows that the power is radiated unequally as a function of direction. The maximum power is at $\theta = \pi/2$ and the least at $\theta = 0$.

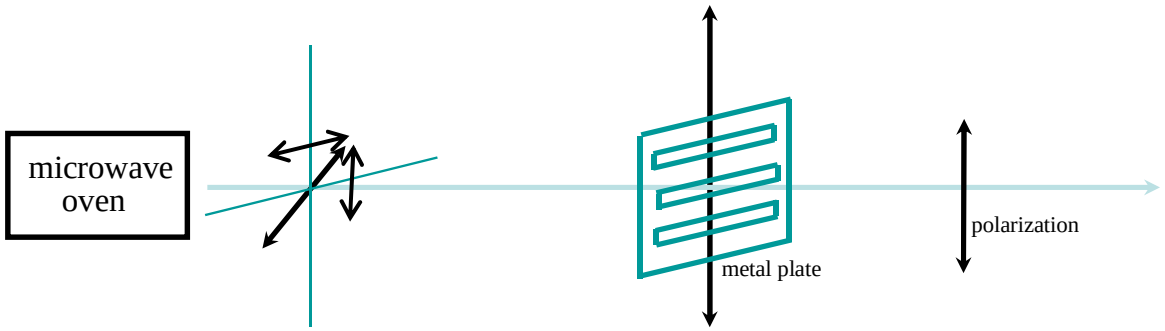
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7. The reception of electromagnetic waves requires an antenna. The incoming wave has an electric field that forces the electrons to run up and down the antenna wire, i.e. it produces a tiny electric current. This current is then amplified (increased in amplitude) electronically. This is schematically indicated below. Here the variable capacitor is used to tune to different frequencies.



8. As we have seen, the electric field of a wave is perpendicular to the direction of its motion. If this is a fixed direction (say, $\hat{x}$), then we say that wave is polarized in the x direction. Most sources - a candle, the sun, any light bulb - produce light that is unpolarized. In this case, there is no definite direction of the electric field, no definite phase between the orthogonal components, and the atomic or molecular dipoles that emit the light are randomly oriented in the source. But for a typical linearly polarized plane electromagnetic wave polarized along $\hat{x}$, $E_x = E_0 \sin(kz - \omega t)$, $B_y = \frac{E_0}{c}\sin(kz - \omega t)$ with all other components zero. Of course, it may be that the wave is polarized at an angle θ relative to $\hat{x}$, in which case $E_x = E_0 \cos\theta \cdot \sin(kz - \omega t)$, $E_y = E_0 \sin\theta \cdot \sin(k z - \omega t)$, $E_z = 0$.

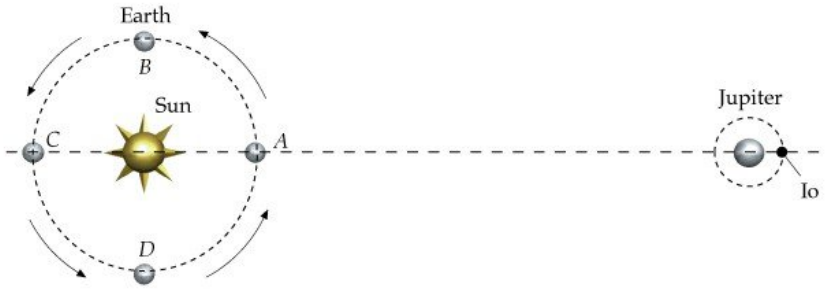
9. Electromagnetic waves from an unpolarized source (e.g. a burning candle or microwave oven) can be polarized by passing them through a simple polarizer of the kind below. A metal plate with slits cut into will allow only the electric field component perpendicular to the slits. Thus, it will produce linearly polarized waves from unpolarized ones.



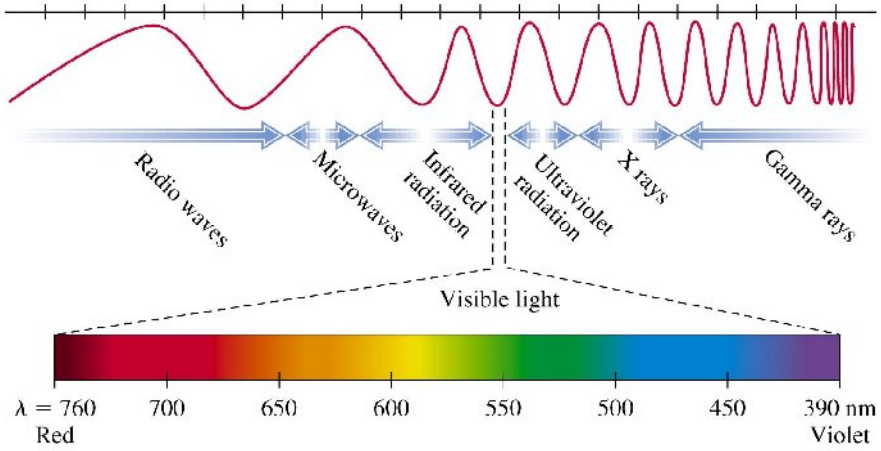
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Summary of Lecture 31 – LIGHT

1. Light travels very fast but its speed is not infinite. Early attempts to measure the speed using earth based experiments failed. Then in 1675 the astronomer Roemer studied timing of the eclipse of one of Jupiter's moon called Io. In the diagram below Io is observed with Earth at A and then at C. The eclipse is 16.6 minutes late, which is the time taken for light to travel AC. Roemer estimated that $c = 3\times10^8$ metres/sec, a value that is remarkably close to the best modern measurement, c = 299792458.6 metres/sec.



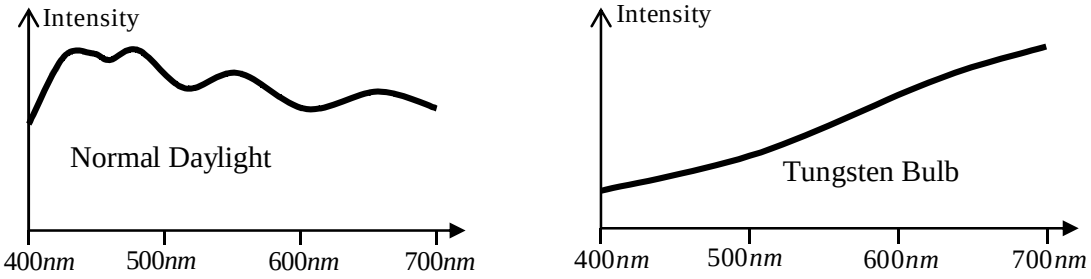
2. Light is electromagnetic waves. Different frequencies ν correspond to different colours. Equivalently, different wavelength λ correspond to different colours. Recall that the product $\nu \times \lambda = c$. In the diagram below you see that visible light is only one small part of the total electromagnetic spectrum. Here nm means nanometres or 10^{-9}metres.



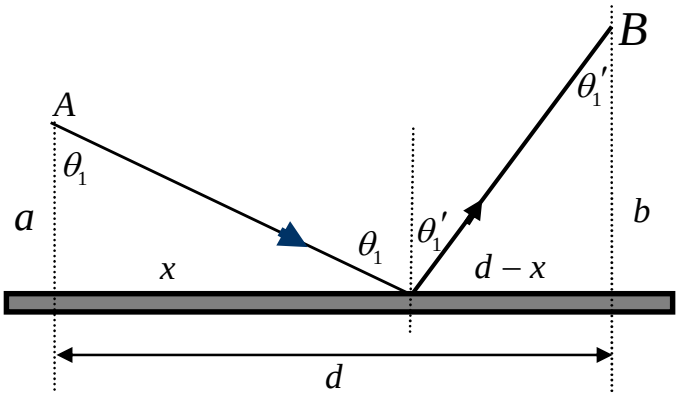
3. If light contained all frequencies with equal strength, it would appear as white to us. course, most things around us appear coloured. That is because they radiate more strongly in one range of frequency than in others. If there is more intensity in the yellow range than the green range, we will see mostly yellow. The sky appears blue to us on a clear day because tiny dust particles high above in the atmosphere reflect a lot of the blue light coming from the sun. In the figure below you can see the hump at smaller wavelengths.

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Compare this with the spectrum of light emitted from a tungsten bulb. You can see that this is smoother and yellow dominates.



4. What path does light travel upon? If there is no obstruction, it obviously likes to travel on straight line which is the shortest path between any two points, say A and B. Fermat's Principle states that in all situations, light will always take that path for which it takes the least time. As an example, let us apply Fermat's Principle to the case of light reflected from a mirror, as below.



The total distance travelled by the ray is $L=\sqrt{a^2+x^2}+\sqrt{b^2+(d-x)^2}$. The time taken is $t=L/c$. To find the smallest time, we must differentiate and then set the derivative to zero,

$$0=\frac{dt}{dx}=\frac{1}{c}\frac{dL}{dx}$$

$$=\frac{1}{2c}\left(a^2+x^2\right)^{-1/2}(2x)+\frac{1}{2c}\left[b^2+(d-x)^2\right]^{1/2}(2)(d-x)(-1)$$

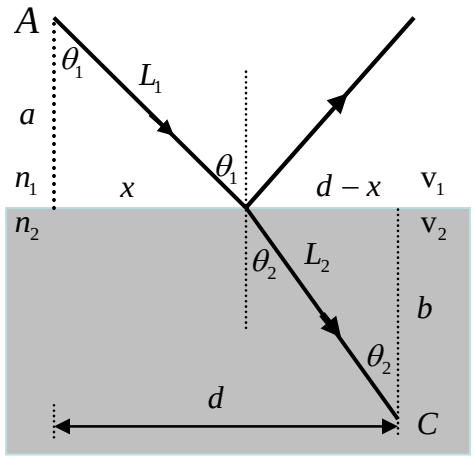
From here we immediately see that $\frac{x}{\sqrt{a^2+x^2}}=\frac{d-x}{\sqrt{b^2+(d-x)^2}}$ From the above diagram, $\sin\theta_1=\sin\theta_1'$. Of course, it is no surprise that the angle of reflection equals the angle of incidence. You knew this from before, and this seems like a very complicated derivation of a simple fact. But it is still nice to see that there is a deeper principle behind it.

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5. The speed of light in vacuum is a fixed constant of nature which we usually call *c*, but in a medium light can travel slower or faster than *c*. We define the "refractive index" of that medium as: $\text{Refractive index} = \dfrac{\text{Speed of light in vacuum}}{\text{Speed of light in material}}$ (or $n = \dfrac{c}{v}$). Usually the values of *n* are bigger than one (e.g. for glass it is around 1.5) but in some special media, its value can be less than one. The value of *n* also depends on the wavelength (or frequency) of light. This is called dispersion, and it means that different colours travel at different speeds inside a medium. This is why, as in the diagram below, white light gets separated into different colours. The fact that in a glass prism blue light travels faster than red light is responsible for the many colours we see here.



6. We can apply Fermat's Principle to find the path followed by a ray of light when it goes from one medium to another. Part of the light is reflected, and part is "refracted", i.e. it bends away or towards the normal. The total time is,



$$t = \frac{L_1}{v_1} + \frac{L_2}{v_2},\ n = \frac{c}{v} \Rightarrow t = \frac{n_1 L_1 + n_2 L_2}{c} = \frac{L}{c}$$

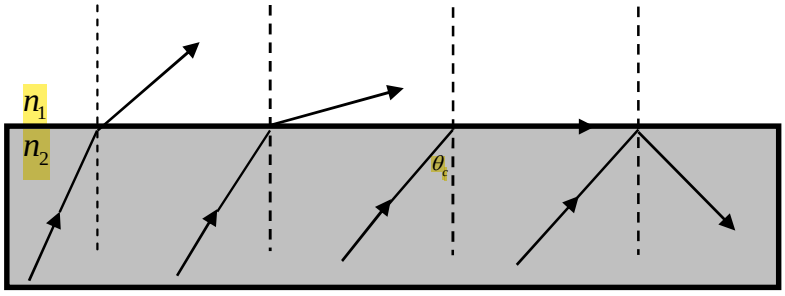
$$L = n_1 L_1 + n_2 L_2 = n_1\sqrt{a^2 + x^2} + n_2\sqrt{b^2 + (d-x)^2}$$

Fermat's Principle says that the time *t* must be minimized: $0 = \dfrac{dt}{dx} = \dfrac{1}{c}\dfrac{dL}{dx} = \dfrac{n_1}{2c}\left(a^2 + x^2\right)^{-1/2}(2x) + \dfrac{n_2}{2c}\left[b^2 + (d-x)^2\right]^{1/2}(2)(d-x)(-1)$

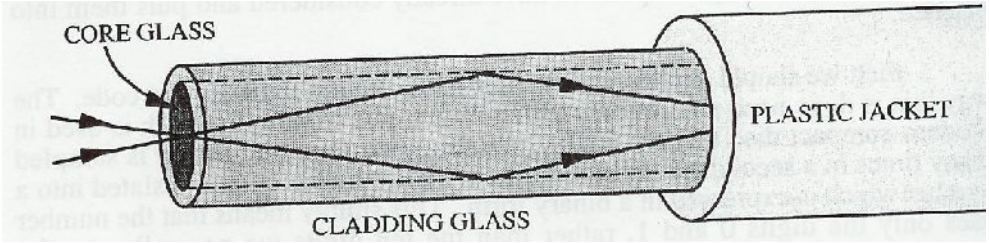
And so we get "Snell's Law", $n_1 \dfrac{x}{\sqrt{a^2 + x^2}} = n_2 \dfrac{d-x}{\sqrt{b^2 + (d-x)^2}}$ or $n_1 \sin\theta_1 = n_2 \sin\theta_2$. This required a little bit of mathematics, but you can see how powerful Fermat's Principle is!

7. Light coming from air into water bends toward the normal. Conversely, light from a source in the water will bend away from the normal. What if you keep increasing the angle with respect to the normal so that the light bends and begins to just follow the surface? This phenomenon is called total internal reflection and θ_c is called the critical

angle. It obeys: $n_1 \sin\theta_c = n_2 \sin 90^0$ from which $\theta_c = \sin^{-1}\frac{n_2}{n_1}$.



8. Fibre optic cables, which are now common everywhere, make use of the total internal reflection principle to carry light. Here is what a fibre optic cable looks like from inside:



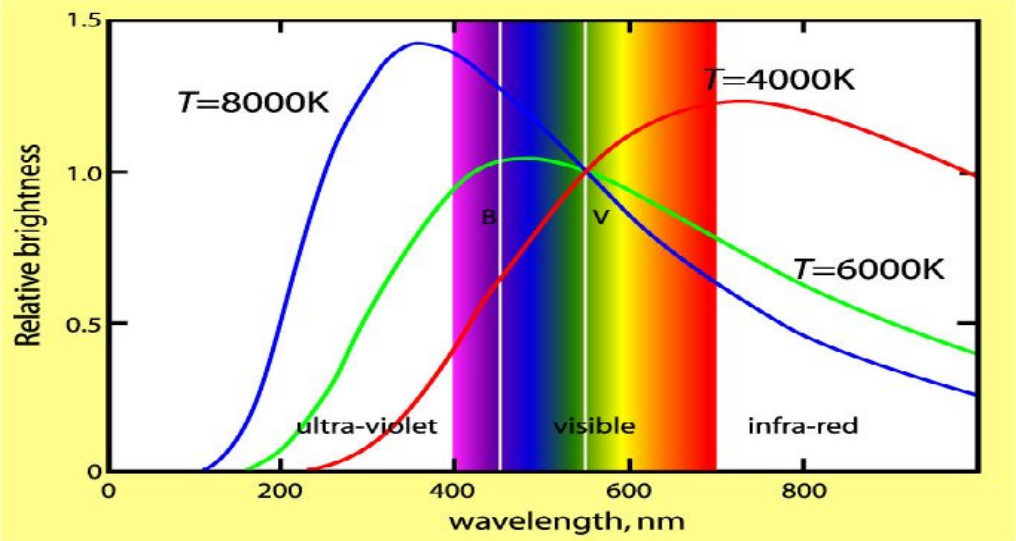
Even if the cable is bent, the light will continue to travel along it. The glass inside the cable must have exceedingly good consistency - if it thicker or thinner in any part, the refractive index will become non-uniform and a lot of light will get lost. Optical fibres now carry thousands of telephone calls in a cable whose diameter is only a little bigger than a human hair!

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Summary of Lecture 32 – INTERACTION OF LIGHT WITH MATTER

1. In this lecture I shall deal with the 4 basic ways in which light interacts with matter:
   a) Emission - matter releases energy as light.
   b) Absorption - matter takes energy from light.
   c) Transmission - matter allows light to pass through it.
   d) Reflection - matter repels light in another direction.

2. When an object (for example, an iron rod or the filament of a tungsten bulb) is heated, it emits light. When the temperature is around 800°C, it is red hot. Around 2500°C it is yellowish-white. At temperatures lower than 800°C, infrared (IR) light is emitted but our eyes cannot see this. This kind of emission is called blackbody radiation. Blackbody radiation is continuous - all wavelengths are emitted. However most of the energy is radiated close to the peak. As you can see in the graph, the position of the peak goes to smaller wavelengths (or higher frequencies) as the object becomes hotter. The scale of temperature is shown in degrees Kelvin (°K). To convert from °C to °K, simply add 273. We shall have more to say about the Kelvin scale later.



Where exactly does the peak occur? Wien's Law states that $\lambda_{max} T = 2.90 \times 10^{-3}$ m K. We can derive this in an advanced physics course, but for now you must take this as given.

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3. In the lecture on electromagnetic waves you had learnt that these waves are emitted when charges accelerate. Blackbody radiation occurs for exactly this reason as well. When a body is heated up, the electrons, atoms, and molecules which it contains undergo violent random motion. Light may emitted by electrons in one atom and absorbed in another. Even an empty box will be filled with blackbody radiation because the sides of the box are made up of material that has charged constituents that radiate energy when they undergo acceleration during their random motion.

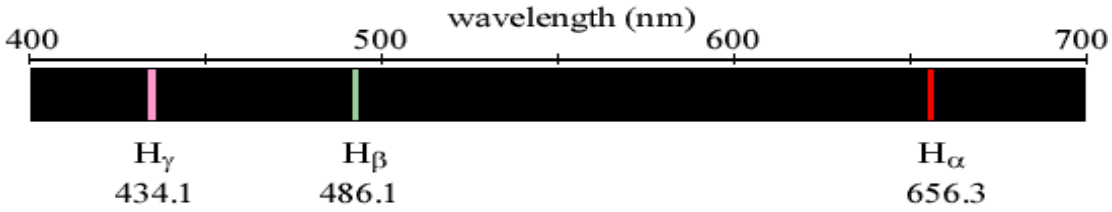
4. When can you use Wien's Law? More generally, when can you expect a body to emit blackbody radiation? Answer: only for objects that emit light, not for those that merely reflect light (e.g. flowers). The Sun and other stars obey Wien's Law since the gases they are composed of emit radiation that is in equilibrium with the other materials. Wien's law allows astronomers to determine the temperature of a star because the wavelength at which a star is brightest is related to its temperature.

5. All heated matter radiates energy, and hotter objects radiate more energy. The famous Stefan-Boltzman Law, which we unfortunately cannot derive in this introductory course, states that the power radiated per unit area of a hot body is $P = \sigma T^4$, where the Stefan-Boltzman constant is $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}.

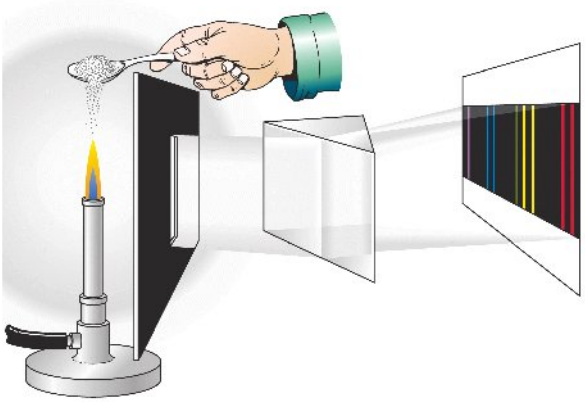
6. Let us apply $P = \sigma T^4$ for finding the temperature of a planet that is at distance R from the sun. The sun has temperature T_{sun} and radius R_{sun}. In equilibrium, the energy received from the sun is exactly equal to the energy radiated by the planet. Now, the total energy radiated by the sun is $\sigma T_0^4 \times 4\pi R_{sun}^2$. But on a unit area of the planet, only $\frac{1}{4\pi R^2}$ of this is received. So the energy received per unit area on the planet is $\sigma T_0^4 \times 4\pi R_{sun}^2 \times \frac{1}{4\pi R^2}$. This must be equal to $\sigma T^4 \Rightarrow \quad T = T_0 \sqrt{\frac{R_{sun}}{R}}$.

7. The above was for blackbody radiation where the emitted light has a continuous spectrum. But if a gas of identical atoms is excited by some mechanism, then only a few discrete wavelengths are emitted. Each chemical element produces a very distinct pattern of colors called an emission spectrum. So, for example, laboratory hydrogen gas lamps emit 3 lines in the visible region, as you can see below. Whenever we see 3 lines spaced apart in this way, we immediately know that hydrogen gas is present. It is as good as the thumbprint of a man!

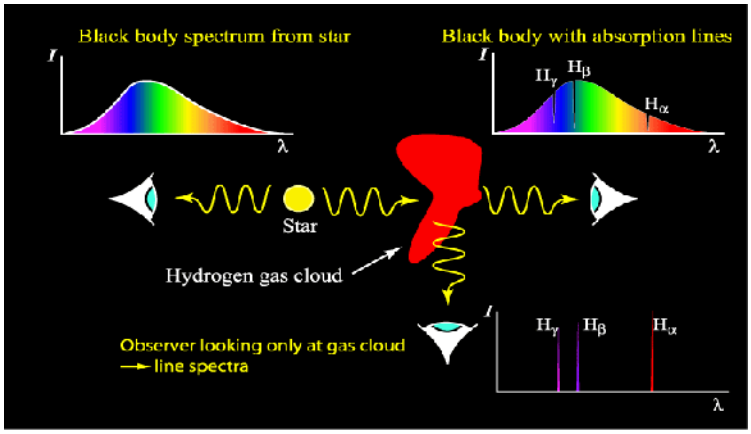
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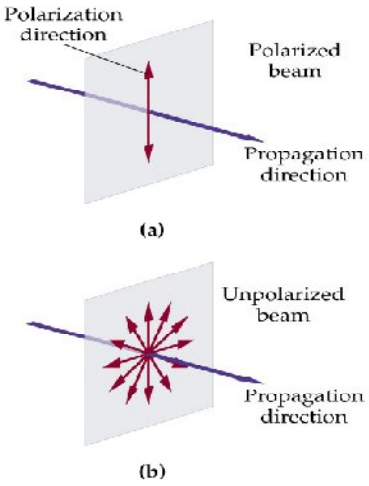


8. But how do we get atoms excited so that they can start revealing their identity? One way is to simply heat material containing those atoms. You saw in the lecture how different colours come from sprinkling different materials on a flame.



9. Everything that I have said about the emission of light applies exactly to the *absorption* of light as well. So, for example, when white light (which has all different frequencies within it) passes through hydrogen gas, you will see that all wavelengths survive except the three on the previous page. So the absorption spectrum looks exactly the same as the emission spectrum - the same lines are emitted and absorbed. This how we know that there are huge clouds of hydrogen floating in outer space. See the diagram below.



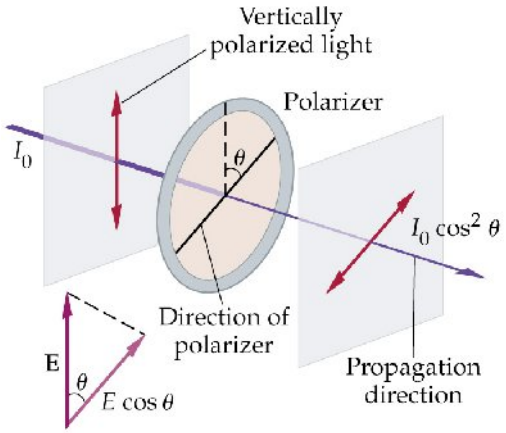


11. As you learned earlier, light is an electromagnetic wave that has an electric field vector. This vector is always perpendicular to the direction of travel, but it can be pointing anywhere in the plane. If it is pointing in a definite direction, we say that the wave is polarized, else it is unpolarized. In general, the light emitted from a source, such as a flame, will be unpolarized and it is equally possible to find any direction of the electric field in the wave. Of course, the magnetic field is perpendicular to both the direction of travel and the electric field.

10. The atmosphere contains various gases which absorb light at many different wavelengths. Molecules of oxygen, nitrogen, ozone, and water have their own absorption spectra, just as atoms have their own.

11. All the beauty of colours we see is due to the selective absorption by molecules of certain frequencies. So, for example, *carotene* is a long, complicated molecule that makes carrots orange, tomatoes red, sarson yellow, and which absorbs blue light. Similarly *chlorophyll* makes leaves green and which absorbs red and blue light.

12. From unpolarized light we can make polarized light by passing it through a polarizer as shown below.



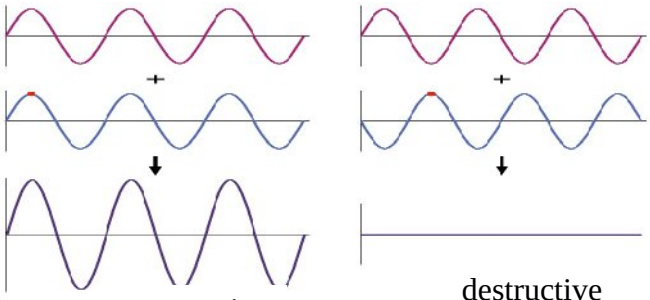
Each wave is reduced in amplitude by $\cos\theta$, and in intensity by $\cos^2\theta$. The wave that emerges is now polarized in the θ direction.

13. We can design materials (crystals or stressed plastics) so that they have different optical properties in the two transverse directions. These are called birefringent materials. They are used to make commonly used liquid crystal displays (LCD) in watches and mobile phones. Birefringence can occur in any material that possesses some asymmetry in its structure where the material is more springy in one direction than another.

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Summary of Lecture 33 – INTERFERENCE AND DIFFRACTION

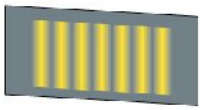
1. Two waves (of any kind) add up together, with the net result being the simple sum of the two waves. Consider two waves, both of the same frequency, shown below. If they start together (i.e. are *in phase* with each other) then the net amplitude is increased. This is called *constructive interference.* But if they start at different times (i.e. are *out of phase* with each other) then the net amplitude is decreased. This is called *destructive interference.*

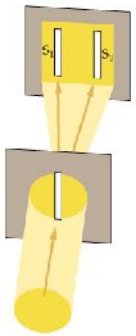


constructive destructive

In the example above, both waves have the same frequency and amplitude, and so the resulting amplitude is doubled (constructive) or zero (destructive). But interference occurs for any two waves even when their amplitudes and frequencies are different.

2. Although any waves from different sources interfere, if one wants to observe the interference of light then it is necessary to have a *coherent* source of light. Coherent means that both waves should have a fixed phase relative to each other. Even with lasers, it is very difficult to produce coherent light from two separate sources. Observing interference usually requires taking two waves from a single source, with each going along a different path. In the figure, an incoherent light source illuminates the first slit. This creates a uniform and coherent illumination of the second screen. Then waves from the slits S_1 and S_2 meet on the third screen and create a pattern of alternating light and dark fringes.

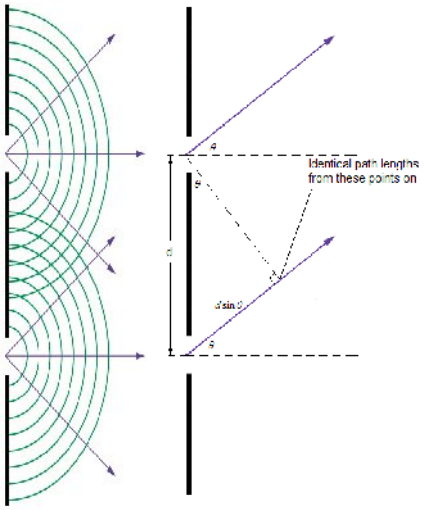




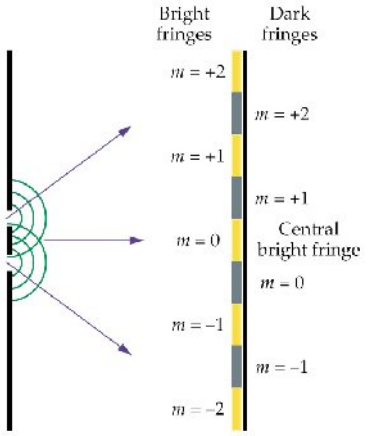
3. Wherever there is a bright fringe, constructive interference has occured, and wherever there is a dark fringe, destructive interference has occured. We shall now calculate where on the third screen the interference is constructive. Take any point on the third screen. Light reaches this point from both S_1 and S_2, but it will take different amounts of time to get there. Hence there will be a phase difference that we can calculate. Look at the diagram below. You can see that light from one of the slits has to travel an extra

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distance equal to $d\sin\theta$, and so the extra amount of time it takes is $(d\sin\theta)/c$. There will be constructive interference if this is equal to $T, 2T, 3T, \cdots$ (remember that the time period is inversely related to the frequency, $T = 1/\nu$, and that $c = \lambda\nu$). We find that $\frac{d\sin\theta}{c} = nT = \frac{n}{\nu} \Rightarrow d\sin\theta = n\lambda$, where $n = 1, 2, 3, \cdots$ What about for destructive interference? Here the waves will cancel each other if the extra amount of time is $\frac{T}{2}, \frac{3T}{2}, \frac{5T}{2}, \cdots$ The condition then becomes $d\sin\theta = (n + \frac{1}{2})\lambda$.



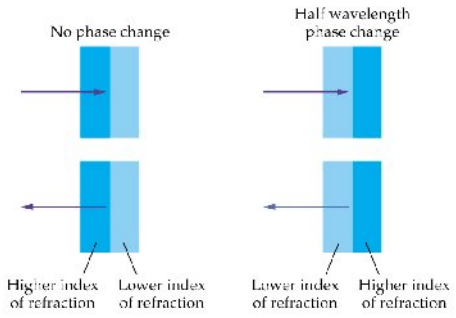
4. Example: two slits with a separation of 8.5×10^{-5}m create an interference pattern on a screen 2.3m away. If the $n = 10$ bright fringe above the central is a linear distance of 12 cm from it, what is the wavelength of light used in the experiment?



Answer: First calculate the angle to the tenth bright fringe using $y = L\tan\theta$. Solving for θ gives, $\theta = \tan^{-1}\left(\frac{y}{L}\right) = \tan^{-1}\left(\frac{0.12m}{2.3m}\right) = 3.0^{\circ}$. From this,

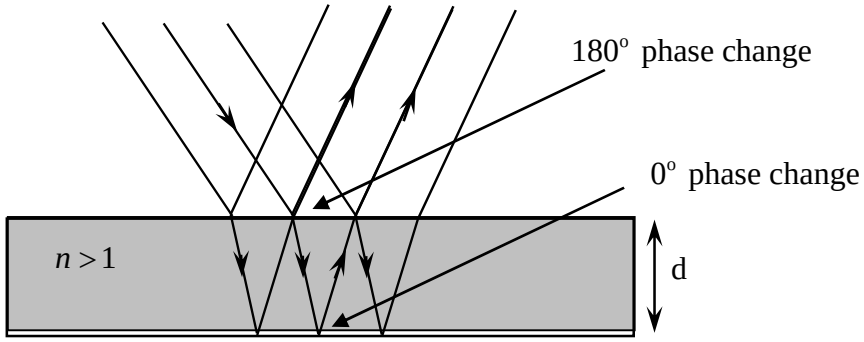
$$\lambda = \frac{d}{n}\sin\theta = \left(\frac{8.5\times10^{-5}}{10}\right)\sin(3.0^{\circ}) = 4.4\times10^{-7}\,m = 440nm \text{ (nanometres).}$$

5. When a wave is reflected at the interface of two media, the phase will not change if it goes from larger refractive index to a smaller one. But for smaller to larger, there will be phase change of a half-wavelength. One can show this using Maxwell's equations and applying the boundary conditions, but this will require some more advanced studies. Instead let's just use this fact below.



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5. When light falls upon a thin film of soapy water, oil, etc. it is reflected from two surfaces. On the top surface, the reflection is with change of phase by π whereas at the lower surface there is no change of phase. This means that when waves from the two surfaces combine at the detector (your eyes), they will interfere.



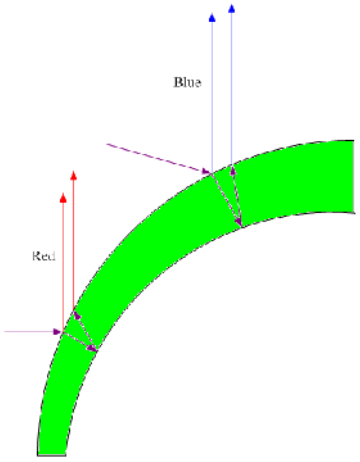
To simplify matters, suppose that you are looking at the thin film almost directly from above. Here n is the index of refraction for the medium. Then,

The condition for destructive interference is: $2nd = m\lambda \quad (m = 0,1,2,...)$

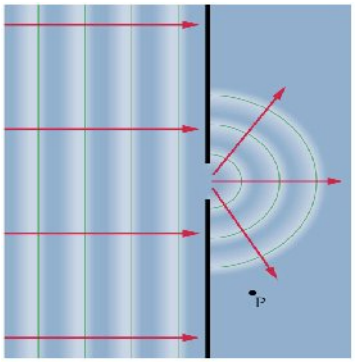
The condition for constructive interference is : $2nd = (m + \frac{1}{2})\lambda \quad (m = 0,1,2,...).$

Prove it!

6. Interference is why thin films give rise to colours. A drop of oil floating on water spreads out until it is just a few microns thick. It will have thick and thin portions. Thick portions of any non-uniform thin film appear blue because the long-wavelength red light experiences destructive interference. Thinner portions appear red because the short-wavelength blue light interferes destructively.



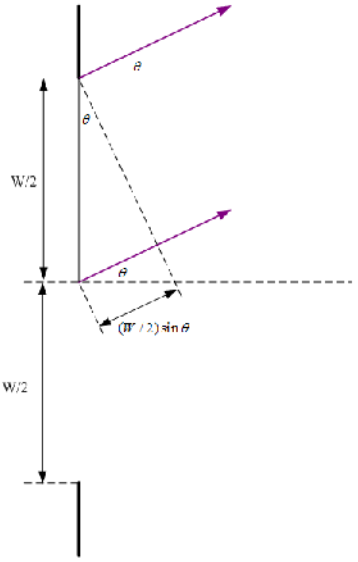
7. **Diffraction :** The bending of light around objects (into what would otherwise be a shadowed region) is known as diffraction. Diffraction occurs when light passes through very small apertures or near sharp edges. Diffraction is actually just interference, with the difference being only in the source of the interfering waves. Interference from a single slit, as in the figure here, is called diffraction.



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We can get an interference pattern with a single slit provided its size is approximately equal to the wavelength of the light (neither too small nor too large).

8. Let's work out the condition necessary for diffraction of light from a single slit. With reference to the figure, imagine that a wave is incident from the left. It will cause secondary waves to be radiated from the edges of the slit. If one looks at angle θ, the extra distance that the wave emitted from the lower slit must travel is $W \sin\theta$. If this is a multiple of the wavelength λ, then constructive interference will occur. So the condition becomes $W \sin\theta = m\lambda$ with $m = \pm 1, \pm 2, \pm 3...$ So, even from a single slit one will see a pattern of light and dark fringes when observed from the other side.

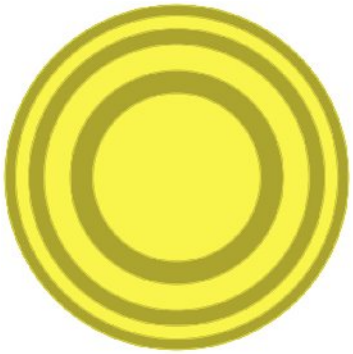


9. Light with wavelength of 511 nm forms a diffraction pattern after passing through a single slit of width 2.2×10^{-6}m. Find the angle associated with (a) the first and (b) the second bright fringe above the central bright fringe.

SOLUTION: For $m = 1$, $\theta = \sin^{-1}\left(\frac{m\lambda}{W}\right) = \sin^{-1}\left(\frac{(1)(511\times10^{-9}m}{2.20\times10^{-6}m}\right) = 13.4^{\circ}$

For $m = 2$, $\theta = \sin^{-1}\left(\frac{m\lambda}{W}\right) = \sin^{-1}\left(\frac{(2)(511\times10^{-9}m}{2.20\times10^{-6}m}\right) = 27.7^{\circ}$

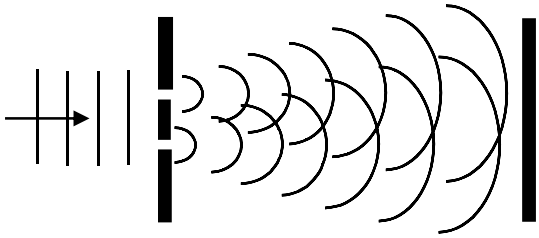
10. Diffraction puts fundamental limits on the capacity of telescopes and microscopes to separate the objects being observed because light from the sides of a circular aperture interferes. One can calculate that the first dark fringe is at $\theta_{\min} = 1.22\frac{\lambda}{D}$, where D is diameter of the aperture. Two objects can be barely resolved if the diffraction maximum of one object lies in the diffraction minimum of the second object. Clearly, the larger D is, the smaller the angular diameter separation. We say that larger apertures lead to better resolution.



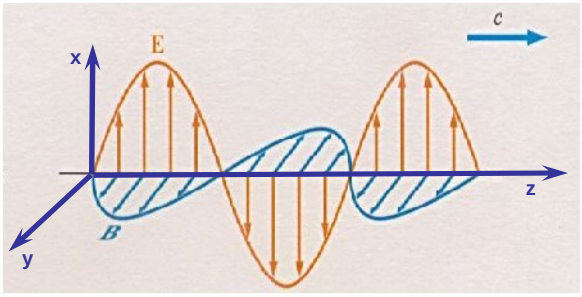
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Summary of Lecture 34 – THE PARTICLE NATURE OF LIGHT

1. In the previous lecture I gave you some very strong reasons to believe that light is waves. Else, it is impossible to explain the interference and diffraction phenomena that we see in innumerable situations. Interference from two slits produces the characteristic pattern.

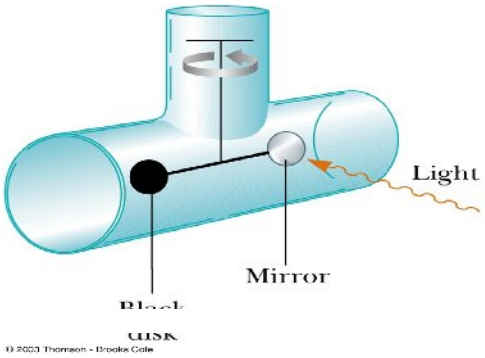


2. Light is waves, but waves in what? of what? The thought that there is some invisible medium (given the name *aether*) turned out to be wrong. Light is actually electric and magnetic waves that can travel through empty space. The electric and magnetic waves



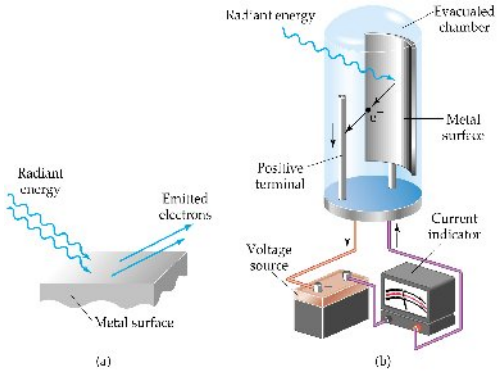
are perpendicular to each other and to the direction of travel (here the z direction).

3. Electromagnetic waves transport linear momentum and energy. If the energy per unit volume in a wave is U then it is carrying momentum p, where $p = U/c$. Waves with large amplitude carry more energy and momentum. For the sun's light on earth the momentum is rather small (although it is very large close to or inside the sun). Nevertheless, it is easily measureable as, for example, in the apparatus below. Light strikes a mirror and rebounds.



Thus the momentum of the light changes and this creates a force that rotates the mirror. The force is quite small - just 5×10^{-6} Newtons per unit area (in metre^2) of the mirror.

4. So strong was the evidence of light as waves that observation of the photoelectric effect came as a big shock to everybody. In the diagram below, light hits a metal surface and

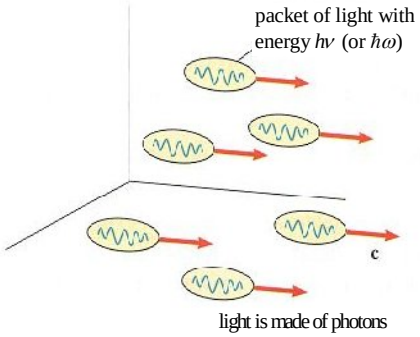


and knocks out electrons that travel to the anode. A current flows only as long as the light is shining. Above the threshold frequency, the number of electrons ejected depends on the intensity of the light. This was called the photoelectric effect. The following was observed:

a) The photoelectric effect does not occur for all frequencies ν; it does not occur at all when ν is below a certain value. *But classically (meaning according to the Maxwell nature of light as an electromagnetic wave) electrons should be ejected at any ν. If an electron is shaken violently enough by the wave, it should surely be ejected!*

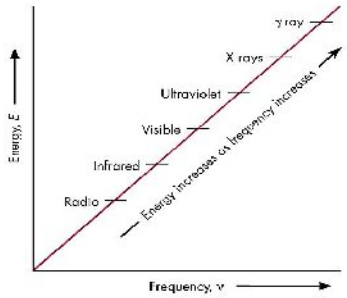
b) It is observed that the first photoelectrons are emitted instantaneously. *But classically the first photoelectrons should be emitted some time after the light first strikes the surface and excites its atoms enough to cause ionization of their electrons.*

5. Explanation of the photoelectric puzzle came from Einstein, for which he got the Nobel Prize in 1905. Einstein proposed that the light striking the surface was actually made of little packets (called quanta in plural, quantum in singular). Each quantum has an energy energy $\varepsilon = h\nu$ (or $\varepsilon = \hbar\omega$ where $\hbar = h/2\pi$ and $\omega = 2\pi\nu$) where h is the famous Planck's constant, 6.626×10^{-34} Joule-seconds. An electron is kicked out of the metal only when a quantum has energy (and frequency) big enough to do the job. It doesn't matter how many quanta of light - called photons - are fired at the metal. No photoelectrons will be released unless ν is large enough. Furthermore the photoelectrons are released immediately when the photon hits an electron.



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6. Microwaves, radio and TV waves, X-rays, γ-rays, etc. are all photons but of very different frequencies. Because Planck's constant h is an extremely small number, the energy of each photon is very small.



7. How many photons do we see? Here is a table that gives us some interesting numbers:

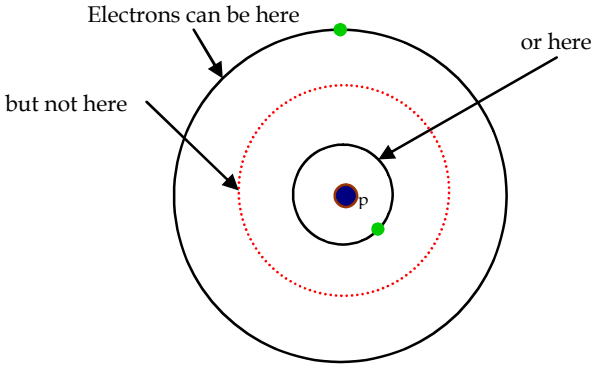
   a) Sunny day (outdoors): 10^{15} photons per second enter eye (2 mm pupil).

   b) Moonlit night (outdoors): 5×10^{10} photons/sec (6 mm pupil).

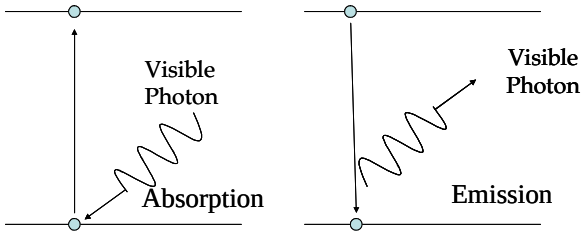
   c) Moonless night (clear, starry sky): 10^{8} photons/sec (6 mm pupil).

   d) Light from dimmest naked eye star (mag 6.5): 1000 photons/sec entering eye.

8. Where do photons come from? For this it is necessary to first understand that electrons inside an atom can only be in certain definite energy states. When an electron drops from a state with higher energy to one with lower energy, a photon is released whose energy is exactly equal to the difference of energies. Similarly a photon is absorbed when a photon of just the right energy hits an electron in the lower state and knocks it into a higher state.

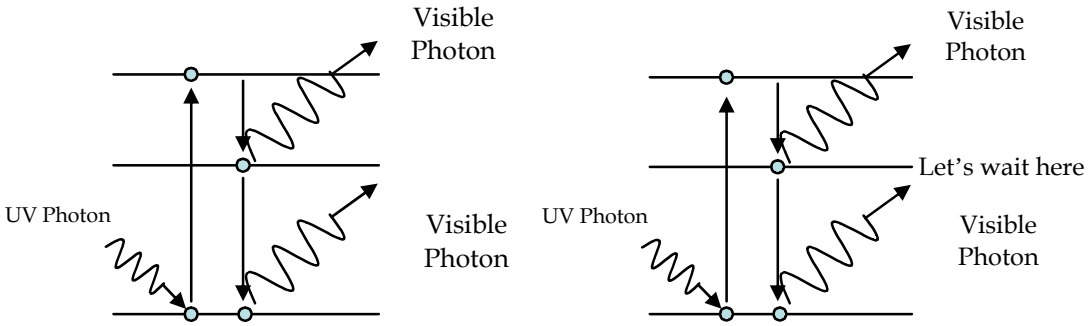


The upper and lower levels can be represented differently with the vertical direction representing energy. The emission and absorption of photons is shown below.



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9. Fluorescence and phosphorence are two phenomena observed in some materials. When they are exposed to a source of light of a particular colour, they continue to emit light of a different colour even after the source has been turned off. So these materials can be seen to glow even in the dark. In phosphorescence, a high-frequency photon raises an electron to an excited state. The electron jumps to an intermediate states, and then drops after a little while to the ground state. This is fluorescence. Phosphorescence is similar to fluorescence, except that phosphorescence materials continue to give off a secondary glow long (seconds to hours) after the initial illumination that excited the atoms.



10. One of the most important inventions of the 20th century is the laser which is short for:

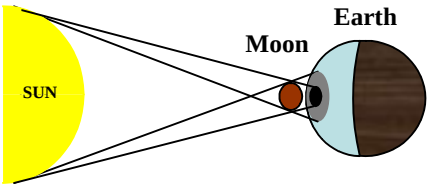
LASER ≡ **L**ight **A**mplification by the **S**timulated **E**mission of **R**adiation

Lasers are important because they emit a very large number of photons all with one single frequency. How is this done? By some means - called optical pumping - atoms are excited to a high energy level. When one atom starts decaying to the lower state, it encourages all the others to decay as well. This is called spontaneous emission of radiation.

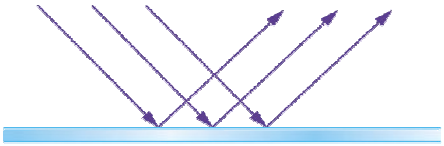
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Summary of Lecture 35 – GEOMETRICAL OPTICS

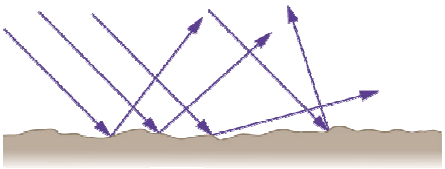
1. In the previous lecture we learned that light is waves, and that waves spread out from every point. But in many circumstances we can ignore the spreading (diffraction and interference), and light can then be assumed to travel along straight lines as rays. This is hown by the existence of sharp shadows, as for case of the eclipse illustrated here.



2. When light falls on a flat surface, the angle of incidence equals the angle of reflection. You can verify this by using a torch and a mirror, or just by sticking pins on a piece of paper in front of a mirror. But what if the surface is not perfectly flat? In that case, as shown in fig. (b), the angle of incidence and reflection are equal at every point, but the normal direction differs from point to point. This is called "diffuse reflection". Polishing a surface reduces the diffusiveness.

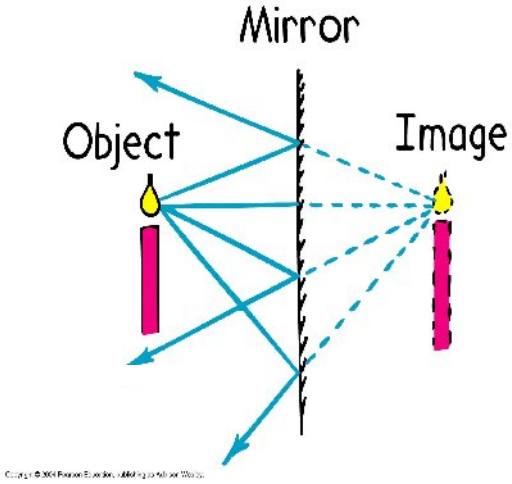


(a) Specular reflection



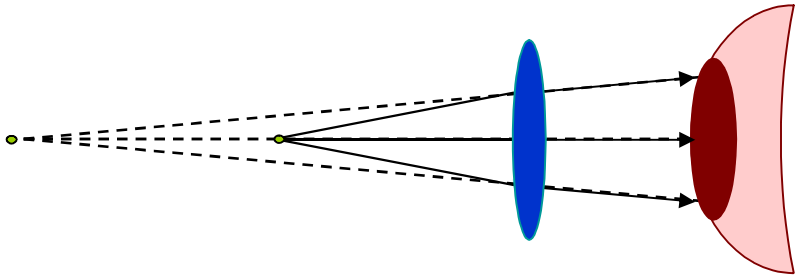
(b) Diffuse reflection

3. If you look at an object in the mirror, you will see its image. It is not the real thing, and that is why it is called a "virtual" image. You can see how the virtual image of a candle is formed in this diagram. At each point on the surface, there is an incident and reflected ray. If we extend each reflected ray backwards, it appears as if they are all coming from the same point. This point is the image of the tip of the flame. If we take other points on the candle, we will get their images in just the same way. This way we will have the image of the whole candle. The candle and its image are at equal distance from the mirror.

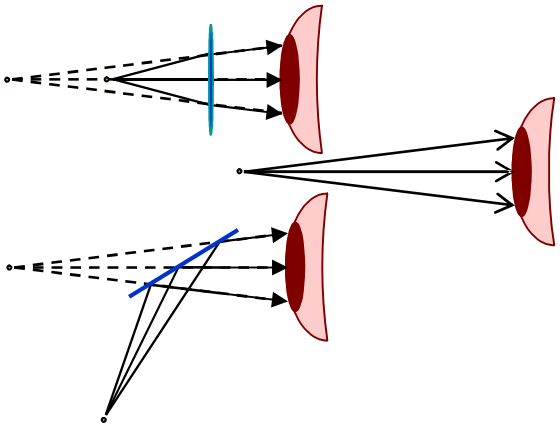


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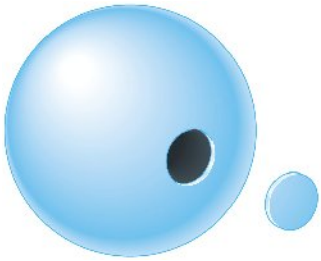
4. Here is another example of image formation. A source of light is placed in front of a bi-convex lens which bends the light as shown. The eye receives rays of light which seem to originate from a positon that is further away than the actual source.

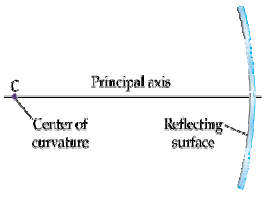


Now just to make the point even more forcefully, in all three situations below, the virtual image is in the same position although the actual object is in 3 different places.

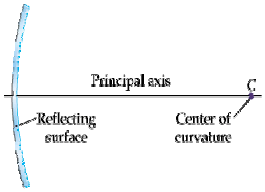


5. Imagine that you have a sphere of radius R and that you can cut out any piece you want. The outside or inside surface can be silvered, as you want. You can make spherical mirrors in this way. These can be of two kinds. In the first case, the silvering can be on the inside surface of the sphere, in which case this is is called a convex spherical mirror. The normal directed from the shiny surface to the centre of the sphere (from which it was cut out from) is called the principal axis, and the radius of curvature is *R*. The other situation is that in which the outside surface is shiny. Again, the principal axis the same, but now the radius of curvature (by definition) is - *R*. What does a negative curvature mean? It means precisely what has been illustrated - a convex surface has a positive and a concave surface has a negative curvature.





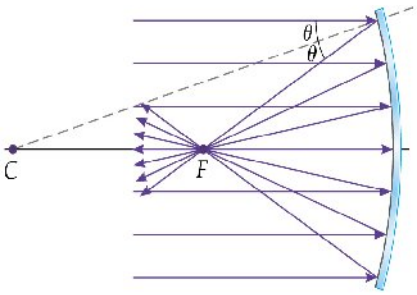
(a)



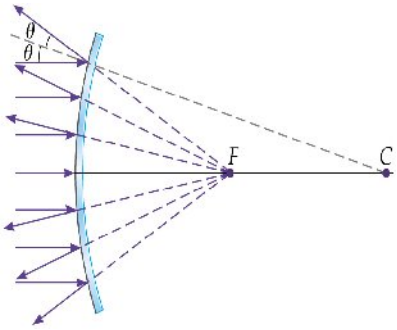
(b)

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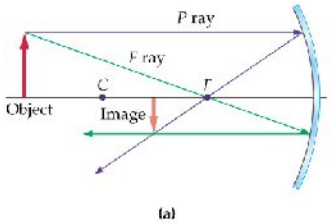
6. Now we come to the important point: a beam of parallel rays will reflect off a convex mirror and all get together (or converge) at one single point, called the focus. Now look at the figure. Since the incident and reflected angles are equal for every ray, you can see that the ray will have to pass through the focus at F, at distance $R/2$. We say that the focal length of such a mirror is $f = R/2$.



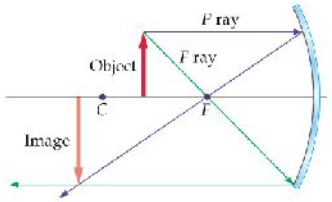
7. The concave mirror is different. A parallel beam incident from the left is reflected and the rays now spread out (diverge). However, if we were to look at any outgoing ray and extend it backward, all the rays would appear to be coming from one single point. This is the *virtual focus* because there is, in fact, no real source of light there. From the diagram you can see the virtual focus F is at distance $R/2$ *behind* the mirror. We say that the focal length of such a mirror is $f = -R/2$.



8. Now let's see how images are formed by concave mirrors. Take two rays that are emitted from the top of an object placed in front of the mirror. Call them the P and F ray, and see what happens to them after reflection. Where the two cross, that is the image point. As you can see the image is inverted and smaller than the actual object if the object is far away (i.e. lies to the left of C) and is larger if close to the mirror. The magnification of size makes this useful as a shaving mirror, among other things.

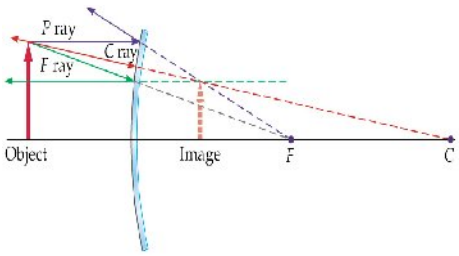


(a)



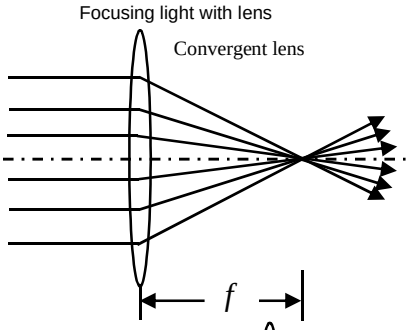
(b)

9. Now repeat for a concave mirror. Here the rays will never actually meet so we can have only a virtual object. Take 3 rays - call them P,F, and C - and see how they are reflected from the mirror's surface. Where they meet is the position of the image. The image is always smaller than the actual size of the object. This is obviously useful for driving a car because you can see a wide area.

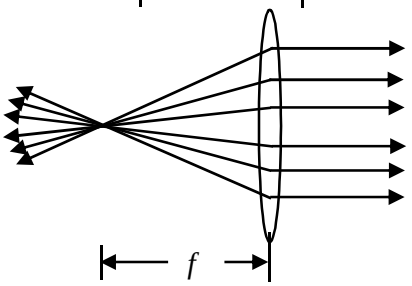


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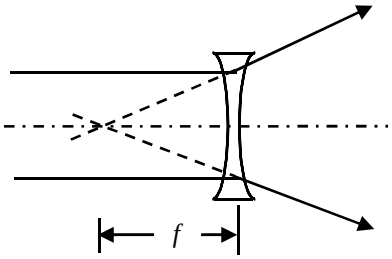
10. A lens is a piece of glass curved in a definite way. Because the refractive index of glass is bigger than one, every ray bends towards the normal. Here you see a double convex lens that focuses a beam of parallel rays. For a perfect lens, all the rays will converge to one single point that is (again) called the focus. The distance f is called the focal length.



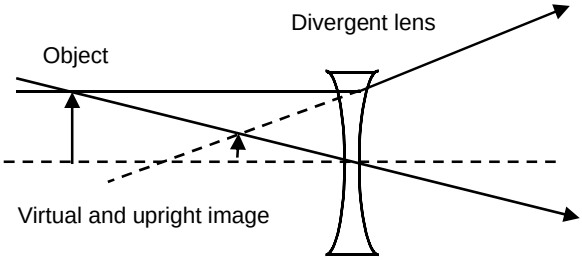
Of course, light can equally travel the other way, so if a point source is placed at the focus of a convex lens then a parallel beam of light will emerge from the other side. This is how some film projectors produce a parallel beam.



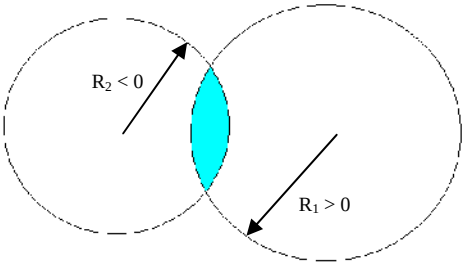
A concave lens does not cause a parallel beam to converge. On the contrary, it makes it diverge, as shown. Note, however, that if the rays are continued backwards, then they appear to come from one single point, which is here the virtual focus. The distance f is the focal length.



12. Here is how a concave (or divergent) lens forms an image. An observer on the side opposite to the object will see the image upright and smaller in size than the object.

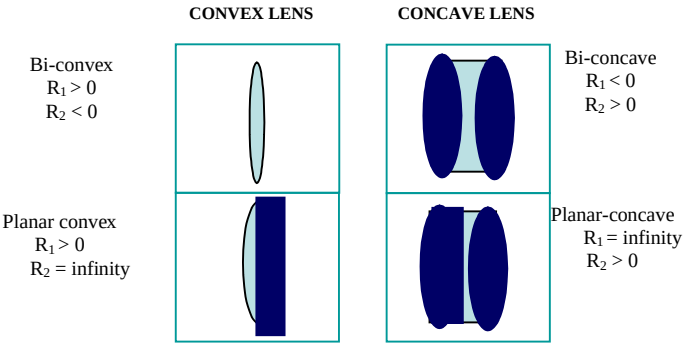


13. A lens can be imagined as cut out of two spheres of glass as shown, with the spheres having radii R_1 and R_2. Note that they tend to bend light in opposite ways. By convention, R_2 is negative. It is possible to show that the focal length of the lens is , $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$.



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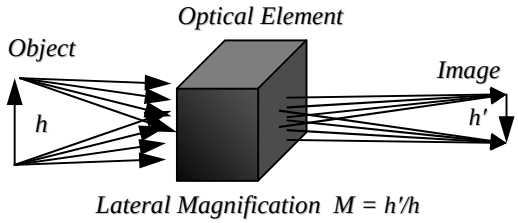
14. The following figure summarizes the shape of some common types of lenses.



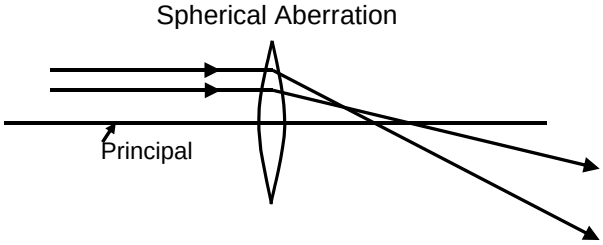
Note that a flat surface has infinite radius of curvature. The focal legth of each can be calculated using the previous formula.

15. The "strength" of a lens is measured in *diopters*. If the focal length of a lens is expressed in metres, the diopter of the lens is defined as $D = 1/f$. If the refractive index of the glass in a lens is n, then the diopters due to the first interface D_1 and the second interface D_2 are, $D_1 = (n-1)/R_1$ and $D_2 = (1-n)/R_2$. The total diopter of the lens is $D = D_1 + D_2$.

16. For any optical system - meaning a collection of lenses and mirrors - we can define a magnification factor as a ratio of sizes -- see the diagram below.



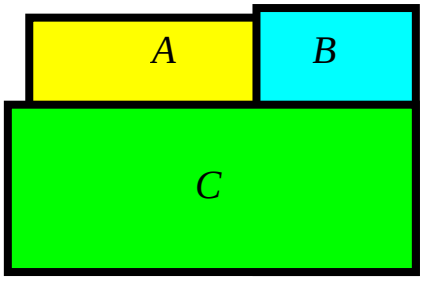
17. The perfect lens will focus a parallel beam of rays all to exactly the same focus. But no lens is perfect, and every lens suffers from aberration although this can be made quite small by following one lens with another. Below you see an example of "spherical aberration". Rays crossing different parts of the lens do not reach exactly the same focus. This distorts the image. Computers can design lens surfaces to minimize this aberration.



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Summary of Lecture 36 – THERMAL PHYSICS I

1. The ancient view was that heat is a colourless, weightless, fluid which occupies no volume has no smell, etc. This imagined substance - called phlogiston - was supposedly stored in objects and transferred between objects. It took a long time to reject this notion. Why is it wrong? Because (as we will see) heat can be created and destroyed, whereas liquids keep their volume and cannot be created or destroyed.
2. We are all familiar with an intuitive notion of temperature. We know that hotter things have higher temperature. But let us try to define temperature more rigorously. In the diagram below, the three bodies A,B,C are in contact with each other. After sufficient time passes, one thing will be common to all three - a quantity that we call temperature.



3. Now let us understand heat. Heat is energy, but it is a very special kind of energy: it is that energy which flows from a system at high temprature to a system at low temperature.

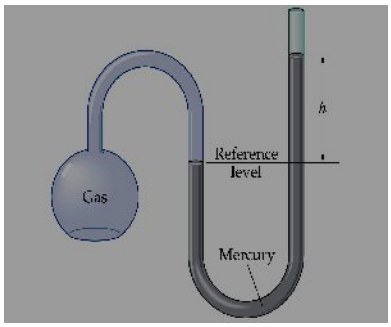


   Stated in a slightly different way: heat is the flow of internal energy due to a temperature difference. (Note that we do not have to know about atoms, molecules, and the internal composition of a body to be able to define heat - all that will come later).
4. The term "thermal equilibrium" is extremely important to our understanding of heat. If some objects (say, a glass of water placed in the open atmosphere) are put in thermal contact but there is no heat exchange, then we say that the objects are in thermal equilibrium. So, if the glass of water is hot or cold initially, after sufficient time passes it will be in thermal equilibrium and will neither receive nor lose heat to the atmosphere.

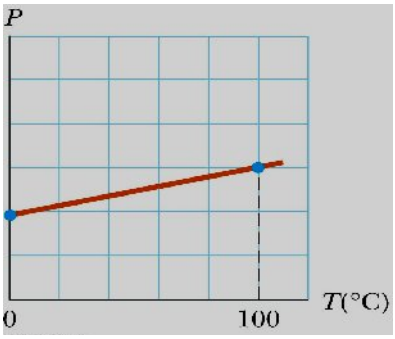
5. We have an intuitive understanding of temperature, but how do we measure it? Answer: by looking at some physical property that changes when the temperature changes. So, for example, when the temperature rises most things expand, the electrical resistance changes, some things change colour, etc. These are called "thermometric properties".

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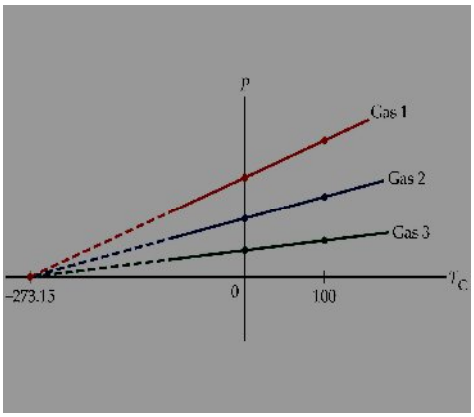
6. Let's take a practical example: the constant volume gas thermometer. Here the reference level is kept fixed by raising or lowering the tube on the right side (the tube below is made of rubber or some flexible material). So the gas volume is fixed. The gas pressure is $\rho \times g \times h$, and so the pressure is known from measuring h.



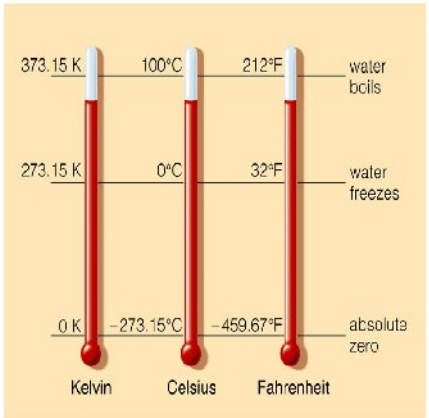
To find the temperature of a substance, the gas flask is placed in thermal contact with the substance. When the temperature is high, the pressure is large. From the graph of pressure versus temperature, you can easily read off the temperature. Note that we are using two fixed points, which we call 0^0C and 100^0C. This is called the Centigrade scale, and the two fixed points correspond to the freezing and boiling of water.



7. There is a temperature below which it is not possible to go. In other words, if you cool and cool there comes a point after which you cannot cool any more! How do we know this? Take different gases and plot how their pressure changes as you cool them down. You can see from the graph that all the lines, when drawn backwards, meet exactly at one point. This point is the absolute zero of temperature and lies at about -273.15^0C. This is also called 0^0K, or zero degrees Kelvin.



8. Temperature Scales. If you had been born 300 years ago, and if you had discovered a reliable way to tell different levels of "hotness", then maybe today there would be a temperature scale named after you! The relation between Centigrade, Fahrenheit, and Kelvin scales is illustrated to your right. You can see that absolute zero corresponds to -460^0K, -273.15^0C, and 0^0K. The Kelvin scale is the most suited for scientific purposes, and you should be careful to use this in all heat related calculations.



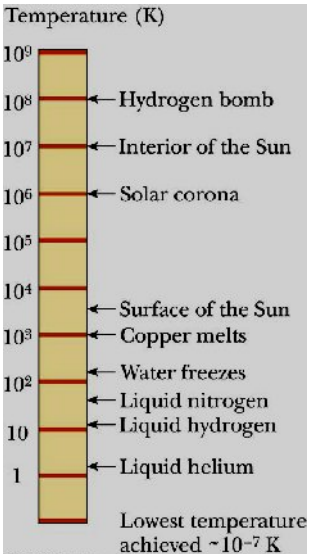
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Conversion between degrees Celsius and degrees Fahrenheit: $T_F = (\frac{9}{5}F^\circ / C^\circ)T_C + 32^\circ F$

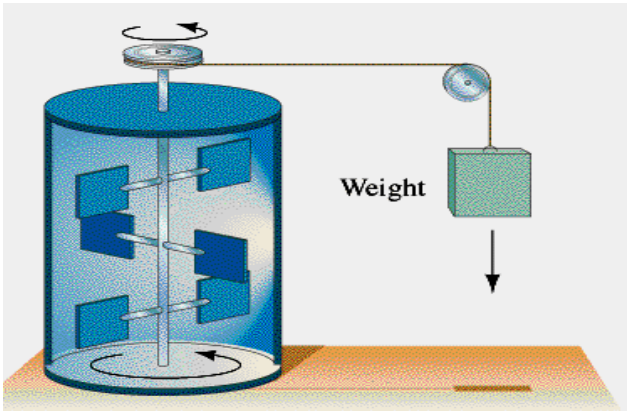
Conversion between degrees Fahrenheit and degrees Celsius: $T_C = (\frac{5}{9}C^\circ / F^\circ)(T_F - 32^\circ F)$

Conversion between Celsius and Kelvin temperatures: $T = T_C + 273.15$

9. How hot is hot, and how cold is cold? Whenever a scientist says something is large or small, it is always relative to something. The hottest thing ever was the universe when it just came into existence (more about this in the last lecture). After this comes the hydrogen bomb (10^8K). The surface of the sun is not so hot, only 5500K. Copper melts around 1000K, water turns to steam at 373K. If you cool further, then all the gases start to solidify. The lowest temperature that has ever been achieved is $10^{-7}K$, which is one tenth of one millionth of one degree! Why not still lower? We shall later why this is not possible.



10. When you rub your hands, they get hot. Mechanical work has been converted into heat. The first person to investigate this scientifically was Joule. In the experiment below, he allowed a weight to drop. This turned a paddle that stirred up the water and caused the temperature to rise. The water got hotter if the weight was released from a greater height.



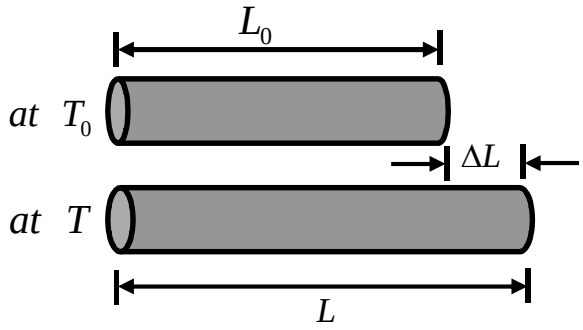
Joule established the units for the mechanical equivalent of heat. The units we use today are: 1 calorie (1 cal) raises the temperature of 1 g of water by 1 °C

1 cal = 4.186 Joule, 1 kilocalorie (1 kcal) = 1000 cal

Remember also that joule is the unit of work: when a force of one newton acts through a distance of one metre, the work done is one joule.

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11. Let's now consider one important effect of heat - most things expand when heated. Of course, our world is 3-dimensional but if there is a thin long rod then the most visible effect of heating it is that the rod increases in length. By how much?



Look at the above diagram. Call the length of the rod at T_0 as L_0. When the rod is heated to T, then the length increases to $L = L_0\left[1+\alpha\left(T-T_0\right)\right]$. The difference of the lengths is, $L - L_0 = \alpha L_0\left(T - T_0\right)$, or $\Delta L = \alpha L_0 \Delta T$. Here α is just a dimensionless number that tells you how a particular material expands. It is called the coefficient of linear expansion. If α was zero, then the material would not expand at all. You can also write it as $\alpha = \dfrac{\Delta L / L_0}{\Delta T}$.

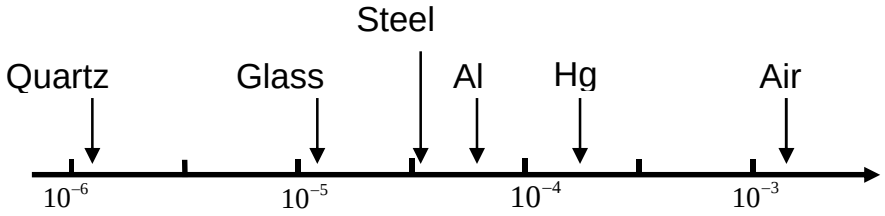
Similarly, define a coefficient of volume expansion – call it β – as $\beta \equiv \dfrac{\Delta V / V_0}{\Delta T}$.

12. There is a relation between α and β, the linear and volume coefficients. Let's look at the change in volume due to expansion: $V' = (L + \Delta L)^3 = (L + \alpha L \Delta T)^3$

$$= L^3 + 3\alpha L^3 \Delta T + 3\alpha^2 L^3 \Delta T^2 + \alpha^3 L^3 \Delta T^3$$

$$\approx L^3 + 3\alpha L^3 \Delta T \ = V + 3\alpha V \Delta T$$

We are only looking for small changes, so the higher terms in ΔT can be safely dropped. Hence $\Delta V = 3\alpha V \Delta T$. From the definition, this we immediately see that $\boxed{\beta = 3\alpha}$. Just to get an idea, here is what β looks like for various different materials:

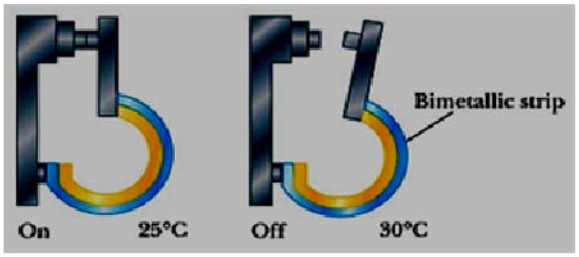


Coefficient of volume expansion β (1/°C)

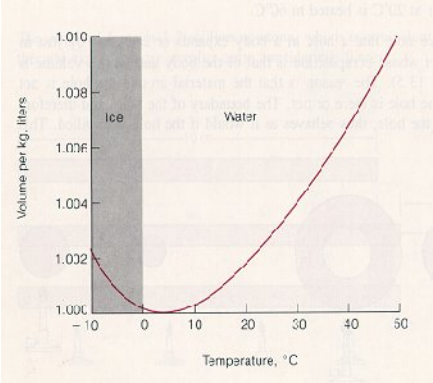
13. We can use the fact that different metals expand at different rates to make thermostats. For example, you need a thermostat to prevent an electric iron from getting too hot or a

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refrigerator from getting too cold. In the diagram below, one metal is bonded to another. When they expand together, one expands less than the other and the shape is distorted. This breaks off a circuit, as shown here.



14. Water behaves strangely - for certain temperatures, it contracts when heated (instead of expanding)! Look at the graph: between 0^0C and 4^0C, it exhibits strange or *anomalous* behaviour. The reason is complicated and has to do with the molecular structure of water. If water behaved normally, it would be very bad for fish in the winter because they rely upon the bottom of the lake or sea to remain liquid even though there the surface is frozen.



15. We define the "heat capacity" of a body as $C = \frac{Q}{\Delta T}$, where ΔT is the increase in temperature when an amount of heat Q is added to the body. Heat capacity is always positive; Q and ΔT have the same sign. The larger the heat capacity, the smaller is the change in the body's temperature when a fixed amount of heat is added. In general, $Q = mc\Delta T$, where $Q =$ heat added, $m =$ mass, $c =$ specific heat, and $\Delta T =$ change in temperature. Water has a very large specific heat c, $c = 1.0\ cal/(^oC\ g)$; this means it takes one calorie to raise the temperature of 1 gm of water by 1 degree Celsius. In joules per kilogram this is the same as 4186. In the table are the specific heats of various common materials. You can see that metals have small c, which means that it is relatively easy to raise or lower their temperatures. The opposite is true of water. Note also that steam and ice have smaller $c's$ than water. This shows that knowing the chemical composition is not enough.

Specific Heats at Atmospheric Pressure

Substance	Specific heat, c [J / (kg·K)]
Water	4186
Ice	2090
Steam	2010
Beryllium	1820
Air	1004
Aluminum	900
Glass	837
Silicon	703
Iron (steel)	448
Copper	387
Silver	234
Gold	129
Lead	128

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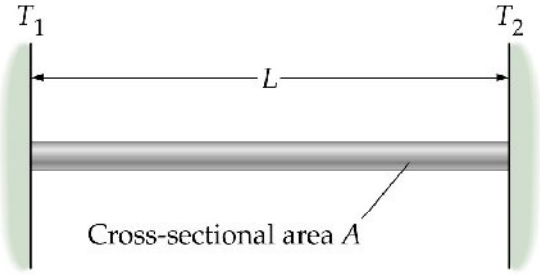
16. A 0.5-kg block of metal with an initial temperature of $30.0^{\circ}C$ is dropped into a container holding 1.12 kg of water at $20.0^{\circ}C$. If the final temperature of the block-water system is $20.4^{\circ}C$, what is the specific heat of the metal?

SOLUTION: Write an expression for the heat flow out of the block $Q_{block} = m_b c_b (T_b - T)$. Do the same for water, $Q_{water} = m_w c_w (T - T_w)$. Now use the fact that all the energy that is lost by the block is gained by the water: $Q_{block} = Q_{water} \Rightarrow T m_b c_b (T_b - T) - m_w c_w (T - T_w) = 0$

From this, $c_b = \dfrac{m_w c_w (T - T_w)}{m_b (T_b - T)} = \dfrac{(1.12kg)[4186J/(kg \cdot K)](20.4^{\circ}C - 20.0^{\circ}C)}{(0.500kg)(30.0^{\circ}C - 20.4^{\circ}C)}$

$= 391J/(kg \cdot K)$

17. When lifting a "daigchee" from a stove, you would be wise to use a cloth. Why? Because metals transfer, or conduct, heat easily whereas cloth does not. Scientifically we define conductivity using experiments and apparatus similar to the following:



Heat flows from the hotter to the colder plate. Let us use the following symbols:

k = thermal conductivity	Q = heat transferred
A = cross sectional area	t = duration of heat transfer
L = length	ΔT = temperature difference

Then the heat transferred in time t is, $Q = kA\left(\dfrac{\Delta T}{L}\right)t$. This formula allows us to measure k if all the other quantities in it are measured.

18. Conduction is one possible way by which heat is transferred from one portion of a system to another. It does not involve physical transport of particles. However, there is another way by which heat can be transferred - convection. In convection, heat is carried by a moving fluid. So when you heat a pot of water, molecules at the bottom move up, and the ones at the top come down - the water has currents inside it that transfer heat. Another mechanism for transferring heat is through radiation. We have already talked about this while discussing blackbody radiation and the Stefan-Boltzman Law.

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Summary of Lecture 37 – THERMAL PHYSICS II

1. Let us agree to call whatever we are studying the "system" (a mixture of ice and water, a hot gas, etc). The state of this system is specified by giving its pressure, volume, temperature, etc. These are called "thermodynamic variables". The relation between these variables is called the "equation of state". This equation relates P,V,T. So, for example:

   $P = f(V,T)$ (knowing V and T gives you P)

   $V = g(P,T)$ (knowing P and T gives you V)

   $T = h(P,V)$ (knowing P and V gives you T)

   For an ideal gas, the equation of state is $PV = Nk_BT$. Here N is the number of molecules in the gas, T is the temperature in degrees Kelvin, and k_B is called the Boltzman cosntant. This EOS is easy to derive, although I shall not do it here. In principle it is possible to mathematically derive the EOS from the underlying properties of the atoms and molecules in the system. In practice, however, this is a very difficult task (except for an ideal gas). However the EOS can be discovered experimentally.

2. Thermodynamics is the study of heat and how it flows. The First Law of Thermodynamics is actually just an acknowledgment that heat is a form of energy (and, of course, energy is always conserved!). Mathematically, the First Law states that:

   $$\Delta E = \Delta q + \Delta w$$

   where: E = internal energy of the system

   q = heat transferred to the system from the surroundings

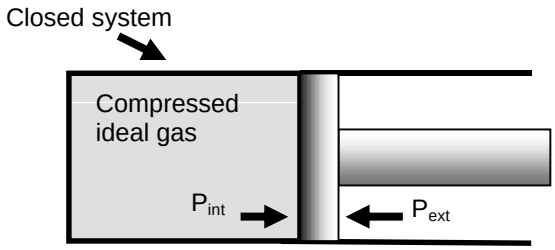
   w = work done on the system by the surroundings.

   In words, what the above formula says is this: if you do an amount of work Δw and also transfer an amount of heat Δq, then the sum of these two quantities will be the additional amount of energy ΔE that is stored in the system. There could be nothing simpler! Note: Δq and Δw are positive if heat is added to or work is done on a system, and negative if heat is removed from the system, or if the system does work on the environment.

3. Work and heat are called *path variables* - their values depend on the steps leading from one state to another. To give an obvious example: suppose you drag a heavy box on the floor in a straight line from point A to point B, and then in a very roundabout way from A to B again. Obviously the work done by you will be very different. On the other hand, the internal energy does not depend upon the path. Example: a gas has internal energy proportional to its temperature. It makes no difference whether the gas had been slowly heated or rapidly; it will have the same internal energy. Since the internal energy does not depend on the path we can write $\int_{State1}^{State2} dE = E_2 - E_1 = \Delta E$.

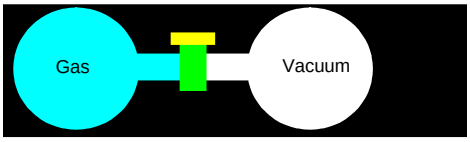
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4. Let us calculate the work done by an expanding gas. In the diagram below, suppose the piston moves out by distance dx, then the work done is $dw = Fdx$ where F is the force exerted on the gas. But $F = P \times A = -P_{ext} \times A$. (The minus sign is important to understand; remember that the force exerted by the gas on the piston is the negative of the force that the piston exerts on the gas). So $dw = Fdx = -P_{ext} \times A \times dx = -P_{ext} dV$. The work done by the expanding gas is $w = -\int_{V_1}^{V_2} P_{ext} dV$ where V_1 and V_2 are the initial and final volumes.

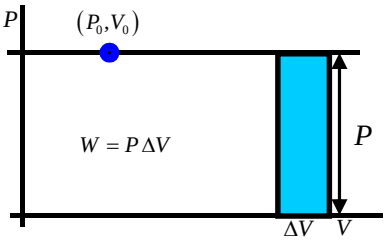


Note that w depends on the path. Let's consider 3 different paths.

a)Suppose that the gas expands into the vacuum, $P_{ext} = 0$. Then $w = 0$. So, if in the figure below, the valve is opened, then gas will flow into the vacuum without doing work.



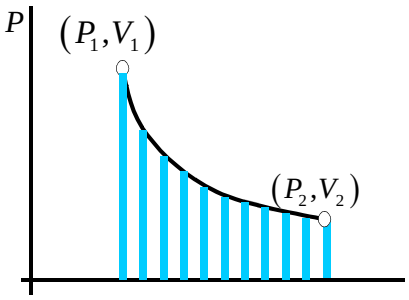
b)Suppose the pressure outside has some constant value. Then, $w = P_{ext} \times (V_2 - V_1)$ or, $w = P_{ext} \Delta V$. You can see that this is just the area of the rectangle below.



c)We can also let the gas expand so that the internal and external pressures are almost the same (this is called reversible expansion). Then the work done *on* the system is: $dw = -P_{ext} dV$. Using $PV = Nk_B T$ gives,

$$w = -\int_{V_1}^{V_2} \frac{Nk_B T}{V} dV = -Nk_B T(\ln V_2 - \ln V_1) = -Nk_B T \ln \frac{V_2}{V_1}$$

Of course, the work done *by* the gas as it expands is positive since V_2 is larger than V_1. Remember that the log function is positive if its argument is bigger than 1.



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5. The internal energy of a gas depends only on the number of molecules it contains and on the temperature, $E = \frac{3}{2} N k_B T$. In a free expansion of gas (it doesn't do any work in this case), the initial and final internal energies are equal, $E_f = E_i$.

6. Imagine that a substance is heated while keeping its volume constant. Obviously, you have to supply more heat as you raise the temperature higher, $\Delta Q_V = C_V \Delta T,$ where C_V is given the name "specific heat at constant volume". No work is done since there is no expansion, $\Delta W = 0$. Hence, using the First Law, $\Delta Q_V = \Delta E + \Delta W = \Delta E,$ and so $\Delta E = C_V \Delta T$.

7. Now take the same system as above, but allow it to expand at constant pressure as it is heated. Then you will have to supply an amount of heat, $\Delta Q_P = C_P \Delta T$. Here, C_P is called the specific heat at constant pressure, $\Delta Q_P = \Delta E + \Delta w = \Delta E + P\Delta V$. Hence, $C_P \Delta T = \Delta U + P\Delta V$. If we supply only a small amount of heat then, $C_P dT = dU + PdV$. This gives, $C_P dT = C_V dT + PdV$. If we consider the special case of an ideal gas, then $PV = Nk_B T$. Hence, $PdV = Nk_B dT$ and $C_P dT = C_V dT + Nk_B dT$. Hence, $C_P = C_V + Nk_B$. Since the internal energy is $E = \frac{3}{2} N k_B T$, it follows that $C_V = \frac{dU}{dT} = \frac{3}{2} N k_B$. From this, the specific heat at constant pressure is: $C_P = \frac{3}{2} N k_B + N k_B = \frac{5}{2} N k_B$. From the equation $C_P = C_V + Nk_B$, it follows that $\frac{Nk_B}{C_V} = \frac{C_P - C_V}{C_V}$. Let us define a new symbol, $\frac{C_P}{C_V} \equiv \gamma$.

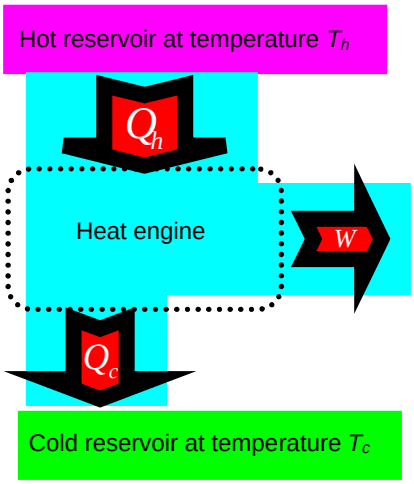
8. We can find the EOS of an ideal gas when it expands without losing any heat (this is called adiabiatic expansion). In this case, $dQ = dE + dW = 0$. Hence, $C_V dT + PdV = 0$, or $C_V dT + Nk_B T \frac{dV}{V} = 0$. Dividing by dT gives, $\frac{dT}{T} + \frac{Nk_B}{C_V} \frac{dV}{V} = 0$ or $\frac{dT}{T} + (\gamma - 1)\frac{dV}{V} = 0.$ Integrating this gives, $\ln T + (\gamma - 1)\ln V = \text{Constant}$. Equivalently, $\ln\left(TV^{\gamma - 1}\right) = \text{Constant}$. A more convenient form is: $TV^{\gamma - 1} = \text{Constant}$. This holds for the entire adiabatic expansion.

9. Almost nothing beats the importance of the Second Law Of Thermodynamics. There are many equivalent ways of stating it. The one I like is: "There can be no process whose only final result is to transfer thermal energy from a cooler object to a hotter object". This seems extremely simple (and almost useless), but in fact it tells you, among other things, that no perpetual motion machine (such as that which generates electricity without any fuel input) can ever be built. If someone could build such a machine, then it could also be used to transfer heat from a cold object to a hot object - a contradiction with the Second Law.

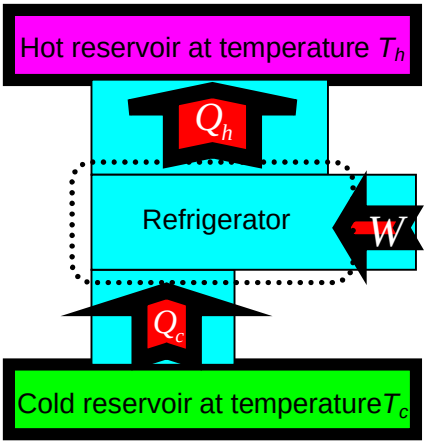
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10. Before we consider some implications of the Second Law, I want to introduce the concept of a *thermal reservoir*. Suppose you put a thermometer in your mouth. You will transfer a small amount of heat to the thermometer, but because your body is so big this will make no measureable difference to your body temperature. Your body therefore is a thermal reservoir as far as the thermometer is concerned. More generally, any large mass of material will act as a thermal reservoir if it is at constant temperature. A *heat engine* (e.g. a car engine, refrigerator, etc) operates between reservoirs at two different temperatures.

11. A heat engine works between a high temperature T_H and a low temperature T_C. It aborbs heat $Q_{in,h}$ from the hot reservoir and rejects heat $Q_{out,c}$ into the cold reservoir. Since the internal energy of the engine does not change, $\Delta U = 0$ and the First Law gives simply, $Q = \Delta U + W = W$. The work done by the engine is $W = Q_{in,h} - Q_{out,c}$. Now define the efficiency of the machine as $\varepsilon = \dfrac{\text{work done by engine}}{\text{heat put in}} = \dfrac{W}{Q_{in,h}}$. Then $\varepsilon = \dfrac{Q_{in,h} - Q_{out,c}}{Q_{in,h}} = 1 - \dfrac{Q_{out,c}}{Q_{in,h}}$. Now, the heat that is transferred out of or into a reservoir is proportional to its absolute temperature. Hence we arrive at the important result that $\varepsilon = 1 - \dfrac{T_C}{T_H}$. This number is always less than one, showing that no machine can convert all the input heat into useful work. As an example, a nuclear reactor has temperature 300^0C at the core and rejects heat into a river at 30^0C. The maximum efficiency it can have is $\varepsilon = 1 - \dfrac{30 + 273}{300 + 273} = 0.471$



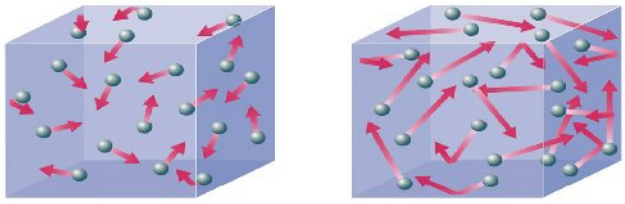
12. A refrigerator is a heat engine working in reverse. Work is done (by an electric motor) to pump heat from a cold reservoir (inside of refrigerator) to the hot exterior. The Second Law can be shown to imply the following: it is impossible for a refrigerator to produce no other effect than the transfer of thermal energy from a cold object to a hot object. Again, the efficiency of a refrigerator is always less than one.



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Summary of Lecture 38 – THERMAL PHYSICS III

1. In the two previous lectures, I concentrated exclusively on heat as a form of energy that flows from a hot to a cold body. I made no mention of the fact that all matter is made of atoms and molecules; the notion of heat would exist even if we did not know this. But, the modern understanding of heat is that it is the random kinetic energy of atoms. So, for example, the difference between a cold and hot gas is illustrated below.



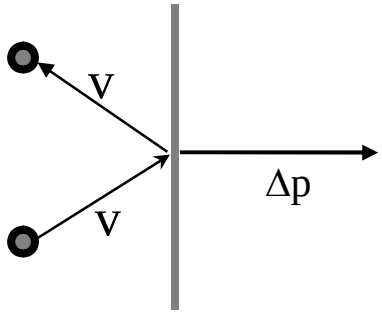
cold gas **hot gas**

Longer arrows denote atoms that are moving faster.
The hotter gas has, on the average, faster moving atoms.

2. The study of heat, considered as arising from the random motion of the basic constituents of matter, is an area of physics called statistical mechanics. Its goal is the understanding, and prediction of macroscopic phenomena, and the calculation of macroscopic properties from the properties of individual molecules. Temperature is the average energy related to the speed of atoms in an object, and heat is the amount of energy transferred from one object to another.

3. Imagine a gas so dilute that atoms rarely collide with each other (this is also called an ideal gas). We can readily understand why the pressure is directly proportional to the temperature for a gas confined to a box: increasing the temperature T makes gas molecules move faster, striking the walls of the container harder and more often, thus giving an increase in pressure P, i.e. $P \propto T$. On the other hand, for an ideal gas at constant pressure, the volume is directly proportional to the temperature : $V \propto T$. So, as the air in a balloon is heated up, its volume will increase in direct proportion because of the impact of the atoms on the balloon walls. Note that this is where the "absolute" or Kelvin scale comes from: at 0° K an ideal gas would have zero volume because the atoms would not be moving.

4. Pressure is related to the outward force per unit area exerted by the gas on the container wall. Newton's 2nd Law is: $F = \frac{\Delta p}{\Delta t}$. Here Δp is to be understood as the momentum destroyed in time Δt. Using this, we can do a little calculation to find the pressure in an ideal gas.

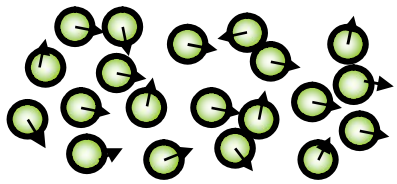


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Suppose there are N atoms in a box of volume, and the average speed of an atom in the x direction is v_x. By symmetry, half are moving in the $+x$ and half in the $-x$ directions. So the number of atoms that hit the wall in time Δt is $\frac{N}{2}\frac{1}{V}v_x \Delta t A$. Hence the change in momentum is $\Delta p = (2mv_x) \times \left(\frac{1}{2}\frac{N}{V}v_x \Delta t A\right) = \frac{N}{V} m v_x^2 A \Delta t$. From this, we can calculate the pressure $P = \frac{F}{A} = \frac{1}{A}\frac{\Delta p}{\Delta t} = \frac{N}{V} m v_x^2$. Hence $PV = Nmv_x^2 = 2N\left(\frac{1}{2}mv_x^2\right)_{av}$.

5. Let us return to our intuitive understanding of heat as the random energy of small particles. Now, $\left(\frac{1}{2}mv_x^2\right)_{av}$ is the average kinetic energy of a particle on account of its motion in the x direction. This will be greater for higher temperatures, so $\left(\frac{1}{2}mv_x^2\right)_{av} \propto T,$ where T is the absolute temperature. So we write $\left(\frac{1}{2}mv_x^2\right)_{av} = k_B T,$ which you can think of as the definition of the Boltzmann constant, k_B. So we immediately get $PV = Nk_B T,$ the equation obeyed by an ideal gas (dilute, no collisions). Now, there is nothing special about any particular direction, so $\left(v_x^2\right)_{av} = \left(v_y^2\right)_{av} = \left(v_z^2\right)_{av}$ and therefore the average total velocity is $\left(v^2\right)_{av} = \left(v_x^2\right)_{av} + \left(v_y^2\right)_{av} + \left(v_z^2\right)_{av} = 3\left(v_x^2\right)_{av}$. So the total average kinetic energy of one atom is, $K_{av} = \left(\frac{1}{2}mv^2\right)_{av} = \frac{3}{2}k_B T,$ and the energy of the entire gas is $K = N\left(\frac{1}{2}mv^2\right)_{av} = \frac{3}{2}Nk_B T.$ From this, the mean squared speed of atoms in a gas at temperature is $\left(v^2\right)_{av} = \frac{3k_B T}{m}$.

6. Heat, which is random kinetic energy, causes changes of phase in matter:
   a) Water molecules attract each other, but if water is heated then they can escape and water becomes steam.
   b) If steam is heated, the water molecules break up and water becomes separated hydrogen and oxygen atoms.
   c) If a gas of atoms gets hot enough, then the atoms collide so violently that they lose electrons (i.e. get ionized).
   d) More heat (such as inside the core of a star) will break up atomic nuclei into protons and neutrons.

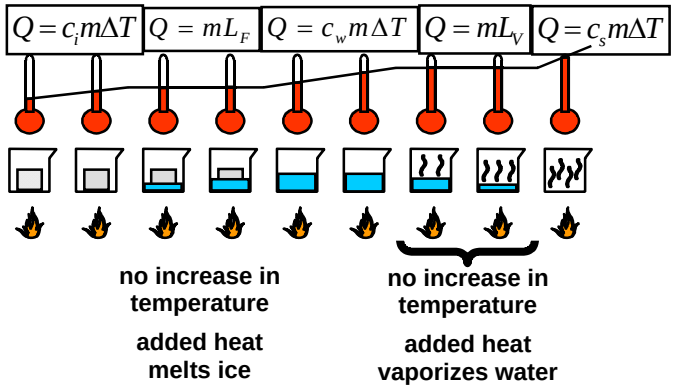


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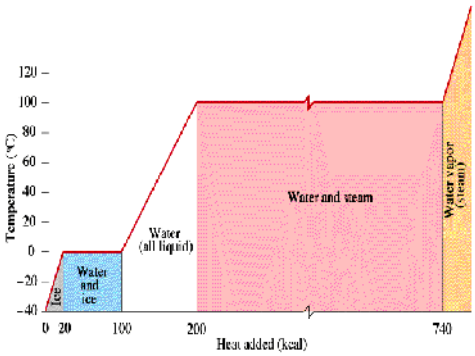
7. Evaporation: if you leave water in a glass, it is no longer there after some time. Why? Water molecules with high enough KE can break free from the water and escape. But how does a water molecule get enough energy to escape? This comes from random inter-molecular collisions, such as illustrated below. Two molecules collide, and one slows down while the other speeds up. The faster molecule might escape the water's surface.



8. Heat needs to be supplied to cause a change of phase:
   a) To melt a solid (ice, wax, iron) we must supply the "latent heat of fusion" L_F for unit mass of the solid. For mass m we must supply an amount of heat $Q = mL_F$ to overcome the attraction between molecules. No temperature change occurs in the melting.
   b) To vapourize a substance (convert to gas or vapour), we must supply the "latent heat of vapourization" L_V for unit mass of the substance. For mass m we must supply an amount of heat $Q = mL_V$. No temperature change occurs in the vapourization .
   c) To raise the temperature of a substance we must supply the "specific heat" C for unit mass of the substance by 1 degree Kelvin. To change the temperature of mass m by ΔT, we must supply heat $Q = mC\Delta T$.



In the above diagram you see what happens as ice is heated. The amount of heat needed is indicated as well. Note that no change of temperature happens until the phase change is completed. The same physical situation is represented to the right as well where the temperature is plotted against Q.



9. A very important concept is that of *entropy*. It was first introduced in the context of thermodynamics. Then, when the statistical nature of heat became clear a century later, it was understood in very different terms. Let us begin with thermodynamics: suppose a system is at temperature T and a small amount of heat dQ is added to it. Then, the small increase in the entropy of the system is $dS = \frac{dQ}{T}$. What if we keep adding little bits of heat? Then the temperature of the system will change by a finite amount, and the change in entropy is got by adding up all the small changes: $\Delta S = \int_{T_1}^{T_2} \frac{dQ}{T}$.

   Now let's work out how much the entropy changes when we add heat dQ to an ideal gas. From the First Law, $dQ = dU + dW = dU + PdV = C_V dT + Nk_B T \frac{dV}{V}$. The entropy change is $dS = \frac{dQ}{T} = C_V \frac{dT}{T} + Nk_B \frac{dV}{V}$, and for a finite change we simply integrate,

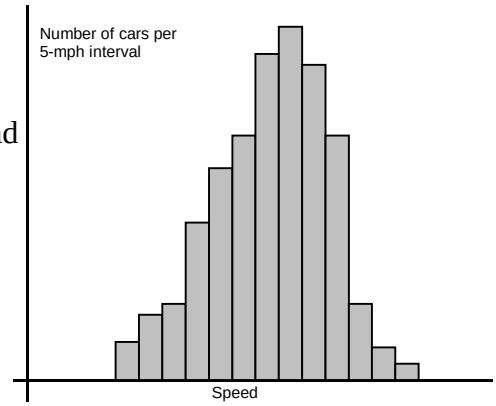
   $\Delta S = \int \frac{dQ}{T} = C_V \ln\frac{T_2}{T_1} + Nk_B \ln\frac{V_2}{V_1} = \frac{3}{2} Nk_B \ln\frac{T_2}{T_1} + Nk_B \ln\frac{V_2}{V_1}$. Note that the following:

   a)The entropy increases if we heat a gas ($T_2 > T_1$).

   b)The entropy increases if the volume increases ($V_2 > V_1$).

10. In statistical mechanics, we interpret entropy as the degree of disorder. A gas with all atoms at rest is considered ordered, while a hot gas having atoms buzzing around in all directions is more disordered and has greater entropy. Similarly, as in the above example, when a gas expands and occupies greater volume, it becomes even more disordered and the entropy increases.

11. We always talk about averages, but how do you define them mathematically? Take the example of cars moving along a road at different speeds. Call $n(\mathrm{v}_i)$ the number of cars with speed v_i , then $N = \sum_i n(\mathrm{v}_i)$ is the total number of cars and the average speed is defined as $\bar{\mathrm{v}} = \frac{\sum_i \mathrm{v}_i n(\mathrm{v}_i)}{\sum_i n(\mathrm{v}_i)}$.

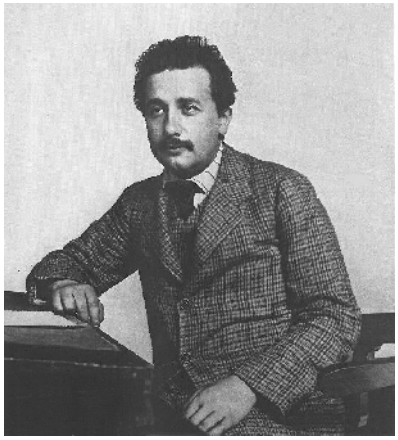


   We define the probability of finding a car with speed v_i as $P(\mathrm{v}_i) \equiv \frac{n(\mathrm{v}_i)}{\sum_i n(\mathrm{v}_i)} = \frac{n(\mathrm{v}_i)}{N}$. We can also write the average speed as, $\bar{\mathrm{v}} = \sum_i \mathrm{v}_i P(\mathrm{v}_i)$.

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Summary of Lecture 39 – SPECIAL RELATIVITY I

1. The Special Theory of Relativity was created by Albert Einstein, when he was a very young man in 1905. It is one of the most solid pillars of physics and has been tested thousands of times. Special Relativity was a big revolution in understanding the nature of space and time, as well as mass and energy. It is absolutely necessary for a proper understanding of fast-moving particles (electrons, photons, neutrinos..). Einstein's General Theory of Relativity - which we shall not even touch here - goes beyond this and also deals with gravity.

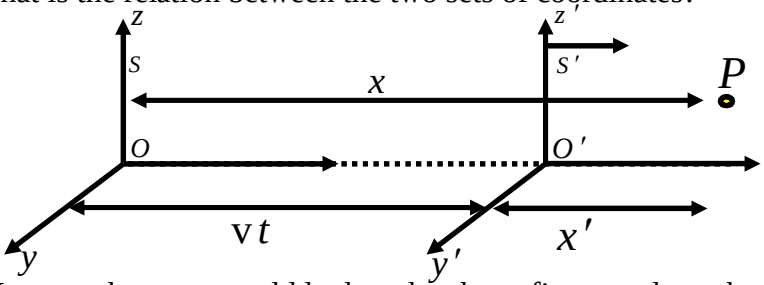


2. Relativity deals with time and space. So let us first get some understanding of how we measure these two fundamental quantities:
   a) *Time* **:** we measure time by looking at some phenomenon that repeats itself. There are endless examples: your heart beat, a pendulum, a vibrating quartz crystal, rotation of the earth around its axis, the revolution of the earth about the sun,... These can all be used as clocks. Of course, an atomic clock is far more accurate than using your heart beat and is accurate to one part in a trillion. Although different systems of measurement have different units it is fortunate that time is always measured in seconds.
   b) *Distance* **:** intuitively we know the difference between short and long. But to do a measurement, we first have to agree on what should be the unit of length. If you use a metre as the unit, then you can use a metre rod and measure any length you want. Of course, sometimes we may use more sophisticated means (such as finding how high a satellite is) but the basic idea is the same: the distance between point A and point B is the number of metre rods (or fractions thereof) that can be made to fit in between the two points.

3. Newton had believed that there was one single time for the entire universe. In other words, time was absolute and could be measured by one clock held somewhere in the centre of the universe. Similarly, he believed that space was absolute and that the true laws of physics could be seen in that particular frame which was fixed to the centre of the universe. (As we shall see, Einstein shocked the world by showing that time and space are not absolute quantities, but depend on the speed of your reference frame. Even more shocking was his proof that time and space are not entirely separate quantities!)

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4. An *event* is something that happens at some point in space at some time. With respect to the frame S below, the event P happened at (x,y,z,t). Now imagine a girl running to the right at fixed speed v in frame S′. According to her, the same event P happened at (x′,y′,z′,t′). What is the relation between the two sets of coordinates?



If you were Newton, then you would look at the above figure and say that obviously it is the following: $x' = x - \mathrm{v}t,\ y' = y,\ z' = z,\ t' = t$. These are called *Galilean coordinate transformations*. Note that it is assumed here that the time is the same in both frames because of the Newtonian belief that there is only one true time in the world.

5. There are certain obvious consequences of using the Galilean transformations. So, for example a rod is at rest in S-frame. The length in S-frame $= x_B - x_A,$ while the length in the S′-frame $= x'_B - x'_A = x_B - x_A - \mathrm{v}(t_B - t_A)$. Since $t_B = t_A,$ the length is the same in both frames: $x'_B - x'_A = x_B - x_A$. (As we shall see later this will not be true in Einstein's Special Relativity).



6. Let us now see what the Galilean transformation of coordinates implies for transformations of velocities. Start with $x' = x - \mathrm{v}t$ and differentiate both sides. Then $\frac{dx'}{dt} = \frac{dx}{dt} - \mathrm{v}$. Since $t = t',$ it follows that $\frac{dx'}{dt} = \frac{dx'}{dt'}$ and so $\frac{dx'}{dt'} = \frac{dx}{dt} - \mathrm{v}$. Similarly, $\frac{dy'}{dt'} = \frac{dy}{dt}$ and $\frac{dz'}{dt'} = \frac{dz}{dt}$.

Now, $\frac{dx'}{dt'} = u'_x$ is the x-component of the velocity measured in S′-frame, and similarly $\frac{dx}{dt} = u_x$ is the x-component of the velocity measured in S-frame. So we have found that:

$$u'_x = u_x - \mathrm{v},\quad u'_y = u_y,\quad u'_z = u_z \text{ (or, in vector form, } \vec{u}' = \vec{u} - \vec{\mathrm{v}})$$

Taking one further derivative, $\frac{du'_x}{dt'} = \frac{d}{dt}(u_x - \mathrm{v}) = \frac{du_x}{dt}$ (remember that v = constant), we find that the components of acceleration are the same in S and S′:

$$\frac{du'_x}{dt'} = \frac{du_x}{dt},\quad \frac{du'_y}{dt'} = \frac{du_y}{dt},\quad \frac{du'_z}{dt'} = \frac{du_z}{dt}.$$

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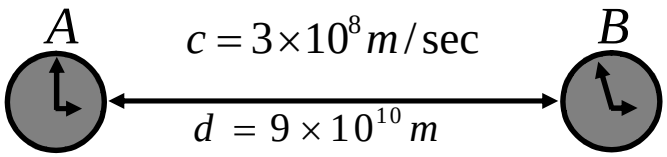
7. We had learned in a previous chapter that light is electromagnetic wave that travels at a speed measured to be $c = 2.907925 \times 10^8 m/\text{sec}$. Einstein, when he was 16 years old, asked himself the question: in which frame does this light travel at such a speed? If I run holding a torch, will the light coming from the torch also move faster? Yes, if we use the formula for addition of velocities derived in the previous section! So if light actually travels at $2.907925 \times 10^8 m/\text{sec}$ then it is with respect to the frame in which the "aether" (a massless fluid which we cannot feel) is at rest. But there is no evidence for the aether!

8. Einstein made the two following postulates (or assumptions).
   1) The laws of physics have the same form in all inertial frames. (In other words, there is no constant-velocity frame which is preferred, or better, than any other).
   2) The speed of light in vacuum has the same value in all inertial systems, independent of the relative motion of source and observer.

   [Note: as stressed in the lecture, no postulate of physics can ever be mathematically proved. You have to work out the consequences that follow from the postulates to know whether the postulates are good ones or not.]

9. Let's first get one thing clear: in any one inertial frame, we can imagine that there are rulers and clocks to measure distances and times. We can synchonize all the clocks to read one time, which will be called the time in that frame S. But how do we do this? If are two clocks, then we can set them to read the same time (i.e. synchronize them) by taking account of the time light takes to travel between them.
   Example: The observer with clock A sees the time on clock B as 2:55pm. But he knows that light took 5 minutes to travel from B to A, and therefore A and B are actually reading exactly the same time.



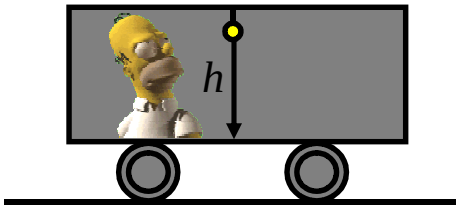


(Time taken by light in going from B to A is equal to $\frac{d}{c} = \frac{9 \times 10^{10} m}{3 \times 10^8 m/\text{sec}} = 300\,\text{sec} = 5\,\text{min}$.)

10. Now I shall derive the famous formula which shows that a moving clock runs slow. This will be something completely different from the older Newtonian conception of time. Einstein derived this formula using a "gedanken" experiment, meaning an experiment which can imagine but not necessarily do. So imagine the following: a rail carriage has a

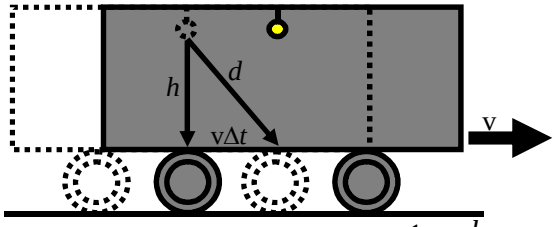
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a bulb that is fixed to the ceiling. The bulb suddenly flashes, and the light reaches the floor. Time taken according to the observer inside the train is $\Delta t' = h/c$.



Now suppose that the same flash is observed by an observer S standing on the ground. According to S, the train is moving with speed v. Let's consider the same light ray. Clearly, the train has moved forward between the time when the light left the ceiling and when it reached the floor. According to S, the time taken is $\Delta t = \frac{d}{c} = \frac{\sqrt{h^2 + (v\Delta t)^2}}{c}$. Now



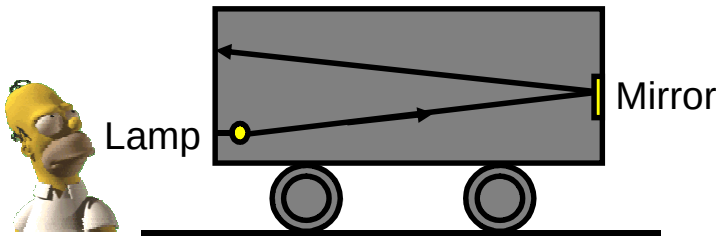


square both sides: $(c\Delta t)^2 = h^2 + (v\Delta t)^2$, which gives $\Delta t = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}\frac{h}{c} = \gamma \Delta t'$. Here γ is the relativistic factor, $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$ and is a number that is always bigger than one. As v gets closer and closer to c, the value of γ gets larger and larger. For v=4c/5, γ=5/3. So, if 1 sec elapses between the ticks of a clock in S′ (i.e. $\Delta t' = 1$), the observer in S will see 5/3 seconds between the ticks. In other words, he will think that the moving clock is slow!

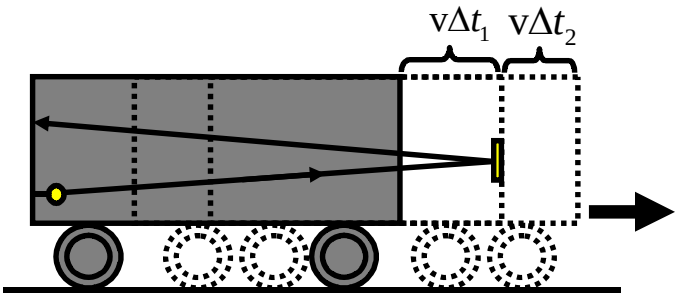
11. The muon is an unstable particle. If at rest, it decays in just 10^{-18} seconds. But if traveling at 3/5 the speed of the light, it will last 25% longer because $\gamma = \frac{1}{\sqrt{1-(3/5)^2}} = \frac{5}{4}$. If it is traveling at v=0.999999c, it will last 707 times longer. We can observe these shifts due to time dilation quite easily, and they are an important confirmation of Relativity.

12. Another amazing prediction of Relativity is that objects are shortened (or contracted) along the direction of their motion. Einstein reached this astonishing conclusion on the basis of yet another gedanken experiment. Again, consider a moving railway carriage with a bulb

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at one end that suddenly flashes. Let Δx be the length of the carriage according to ground observer S, and $\Delta x'$ be the length according to the observer S′ inside the carriage. So, $\Delta t' = \dfrac{2\Delta x'}{c}$ is the time taken for the light to go from one end to the other, and then return after being reflected by a mirror. Now let's look at this from the point of view of S, who is fixed to the ground. Let Δt_1 be the time for the signal to reach the front end. Then, because the mirror is moving forward, $\Delta t_1 = \dfrac{\Delta x + \mathrm{v}\Delta t_1}{c}$. Call Δt_2 the return time. Then,

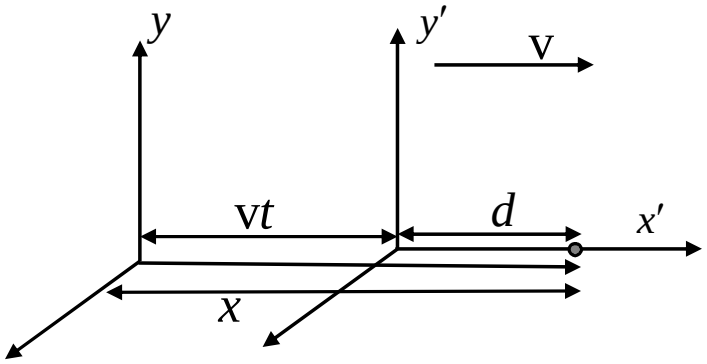


$\Delta t_2 = \dfrac{\Delta x - \mathrm{v}\Delta t_2}{c}$. Solving for Δt_1 and Δt_2: $\Delta t_1 = \dfrac{\Delta x}{c - \mathrm{v}}$ and $\Delta t_2 = \dfrac{\Delta x}{c + \mathrm{v}}$. The total time is therefore $\Delta t = \Delta t_1 + \Delta t_2 = \dfrac{2\Delta x / c}{\left(1 - \mathrm{v}^2 / c^2\right)}$. Now, from the time dilation result derived earlier, $\Delta t' = \sqrt{1 - \mathrm{v}^2 / c^2}\,\Delta t,$ and so $\Delta t' = \sqrt{1 - \mathrm{v}^2 / c^2}\left(\dfrac{2\Delta x / c}{\left(1 - \mathrm{v}^2 / c^2\right)}\right) = \dfrac{2\Delta x / c}{\sqrt{1 - \mathrm{v}^2 / c^2}}$. This gives, $\Delta x' = \dfrac{\Delta x}{\sqrt{1 - \mathrm{v}^2 / c^2}}$, or $\Delta x = \Delta x' / \gamma$. This an astonishing result! Suppose that there is a metre rod. Then the observer riding with it has $\Delta x' = 1$. But according to someone who sees the metre rod moving towards/away from him, the length is $1/\gamma$. This is less than 1 metre!

13. Although an object shrinks in the direction of motion (both while approaching and receding), the dimensions perpendicular to the velocity are not contracted. It is easy to conceive of a gedanken experiment that will demonstrate this. One of the exercises will guide you in this direction.

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14. We shall now derive the "Lorentz Transformation", which is the relativistic version of the Galilean transformation discussed earlier. Consider an event that occurs at position x (as measured in S) at a distance d (again, as seen in S) from the origin of S′. Then, $x = d + \mathrm{v}t$. From the Lorentz contraction formula derived earlier, $d = \frac{x'}{\gamma}$ where x' is the distance measured in S′. This gives $x' = \gamma\left(x - \mathrm{v}t\right)$. Now, by the same logic, $x' = d' - \mathrm{v}t'$.



Here $d' = \frac{x}{\gamma}$. This gives $x = \gamma\left(x' + \mathrm{v}t'\right) = \gamma\left(\gamma\left(x - \mathrm{v}t\right) + \mathrm{v}t'\right)$. We find that the time in S′ is related to the time in S by $t' = \gamma\left(t - \frac{\mathrm{v}}{\mathrm{c}^2}x\right)$. Note that if we make c very large, then $t' = t$.

To summarize:

LORENTZ TRANSFORMATION

$$x' = \gamma\left(x - \mathrm{v}t\right)$$
$$y' = y$$
$$z' = z$$
$$t' = \gamma\left(t - \frac{\mathrm{v}}{c^2}x\right).$$

(Note: in various books you will find slightly different derivations of the above Lorentz transformation. You should look at one of your choice and understand that as well.)

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Summary of Lecture 40 – SPECIAL RELATIVITY II

1. Recall the Lorentz Transformation: $x' = \gamma(x - \mathrm{v}t)$ and $t' = \gamma\left(t - \frac{\mathrm{v}}{c^2}x\right)$. Suppose we take the space interval between two events $\Delta x = x_1 - x_2$, and the time interval $\Delta t = t_1 - t_2$. Then, these intervals will be seen in S′ as $\Delta t' = \gamma\left(\Delta t - \frac{\mathrm{v}}{c^2}\Delta x\right)$ and $\Delta x' = \gamma(\Delta x - \mathrm{v}\Delta t)$.

   Now consider two particular cases:

   a) Suppose the two events occur at the same place (so $\Delta x = 0$) but at different times (so $\Delta t \neq 0$). Note that in S′ they do not occur at the same point: $\Delta x' = \gamma(0 - \mathrm{v}\Delta t)$!

   b) Suppose the two events occur at the same time (so $\Delta t = 0$) but at different places (so $\Delta x \neq 0$). Note that in S′ they are not simultaneous: $\Delta t' = \gamma\left(0 - \frac{\mathrm{v}}{c^2}\Delta x\right)$.

2. As seen in the frame S, suppose a particle moves a distance dx in time dt. Its velocity u is then $u = \frac{dx}{dt}$ (in S-frame). As seen in the S′-frame, meanwhile, it has moved a distance dx' where $dx' = \gamma(dx - \mathrm{v}dt)$ and the time that has elapsed is $dt' = \gamma\left(dt - \frac{\mathrm{v}}{c^2}dx\right)$. The velocity in S′-frame is therefore $u' = \frac{dx'}{dt'} = \frac{\gamma(dx - \mathrm{v}dt)}{\gamma\left(dt - \frac{\mathrm{v}}{c^2}dx\right)} = \frac{dx/dt - \mathrm{v}}{1 - \frac{\mathrm{v}}{c^2}dx/dt} = \frac{u - \mathrm{v}}{1 - \frac{u\mathrm{v}}{c^2}}$. This is the Einstein velocity addition rule. It is an easy exercise to solve this for u in terms of u',

$$u = \frac{u' + \mathrm{v}}{1 + \frac{u'\mathrm{v}}{c^2}}.$$

3. Note one very interesting result of the above: suppose that a car is moving at speed v and it turns on its headlight. What will the speed of the light be according to the observer on the ground? If we use the Galilean transformation result, the answer is v+c (wrong!). But using the relativistic result we have $u' = c$ and $u = \frac{c + \mathrm{v}}{1 + \frac{c\mathrm{v}}{c^2}} = \frac{c + \mathrm{v}}{1 + \frac{\mathrm{v}}{c}} = c\frac{c + \mathrm{v}}{c + \mathrm{v}} = c$. In other words, the speed of the source makes no difference to the speed of light in your frame. Note that if either u or v is much less than c, then u' reduces to the familiar result:

   $u' = \frac{u - \mathrm{v}}{1 - \frac{u\mathrm{v}}{c^2}} \to u - \mathrm{v}$, which is the Galilean velocity addition rule.

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4. The Lorentz transformations have an interesting property that we shall now explore. Take the time and space intervals between two events as observed in frame S, and the corresponding quantities as observed in S′. We will now prove that the quantities defined respectively as $I = (c\Delta t)^2 - (\Delta x)^2$ and $I' = (c\Delta t')^2 - (\Delta x')^2$ are equal. Let's start with I':

$$I' = (c\Delta t')^2 - (\Delta x')^2 = \gamma^2\left((c\Delta t)^2 + (\Delta x)^2 v^2/c^2 - \cancel{2v\Delta t\Delta x} - (\Delta x)^2 - (v\Delta t)^2 + \cancel{2v\Delta t\Delta x}\right)$$

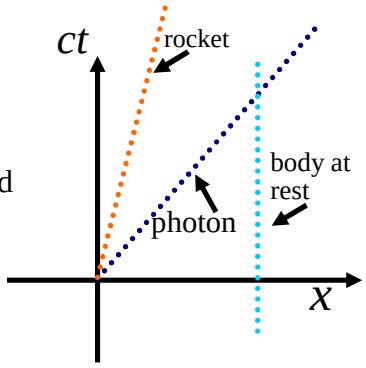
$$= \frac{1}{(1 - v^2/c^2)}\left((c^2 - v^2)(\Delta t)^2 - (\Delta x)^2(1 - v^2/c^2)\right)$$

$$= (c\Delta t)^2 - (\Delta x)^2 = I$$

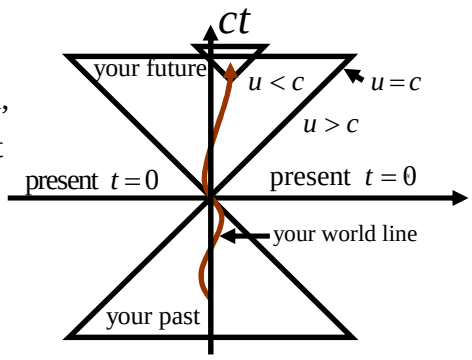
This guarantees that all inertial observers measure the same speed of light !!

5. If the time separation is large, then $I > 0$ and we call the interval *timelike.*
If the space separation is large, then $I < 0$ and we call the interval *spacelike.*
If $I = 0$ and we call the interval *lightlike.*
Note : a) If an interval is timelike in one frame, it is timelike in all other frames as well.
b) If interval between two events is timelike, their time ordering is absolute.
c) If the interval is spacelike the ordering of two events depends on the frame from which they are observed.

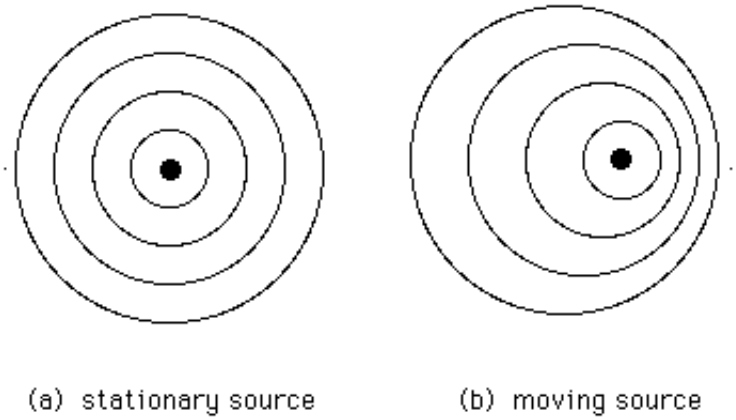
6. It is sometimes nice to look at things graphically. Here is a graph of position versus time for objects that move with different speeds along a fixed direction. First look at an object at rest. Its position is fixed even though time (plotted on the vertical axis) keeps increasing. Then look at the rocket moving at constant speed (which has to be less than c), and finally a photon (which can only move at c).



7. The trajectory of a body as it moves through space-time is called its world-line. Let's take a rocket that is at $x = 0$ at $t = 0$. It moves with non-constant speed, and that is why its world-line is wavy. A photon that moves to the right will have a world line with slope equal to +1, and that to the right with slope -1. The upper triangle (with t positive) is called the future light cone (don't forget we also have y and x). The lower light cone consists of past events.



8. Earlier on we had discussed the Doppler effect in the context of sound. Now let us do so for light. Why are they different? Because light always moves with a fixed speed while the speed of sound is different according to a moving and a fixed observer. Consider the case of a source of light at rest, and one that is moving to the right as shown below:



Let $\nu_0 = \frac{1}{T_0}$ be the frequency measured in the source's rest frame S, where T_0 is the time for one complete cycle in S. We want to calculate ν, the frequency as seen by the observer in S′ moving to the right at speed v. Call λ the distance between two successive wave crests (i.e. the wavelength according to the observer). In time T the crests ahead of the source move a distance cT, even as the source moves a shorter distance vT in the same direction. Hence $\lambda = (c - \mathrm{v})T$ and so $\nu = \frac{c}{\lambda} = \frac{c}{(c-\mathrm{v})T}$. Now, as discussed earlier, the time measured by observer will not be T_0 because of time dilation. Instead,

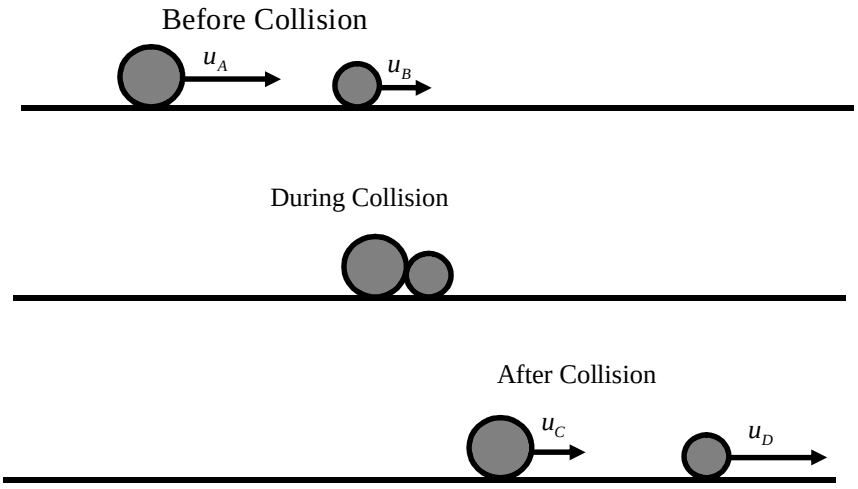
$T = \frac{T_0}{\sqrt{1-\mathrm{v}^2/c^2}} = \frac{cT_0}{\sqrt{c^2-\mathrm{v}^2}}$. Hence, $\nu = \frac{c}{(c-\mathrm{v})T} = \left(\frac{c}{c-\mathrm{v}}\right)\frac{\sqrt{c^2-\mathrm{v}^2}}{c}\nu_0 = \sqrt{\frac{c+\mathrm{v}}{c-\mathrm{v}}}\nu_0.$

(If source moves away from the observer just change the sign of v: $\nu = \sqrt{\frac{c-\mathrm{v}}{c+\mathrm{v}}}\nu_0$.)

This is the famous Doppler effect formula. In the lecture I discussed some applications such as finding the speed at which stars move away from the earth, or finding the speed of cars or aircraft.

9. We now must decide how to generalize the concept of momentum in Relativity theory. The Newtonian definition of momentum is $\vec{p} = m\vec{u}$. The problem with this definition is that we are used to having momentum conserved when particles collide with each other, and this old definition will simply not work when particles move very fast. Consider the collision of two particles as in the diagrams below:

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In the frame fixed to the lab (S-frame) conservation of momentum implies: $m_A u_A + m_B u_B = m_C u_C + m_D u_D$. Mass is also conserved: $m_A + m_B = m_C + m_D$. Now suppose we wish to observed the collision from a frame S′ moving at speed v. Then, from the relativistic addition of velocities formula, $u' = \dfrac{u - \mathrm{v}}{1 - \dfrac{u\mathrm{v}}{c^2}}$ and, from it, $u = \dfrac{u' + \mathrm{v}}{1 + \dfrac{u'\mathrm{v}}{c^2}}$. Insert this into the equation of conservation of momentum (using $p = m\mathrm{v}$ as the definition):

$$m_A\left(\frac{u'_A + \mathrm{v}}{1 + u'_A\mathrm{v}/c^2}\right) + m_B\left(\frac{u'_B + \mathrm{v}}{1 + u'_B\mathrm{v}/c^2}\right) = m_C\left(\frac{u'_C + \mathrm{v}}{1 + u'_C\mathrm{v}/c^2}\right) + m_D\left(\frac{u'_D + \mathrm{v}}{1 + u'_D\mathrm{v}/c^2}\right)$$

This is clearly not the equation $m_A u'_A + m_B u'_B = m_C u'_C + m_D u'_D$ So momentum will not be conserved relativistically if we insist on using the old definition!!

10. Can we save the situation and make the conservation of momentum hold by finding some suitable new definition of momentum? The new definition must have two properties:

    1) At low speeds it must reduce to the old one.

    2) At all speeds momentum must be conserved.

    Let's see if the definition $\vec{p} = \dfrac{m\vec{u}}{\sqrt{1 - u^2/c^2}} = \gamma m\vec{u}$ will do the job. Obviously if u<<c we get $\vec{p} = m\vec{u},$ so requirement 1 is clearly satisfied. . Let's now see if the conservation of momentum equation will hold if the new definition of momentum is used:

    $$m_A\gamma_A u_A + m_B\gamma_B u_B = m_C\gamma_C u_C + m_D\gamma_D u_D.$$

    After doing some algebra you find,

    $$m_A\left(\frac{1}{\gamma}\gamma'_A u'_A + \gamma_A\mathrm{v}\right) + m_B\left(\frac{1}{\gamma}\gamma'_B u'_B + \gamma_B\mathrm{v}\right) = m_C\left(\frac{1}{\gamma}\gamma'_C u'_C + \gamma_C\mathrm{v}\right) + m_D\left(\frac{1}{\gamma}\gamma'_D u'_D + \gamma_D\mathrm{v}\right)$$

    This gives $m_A\gamma'_A u'_A + m_B\gamma'_B u'_B = m_C\gamma'_C u'_C + m_D\gamma'_D u'_D,$ which is just what we want.

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11. We shall now consider how energy must be redefined relativistically. The usual expression for the kinetic energy $K = \frac{1}{2}mu^2$ is not consistent with relativistic mechanics (it does not satisfy the law of conservation of energy in relativity). To discover a new definition, let us start from the basics: the work done by a force F when it moves through distance dx adds up to an increase in knetic energy, $K = \int F dx = \int \frac{dp}{dt} dx = \int \frac{dx}{dt} dp = \int u dp$. Now use:

$$dp = md\left(\frac{u}{\sqrt{1-u^2/c^2}}\right) = \frac{mdu}{\sqrt{1-u^2/c^2}} + \frac{m\left(u^2/c^2\right)du}{\left(1-u^2/c^2\right)^{3/2}} = \frac{mudu}{\left(1-u^2/c^2\right)^{3/2}}$$

This give $K = \int_0^u udp = m\int_0^u \frac{udu}{\left(1-u^2/c^2\right)^{3/2}} = \frac{mc^2}{\sqrt{1-u^2/c^2}} - mc^2$

Note that K is zero if there is no motion, or $K = E - E_0$ where $E = \frac{mc^2}{\sqrt{1-u^2/c^2}}$ and $E_0 = mc^2$. Now expand $\frac{1}{\sqrt{1-u^2/c^2}} = \left(1-u^2/c^2\right)^{-1/2} = 1 + \frac{u^2}{2c^2} + \cdots$

Hence, $K = mc^2\left(1 + \frac{u^2}{2c^2} + \cdots\right) - mc^2 \to \frac{1}{2}mu^2$ as $u/c \to 0$.

12. Now that we have done all the real work, let us derive some alternative expressions using our two main formulae: $\vec{p} = \frac{m\vec{u}}{\sqrt{1-u^2/c^2}} = \gamma m\vec{u}$ and $E = \frac{mc^2}{\sqrt{1-u^2/c^2}} = \gamma mc^2$.

a) $\vec{u} = \frac{\vec{p}}{\gamma m} = \frac{\vec{p}c^2}{E}$ or $pc = E\left(\frac{u}{c}\right)$

b) Clearly $(pc)^2 = E^2\left(\frac{u}{c}\right)^2 = \gamma^2 m^2 c^4\left(1 - 1/\gamma^2\right) = \gamma^2 m^2 c^4 - m^2 c^4 = E^2 - m^2 c^4$. Hence,

$E^2 = p^2 c^2 + m^2 c^4$.

c) For a massless particle ($m = 0$), $E = pc$.

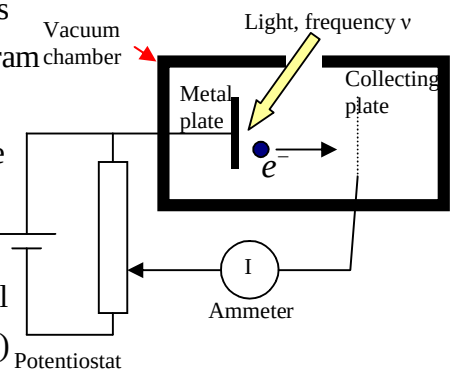
13. A particle with mass m has energy mc^2 even though it is at rest. This is called its rest energy. Since c is a very large quantity, even a small m corresponds to a very large energy. We interpret this as follows: suppose all the mass could somehow be converted into energy. Then an amount of energy equal to mc^2 would be released.

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Summary of Lecture 41 – WAVES AND PARTICLES

1. We think of particles as matter highly concentrated in some volume of space, and of waves as being highly spread out. Think of a cricket ball, and of waves in the ocean. The two are completely different! And yet today we are convinced that matter takes the form of waves in some situations and behaves as particles in other situations. This is called wave-particle duality. But do not be afraid - there is no logical contradiction here! In this lecture we shall first consider the evidence that shows the particle nature of light.

2. The photoelectric effect, noted nearly 100 years ago, was crucial for understanding the nature of light. In the diagram shown, when light falls upon a metal plate connected to the cathode of a battery, electrons are knocked out of the plate. They reach a collecting plate that is connected to the battery's anode, and a current is observed. A vacuum is created in the apparatus so that the electrons can travel without hindrance. According to classical (meaning old!) physics we expect the following:



a)As intensity of light increases, the kinetic energy of the ejected electrons should increase.

b)Electrons should be emitted for any frequency of light ν, so long as the intensity of the light is sufficiently large.

But the actual observation was completely different and showed the following:

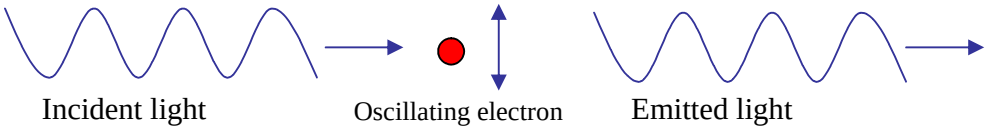
a)The maximum kinetic energy of the emitted electrons was completely independent of the light intensity, but it did depend on ν.

b)For $\nu < \nu_0$ (i.e. for frequencies below a cut-off frequency) no electrons are emitted no matter how large the light intensity was.

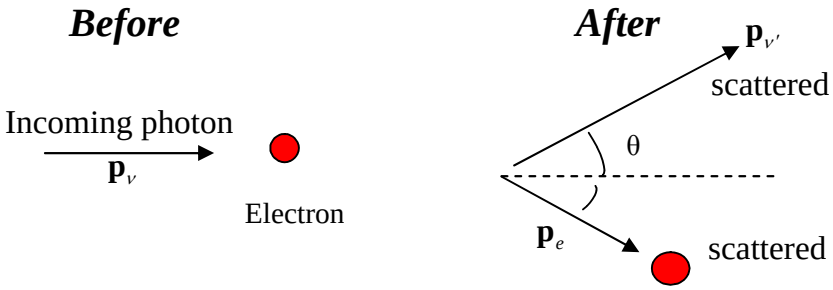
3. In 1905, Einstein realized that the photoelectric effect could be explained if light actually comes in little packets (or quanta) of energy with the energy of each quantum $E = h\nu$. Here h is a universal constant of nature with value $h = 6.63 \times 10^{-34}$ Joule-seconds, and is known as the Planck Constant. If an electron absorbs a single photon, it would be able to leave the material if the energy of that photon is larger than a certain amount W. W is called the work function and differs from material to material, with a value varying from 2-5 electron volts. The maximum KE of an emitted electron is then $K_{max} = h\nu - W$. We visualize the photon as a particle for the purposes of this experiment. Note that this is completely different from our earlier understanding that light is a wave!

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4. That light is made of photons was confirmed by yet another experiment, carried out by Arthur Compton in 1922. Suppose an electron is placed in the path of a light beam. What will happen? Because light is electromagnetic waves, we expect the electron to oscillate with the same frequency as the frequency of the incident light ν. But because a charged particle radiates em waves, we expect that the electron will also radiate light at frequency ν. So the scattered and incident light have the same ν. But this is not what is observed!



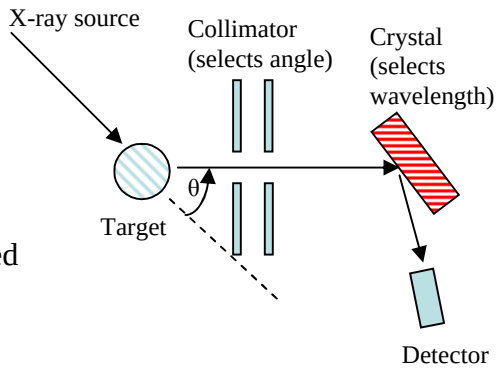
To explain the fact that the scattered light has a different frequency (or wavelength), Compton said that the scattering is a collision between particles of light and electrons. But we know that momentum and energy is conserved in scattering between particles. Specifically, from conservation of energy $h\nu + m_e c^2 = h\nu' + \left(p_e^2 c^2 + m_e^2 c^4\right)^{1/2}$. The last term is the energy of the scattered electron with mass m_e. Next, use the conservation



of momentum. The initial momentum of the photon is entirely along the $\hat{z}$ direction, $\mathbf{p}_\nu = \frac{h}{\lambda}\hat{z} = \mathbf{p}_{\nu'} + \mathbf{p}_e$. By resolving the components and doing a bit of algebra, you can get the change in wavelength $\lambda' - \lambda = \frac{h}{m_e c}(1-\cos\theta) = \lambda_c(1-\cos\theta)$, where the Compton wavelength $\lambda_c = \frac{h}{m_e c} = 2.4\times10^{-12}$ m. Note that $\lambda' - \lambda$ is always positive because $\cos\theta$ has magnitude less than 1. In other words, the frequency of the scattered photon is always less than the frequency of the incoming one. We can understand this result because the incoming photon gives a kick to the (stationary) electron and so it loses energy. Since $E = h\nu$, it follows that the outgoing frequency is decreased. As remarked earlier, it is impossible to understand this from a classical point of view. We shall now see how the Compton effect is actually observed experimentally.

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5. In the apparatus shown, X-rays are incident upon a target which contains electrons. Those X-rays which are scattered in a particular angle θ are then selected by the collimator and are incident upon a crystal. The crystal diffracts the X-rays and, as you will recall from the lecture on diffraction, $2d\sin\theta_D = n\lambda$ is the necessary condition. It was thus determined that the changed wavelength of the scattered follows $\lambda' - \lambda = \lambda_c(1-\cos\theta)$.



6. Here is a summary of photon facts:
   a)The relation between the energy and frequency of a photon is $E = h\nu$.
   b)The relativistic formula relating energy and momentum for photons is $E = pc$. Note that in general $E^2 = p^2c^2 + m^2c^4$ for a massive particle. The photon has $m = 0$.
   c)The relation between frequency, wavelength, and photon speed is $\lambda\nu = c$.
   d)From the abbove, the momentum-wavelength relation for photons is $p = \frac{h\nu}{c} = \frac{h}{\lambda}$.
   e)An alternative way of expressing the above is: $E = \hbar\omega$ and $p = \hbar k$. Here $\omega = 2\pi\nu$ and $k = \frac{2\pi}{\lambda}$, $\hbar \equiv \frac{h}{2\pi}$ ($\hbar$ is pronounced h-bar).
   f)Light is always detected as packets (photons); if we look, we never observe half a photon. The number of photons is proportional to the energy density (i.e. to square of the electromagnetic field strength).

7. So light behaves as if made of particles. But do all particles of matter behave as if they are waves? In 1923 a Frenchman, Louis de Broglie, postulated that ordinary matter can have wave-like properties with the wavelength λ related to the particle's momentum p in the same way as for light, $\lambda = \frac{h}{p}$. We shall call λ the de Broglie (pronounced as Deebrolee!) wavelength.

8. Let us estimate some typical De Broglie wavelengths:
   a) The wavelength of 0.5 kg cricket ball moving at 2 m/sec is:

$$\lambda = \frac{h}{p} = \frac{6.63\times10^{-34}}{0.5\times2} = 6.63\times10^{-34}\ \text{m}$$

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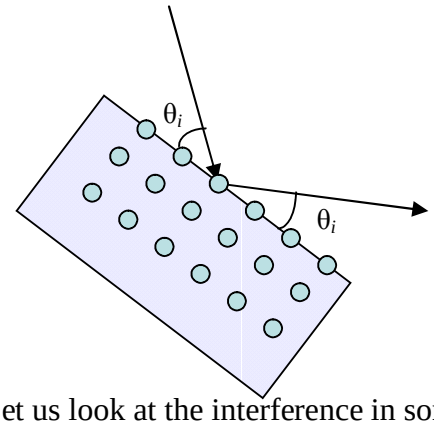
This is extremely small even in comparison to an atom, 10^{-10} m.

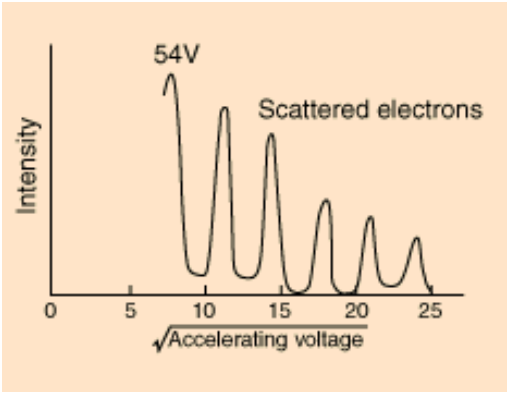
b) The wavelength of an electron with 50eV kinetic energy is calculated from:

$$K = \frac{p^2}{2m_e} = \frac{h^2}{2m_e\lambda^2} \Rightarrow \lambda = \frac{h}{\sqrt{2m_eK}} = 1.7\times10^{-10}\,\text{m}$$

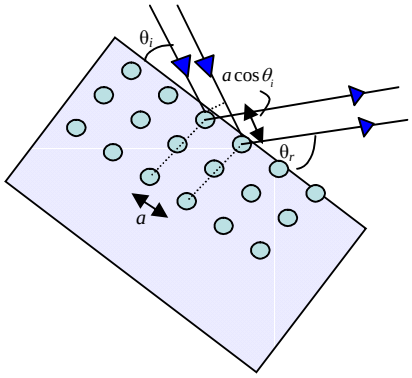
Now you see that we are close to atomic dimensions.

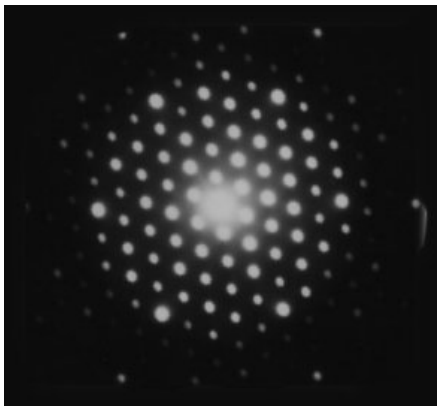
9. If De Broglie's hypothesis is correct, then we can expect that electron waves will undergo interference just like light waves. Indeed, the Davisson-Germer experiment (1927) showed that this was true. At fixed angle, one find sharp peaks in intensity as a function of electron energy. The electron waves hitting the atoms are re-emitted and reflected, and waves from different atoms interfere with each other. One therefore sees the peaks and valleys that are typical of interference (or diffraction) patterns in optical experiments.





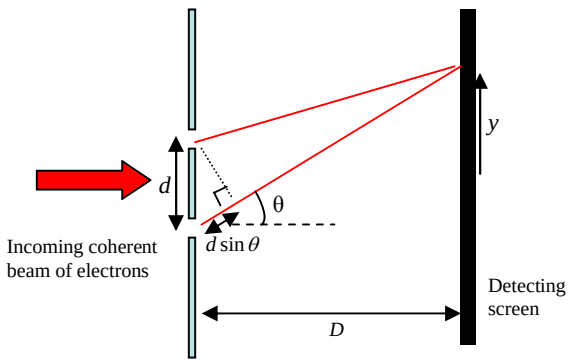
10. Let us look at the interference in some detail. When electrons fall on a crystalline surface, the electron scattering is dominated by surface layers. Note that the identical scattering planes are oriented perpendicular to the surface. Looking at the diagram, we can see that constructive interference happens when $a(\cos\theta_r - \cos\theta_i) = n\lambda$. When this condition is satified, there is a maximum intensity spot. This is actually how we find a and determine the structure of crystals.

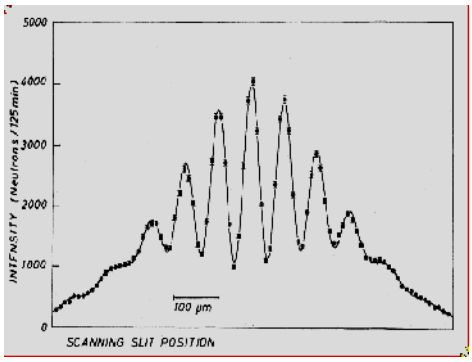




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11. Let's take a still simpler situation: electrons are incident upon a metal plate with two tiny holes punched into it. The holes - spearated by distance d - are very close together. A screen behind, at distance D, is made of material that flashes whenever it is hit by an electron. A clear interference pattern with peaks and valleys is observed. Let us analyze: there will be a maximum when $d\sin\theta = n\lambda$. If the screen is very far away, i.e. $D >> d$, then θ will be very small and $\sin\theta \approx \theta$. So we then have $\theta \approx \frac{n\lambda}{d}$, and the angular separation between two adjacent minima is $\Delta\theta \approx \frac{\lambda}{d}$. The position on the screen is y, and $y = D\tan\theta \approx D\theta$. So the separation between adjacent maxima is $\Delta y \approx D\Delta\theta$ and hence $\Delta y = \frac{\lambda D}{d}$. This is the separation between two adjacent bright spots. You can see from the experimental data that this is exactly what is observed.





12. The double-slit experiment is so important that we need to discuss it further. Note the following:

a) It doesn't matter whether we use light, electrons, or atoms - they all behave as waves. in this experiment. The wavelength of a matter wave is unconnected to any internal size of particle. Instead it is determined by the momentum, $\lambda = \frac{h}{p}$.

b) If one slit is closed, the interference disappears. So, in fact, each particle goes through both slits at once.

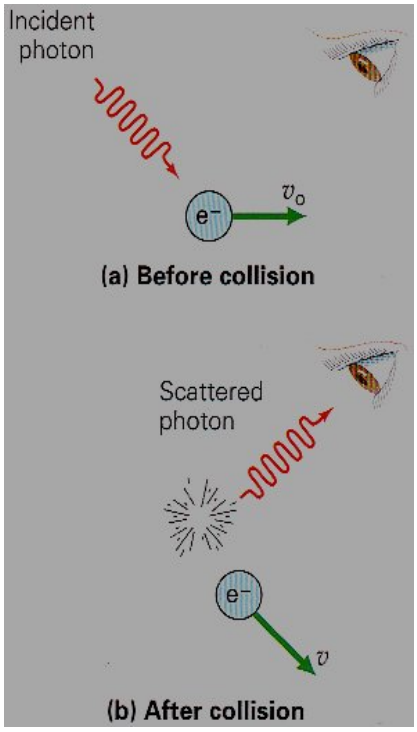
c) The flux of particles arriving at the slits can be reduced so that only one particle arrives at a time. Interference fringes are still observed !Wave-behaviour can be shown by a single atom. In other words, a matter wave can interfere with itself.

d) If we try to find out which slit the particle goes through the interference pattern vanishes! We cannot see the wave/particle nature at the same time.

All this is so mysterious and against all our expectations. But that's how Nature is!

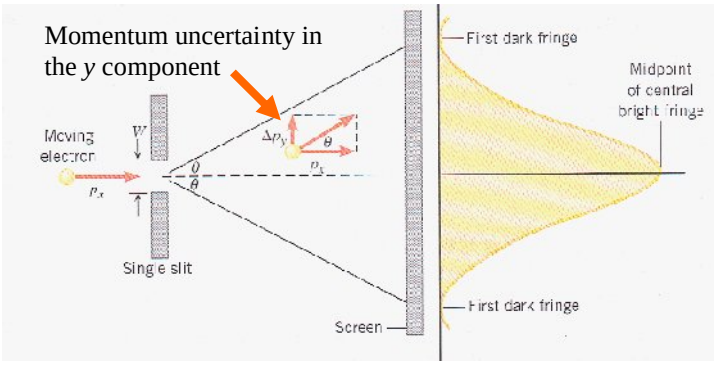
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13. *Heisenberg Uncertainty Principle.* In real life we are perfectly familiar with seeing a cricket ball at rest - it has both a fixed position and fixed (zero) momentum. But in the microscopic world (atoms, nuclei, quarks, and still smaller distance scales) this is impossible. Heisenberg, the great German physicist, pointed out that if we want to see an electron, then we have to hit it with some other particle. So let's say that a photon hits an electron and then enters a detector. It will carry information to the detector of the position and velocity of the electron, but in doing so it will have changed the momentum of the electron. So if the electron was initially at rest, it will no longer be so afterwards. The act of measurement changes the state of the system! This is true no matter how you do the experiment. The statement of the principle is:



"If the position of a particle can be fixed with accuracy Δx, then the maximum accuracy with which the momentum can be fixed is Δp, where $\Delta x \Delta p \geq \hbar/2$." We call Δx the position uncertainty, and Δp the momentum uncertainty. Note that their product is fixed. Therefore, if we fix the position of the particle (make Δx very small), then the particle will move about randomly very fast (and have Δp very large).

14. By using the De Broglie hypothesis in a simple gedanken experiment, we can see how the uncertainty principle emerges. Electrons are incident upon a single slit and strike a screen far away. The first dark fringe will be when $W \sin\theta = \lambda$. Since θ is small, we can use $\sin\theta \approx \theta$, and so $\theta \approx \dfrac{\lambda}{W}$. But, on the other hand, $\tan\theta = \dfrac{\Delta p_y}{p_x}$ and $\theta \approx \dfrac{\Delta p_y}{p_x}$. So, $\dfrac{\Delta p_y}{h/\lambda} = \dfrac{\lambda}{W}$. But W is really the uncertainty in the y position and we should call it Δy.



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Thus we have found that $\Delta p_y \Delta y \approx h$. The important point here is that by localizing the y position of the electron to the width of the slit, we have forced the electron to acquire a momentum in the y direction whose uncertainty is Δp_y.

15. In a proper course in quantum mechanics, one can give a definite mathematical meaning to Δx and Δp_x etc, and derive the uncertainty relations:

$$\Delta x \Delta p_x \geq \hbar/2, \quad \Delta y \Delta p_y \geq \hbar/2, \quad \Delta z \Delta p_z \geq \hbar/2$$

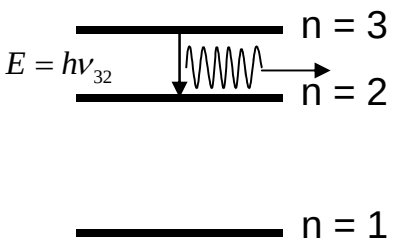
Note the following:

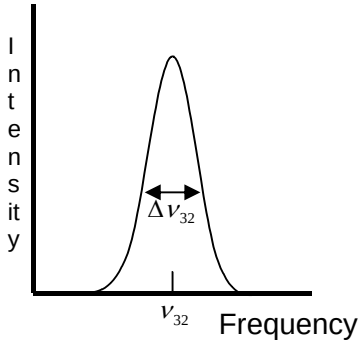
a) There no uncertainty principle for the product $\Delta x \Delta p_y$. In other words, we can know in principle the position in one direction precisely together with the momentum in another direction.

b) The thought experiment I discussed seems to imply that, while prior to experiment we have well defined values, it is the act of measurement which introduces the uncertainty by disturbing the particle's position and momentum. Nowadays it is more widely accepted that quantum uncertainty (lack of determinism) is intrinsic to the theory and does not come about just because of the act of measurement.

16. There is also an Energy-Time Uncertainty Principle which states that $\Delta E \Delta t \geq \hbar/2$. This says that the principle of energy conservation can be violated by amount ΔE, but only for a short time given by Δt. The quantity ΔE is called the uncertainty in the energy of a system.

17. One consequence of $\Delta E \Delta t \geq \hbar/2$ is that the level of an atom does not have an exact value. So, transitions between energy levels of atoms are never perfectly sharp in frequency. So, for example, as shown below an electron in the $n = 3$ state will decay to a lower level after a lifetime of order $t \approx 10^{-8}$s. There is a corresponding "spread" in the emitted frequency.

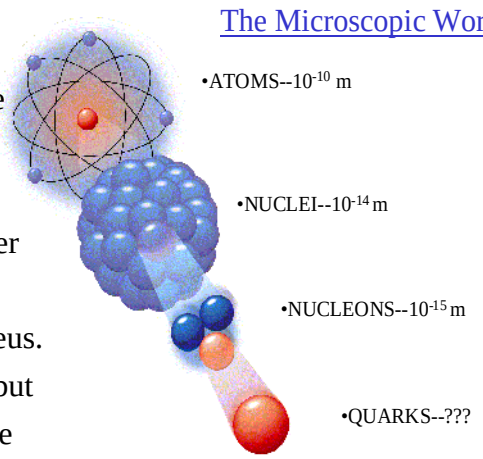




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Summary of Lecture 42 – QUANTUM MECHANICS

1. The word "quantum" means packet or bundle. We have already encountered the quantum of light - called photon - in an earlier lecture. Quanta (plural of quantum) are discrete steps. Walking up a flight of stairs, you can increase your height one step at a time and not, for example, by 0.371 steps. In other words your height above ground (and potential energy can take discrete values only).
2. Quantum Mechanics is the true physics of the microscopic world. To get an idea of the sizes in that world, let us start from the atom which is normally considered to be a very small object. But, as you can see, the atomic nucleus is 100,000 times smaller than the atom. The neutron and proton are yet another 10 times smaller than the nucleus. We know that nuclei are made of quarks, but as yet we do know if the quarks have a size or if they are just point-like particles.



3. It is impossible to cover quantum mechanics in a few lectures, much less in this single lecture. But here are some main ideas:
   a) Classical (Newtonian) Mechanics is extremely good for dealing with large objects (a grain of salt is to be considered large). But on the atomic level, it fails.The reason for failure is the uncertainty principle - the position and momenta of a particle cannot be determined simultaneously (this is just one example; the uncertainty principle is actually more general). Quantum Mechanics properly describes the microscopic - as well as macroscopic - world and has always been found to hold if applied correctly.
   b) Atoms or molecules can only exist in certain energy states. These are also called "allowed levels" or quantum states. Each state is described by certain "quantum numbers" that give information about that state's energy, momentum, etc.
   c) Atoms or molecules emit or absorb energy when they change their energy state. The amount of energy released or absorbed equals the difference of energies between the two quantum states.
   d) Quantum Mechanics always deals with probabilities. So, for example, in considering the outcome of two particles colliding with each other, we calculate probabilities to scatter in a certain direction, etc.

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4. What brought about the Quantum Revolution? By the end of the 19th century a number of serious discrepancies had been found between experimental results and classical theory. The most serious ones were:

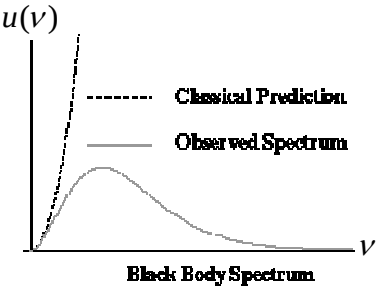
A) The blackbody radiation law

B) The photo-electric effect

C) The stability of the atom, and puzzles of atomic spectra

In the following, we shall briefly consider these discrepancies and the manner in which quantum mechanics resolved them.

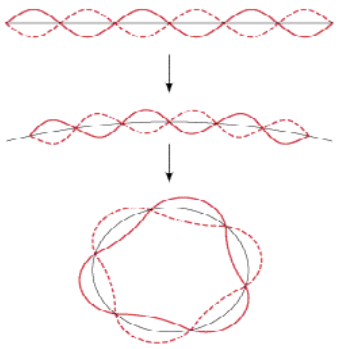
A)Classical physics gives the wrong behaviour for radiation emitted from a hot body. Although in this lecture it is not possible to do the classical calculation, it is not difficult to show that the electromagnetic energy $u(\nu)$ radiated at frequency ν increases as ν^3 (see graph). So the energy radiated over all frequencies is infinite. This is clearly wrong. The correct calculation was done by Max Planck.



Planck's result is shown above, and it leads to the sensible result that u(ν) goes to zero at large ν. He assumed that radiation of a given frequency ν could only be emitted and absorbed in quanta of energy $\varepsilon = h\nu$. If the electromagnetic field is thought of as harmonic oscillators, Planck assumed that the total energy of this large number of oscillators is made of finite energy elements $h\nu$. With this assumption, he came up with a formula that fitted well with the data. But he called his theory "an act of desperation" because he did not understand the deeper reasons.

B) I have already discussed the photoelectric effect in the previous lecture. Briefly, Einstein (1905) postulated a quantum of light called photon, which had particle properties like energy and momentum. The photon is responsible for knocking electrons out of the metal - but only if it has enough energy.

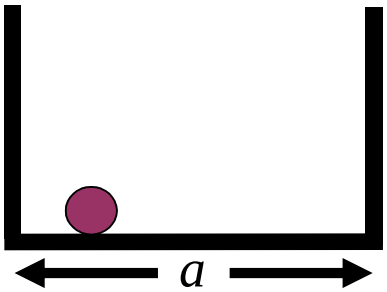
C) Classical physics cannot explain the fact that atoms are stable. An accelerating charge always radiates energy if classical electromagnetism is correct. So why does the hydrogen atom not collapse? In 1921 Niels Bohr, a great Danish physicist, made the following hypothesis: if an electron moves around a nucleus so that its angular momentum is $\hbar, 2\hbar, 3\hbar, \cdots$ then it will not radiate energy. In the next lecture we explore the consequences of this hypothesis. Bohr's hypothesis called for the quantization



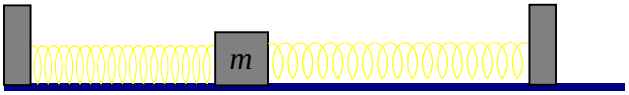
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of angular momentum. If the electron is regarded as a De Broglie wave, then there must be an integer number of waves that go around the centre. Of course, this is not proper quantum mechanics and cannot be taken too seriously, but it definitely was a major step in the ultimate development of the subject. Even if one's understanding was imperfect, it was now possible to understand why atoms had certain energies only, and why the light radiated by atoms was of discrete frequencies only. In contrast, classical physics predicted that atoms could radiate at any frequency.

5. One consequence of quantum mechanics is that it explains through the uncertainty principle the stability of the atom. But before talking of that, let us consider a particle moving between two walls. Each time it hits a wall, a force pushes it in the opposite direction. There is no friction, so classically the particle just keeps moving forever between the two potential walls. In QM the uncertainty of the particle's position is $\Delta x = a$ and so, from $\Delta x \Delta p \geq \hbar/2,$ we have $\Delta p\, a \approx \frac{\hbar}{2}$. From this we learn that the kinetic energy $\frac{(\Delta p)^2}{2m} \approx \frac{\hbar^2}{8ma^2}$. This is telling us that as we squeeze the particle into a tighter and tighter space, the kinetic energy goes up and up!



6. The story repeats for a harmonic oscillator. So imagine that a mass moves in a potential of the type shown below, and that its frequency of oscillation is ω. Classically, the lowest energy would be that in which the mass is at rest (no kinetic energy) and it is at the position where the potential is minimum. But this means that the body has both a

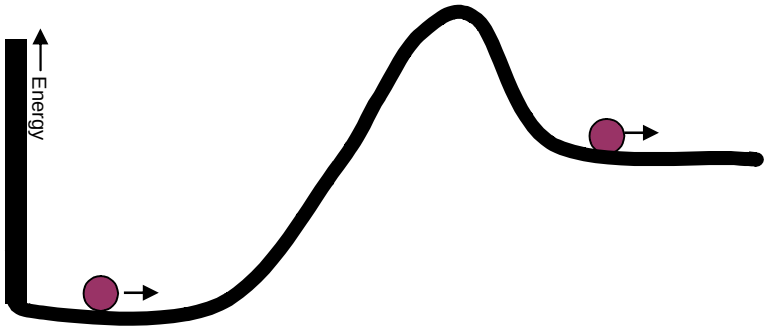


well-defined momentum and position. This is forbidden by the Heisenberg uncertainty principle. A proper quantum mechanical calculation shows that the minimum energy is actually $\frac{1}{2}\hbar\omega$. This is called the zero-point energy and comes about because it is not possible for the mass to be a rest. The oscillator's other energy states have energies $\frac{1}{2}\hbar\omega, \frac{3}{2}\hbar\omega, \frac{5}{2}\hbar\omega, \frac{7}{2}\hbar\omega, \cdots$ We say that the energies are quantized. This is completely impossible to understand from classical mechanics where we know that we can excite an oscillator to have any energy we want. Quantized energy levels are indeed what we observe in so many physical systems: atoms, rotating or vibrating molecules, nuclei, etc.

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7. We can understand from both the previous examples why the hydrogen atom does not collapse even if the electron does not have any orbital angular momentum around the proton. The uncertainty principle essentially forces the electron to stay away from the proton - if it tried to get too close, the kinetic energy would rise enormously because $\Delta x \Delta p \geq \hbar/2$ says that if Δx becomes small then Δp must become big to compensate.

8. Quantum mechanics predicts what is called "tunneling" of particles through a potential barrier. Again, we shall use the uncertainty principle but this time the energy-time one. Imagine a particle that moves in a potential of the shape shown below. Suppose it does not have enough energy to go over the peak and on to the other side. In that case, we



know from our experience that it will just keep oscillating - moving first towards the hill and then down again, etc. But quantum mechanically, it can "steal" energy ΔE for a time Δt and this may be enough to surmount the hill. Of course, the particle must respect $\Delta E \Delta t \geq \hbar/2$ so the time is small if it needs a large amount of energy to cross over. Again, I have given only a rough argument here, but in quantum mechanics we can do proper calculations to find tunneling probabilities.

9. Without tunneling our sun would go cold. As you may know, it is powered by hydrogen fusion. The protons must somehow overcome electrostatic repulsion to get close enough so that they can feel the nuclear force and fuse with each other. But the thermal energy at the core of the sun is not high enough. It is only because the tunneling effect allows protons to sometimes get close enough that fusion happens.

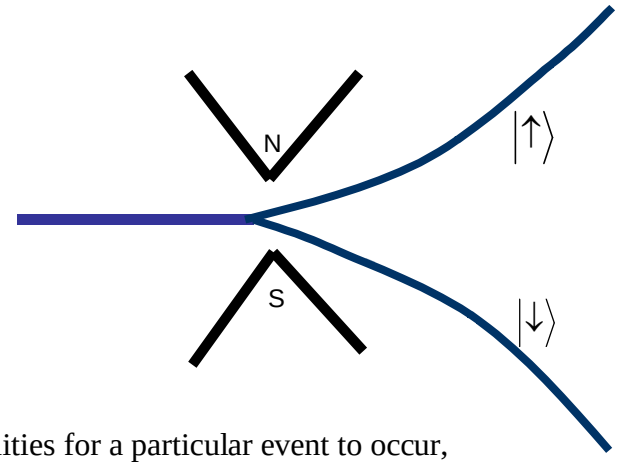
10. Quantum mechanics is all about probabilities.What is probability? Probability is a measure of the likelihood that an event will occur. Probability values are assigned on a scale of zero (the event can never occur) to one (the event definitely occurs). More precisely, suppose we do an experiment (like rolling dice or flipping a coin) N times where N is large. Then if a certain outcome occurs n times, the probability is $P = n/N$.

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11. The simplest system for discussing quantum mechanics is one that has only two states. Let us call these two states "up" and "down" states, $|\uparrow\rangle$ and $|\downarrow\rangle$. They could denote an electron with spin up/down, or a switch which is up/down, or an atom which can be only in one of two energy states, etc. Things like $|\uparrow\rangle$ were called kets by their inventor, Paul Dirac.

For definiteness let us take the electron example. If the state of the electron is known to be $|\Psi\rangle = \sqrt{\frac{2}{3}}|\uparrow\rangle + \sqrt{\frac{1}{3}}|\downarrow\rangle$, then the probability of finding the electron with spin up is $P(\uparrow) = \frac{2}{3}$, and with spin down is $P(\downarrow) = \frac{1}{3}$. More generally: $|\Psi\rangle = c_1|\uparrow\rangle + c_2|\downarrow\rangle$ denotes an electron with $P(\uparrow) = |c_1|^2$ and $P(\downarrow) = |c_2|^2$. This means that if we look at look at a large number of electrons N all of which are in state $|\Psi\rangle$, then the number with spin up is $N|c_1|^2$ and with spin down is $N|c_2|^2$. We sometimes call $|\Psi\rangle = c_1|\uparrow\rangle + c_2|\downarrow\rangle$ a *quantum state*, and c_1 and c_2 *quantum amplitudes.*

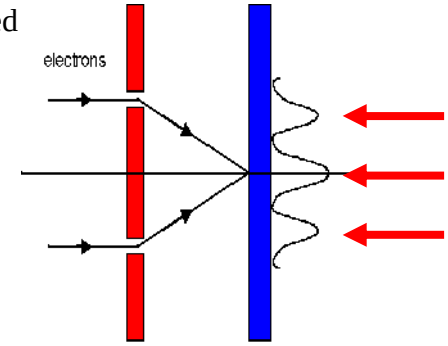
12. The Stern-Gerlach experiment illustrated here shows an electron beam entering a magnetic field. The electrons can be pointing either up or down relative to any chosen axis. The field forces them to choose one of the two states. That the beam splits into only two parts shows that the electron has only two states. Other particle beams might split into 3,4, $\cdots$



13. If a_1 and a_2 are the amplitudes of the two possibilities for a particular event to occur, then the amplitude for the total event is $A = a_1 + a_2$. Here a_1 and a_2 are complex numbers in general. But the probability for the event to occur is given by $P = |A|^2 = |a_1 + a_2|^2$. In daily experience we add probabilities, $P = P_1 + P_2$ but in quantum mechanics we add amplitudes: $P = |a_1 + a_2|^2 = a_1^* a_1 + a_2^* a_2 + a_1^* a_2 + a_2^* a_1 = P_1 + P_2 + a_1^* a_2 + a_2^* a_1$. The cross terms $a_1^* a_2 + a_2^* a_1$ are called interference terms. They are familiar to us from the lecture on light where we add amplitudes first, and then square the sum to find the intensity. Of course, if we add all possible outcomes then we will get 1. So, for example, in the electron case $P(\uparrow) + P(\downarrow) = |c_1|^2 + |c_2|^2 = 1$. Note that amplitudes can be complex but probabilities are always real.

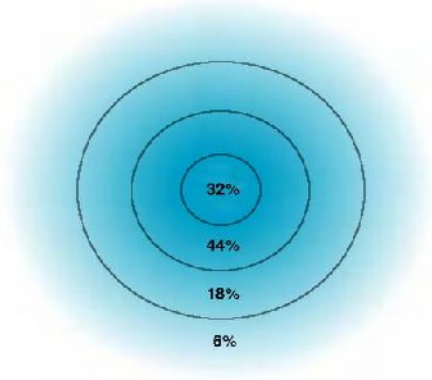
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14. Let us return to the double slit experiment discussed earlier. Here the amplitude for an electron wave coming from one slit interferes with the amplitude for an electron wave coming from the other slit. This is what causes a pattern to emerge in which electron are completely absent in certain places (destructive interference) and are present in large numbers where there is constructive interference. So what we must deal with are matter waves. But how to treat this mathematically?



15. The above brings us to the concept of a "wave function". In 1926 Schrödinger proposed a quantity that would describe electron waves (or, more generally, matter waves).
    - The wavefunction $\Psi(x,t)$ of a particle is the amplitude to be at position x at time t.
    - The probability of finding the particle at position x between x and $x+dx$ (at time t) is $|\Psi(x,t)|^2\,dx$. Since the particle has to be somewhere, if we add up all possibilities then we must get one, i.e. $\int_{-\infty}^{+\infty} |\Psi(x,t)|^2\,dx = 1$.
    - $\Psi(x,t)$ is determined by solving the "Schrodinger equation" which, unfortunately, I shall not be able to discuss here. This is one of the most important equations of physics. If it is solved for the atom then it tells you all that is possible to know: energies, the probability of finding an electron here or there, the momenta with which they move, etc. Of course, one usually cannot solve this equation in complicated situations (like a large molecule, for example) and this is what makes the subject both difficult and interesting.

16. For an electron moving around a nucleus, one can easily solve the Schrodinger equation and thus find the wavefunction $\Psi(x,t)$. From this we compute $|\Psi(x,t)|^2$, which is large where the electron is more likely to be found. In this picture, the probability of finding the electron inside the first circle is 32%, between the second and first is 44%, etc.



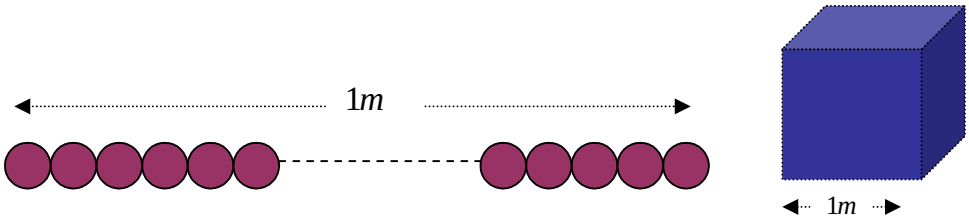
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Summary of Lecture 43 – INTRODUCTION TO ATOMIC PHYSICS

1. • About 2500 years ago, the ancient Greek philosopher Democritus asked the question: what is the world made of? He conjectured that it is mostly empty, and that the remainder is made of tiny "atoms". By definition these atoms are indivisible.
   • Then 300 years ago, it was noted by the French chemist Lavoisier that in all chemical reactions the total mass of reactants before and after a chemical reaction is the same. He demonstrated that burning wood caused no change in mass. This is the Law of Conservation of Matter.
   • A major increase in understanding came with Dalton (1803) who showed that:
   1) Atoms are building blocks of the elements.
   2) All atoms of the same element have the same mass.
   3) Atoms of different elements are different.
   4) Two or more different atoms bond in simple ratios to form compounds.

2. Avogadro made the following hypothesis : "Equal volumes of all gases, under the same conditions of temperature and pressure, contain equal numbers of molecules". Why? Because we know that pressure is caused by molecules hitting the sides of the containing vessel. If the temperature of two gases is the same, then their molecules move with the same speeds, and so Avogadro's hypothesis follows for ideal gases. The famous number $N_0 = 6.023 \times 10^{26}$ per kilogram-mole is called Avogadro's Number.

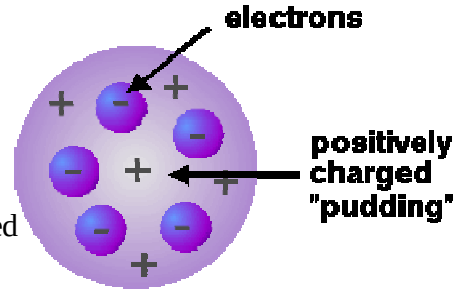
3. Let's get an idea of the size of atoms. Amazingly, we do not need high-powered particle accelerators to do so. Consider a cube of $1m \times 1m \times 1m$. If the radius of an atom is r, then we have $(1/2r)^3$ atoms in the cube. Now in 1kg.atom we have $N_0 = 6 \times 10^{26}$ atoms and each atom occupies a volume (A/ρ) m^3, where A = atomic weight and ρ = density.



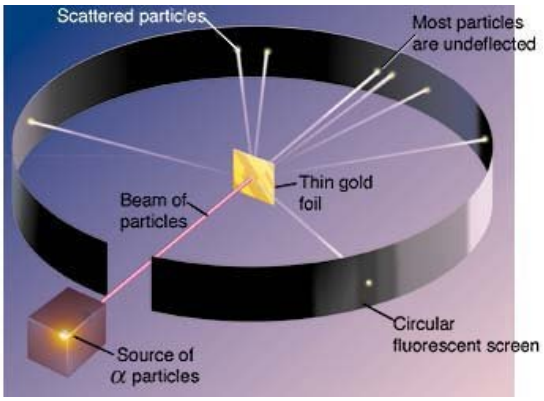
Hence $N_0 = (1/2r)^3 \times A/\rho$. This give $r = \frac{1}{2}\left(\frac{A}{\rho N_0}\right)^{1/3}$. Putting in some typical densities, we find that $r_{Ag} \approx r_{Be} \approx 10^{-10} m$. This shows that atoms are mostly of the same size. This is quite amazing because one expects a Be atom to be much smaller than an Ag atom.

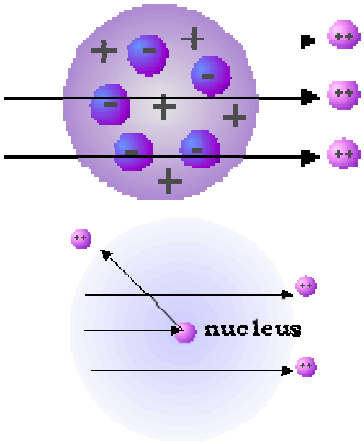
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4. Even if you know how big an atom is, this does not mean that its internal structure is known. In 1895 J.J.Thomson proposed the "plum pudding" model of an atom. Here the atom is considered as made of a positively charged material with the negatively charged electrons scattered through it.



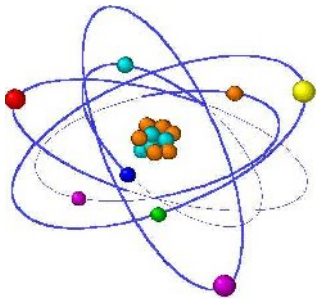
5. But the plum-pudding model was wrong. In 1911, Rutherford carried out his famous experiment that showed the existence of a small but very heavy core of the atom. He arranged for a beam of α particles to strike gold atoms in a thin foil of gold.





If the positive and negative charges in the atom were randomly distributed, all $\alpha's$ would go through without any deflection. But a lot of backscattering was seen, and some α's were even deflected back in the direction of the incident beam. This was possible only if they were colliding with a very heavy object inside the atom. Rutherford had discovered the atomic nucleus.

6. The picture that emerged after Rutherford's discovery was like that of the solar system - the atom was now thought of as mostly empty space with a small, positive nucleus that contained protons. Negative electrons moved around the outside in orbits hat resembled those of planets, attracted towards the centre by a coulomb force.

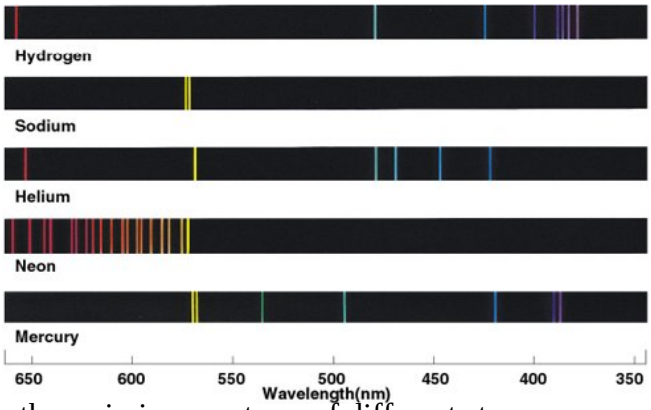


7. This sounds fine, but there is a serious problem: we know that a charge that accelerates radiates energy. In fact the power radiated is $P \propto e^2a^2$, where e is the charge and a is the acceleration. Now, a particle moving in a circular orbit has an acceleration even if it is moving at constant speed because it is changing its direction all the time. So this means

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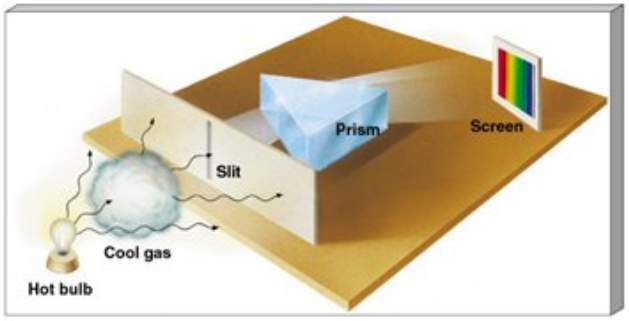
that the electron will be constantly radiating power and thus will slow down, collapsing eventually into the nucleus.

8. This is not the only thing wrong with the solar system model. If you look at the light emitted by any atom, you do not see a continuous distribution of colours (frequencies). Instead, a spectroscope will easily show that light is emitted at only certain discrete frequencies.



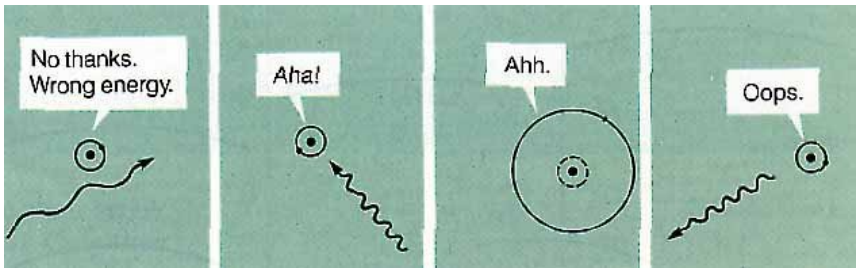
In the above you see the emission spectrum of different atoms.

9. Similarly, if white light is passed through a gas of atoms of a certain type, only certain colours are absorbed, and the others pass through without a hindrance.



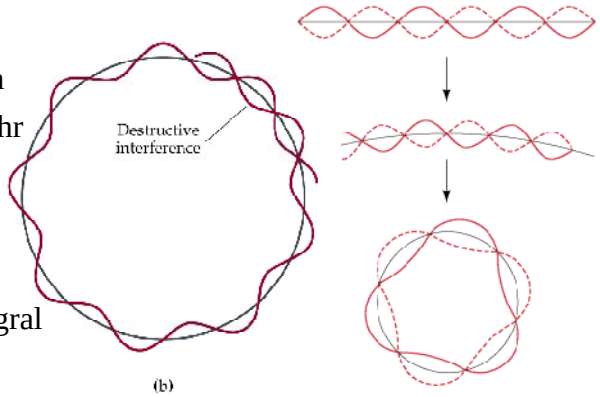


The above shows the absorption spectrum of a certain atom. The wavelengths for both emission and absorption lines are exactly equal. Classical physics and the Rutherford model have no explanation for the spectrum.



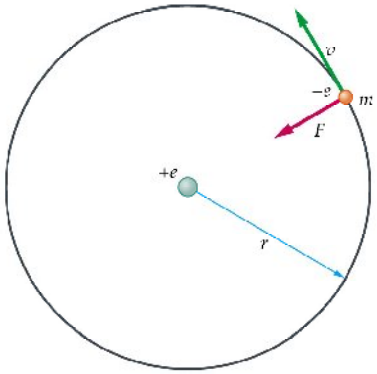
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10. Then came Niels Bohr. By this time it was known that electrons had a dual character as waves (De Broglie relation and Davisson-Germer experiment). Bohr said: suppose I bend a standing wave into a circle. If the wavelength is not exactly correct, wave interference will make the wave disappear. So only integral numbers of wavelengths can interfere constructively.



11. Let us pursue this idea further. The electron has a wavelength and forms standing waves in its orbit around the nucleus. An integral number of electron wavelengths must fit into the circumference of the circular orbit. Hence $n\lambda = 2\pi r$ with $n = 1, 2, 3\cdots$ The momentum is $p = m\mathrm{v} = \frac{h}{\lambda} = \frac{h}{(2\pi r/n)} = \frac{n\hbar}{r}$. The angular momentum $L = rm\mathrm{v} = n\hbar$ is therefore quantized in units of $\hbar$.

12. Now let us suppose that the electron moves in an orbit of radius r when it has $L_n = n\hbar$. Equilibrium demands that the centrifugal force be equal to the coulomb attraction: $\frac{m\mathrm{v}_n^{\,2}}{r} = k\frac{e^2}{r^2}$. From $\mathrm{v}_n = \frac{n\hbar}{mr_n}$ we find that the radius $r_n = \left(\frac{\hbar^2}{mke^2}\right)n^2$.



- For $n = 1$ the electron orbit which is closest to the nucleus, $r_n = a_0 \equiv 0.53\times10^{-8}cm$ (this is called the Bohr radius).
- For higher n, $r_n = a_0 n^2$. The atom becomes huge for n ≈ 100, the so-called Rydberg atom. Such atoms are of experimental interest these days.
- Note that the speed of the electron is smaller in orbits farther from the nucleus, $\mathrm{v}_n = \frac{ke^2}{n\hbar}$. As n becomes very large, the electron is very far out and very slow.
- In the above $n = 0$ is strictly not allowed. As you can see, none of the formulae make any sense for this case. The minimum angular momentum that the electron can have is $\hbar$. (In proper quantum mechanics the minimum is $0\hbar$ and this is a big difference with the Bohr model of the atom).

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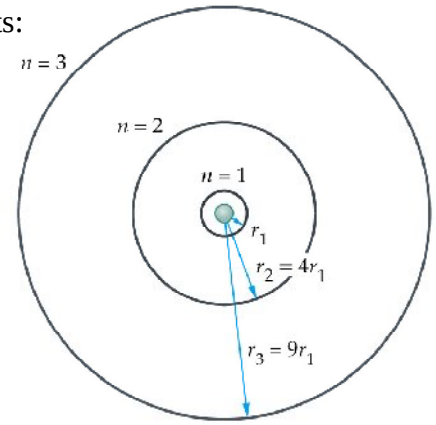
13. We can compute the energies of the various orbits:

$$E = K + U = \tfrac{1}{2}mv^2 + U$$

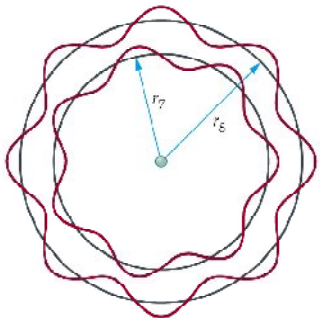
$$= \tfrac{1}{2}\left(\frac{ke^2}{r}\right) - \frac{ke^2}{r} = -\frac{ke^2}{2r}$$

Hence, $E_n = -\left(\frac{ke^2}{2}\right)\left(\frac{mke^2}{\hbar^2}\right)\frac{1}{n^2}$

$$= -\left(\frac{mk^2e^4}{2\hbar^2}\right)\frac{1}{n^2} = -\frac{13.6 \text{ eV}}{n^2}$$



14. In the Bohr model, electrons can jump between different orbits due to the absorption or emission of photons. Dark lines in the absorption spectra are due to photons being absorbed, and bright lines in the emission spectra are due to photons being emitted. The energy of the emitted or absorbed photon is equal to the difference of the initial and final energy levels, $h\nu = E_f - E_i$. The picture below shows the electron in the $n = 7$ and $n = 8$ levels. The photon emitted has $h\nu = E_8 - E_7 = -13.6\left(\frac{1}{8^2} - \frac{1}{7^2}\right)$ ev.

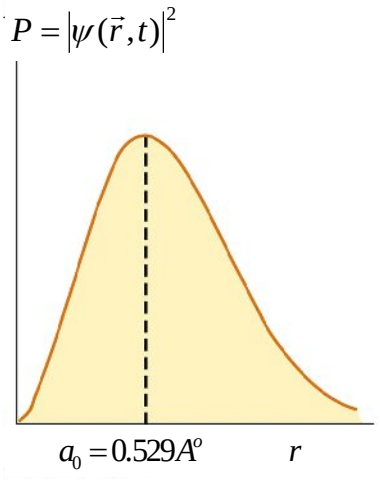


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15. The Bohr model gave wonderful results when compared against the hydrogen spectrum. It was the among the first indications that some "new physics" was needed at the atomic level. But this model is not to be taken too seriously - it fails to explain many atomic properties, and fails to explain why the H atom can exist even when the electron has no orbital angular momentum (and hence no centrifugal force to balance against the Coulomb attraction). It cannot predict all the lines observed for H, much less for multi-electron atoms such as Oxygen. The real value of this model was that it showed the way forward towards developing quantum mechanics, which is the true physics of the world, both microscopic and macroscopic. I have discussed some elements of QM in the last lecture, in particular the wavefunction $\psi(\vec{r},t)$ of the electron.

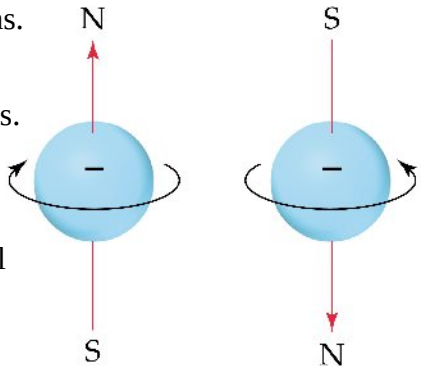
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16. Quantum mechanics gives a picture that is quite different from the solar-system model. We solve the "Schrodinger Equation" to find the wavefunction $\psi(\vec{r},t)$, whose square gives the probability of finding the electron at the point $\vec{r}$. In the lowest energy state, the electron can be viewed as a spherical cloud surrounding the nucleus. The densest regions of the cloud represent the highest probability for finding the electron.



- The principal quantum number $n = 1,2,3\cdots$ determines the allowed energy levels. An electron can only have energy $E_n = -\dfrac{13.6}{n^2}$ eV. Miraculously this is the same result as in the Bohr model.
- The orbital angular momentum is determined by the number l, and $L = \sqrt{l(l+1)}\hbar$. Allowed values of l are, $l = 0,1,2,3\cdots, n-1$.
- The magnetic quantum number m_l determines the projection (or component) of the vector angular momentum $\vec{L}$ on to any fixed axis, $L_z = m_l\hbar$. Allowed values are, $m_l = -l,\cdots -2,-1,0,1,2,\cdots,l$.

17. Electrons can be thought of as little spinning balls of charge. All electrons spin at the same speed (more accurately, their spin angular momentum is the same and equals $\hbar/2$). An electron can spin in only one of two possible directions. When charges move around in a circle, that constitutes a current. As you know, currents give rise to magnetic fields. This is why electrons are also little magnets that interact with other magnets. Now go back to the Stern-Gerlach experiment described in the previous lecture, and you will understand better why the electrons were deflected by the applied magnetic field. Now that we have learned that the electron has spin, we can describe the two spin states by giving the "magnetic" quantum number m_s where $m_s = -\frac{1}{2},\frac{1}{2}$. These two states have the same energy except when there is some magnetic field present.



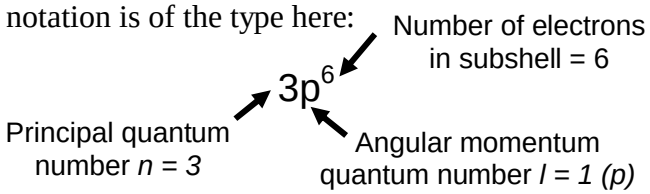
18. States (or orbitals) having $l = 0,1,2,\cdots$ are called s,p,d,$\cdots$. The s-states are spherical. As n increases, the s-orbitals get larger and the wavefunction is larger away from the nucleus.

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19. In the world we are used to, we can always tell apart identical particles (same mass, charge, spin,...) by simply watching them. But in QM, their idetities can get confused and identical particles are indistinguishable. Suppose that A and B are particles that are identical in every possible way, and we exchange them. Of course, the probability of finding one or the other must remain unchanged. In other words, $|\Psi(1,2)|^2 = |\Psi(2,1)|^2$, where 1 and 2 denote the positions of the first and second particles. But something very interesting happens now because either one of two possibilities can be true: $\Psi(1,2) = +\Psi(2,1)$ or $\Psi(1,2) = -\Psi(2,1)$. Particles obeying the first are called *bosons*, while those obeying the second are called *fermions*. What if we bring two fermions to same point in space? Then: $\Psi(1,1) = -\Psi(1,1)$. This means that $\Psi(1,1) = 0$! In other words, two identical fermions will never be at the same point or in the same quantum state. This is the famous Pauli Exclusion Principle.

20. Let us apply the Pauli Exclusion Principle to the multi-electron atom where each electron has the quantum numbers $\{n, l, m_l, m_s\}$. Only one electron at a time may have a particular set of quantum numbers. Now for some definitions:
    - Shell - electrons with the same value of n
    - Subshell - electrons with the same values of n and l
    - Orbital - electrons with the same values of n, l, and m_l

    Once a particular state is occupied, other electrons are excluded from that state. The electron configuration is how the electrons are distributed among the various atomic orbitals in an atom. A common notation is of the type here:



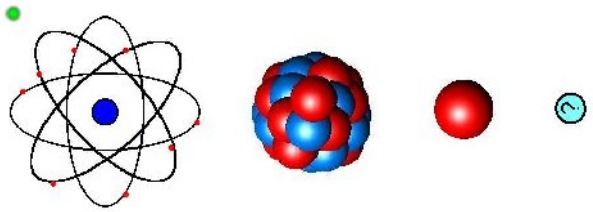
21. Building the shell structure of multi-electron atoms through $n = 4$ using the Pauli Principle.

n	Possible Values of l	Subshell Designation	Possible Values of m_l	Number of Orbitals in Subshell	Total Number of Orbitals in Shell
1	0	1*s*	0	1	1
2	0	2*s*	0	1	
	1	2*p*	1, 0, −1	3	4
3	0	3*s*	0	1	
	1	3*p*	1, 0, −1	3	
	2	3*d*	2, 1, 0, −1, −2	5	9
4	0	4*s*	0	1	
	1	4*p*	1, 0, −1	3	
	2	4*d*	2, 1, 0, −1, −2	5	
	3	4*f*	3, 2, 1, 0, −1, −2, −3	7	16

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Summary of Lecture 44 – INTRODUCTION TO NUCLEAR PHYSICS

Q.1 In the previous lecture you learned how it was discovered that the atom is mostly empty space with a cloud of electrons. At the centre is a small but very heavy nucleus that has protons and neutrons. The word "nucleon" refers to both of these. So you can think of



the neutron or proton as being two different varieties of the nucleon. The masses of the two are very similar, and they are roughly 2000 times heavier than the electron.

$$\text{proton mass} = M_p = 1.672 \times 10^{-27}\, kg$$

$$\text{neutron mass} = M_n = 1.675 \times 10^{-27}\, kg$$

$$\text{electron mass} = M_e = 9.109 \times 10^{-31}\, kg$$

The neutron is neutral, of course, but the charge on the proton is $1.6 \times 10^{-19} C$ while the charge on the electron is the negative of this, $-1.6 \times 10^{-19} C$.

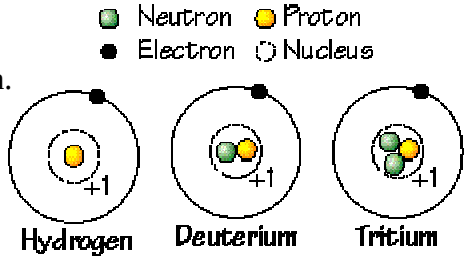
2. Using kilograms is very awkward if you are dealing with such small particles. Instead we we use $E = mc^2$ to write the mass of a particle in terms of its rest energy, $m = E/c^2$. So mass is measured in units of MeV/c^2.

$$\text{proton mass} = M_p = 938\ MeV/c^2$$

$$\text{neutron mass} = M_n = 940\ MeV/c^2$$

$$\text{electron mass} = M_e = 0.5\ MeV/c^2$$

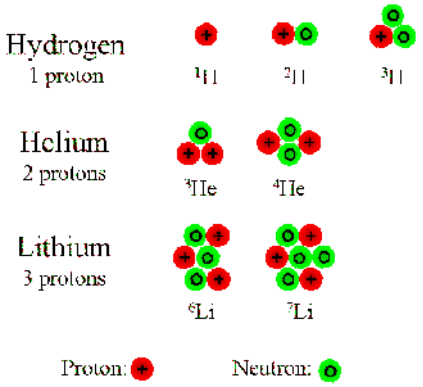
3. a) A hydrogen nucleus is just one proton.
   b) A deuteron has a proton plus one neutron.
   c) A triton (or tritium nucleus) has a proton plus two neutrons.
   All three atoms have one electron only, and thus completely identical chemical properties.



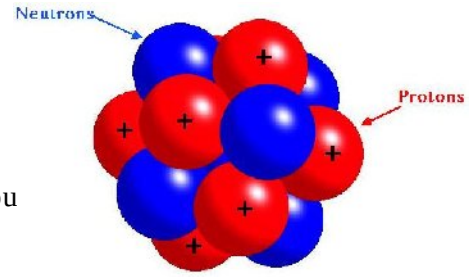
4. Hydrogen, deuterium, and tritium are called isotopes. If a nucleus with Z protons has N neutrons then its isotopes will have fewer, or more, neutrons. Since the number of electrons is also Z, the chemical properties of all isotopes are exactly the same. But for any given element, at most there is only one stable isotope.

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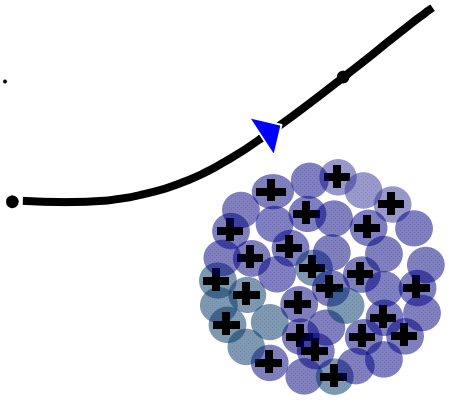
4. A commonly used notation is ${}^{A}_{Z}X$ where X is the element, and $A = Z + N$. Of the elements that you see on the right, the most stable ones are ^{1}H, ^{3}He, and ^{6}Li. Now consider oxygen. The most stable isotope is ^{16}O. When you breathe in oxygen from the atmosphere, 99.8% is ^{16}O, 0.037% is ^{17}O, and 0.163% is ^{18}O. We shall see later what unstable means and how nuclei decay.



5. The diameter of the nucleus is about 10 million times smaller that the overall diameter of the atom. Nuclei follow an approximate rule for the radius, $r \approx r_0 A^{1/3}$ where $r_0 = 1.2\ fm$ (remember, 1 fermi $= 10^{-13}$ cm) and $A = Z + N$. Now, $A^{1/3}$ increases very slowly with A. You can check that $16^{1/3} = 2.52$ while $208^{1/3} = 5.93$. This means that a very heavy lead nucleus ${}^{208}Pb$ is only about 2.4 times the size of the much lighter ${}^{16}O$ nucleus.



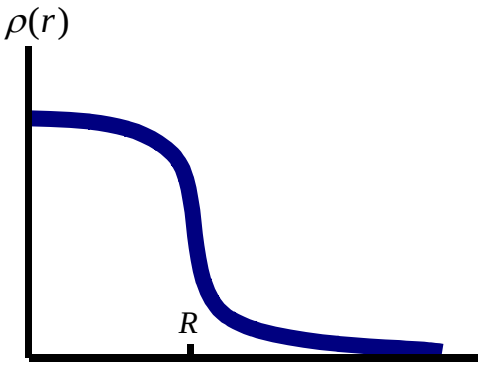
6. To learn about how protons are distributed inside a nucleus, we send a beam of electrons at a nucleus and observe how they scatter in different directions. The negatively charged electrons interact with the positively charged protons, but they obviously will not see the neutrons. The scattered electrons are captured in a detector which can be moved around to different angles. In this way one can reconstruct the charge distribution which caused the electrons to be scattered in that particular way.



7. What energy should electrons have in order to see a nucleus? We know that electrons are waves with $\lambda = h/p$ (the De Broglie relation). To see something as small as $1\,fm$ requires a wave with wavelength at least $\lambda \approx 1\,fm$. A wave with longer wavelength would simply pass over the nucleus without being disturbed. So the minimum electron energy is, $E = \dfrac{p^2}{2m} = \dfrac{h^2}{2m\lambda^2}$. Evaluation gives this to be a few MeV, requiring an electron accelerator of more than this minimum energy.

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8. From electron scattering we see that the proton density is almost constant throughout the nucleus and falls sharply at the surface. So nuclei should be thought of as rather fuzzy balls. A typical plot of density versus distance from the centre looks like this. Here the distance $R \approx r_0 A^{1/3}$ can be called the nuclear radius. The distribution of neutrons is very similar to this plot, but requires other techniques.



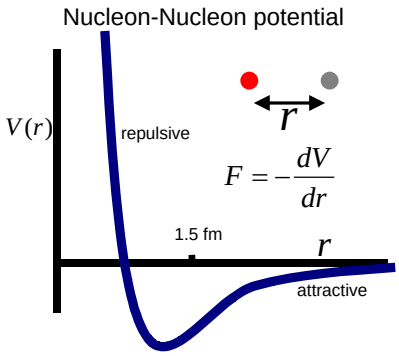
9. Let us consider the implication of the approximate formula for the nuclear radius, $r \approx r_0 A^{1/3}$ where $r_0 = 1.2\ fm$. The volume of the nucleus is: $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi r_0{}^3 A$. From this $\frac{A}{V} = \frac{3}{4\pi r_0{}^3} \approx 0.14$ nucleons/fm^3. This is the number of nucleons per cubic fermi, and is independent of the nucleus considered. This is the density you would find at the centre of any nucleus. Of course, this is approximately true only but it is quite remarkable.

10. Protons repel protons through the electrostatic force. So why does the nucleus not blow apart. Obviously there must be some attractive force that is stronger than this repulsion. It is, in fact, called the strong force. From what we have learned so far, we can guess some of its important features:
    a) Since neutron and proton distributions are almost the same, the N-P force cannot be very different from the N-N or P-P force.
    b) Since the density of nucleons in large nuclei is the same as in lighter nuclei, this means that a given nucleon feels only the force due to its immediate neighbours, and does not interact much with nucleons on the other side of the nucleus. In other words, the range of the nucleon-nucleon force is very short and of the order of 1-2 fm only.

11. The force between two charges is always of one sign - repulsive if the signs are the same, and attractive if they are opposite. In the early years of quantum theory, people realized that this force comes about because of the exchange of photons between charges. The nucleon-nucleon force is different. It has to be attractive to keep the nucleus together, and has to short range (as discussed above). But, to prevent nucleons from sticking to each other, it must be repulsive at short distances. Now here, "short" and long means distances on the scale of fermis. Typically the distances between nucleons is on this scale as well.

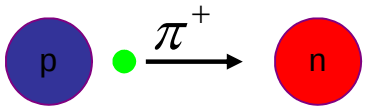
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Here is roughly what the N-N potential looks like. Since the force is the negative of the slope, you can see that for large value of r, the force is attractive (negative means directed towards smaller values of r). At roughly 1.4 *fm* the potential reaches its most attractive point. For values of r smaller than this, the force is repulsive. In fact there is a very strong core that almost completely forbids the nucleons from getting closer than about 0.5 *fm*.

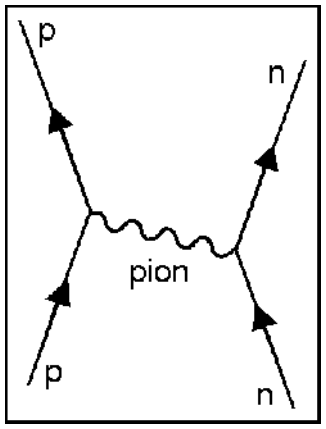


12. In 1935, the Japanese physicist Hideki Yukawa made an astonishing breakthrough in understanding the basis for the attractive N-N force. He assumed that, just as between charges, there must be a particle that is emitted by one nucleon and then captured by the other. He called this a "pi-meson" or "pion", and made a good guess for what its mass should be. His argument uses the time-energy uncertainty principle discussed earlier:

- Let Δt be the time that the pion takes between emission and capture. In this time it will have travelled a distance approximately equal to $c\Delta t$ because light particles can travel no faster than light.
- Creating a pion from "nothing" means that energy conservation has been violated. The amount of violation is ΔE = minimum energy of pion $\approx mc^2$ and this must obey $\Delta E\,\Delta t \approx \hbar/2$. Hence, $\Delta t \approx \dfrac{\hbar}{2mc^2}$ and the distance travelled by the pion $\approx$ c $\Delta t \approx \dfrac{\hbar}{2mc}$.



- Put $\dfrac{\hbar}{2mc} \approx 1.2\ fm$ (range of nuclear force). This gives the mass of the pion as close to $mc^2 \approx 124\ MeV$. This was an amazing prediction - the first time a particle had been predicted to exist on the basis of a theoretical argument. When experimentalists searched for it in 1947, they indeed found a particle of mass rather close to it, with $mc^2 \approx 138\ MeV$. It was a very dramatic confirmation for which Yukawa got the Nobel prize.



- Pions can rightfully be called the carriers of the strong nuclear force. They have 3 possible charge states: π^+, π^0, π^-. They belong to a larger family of particles called mesons. Other family members are rho-mesons, omega-mesons, K-mesons,... Today we can produce mesons in huge amounts by smashing nucleons against each other in an accelerator.

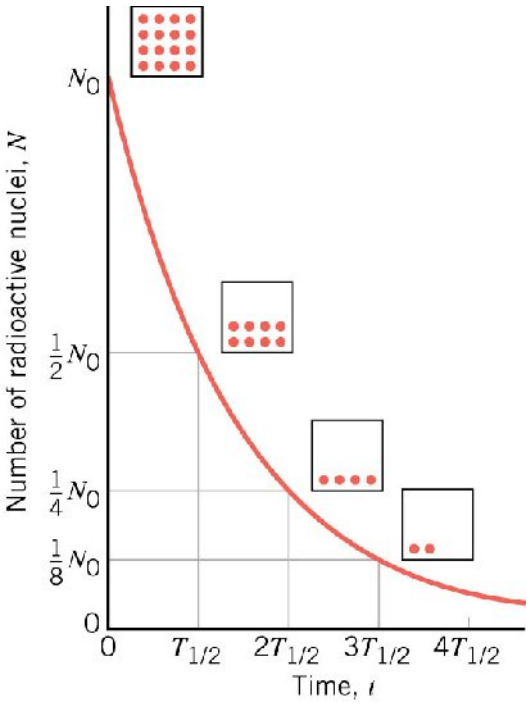
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13. A few nuclei are stable, most decay. The decay law is simply derived: if the number of nuclei decreases by dn in time dt, then dn must be proportional to both the the number of nuclei n that are decaying and dt, so $dn \propto -ndt$ (minus sign for decrease). with the proportionality constant λ, we have $dn = -\lambda n dt$, or $\frac{dn}{dt} = -\lambda n$. We have encountered the solution of this type of equation before, $n = n_0 e^{-\lambda t}$. You can see that at $t = 0$, $n = n_0$. Taking the log, we have $\ln \frac{n}{n_0} = -\lambda t$. We define the half-life $T_{\frac{1}{2}}$ as the time it takes for half the original sample to decay. If $n = n_0 / 2$ then $\log \frac{n_0}{2n_0} = -\lambda t$, from which the half life is related to λ by, $T_{\frac{1}{2}} = \frac{\log 2}{\lambda} = \frac{0.693}{\lambda}$. The larger λ, the more radioactively unstable a nucleus is. Some typical half-lives are:

Polonium	$^{214}_{84}P$	$1.64 \times 10^{-4} s$
Krypton	$^{89}_{36}$K	3.16 minutes
Strontium	$^{90}_{38}$Sr	28.5 years
Radium	$^{226}_{88}$Ra	1600 years
Carbon	$^{14}_{6}$C	5730 years
Uranium	$^{238}_{92}$U	4.5×109 years

You can see how hugely different the lifetimes of different nuclei are!

14. Here is a plot of the number of unstable nuclei left as a function of time. After each half-life, the number of nuclei decreases in number by half of the previous. Eventually there is only one nucleus left, and that too will eventually decay. So how can the derivation for the decay law be correct? Strictly speaking, we are not allowed to write down, or solve, a differential equation like $\frac{dn}{dt} = -\lambda n$ because this assumes that $n(t)$ is a continuous function. But this is almost true because in real life we deal with very large numbers of nuclei and so it makes a lot of sense to think of $n(t)$ as continuous.



15. Just to get an idea, consider the decay of $^{222}_{86}\text{Rn}$ (Radon, a very dangerous gas that is found underground) into $^{218}_{84}\text{Po}$ (Polonium, another terrible poison) and $^{4}_{2}\text{He}$ (harmless, fortunately!). The half life is 3.8 days. So, if we started with 20,000 atoms of $^{222}_{86}\text{Rn}$, then in 3.8 days we would have 10,000 atoms of $^{222}_{86}\text{Rn}$ and 10,000 atoms of $^{218}_{84}\text{Po}$ In 7.6 days we would have 5000 atoms of $^{222}_{86}\text{Rn}$, in 11.4 days, 2500 , $^{222}_{86}\text{Rn}$ etc.

16. The decay law can be used to see how old things are. This is called radioactive dating. Carbon dating is widely used for living things that died a few hundred or few thousand years ago. How does it work?This uses the decay of the unstable isotope, $^{14}_{6}C$. Of course, the stable isotope of carbon is $^{12}_{6}C$.
    - When a living organism dies, CO_2 is no longer absorbed. Thus the ratio of carbon 14:12 decreases by half every 5730 years. We can measure the rate of decrease through $N = N_o e^{-\lambda t}$ or the "activity" $A = A_o\ e^{-\lambda t}$ with $A_o = 0.23\ Bq/g$. (The becquerel Bq is the unit of radioactivity, defined as the activity of a quantity of radioactive material in which one nucleus decays per second.)
    - The amount of isotopes in the atmosphere is approximately constant, despite a half-life of 5730 y because there is a constant replenishment of $^{14}_{6}C$ through the reaction, $^{14}_{7}N + n \rightarrow {}^{14}_{6}C + p$

17. Let us use the above idea to find the time when this man died. His body was found a few years ago buried under deep snow in a mountain pass, so it it did not decay as usual. By looking at the radioactivity in his body, it was found that that the activity of $^{14}_{6}C$ was $0.121\ Bq/g$ of body tissue. This is less than the normal activity $0.23\ Bq/g$. because $^{14}_{6}C$ has been decaying away. First find λ, $\lambda = \dfrac{0.693}{T_{1/2}} = \dfrac{0.693}{5730} = 1.21\times10^{-4}\ \ y^{-1}$



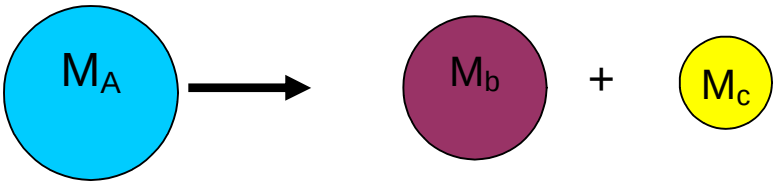
Then use, $0.121 = 0.23\ e^{-1.21\times10^{-4}\times t}$ which gives,

$\ln\dfrac{0.121}{0.23} = -1.21\times10^{-4}\times t$ and so $t = 5300$ years is when this poor man was killed (or died somehow)!

18. The most famous formula of physics, $E = mc^2$, is the basis for nuclear energy. In 1935, it was discovered by two German physicists, Otto Robert Frisch and Lise Meitner, that a heavy nucleus can fission (or break up) into two or more smaller nuclei. The total energy is, of course, conserved but the mass is not. This is completely different from the

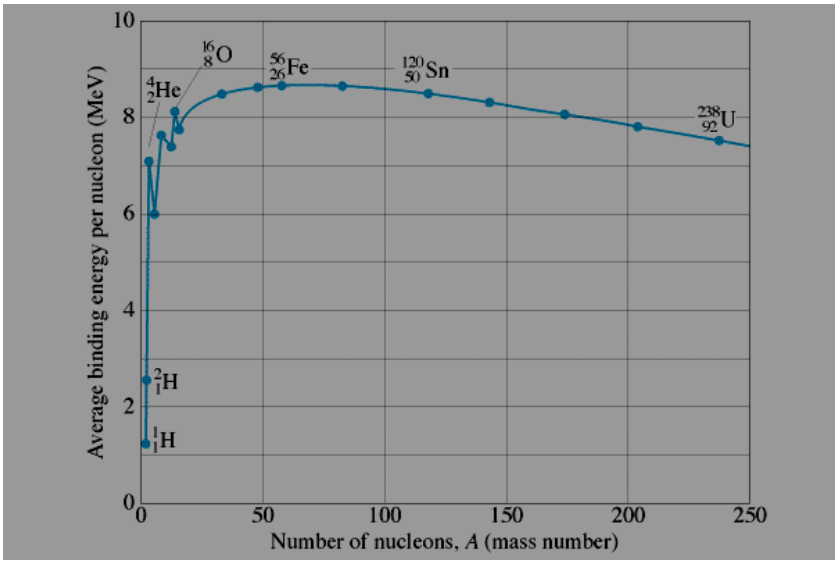
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usual situation. In the picture below you see an example of fission.



The masses of the two nuclei add up to less than the mass of the parent nucleus, and the energy released is $Q = (M_A - M_b - M_c)c^2$. This goes into kinetic energy and sends the two daughter nuclei flying apart at a large velocity. There happen to be NO completely stable nuclei above $Z = 82,$ and no naturally occurring nuclei above $Z = 92$. Above these limits the nuclei decay or fall apart in some fashion to get below these limits.

19. A very useful concept is "binding energy". Suppose you want to take a nucleon out of a nucleus. The binding energy is the amount of energy that you would have to provide to pull it on the average. Nuclei with the largest BE per nucleon are the most stable. As you

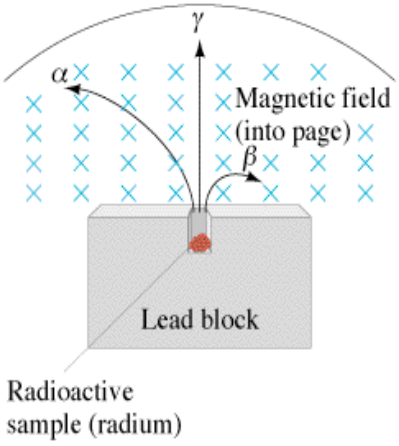


can see from the graph below, the most stable element is iron, ^{56}Fe with a BE per nucleon of about 8.6 MeV. This is why iron is the heavy element found in the largest quantity on earth and inside stars. The curve is not smmoth and you see that a ^{4}He nucleus (i.e. an α parrticle, has a relatively high binding energy and so is relatively stable. In contrast, Uranium ^{235}U or deuterium ^{2}H are much less bound and they decay.

20. Nuclei can be unstable in different ways. A nucleus can emit $\alpha, \beta,$ and γ radiations. Usually a nucleus will emit one of these three, but it is possible to emit two, or even all three of these. (In addition, as we have discussed above, a nucleus can break up into

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two or more smaller fragments through the process of fission.) What is the nature of α, β, γ radiations? In the experiment shown here, you see that a piece of radium has been put in a lead block (as shielding). A magnetic field bends the charged particles emitted during the decay. We find that they are of three types:



a)Heavy positively charged particles are bent to one side by the magnetic field. The amount of bending shows that they are α particles.

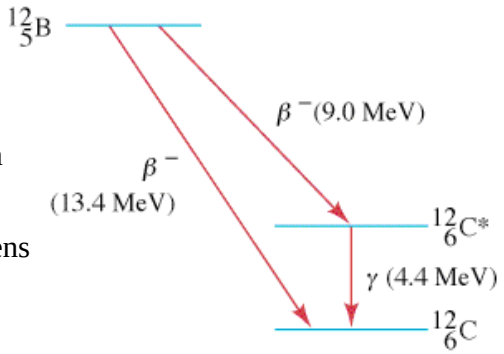
b)Other particles are bent much more, and in the other direction. They are electrons (β particles).

c)Some particles are not bent at all, hence must be neutral. They are γ particles (or rays, or γ photons, same thing!).

21. Let's first consider alpha decay, ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}X' + {}^{4}_{2}He$. As you see A changes by 4 and Z by 2. You can think of α particles as a gang of 4 particles that always stays together in a big nucleus. Unstable nuclei simply cannot overcome the proton repulsion and an α particle ultimately succeeds in esaping the nucleus. indefinitely.

22. The simplest beta decay reaction is when a neutron decays, $n \rightarrow p + e^{-} + \bar{\nu}$. Other than a proton and electron, an anti-neutrino is also emitted. As discussed in the lecture, the (anti) neutrino is a neutral particle with a very tiny mass that interacts very weakly with matter. A nucleus can undergo beta decay with either an electron being produced, i.e ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}X' + e^{-} + \bar{\nu}$ (called β^{-} decay) or with an anti-electron (positron, or positive electron), ${}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}X' + e^{+} + \nu$, (called β^{+} decay). Beta decay involves the weak nuclear force. This is one of the four fundamental forces in the world, and its small strength means that the decay happens much more slowly than most other reactions.

23. Just as for an atom, a nucleus can only exist in certain definite energy states. When a nucleus goes from one state to the other, it can emit a photon (γ ray). Because the spacing between nuclear levels is of the order of one MeV (i.e. a million times more than in atoms), the photon is much more energetic than an optical photon. In going between levels, β emission also happens as can be seen in this diagram.



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Summary of Lecture 45 – PHYSICS OF THE SUN

1. In this lecture I shall pull together the different parts of physics that you have learned in this course and apply them to understanding the source of all life on earth - our sun. We will learn that the sun operates according to principles that we can understand, and on the basis of this we can even predict the manner in which it will eventually die.

2. Basic solar facts:



a) Mass of sun $= 2\times10^{30}$ kg = 333,000 Earth's

b) Diameter of sun $=$ 1,392,000 km = 10^9 Earth's

c) Age of sun = 4.6 billion years

d) Rotation Period = 25 days at equator, 36 at poles (surface)

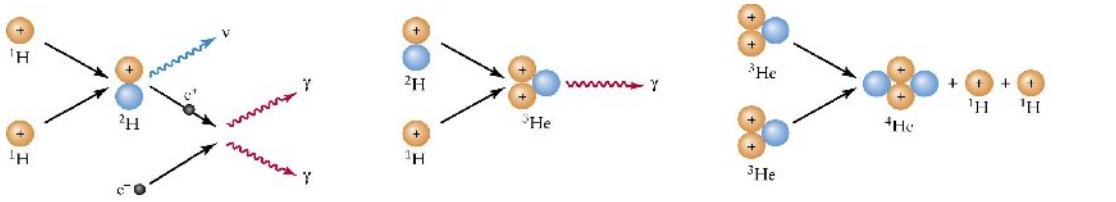
e) Temperature = 15 million ^{0}K at core, 5770 ^{0}K at surface

f) Density = 8 × gold at the core, average is ~ 1.5 water

g) Composition: 72% H, 25% He, rest is metals

3. The sun puts out a huge amount of energy. In quantitative terms we measure this in by its luminosity, 3.83×10^{27} joules per second. The power output is 3.83×10^{24} kilowatts. This is equal to 8×10^{16} of the largest power plants on Earth, meaning those which produce ~5000 MW of power. Another way of expressing this: every second the sun puts out as much energy as 2.5×10^9 (2 billion) such power plants would put out every year.

4. What powers the sun? The earth is very old (billions of years). If there was a chemical fuel (say, coal or oil) at most that would last a few million years. But the sun is many thousands of tons older than that. Only after the discovery of $E = mc^2$ did we know the real secret. The sun gets its energy from the fusion (the coming together and combining of atomic nuclei. For this extremely high temperatures, density, and pressure is needed.



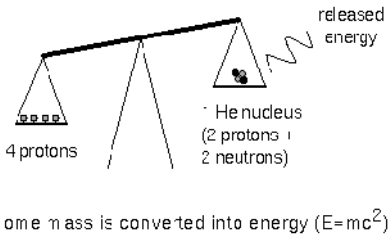
a Step 1:
- Two protons (hydrogen nuclei, ^{1}H) collide.
- One of the protons changes into a neutron (shown in blue), a neutral, nearly massless neutrino (ν), and a positron (e^+), a positively charged electron.
- The proton and neutron form a hydrogen isotope (^{2}H).
- The positron encounters an ordinary electron (e^-), annihilating both particles and converting them into gamma-ray photons (γ).

b Step 2:
- The ^{2}H nucleus from the first step collides with a third proton.
- A helium isotope (^{3}He) is formed and another gamma-ray photon is released.

c Step 3:
- Two ^{3}He nuclei collide.
- A different helium isotope with two protons and two neutrons (^{4}He) is formed and two protons are released.

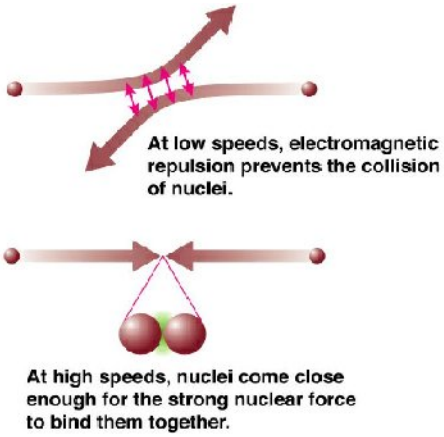
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4. The basic point is that, as you can see in the picture, the combined mass of 4 protons is higher than than that of the helium nuclei into which they convert through the process shown earlier. The difference then appears in the form of kinetic energy of the released particles, which in random form is heat.

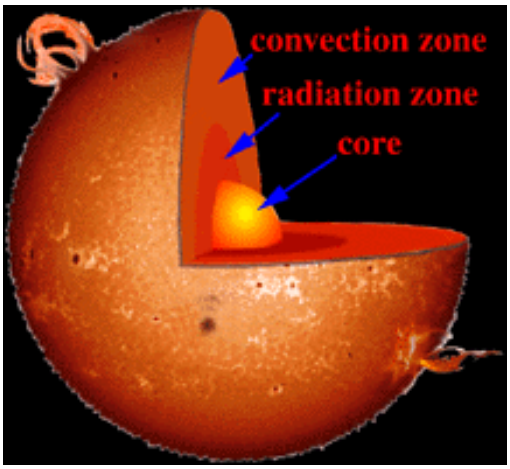


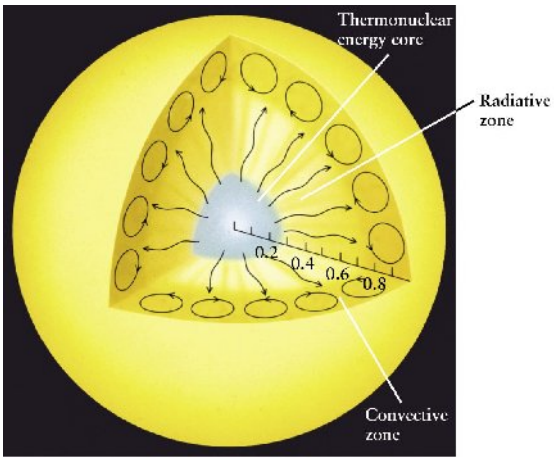
Some mass is converted into energy ($E=mc^2$)

5. Protons repel protons, but the only way in which they will participate in a fusion reaction is when they can come sufficiently close. This requires that they smash into each other at sufficiently high speeds, and hence nuclear fusion in the sun requires core temperatures greater than about 8 million ^{0}K. He nuclei can also fuse with each other to release energy, but because they are heavier much higher temperatures & densities (about 100 million ^{0}K for helium fusion) are required. Thus, stars fuse hydrogen first. Each second, the sun turns 4 million tons of hydrogen into energy.



6. Fusion takes place only in the core of the sun (see diagrams below) because it is only there that the hydrogen gas is hot enough. From the core, the heat gets out by the emission of photons (radiation zone). The hot gas then exchanges heat with the sun's relatively cold exterior through convective currents. Huge columns of hot gas move from the inside towards the surface. After giving up most of the heat, the "cold" gas sinks towards the centre and the cycle goes on.

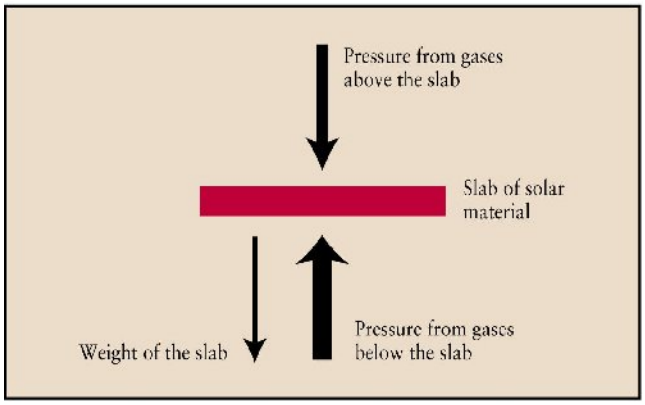


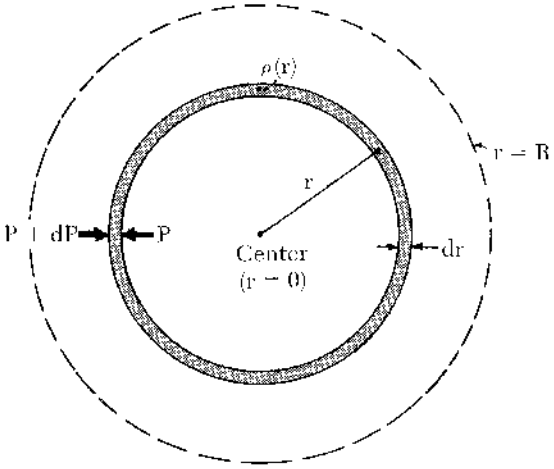


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7. For over 4 billion years the sun has been nearly steady in maintaining its size. It is in a state of equilibrium between two big forces acting oppositely to each other:
   a) The hot hydrogen gas seeks to expand outwards because the thermal velocity of H atoms leads to a pressure directed outwards.
   b) Gravity tries to squeeze the star inwards because every piece of matter attracts every other piece. So the gravitational pressure is inward, increasing toward the core.

   As in the diagrams below, imagine a piece of solar material at some distance from the centre of the sun. The two forces acting upon it must exactly balance in equilibrium. If for some reason the sun cools down, the pressure of gas will decrease and the sun will contract to a new equilibrium point. Conversely, the sun will expand if the rate of fusion were to increase.

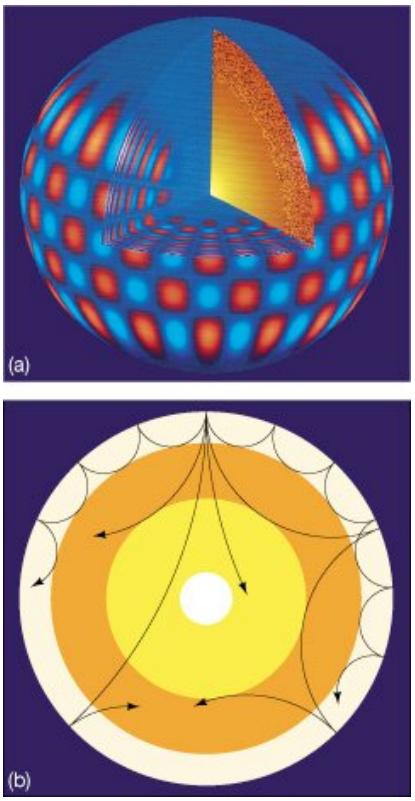




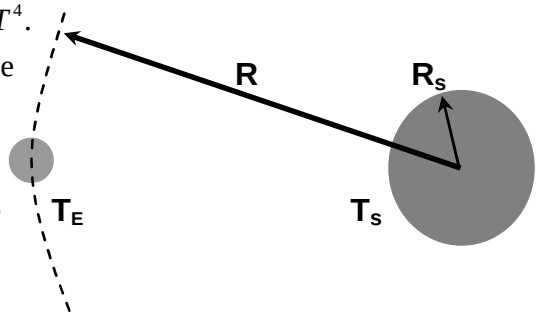
Look at the second diagram. Let $M(r)$ be the mass contained within radius r. We will not assume that the density is constant in r. First find the inward directed gravitational force on a shell of matter at radius r and thickness dr, $dF = -\dfrac{GM(r)\times\rho(r)4\pi r^2 dr}{r^2}$. There is a net pressure as shown, and we will call dP the difference of pressures. Then obviously $dF = dP\times 4\pi r^2$. Hence, $dP = -\dfrac{GM(r)\rho(r)dr}{r^2}$ or, $\dfrac{dP}{dr} = -\dfrac{GM(r)\rho(r)}{r^2}$. The total mass upto radius r is, $M(r) = \int_0^r \rho(r')4\pi r'^2 dr'$. If we knew $\rho(r)$ then $M(r)$ would also be known. By solving the differential equation, we would also then know $P(r)$. To give us a better understanding, suppose for simplicity that the sun is approximately uniform. Then the density is, $\rho \approx \dfrac{M}{(4\pi/3)R^3} \approx 1.4\text{gm cm}^{-3}$. Hence the pressure at the centre of the sun can be computed, $P_{centre} \approx \dfrac{GM\rho}{R} \approx 3\times 10^9$ atmospheres.

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8. The equilibrium of forces is a nearly perfect one, but there are small disturbances and these cause "earthquakes" (actually sun-quakes) to occur on the surface. The study of the surface of the sun and sun-quakes is an area known as "helioseismology" (helio=sun, seismology is the study of vibrations and earthquakes). The pulsating motions of the sun can be seen by measuring the Doppler shifts of hydrogen lines across the face of the sun. Some parts are expanding towards the earth while adjacent regions contract away. This is like the modes of a ringing bell. Vibrations propagate inside the sun, and the waves travel through a hot dense gas. They experience reflection and refraction since the speed at which a wave travels changes when it goes from a region of high to low density or vice-versa. All this is being studied by astrophysicists today as they map out the sun's interior.

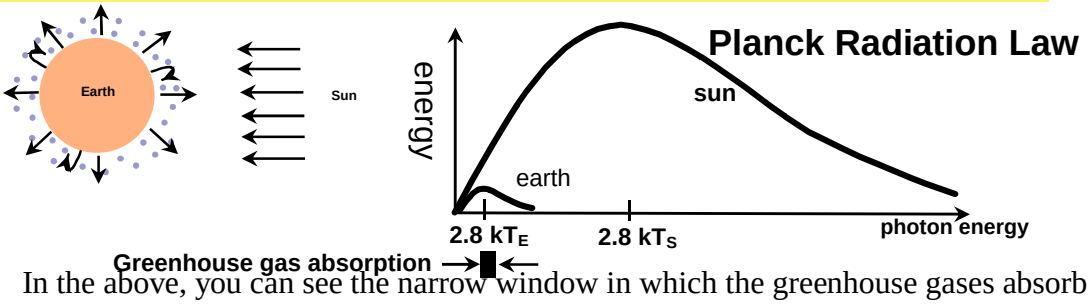


9. Let us calculate the surface temperature of a planet circulating the sun. We shall use the Stefan Boltzman law that you studied in the lecture of heat: the power radiated by a black body per unit surface area at temperature T is σT^4. For thermal equilibrium, we must have that all the power absorbed from the sun is re-radiated by the planet. Let P_S be the sun's flux at its surface, Then, $P_S = \sigma T_S^4$. Since radiation decreases by the distance squared, P_R = flux at earth $= \sigma T_S^4 \times \left(\frac{R_S}{R}\right)^2$ This must be multiplied by πR_E^2, which is the crossectional area of the planet. On the other hand, for emission from the planet, P_E = flux at its surface $= \sigma T_E^4$. We must multiply this by $4\pi R_E^2$, the area of planet, to get the total radiated power. Now impose the equilibrium condition: $P_R \times \pi R_E^2 = P_E \times 4\pi R_E^2$. This gives $\sigma T_S^4 \times \left(\frac{R_S}{R}\right)^2 = 4\sigma T_E^4$, and hence $T_E = \left(\frac{R_S}{2R}\right)^{1/2} T_S$. This is true for any planet, so let's see what this gives for the earth using $R_S = 7 \times 10^8$ metres, $R = 1.5 \times 10^{11}$ metres, and $T_S = 5800^o K$. This gives $T_E = 280K$, which is very sensible! Of course, we have assumed that the earth absorbs and emits as a black body. True?



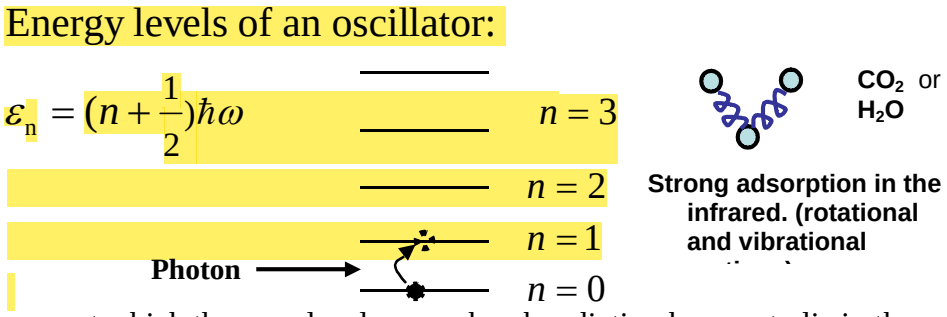
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10. Actually, the black body assumption is only approximately true. About 30% of the sun's radiation reflects off our atmosphere! In fact the average surface temperature of the earth is about 290K, or about 13 ^{0}C. The extra warming is mainly due to the "greenhouse effect". Certain gases trap the sun's radiation and it is not able to get out, thus leading to greater absorption. The main greenhouse gases are CO_2, CH_4, H_2O. (N_2, O_2 and Ar are transparent to solar and earth radiation.) CO_2 is now 360 ppm (parts per million) of the atmosphere, up from 227 ppm in 1750, before the industrial revolution. Plants turn H_2O + CO_2 into O_2 plus organics. Some estimates predict a 3 to 10 ^{0}C rise in the earth's surface temperature over the next 100 years due to the increased CO_2 greenhouse effect.



In the above, you can see the narrow window in which the greenhouse gases absorb the sun's radiation. But why are only some gases responsible, and not others?

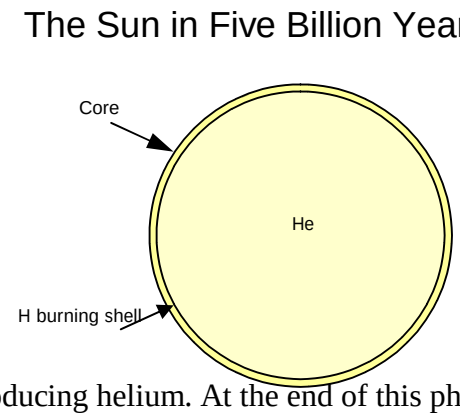
11. The answer lies in quantum mechanics. In a previous lecture you learned that molecules can exist only in certain states that have very specific values of energy. For example, a molecule of CO_2 can oscillate and have equally spaced energy levels as shown below:



The energy at which these molecules can absorb radiation happen to lie in the spectrum of the re-radiated energy from the earth's surface. Thus, they effectively trap the out-going radiation, leading to enhanced earth temperatures. For any molecule, we can both calculate (using quantum mechanics) the frequencies at which radiation is absorbed or emitted. We can also experimentally measure the frequencies. Both of these are in very good agreement, and this is one of the reasons why we have such confidence in the correctness of quantum mechanics.

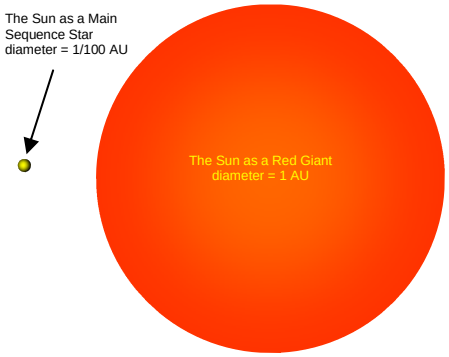
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12. Finally, I have some bad news: the sun is going to die because the hydrogen supply is eventually going to run out. There is, of course, no immediate danger - the Sun can last another 5 billion years on core hydrogen fusion. During this time it will be consuming

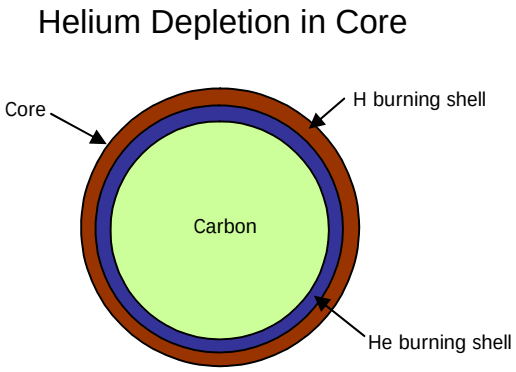


hydrogen and producing helium. At the end of this phase, the core of the sun will have mostly helium with the little bit of hydrogen left almost entirely outside in a thin layer. The reaction rate will fall, and the core will no longer be able to balance the pull due to gravity. This will cause a shrinkage of the core. As a consequence the temperature will increase. A new phase of the sun is about to start.

13. At this point a fusion reaction in next core zone begins. This lifts the envelopes and the sun will brighten up significantly. It is now a Red Giant star, and as you can see below, it is huge! The earth will be swallowed up by the expanded sun.



14. As remarked earlier, helium can also "burn" to produce carbon, but it needs a much higher temperature. Eventually even the He will be depleted, and the reaction rate will fall. Again the small reaction rate means that the core will shrink, and heat up to the point that it crosses the carbon fusion threshold of 600 million K. Low mass stars cannot reach this temperature. Envelope expands to a supergiant, many times larger than even a Red Giant.



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